

MINDCRAFT

Research Constitution

How do humans become confident mathematical thinkers?

Identity transformation · Evidence · Red Team · Product systems · Experiments

This is not a pitch deck. This is the company's research operating system.
Challenge assumptions. Label uncertainty. Kill weak arguments.
Source corpus: ~87,999 words · multi-chapter living lab

MindCraft Research Lab · ongoing evidence program

MindCraft Research Constitution v1

Status: Living operating document — not a pitch deck

Edition: v1.8 (Synthesizer consolidation of Parts L–LVIII into surviving doctrine)

Research question: How do humans become *confident mathematical thinkers*?

Product thesis under audit: The product is identity transformation, not mathematics delivery.

Last updated: 2026-08-02 (Researcher — Part LXII Longitudinal identity change / SAFE-LONGID)

Growth model: Core OS (this file) + chapters/*.md via CHAPTER_MANIFEST.txt → PDF

Scale intent: Multi-month densification toward 150–300 pages of *evidenced* material — never fluff

Epistemic rule: Every claim is labeled FACT / HYPOTHESIS / FOUNDER BELIEF / SPECULATION.

Deep-dive chapters currently mounted

Part	File	Focus
XXIV	chapters/24_affect_anxiety_belonging.md	Anxiety, WM, stereotype threat, belonging
XXV	chapters/25_self_efficacy_narrative_identity.md	Bandura sources, narrative identity, habit≠identity
XXVI	chapters/26_cognitive_load_mastery_tutoring.md	CLT, fading, mastery war, tutoring calibration
XXVII	chapters/27_markets_parents_competition.md	WTP, parents, competitive teardown
XXVIII	chapters/28_history_meaning_math.md	Invention-story thesis under trial
XXIX	chapters/29_spacing_retrieval.md	Spacing, retrieval, memory of competence
XXX	chapters/30_attribution_helplessness.md	Weiner attributions, helplessness, feedback language
XXXI	chapters/31_flow_challenge_skill.md	Flow, challenge–skill, difficulty design
XXXII	chapters/32_parent_anxiety_transmission.md	Maloney/Beilock parent anxiety pathway
XXXIII	chapters/33_ai_tutors_trust_sycophancy.md	AI trust / sycophancy; RT: Bastani■PWC, guards≠gain
XXXIV	chapters/34_formal_causal_dag_identification.md	FEI causal DAG; L0–L4 claim ladder; confounders
XXXV	chapters/35_competitive_session_audits.md	Khan/Duo/Brilliant/ChatGPT session mechanism audits
XXXVI	chapters/36_equity_audit_story_worlds.md	Equity audit of story worlds; whose history; tokenism kill

XXXVII	chapters/37_expectancy_value_eccles.md	Eccles SEVT; utility/cost/attainment → choice; EVT experiments
XXXVIII	chapters/38_goal_orientation.md	Mastery vs performance goal structures; appearance ban; GO experiments
XXXIX	chapters/39_interleaving_vs_blocking.md	Interleave vs block; strategy selection; IL experiments
XL	chapters/40_self_explanation_prompts.md	Self-explanation prompts; Chi/Renkl → coach UX; SE experiments
XLI	chapters/41_desirable_difficulties_x_anxiety.md	Desirable difficulties × anxiety; SAFE-DD; Bjork vs Ashcraft
XLII	chapters/42_social_comparison_leaderboards.md	Social comparison & leaderboards; SAFE-COMPARE; ranks help/hurt
XLIII	chapters/43_habit_formation_science.md	Habit formation (Wood/Clear); SAFE-HABIT; cue≠colonization
XLIV	chapters/44_intrinsic_motivation_killers.md	Intrinsic motivation killers (Deci); SAFE-REWARD; controlling≠informational
XLV	chapters/45_mathematical_resilience.md	Mathematical resilience (JW/Lee); SAFE-RESILIENCE; grit theater kill
XLVI	chapters/46_teacher_tutor_expectancy.md	Tutor expectancy (Pygmalion/Golem); SAFE-EXPECTANCY; trait-label ban
XLVII	chapters/47_sleep_stress_learning.md	Sleep/stress/learning; SAFE-TIMING; all-nighter & sleep-app kill
XLVIII	chapters/48_transfer_variation_theory.md	Transfer & variation theory; SAFE-TRANSFER; story≠far-transfer kill
XLIX	chapters/49_misconceptions_as_productive.md	Misconceptions as productive; SAFE-MISCON; soft-wrong / diagnostic wrongs
L	chapters/50_confidence_calibration.md	Confidence calibration; SAFE-CALIB; over/underconfidence; C4 hide-correctness

LI	chapters/51_deliberate_practice_tutoring.md	Deliberate practice in tutoring; SAFE-DP; Ericsson vs Hambrick; 45-min spine
LII	chapters/52_community_of_practice.md	Community of practice; SAFE-CoP; Lave/Wenger LPP; careful school transfer
LIII	chapters/53_religion_ritual_light_touch.md	Religion/ritual light-touch; SAFE-RITUAL; meaning without cult dynamics
LIV	chapters/54_military_aviation_brief_debrief.md	Military/aviation brief-debrief; SAFE-AAR; after-action reviews for sessions
LV	chapters/55_sports_film_study_pedagogy.md	Sports film-study pedagogy; SAFE-FILM; error clips → coach cards
LVI	chapters/56_music_pedagogy_ladders.md	Music pedagogy ladders; SAFE-MUSIC; scales → repertoire → recital
LVII	chapters/57_chess_annotation_metacognition.md	Chess annotation & metacognition; SAFE-ANNOT; postmortem UX
LVIII	chapters/58_therapy_graded_exposure.md	Therapy graded exposure; SAFE-EXPOSE; math anxiety ladders
LIX	chapters/59_parent_wtp_experiments.md	Parent WTP experiments; SAFE-WTP; CBC/conjoint + message tests
LX	chapters/60_trust_after_ai_hallucination.md	Trust after AI hallucination; SAFE-REPAIR; escalate-to-human
LXI	chapters/61_measurement_of_math_identity.md	Measurement of math identity; SAFE-IDMEASURE; scales + demand effects
LXII	chapters/62_longitudinal_identity_change.md	Longitudinal identity change; SAFE-LONGID; 8/26/52 week horizons

Queued next: see NEXT_LAB.md (Part LXIII ACT/exam pressure, then QUEUE_EXTENDED).

Synthesizer note (v1.8): Nine researcher chapters (L–LVIII) landed after v1.7 without a merge pass — calibration, deliberate-practice spine, CoP caution, ritual light-touch, AAR, film-study, music ladder, annotation, and graded exposure lived only in deep dives. This edition promotes each stack to its own I.4 law, adds Red Team kills #15–#17, mounts experiment families CAL/DP/CoP/RIT/AAR/FILM/MUSIC/ANNOT/EXP-O in Part IX, and refreshes metrics/glossary. Deep-dive files remain authoritative for citations; the OS keeps only *surviving* product rules.

0. How to read this document

This Constitution is the research OS for MindCraft. It exists to prevent the company from building features that feel right and fail quietly.

Rules of engagement:

1. Founder beliefs enter as **FOUNDER BELIEF**, not as law.
2. The Red Team’s job is destruction. Unchallenged claims are invalid.
3. Page count is not quality. A shorter true section beats a longer vague one.
4. “AI will make explanations free” is a hypothesis about markets, not a moral slogan.
5. Product implications must map to experiments with falsifiers.

This v1 is dense and incomplete on purpose. Expanding toward 150–300 pages is allowed only when new evidence, frameworks, or contradicted priors justify the ink.

Part I — Executive Summary

I.1 The compressed answer

HYPOTHESIS (confidence: medium–high): Humans become confident mathematical thinkers when three conditions compound:

1. **Affective permission** — the nervous system stops treating math as social/physical threat.
2. **Competence evidence** — the learner accumulates undeniable, self-attributed successes at the right grain of difficulty — including *strategy selection* and *self-accounted why*, not only blocked accuracy.
3. **Identity re-storying** — the person revises the narrative “I am not a math person” into a workable self-story compatible with struggle — via expectancy $\times$ value $\times$ cost (Eccles), not posters.

Explanations alone rarely produce (1)–(3). Fluent AI explanations can *accelerate wrong confidence* (Bastani unguarded GPT; Part XXXIII). Explanations help (2) only *after* (1) and when the student still does the decisive generative work (Parts XXVI, XL).

I.2 Verdict on the scarcity thesis

FOUNDER BELIEF under audit: AI commoditizes knowledge; scarcity shifts to motivation, confidence, identity, trust, curiosity, persistence, emotional safety, meaning.

Red Team verdict (updated v1.8):

Claim	Status	Notes
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Explanations are getting cheaper/faster	FACT (directionally)	Generative AI + open content lowers marginal cost of explanation
“Tutoring is free”	FALSE as stated	Attention, accountability, diagnosis, calibrated trust remain scarce
Content is free	MOSTLY FACT for commodity content	Differentiation = curation, sequencing, trust, outcomes
Motivation/identity become scarce	HYPOTHESIS	Product differentiator; not proven sole moat — instrument $E \times V \times \text{Cost}$ separately (XXXVII)
Emotional safety is first-order	HYPOTHESIS with supportive evidence	Ashcraft WM tax; segment, don’t romanticize softness
Unguarded AI chat improves solo learning	KILLED	Bastani: concurrent $\uparrow$ then solo exam $\downarrow$; guards neutralize harm $\neq$ prove gain
Bastani “GPT Tutor” $\equiv$ MindCraft ontology wrap	KILLED	Category error (XXXIII RT)
Streaks / blocked fluency = mastery	KILLED as North Star	HID + XXXVIII/XXXIX; measure delayed mix + transfer
Harder-always / anxiety-blind “desirable difficulty”	KILLED	SAFE-DD (XLI); equip $\rightarrow$ destake $\rightarrow$ dose before Bjork dose
Ranks / XP / grit meters as identity engines	KILLED	SAFE-COMPARE / REWARD / RESILIENCE (XLII–XLV)
Story immersion $\equiv$ far transfer; mistake posters $\equiv$ learning	KILLED	SAFE-TRANSFER / MISCON (XLVIII–XLIX)
Raise confidence / Belief Score™ / CA $\rightarrow$ grades	KILLED	SAFE-CALIB (L); Foster: appropriate confidence, not inflate
Hours / forums / ritual $\rightarrow$ score / domain cosplay	KILLED	SAFE-DP / CoP / RITUAL (LI–LIII); borrow method not costume
Tip-flood / engine-dump / flooding $\equiv$ pedagogy	KILLED	SAFE-AAR / FILM / MUSIC / ANNOT / EXPOSE (LIV–LVIII)

Implication: Do not bet on “better explanations,” “AI tutor warmth,” streak theater, grit posters, therapy cosplay, or academy/bootcamp branding. Bet on **FEI + pedagogy wrap + SAFE-* stack**: fear $\rightarrow$ evidence $\rightarrow$ identity, with AI as guarded infrastructure and student generation as the spine.

I.3 What to optimize (North Star debate)

Metric	Predicts short-term engagement?	Predicts durable math identity?	Risk
Correct answers (esp. blocked)	Medium	Low–medium	Fluency illusion; gaming
Time spent / chat tokens	High	Low	Engagement / AI-monologue theater
Streaks / DAU	High	Uncertain	Habit ≠ identity (HID)
Confidence (self-report)	Medium	Medium	Cheap talk; overtrust after fluent AI
Challenge-seeking (challenge_accept + why)	Medium	High	Must code mastery vs appearance/normative motive (XXXVIII)
Voluntary return after failure (retry_120s)	Medium	High	Needs coach content discipline (SE prompts)
Delayed mixed / transfer_pass / solo_transfer_pass	Medium	High	Anti-fake-mastery; anti-AI-crutch
“I am a math person” + behavior	Medium	Highest target	Slow, noisy; L3+ claim only

Working North Star (HYPOTHESIS):

Challenge-seeking under safety with transfer — the student chooses a harder problem for a *mastery* reason, returns after a miss, can pick the strategy on a mixed delayed set, and attributes success to strategy — without needing the model to finish the thought.

I.4 Surviving doctrine stack (Synthesizer merge through LVIII)

Duplicate frameworks collapsed. Deep dives own citations; this table is **company law until killed**.

Layer	Surviving rule	Source parts	Commercial use
FEI / SAFE-CRAFT	Safety → attempt → informative miss → earned win → attribution → transfer	XVI, VIII	Session demo in marketing; not feature matrix
Claim ladder L0–L4	Do not sell identity (L3) from session A/Bs (L1)	XXXIV	Parent/copy hygiene; pre-reg estimands
PWC (pedagogy wrap constraint)	Guarded AI + ontology spine; success = solo_transfer_pass ≥ no-AI	XXXIII, XXXV	Ban “we’re Bastani’s Tutor”; ban unguarded Solver hero

E×V×Cost	Instrument expectancy, value, cost separately; student-authored utility > staff utility dump	XXXVII	Kill single “motivation” KPI
TARGET / mastery climate	Default-ban appearance UX + streak-as-North-Star; soft-wrong = mastery evaluation structure	XXXVIII	CTA copy audit; appearance climate kill
SAFE-DD	Equip → destake → dose → frame → measure; desirable difficulty only after WM/threat capacity	XLI	Ban hardness-as-virtue; gate IL/SE by anxiety segment
SAFE-COMPARE	Criterion feedback default; peer ranks opt-in; no forced infinite named leaderboards for Maya	XLII	Leaderboard / league kill as growth engine
SAFE-HABIT	Cue practice not the counter; miss ≠ identity failure; habit metrics subordinate to FEI	XLIII	Streak colonization kill; if-then onboarding OK
SAFE-REWARD	No expected tokens for mere engagement; informational feedback > inducement; transfer-gated unlocks	XLIV	XP-as-love / coin-for-open kill
SAFE-RESILIENCE	Growth-zone challenge + support recruitment; ban grit NS, danger-zone-as-DD, Resilience Score™	XLV	Grit-poster / character-meter kill
SAFE-EXPECTANCY	Task-only tutor briefs; ban trait/weakness labels; CIOF mediation; cold-start scripts	XLVI	Pygmalion marketing kill; tutor ethics
SAFE-TIMING	Encode → sleep → transfer claim; never reward midnight volume; stress/restriction-aware load; pressure ≠ DD	XLVII	All-nighter / sleep-app brand kill

SAFE-TRANSFER	Name the hop; VT vary-on-purpose; hug exam + bridge abstraction; story ≠ transfer; blocked ≠ ready	XLVIII	Narrative-theater / blocked-accuracy vanity kill
SAFE-MISCON	Elicit → classify → destake → springboard → route; soft-wrong ≠ empty kindness; confidence-tiered feedback	XLIX	Mistake-poster / red-X-as-rigor / equal-wrong graph kill
SAFE-CALIB	Item-level confidence → miss class → tiered feedback; raise <i>appropriate</i> confidence, not Belief Score™	L	Hide-correctness diagnostic; ban CA→grades marketing
SAFE-DP	Diagnose → isolate → goal → stretch → attempt → method feedback → revise → prove → stop	LI	Ban 10k-hours / minutes / coverage theater as DP
SAFE-CoP	Name the practice; peripheral = real work; old-timer required; trajectory > community tab	LII	Ban Discord/streak ≡ LPP; “join the practice” not “join the community”
SAFE-RITUAL	Light, optional, practice-pointing marks; doctrinal default; anti-totalism; no ritual→score	LIII	Ban streak-as-liturgy; family/order milieu copy
SAFE-AAR	Brief → attempt with traces → four-question student-first close → one improve-how	LIV	Ban bootcamp brand; +25% ACT from debrief metas
SAFE-FILM	Objective clip + ≤3 cues + learner-timed review + self∪model; next-attempt commitment	LV	Ban tip-flood film room; sports-academy cosplay
SAFE-MUSIC	Scales (goalful ingredients) → chosen repertoire → transfer-as-recital proof	LVI	Ban virtuoso cosplay; hours KPI; bare-pass theater

SAFE-ANNOT	Annotate-before-reveal ; principle tags; forced critical restudy; classify→drill; annotate wins	LVII	Ban GM cosplay; engi ne-dump≡coaching; rating NS
SAFE-EXPOSE	Situation hierarchy; destaked approach; expectancy violation; fade safety behaviors; multi-context	LVIII	Ban therapy branding; flooding; Calm Score™; one-session cure
Block → near-miss interleave → spaced mix	Blocking = acquisition scaffold; delayed mixed accuracy = readiness signal	XXIX, XXXIX	Ban “shuffle = science”; ban blocked-accuracy vanity
Student-generated why	Faded examples + structured principle/mi sconception prompts <i>before</i> AI wrap	XXVI, XL	Ban AI-monologue≡SE; ban explain-own-wron g-first default
Competitive wedge	Do not out-content Khan, out-streak Duo, out-delight Brilliant, or out-fluency ChatGPT	XXXV, XX	Sell recoverable struggle + competence evidence + solo transfer
Equity of worlds	Story wrap is identity technology; tokenism and stereotype-cueing copy are kills	XXXVI, XXVIII	HIST-EQ; belonging without “even you can”

Merged / demoted (do not treat as separate products): “AI tutor,” “mastery path,” “growth mindset,” “engagement,” “grit,” “habit streak,” “celebrate mistakes,” “community of practice,” “deliberate practice brand,” “film study,” “recital,” “grandmaster review,” and “exposure therapy” are *not* independent North Stars — they are subordinate UX under FEI + the rows above (borrow *method*, never *costume*).

Next research bottleneck: Part LXIII ACT/exam pressure special case. Experiment families CAL/DP/CoP/RIT/AAR/FILM/MUSIC/ANNOT/EXP-O/WTP/REPAIR/IDM/LONG gate confidence-tiered, session-architecture, participation, ritual, brief–debrief, review-media, ladder, postmortem, anxiety-approach, parent-pricing, post-hallucination repair, identity measurement, and longitudinal-horizon claims above L1. Researcher count since v1.8 synthesizer = 4.

Part II — Key Insights (challenged)

Insight 1 — Fear is a performance bottleneck, not a personality trait

FACT: Math anxiety is associated with reduced working-memory availability during math tasks; anxious learners underperform relative to ability (Ashcraft & Kirk line of work; broader anxiety–WM literature).

HYPOTHESIS: For a large subset of MindCraft’s Maya archetype, the binding constraint is not “missing explanation,” it is **threat appraisal** (“I will look stupid”).

Implication: Soft-wrong UX, hide-correctness diagnostics, wizard coaching, and narrative framing are not polish — they are load-bearing if the thesis holds.

Red Team: Many high performers succeed under stress. Anxiety interventions have heterogeneous effects. Do not medicalize ordinary difficulty.

Insight 2 — Growth mindset is real, small, and context-dependent

FACT: Yeager et al. (2019, *Nature*) — brief online growth-mindset intervention improved grades for lower-achieving students and increased advanced math enrollment in a nationally representative U.S. sample; effects stronger when peer norms supported challenge-seeking.

FACT / REVIEW: Domain-general mindset messaging shows mixed results; math-*specific* mindset interventions more often report positive outcomes (2023 systematic review in *Educational Research Review*).

FACT / CAUTION: “False growth mindset” (praise effort without teaching strategy / changing environment) is harmful (Dweck’s own caution; Kohn’s critique).

Implication: MindCraft must not ship empty “believe in yourself” copy. Mindset must be embodied in **task design, feedback, and tutor norms**.

Insight 3 — Autonomy, competence, relatedness are the psychological tripod

FACT: Self-Determination Theory (Ryan & Deci) — sustained intrinsic motivation requires autonomy, competence, relatedness.

FACT (meta): SDT-based education interventions show reliable gains for autonomy and competence support; relatedness effects are less consistent (Wang et al. 2024 meta-analysis).

Implication: Product loops must deliver:

- Autonomy: meaningful choice (path, story world, pace)
- Competence: mastery grain that produces earned wins
- Relatedness: tutor presence, parent signal, peer-safe identity (hardest; do not fake with empty social)

Insight 4 — Mastery learning works — until you measure the wrong thing

FACT: Kulik, Kulik & Bangert-Drowns (1990) meta-analysis — mastery programs show positive exam effects (overall ~0.5 SD in their synthesis), stronger for weaker students; can increase time; self-paced college mastery can reduce completion.

FACT / CONTRADICTION: Slavin’s “Mastery Learning Reconsidered” (1987) — little support for group-based mastery on *standardized* measures; experimenter-made tests show moderate effects; coverage vs mastery tradeoff is real.

Implication: MindCraft’s mastery graph is promising **if** mastery means transferable competence, not local quiz familiarity. Measure far transfer, not only same-item success.

Insight 5 — Habit products ≠ identity products

FACT / INDUSTRY: Duolingo’s public method emphasizes gamification for motivation and return (streaks, leagues, variable rewards) (Duolingo Method whitepaper, 2023). Streaks exploit loss aversion; they drive DAU.

HYPOTHESIS: Streaks can increase practice frequency without changing “I am a math person.” Habit without identity yields brittle engagement (stop streak → stop learning).

Implication: Use habit mechanics as **on-ramps**, not as the destination. Pair with identity-marking rituals (map fill, story progression, tutor recognition of growth).

Insight 6 — Narrative is not decoration; it is encoding + meaning

FOUNDER BELIEF (WORLD_VISION): Math stripped of invention-story becomes alienating machinery.

HYPOTHESIS: Historical/human invention frames increase curiosity and reduce “arbitrary rule” appraisals, improving persistence.

Evidence status: Strong tradition in math education (history of math pedagogy); fewer large RCTs than mindset literature. Treat as **design prior + research program**, not settled science.

Insight 7 — AI commoditizes *answers*, not *becoming*

HYPOTHESIS: As answer generation approaches free, willingness-to-pay migrates to:

- 1. Diagnosis of the *actual* break (gap map)
- 2. Emotional containment during struggle
- 3. Accountability relationships (tutor)
- 4. Identity-consistent practice worlds
- 5. Trusted outcome signaling to parents

SPECULATION: In 30 years, “show me the steps” is ambient. “Help me become the kind of person who stays with hard problems” remains a human+system problem.

Part III — Research Log (v1 seed)

Date	Finding	Type	Action
2026-08-01	Synthesizer v1.8: merge L–LVIII SAFE-* into I.4; kills #15–#17	Synthesis	Surviving commercial law; next = LIX parent WTP

2026-08-01	SAFE-EXPOSE: graded approach; kill therapy cosplay / flooding / Calm Score™	Evidence	EXP-O1...5; expectancy violation
2026-08-01	SAFE-ANNOT / MUSIC / FILM: postmortem + ladder + sparse clips	Evidence	ANNOT/MUSIC/FIL M-1...5
2026-07-31	Synthesizer v1.7: merge XLI–XLIX SAFE-* into I.4; kills #10–#14	Synthesis	Surviving commercial law; next = L confidence calibration
2026-07-31	Misconceptions productive: SAFE-MISCON; soft-wrong ≠ empty kindness	Evidence	MIS-1...4; springboard + classify
2026-07-30	Transfer/VT: story ≠ far transfer; blocked ≠ ready	Evidence	TR-1...4; SAFE-TRANSFER
2026-07-30	Sleep/stress: encode→sleep→claim; pressure ≠ DD	Evidence	TIM-1...4; SAFE-TIMING
2026-07-29	Synthesizer v1.6: merge XXXIII–XL into I.4 doctrine stack	Synthesis	Surviving commercial law; next was XLI DD×anxiety
2026-07-29	SE prompts: AI monologue ■ SE; structure before wrap	Evidence	SE-1...4; densify Exp A/D
2026-07-29	Interleaving: block→n ear-miss→spaced mix; kill shuffle=science	Evidence	IL-1...4; ban blocked vanity
2026-07-29	Goal climate: appearance/streak NS killed; TARGET audit	Evidence	GO-1...4
2026-07-29	Eccles E×V×Cost: kill single motivation KPI	Evidence	EVT-1...4
2026-07-25	Constituted Research Lab + Constitution v1	Process	Create OS
2026-07-25	Scarcity thesis partially false as slogans; true as product wedge	Analysis	Rewrite pitch language

2026-07-25	Yeager 2019 constrains mindset claims	Evidence	Ban empty mindset copy
2026-07-25	Soft-wrong + wizard coach shipped in product	Product	Instrument coach → retry rate
2026-07-25	Mastery literature split (Kulik vs Slavin)	Evidence	Define mastery measurement carefully
2026-07-25	Bloom 2-sigma overstated vs VanLehn / modern tutoring metas	Evidence	Stop marketing 2-sigma
2026-07-25	Founder sequence wounded → SAFE-CRAFT	Synthesis	Attempt earlier; invention-story optional
2026-07-25	Deliberate practice necessary≠sufficient (Hambrick line)	Evidence	Progress-vs-self framing

Part IV — Evidence Table (selected)

Claim	Best supporting evidence	Contradicting / limiting evidence	Confidence	MindCraft implication
Brief mindset intervention can move grades + advanced math taking	Yeager et al., Nature 2019	Effects heterogeneous; peer norms matter; replication debates in broader mindset meta-analyses	Medium–High	Mindset must be ecological, not banner text
Math-domain mindset > generic	2023 Ed Research Review systematic review	Study quality varies; publication bias possible	Medium	Math-specific identity language
SDT need support raises motivation	Wang et al. 2024 meta	Relatedness effects weaker	Medium–High	Design for autonomy + competence first
Math anxiety impairs performance	Cognitive literature (Ashcraft tradition)	Not all low performers are anxious	Medium–High	Affective onboarding is first-class

Mastery learning raises achievement	Kulik et al. 1990	Slavin 1987 on standardized tests	Medium	Mastery graph + transfer tests
Gamified habits raise return	Duolingo method / industry practice	Habit ≠ deep learning; extrinsic traps (Deci)	Medium	Streaks as on-ramp only
1:1 tutoring is uniquely powerful	Bloom “2 sigma” framing (1984)	Often overstated; quality varies; VanLehn etc. nuance tutoring effects	Medium	Keep human tutors; AI amplifies, not replaces

Part V — Contradictory Evidence (do not skip)

1. **Mindset skepticism:** Some metas find small average effects; school implementations often degrade into posters.
2. **Grit / resilience hype:** Duckworth-style grit is contested; structural inequality and curriculum quality matter more than character slogans.
3. **Discovery learning failures:** Pure constructivism without guidance often fails (Kirschner, Sweller, Clark critiques). Narrative worlds must still teach.
4. **Engagement ≠ learning:** Time-on-app can rise while understanding falls (edtech’s original sin).
5. **AI may increase explanation value temporarily:** Novices may trust fluent wrongness; trust becomes the scarce resource, not explanation volume.
6. **Some students want speed and drills:** Identity-through-story is not universal. Offer multiple vessels.

Part VI — Mental Models

VI.1 The Identity Transformation Cascade (ITC)

HYPOTHESIS — MindCraft original synthesis

Threat ↓ → Attention available → Micro-success → Attribution ("I figured it")
→ Willingness to re-enter challenge → Accumulated competence evidence
→ Narrative update ("maybe I can") → Identity claim ("I do math")
→ Environment selection (harder courses, peers, tutors)
→ Reinforcing loop

Balancing loop: Premature hard problems → threat ↑ → avoidance → identity freeze.

Product translation: Gap scan (no shame) → soft-wrong → tiny earned wins → map fill → tutor recognition → advanced path.

VI.2 The Three Doors (what students actually need in a stuck moment)

Door	Need	Wrong product response	Right response
A	Safety	“Here’s a longer explanation”	Contain affect; normalize miss
B	Clarity	Pep talk	Precise model / worked example at right grain
C	Agency	Take over and solve for them	Scaffolded choice; student does the decisive step

FOUNDER BELIEF: Tutors often open Door B first. Many students are still behind Door A.

VI.3 Knowledge commodity stack (post-AI)

Layer 0: Facts & procedures → commoditizing fast
 Layer 1: Explanations → commoditizing fast
 Layer 2: Diagnosis of misconceptions → partially automatable; trust-sensitive
 Layer 3: Adaptive sequencing → automatable with good ontology
 Layer 4: Emotional co-regulation → scarce (human + careful UX)
 Layer 5: Identity & meaning → scarce
 Layer 6: Accountability relationship → scarce

MindCraft’s durable stack lives in **Layers 2–6**, with 0–1 as utilities.

VI.4 Transfer from outside education

Domain	Mechanism	Transfer to MindCraft (surviving law)
Games (Celeste, FromSoftware-lite design)	Fair difficulty + death as information	Soft-wrong as physics (SAFE-MISCON / FEI)
Sports coaching	Film study of <i>specific</i> error	SAFE-FILM: ≤3-cue attempt clips + self↪model
Music	Scales → repertoire → recital	SAFE-MUSIC: ingredient drills → chosen pieces → transfer proof
Therapy (exposure)	Graded approach + expectancy violation	SAFE-EXPOSE: situation hierarchy under destaked stakes
Aviation / military	Brief → attempt → structured debrief	SAFE-AAR: four-question student-first close
Chess annotation	Postmortem before engine; principle tags	SAFE-ANNOT: annotate→classify→drill; wins too

Martial arts belts	Visible competence ladder	Mastery pips / map regions (criterion, not league)
Language apps	Habit loops	SAFE-HABIT cue practice — without Duo's extrinsic ceiling
Religion / ritual	Shared meaning + belonging	SAFE-RITUAL: light optional marks; anti-totalism
Craft apprenticeship	Legitimate peripheral participation	SAFE-CoP: real peripheral work + old-timer; not Discord KPIs

Part VII — Universal Learning Principles (provisional)

1. **Emotion gates cognition** in threatened domains.
2. **Earned difficulty** beats random difficulty.
3. **Feedback should be information, not identity judgment.**
4. **Attribution matters:** strategy > talent for growth; luck attributions kill agency.
5. **Spaced retrieval** beats massed re-reading (cognitive science consensus).
6. **Worked examples** → **fading guidance** for novices (cognitive load).
7. **Belonging uncertainty** destroys persistence for stereotyped groups (Walton/Cohen tradition).
8. **Transfer requires varied practice**, not identical clones.
9. **Human accountability** multiplies AI tools.
10. **Identity is a lagging indicator**; design leading indicators (challenge-seeking, return-after-fail).

Part VIII — Product Implications (systems, not features)

VIII.1 Core system: Fear → Evidence → Identity (FEI Loop)

Reinforcing loop R1 (desired):

Safety UX → attempt → informative miss → coach → success → map fill → identity → more attempts.

Balancing loop B1 (threat):

Public shame / red buzz → avoidance → less practice → skill lag → more shame.

MindCraft already partially implements FEI: hide-correctness diagnostic, soft-wrong, wizard coach, story chapters, knowledge map. **Missing instrumentation:** does coach raise retry rate and later challenge-seeking?

VIII.2 What to stop optimizing

- Vanity DAU without learning transfer

- Explanation length as quality proxy
- Points that do not mark identity-relevant milestones

VIII.3 What to instrument next (minimum viable science)

1. **Retry rate within 2 minutes of soft-wrong** (with vs without wizard coach)
2. **Challenge-seeking**: % choosing harder level when offered
3. **Return after fail session** (D1 retention after a loss session)
4. **Self-identity item** (pre/post, 1–2 items, math-specific)
5. **Advanced course intent / enrollment** (slow, gold)

VIII.4 Tutor system role

Tutors are not content pipes. They are:

- Affect regulators
- Attribution coaches
- Bridge diagnosticians (concept A known, link A→B broken)
- Identity witnesses (“I saw you become someone who stays”)

AI should brief tutors, not replace witnessing.

Part IX — Experiments (pre-registered style)

Core A–D remain. Chapter ticks densified them into families — prefer the densified IDs when shipping.

Experiment A — Wizard coach on soft-wrong

- **Question**: Does the under-Graph wizard increase productive retry vs soft-wrong alone?
- **Design**: A/B, chapter + practice
- **Primary**: retry within 120s; secondary: eventual correct without hint binge
- **Falsifier**: no lift; or lift only in engagement with worse accuracy
- **Densified content (v1.6)**: Coach turns must be SE-constrained (principle / misconception contrast), not pep talk — see SE-1...4 / SE-2 as operational pairs with Experiment D.

Experiment B — Story-first vs bare stem

- **Question**: Does invention-story framing raise persistence on first miss?
- **Design**: within-concept crossover
- **Primary**: time-to-abandon after first miss
- **Falsifier**: story slows without improving retention/transfer
- **Equity densify**: HIST-EQ / Part XXXVI — whose story, tokenism kill.

Experiment C — North Star validation

- **Question:** Which early metric best predicts 8-week challenge-seeking?
- **Candidates:** streak length, soft-wrong retry, identity item, map mastery Δ , delayed mix accuracy
- **Method:** observational + lagged prediction
- **v1.6 prior:** Expect streaks to lose; retry_120s + transfer_pass + mastery-motive challenge_accept to win.

Experiment D — Explanation timing

- **Question:** Is explanation-before-attempt worse than attempt-with-safety for anxious students?
- **Design:** screen with anxiety pretest; randomize explanation-first vs attempt-first
- **Falsifier:** explanation-first wins for all segments
- **Densified:** Prefer SE-2/SE-3 (contrastive / structured generation) over monologue-first.

Experiment families mounted in chapters (do not orphan)

Family	Identifies (max claim without follow-up)	Parts
AIT-*	Guarded AI → solo transfer / anti-sycophancy	XXXIII
DAG-*	Estimand hygiene; covariate vs mediator	XXXIV
CSA-*	Session preference vs competitors; paste under guards	XXXV
HIST-EQ / equity probes	Story belonging without stereotype cue	XXXVI
EVT-*	Expectancy / value / cost levers	XXXVII
GO-*	Mastery vs performance CTA climate	XXXVIII
IL-*	Schedule → discrimination / delayed mix	XXXIX
SE-*	Prompt regime → transfer / misconception recurrence	XL
DD-*	Equip/destake/dose × anxiety → retry + delayed mix	XLI
SC-*	Criterion vs peer rank climate → affect + challenge	XLII
HAB-*	Cue/if-then vs streak colonization → return without identity harm	XLIII
IM-*	Informational vs engagement-loot rewards → free-choice + transfer	XLIV

RES-*	Growth-zone + support recruitment vs grit theater	XLV
EXP-*	Tutor brief / CIOF / AI expectancy → talk ratio + solo transfer	XLVI
TIM-*	Practice timing × sleep/stress → delayed transfer	XLVII
TR-*	VT / hug / bridge sequence → format-hop + solo transfer	XLVIII
MIS-*	Soft-wrong springboard / mis-mapped distractors / confidence tier	XLIX
CAL-*	Confidence-tiered feedback / hide-correctness / underconf unlock	L
DP-*	Session spine / edge targeting / method feedback / effort cap	LI
CoP-*	Named practice / peripheral work / old-timer / trajectory metrics	LII
RIT-*	Light ritual vs silent / recovery marker / anti-streak liturgy	LIII
AAR-*	Brief + four-question close / student-first / trace-backed debrief	LIV
FILM-*	Sparse clip cues / self-controlled review / self∪model pairing	LV
MUSIC-*	Ladder UX / repertoire choice / identified technique / transfer recital	LVI
ANNOT-*	Annotate-before-reveal / forced restudy / classify→drill / annotate wins	LVII
EXP-O-*	Situation hierarchy / expectancy check / fade safety / multi-context	LVIII
WTP-*	Parent CBC / FEI report vs score / message arms / cash pilot	LIX
REPAIR-*	Post-hallucination repair / reattempt / packetized escalate	LX

Part X — Weekly Priorities (research × product)

1. Instrument FEI loop events in analytics (retry, coach shown, write-mode exit).
 2. Run Experiment A for 2 weeks.
 3. Red Team the WORLD_VISION narrative claims with one outside learning scientist.
 4. Interview 10 Mayas: “When did math stop feeling dangerous?” (qual).
 5. Define “mastery” operationally so Slavin’s critique cannot gut the graph.
-

Part XI — Founder Questions (do not dodge)

1. If ChatGPT tutors become “good enough,” what *relationship* do parents still buy?
 2. Are we building for Maya’s identity — or for parents’ anxiety about scores? Both? Tradeoff?
 3. Would we accept lower short-term accuracy for higher long-term challenge-seeking?
 4. Is the story world load-bearing, or is gap+tutor enough?
 5. What would falsify the identity-transformation thesis in 90 days?
-

Part XII — Long-term Roadmap (research horizons)

H1 (now): Prove FEI loop moves retry + challenge-seeking.

H2: Prove identity language + map produce durable self-concept change.

H3: Prove human tutor + AI diagnosis beats AI-alone on persistence and transfer.

H4: Generalize beyond math (science/identity) only after math identity mechanism is solid.

Part XIII — Open Problems

1. Causal path from narrative history-of-math to identity (under-identified).
 2. Relatedness at scale without creepy social.
 3. Measuring identity without demand effects.
 4. Preventing gamification from colonizing meaning.
 5. AI fluent nonsense vs trust calibration for teens.
 6. Equity: does story-world privilege culturally specific narratives?
-

Part XIV — Red Team Dossier (destroy weak arguments)

Kill #1: “Explanations are free, therefore MindCraft should not explain”

Destroyed form: Absolutism.

Surviving form: Explanations are *necessary infrastructure* with collapsing *willingness-to-pay*; differentiation is diagnosis + affect + identity + accountability.

Kill #2: “Just fix mindset”

Destroyed: Poster mindset.

Surviving: Ecological mindset (task + feedback + peer/tutor norms), math-specific — TARGET structures (XXXVIII).

Kill #3: “Engagement is the goal”

Destroyed: Engagement without transfer.

Surviving: Engagement as oxygen for the FEI loop, not the fire.

Kill #4: “Mastery graph = Bloom 2-sigma”

Destroyed: Inflated tutoring mythology.

Surviving: Mastery *can* help weaker students if mastery is real (transfer_pass) and time costs are managed.

Kill #5: “Identity transformation is unmeasurable, so ship vibes”

Destroyed: Vibes-as-strategy.

Surviving: Leading indicators (retry, challenge-seeking, delayed mix) + lagging identity items; L0–L4 claim ladder (XXXIV).

Kill #6 (v1.6): “AI tutor / Bastani GPT Tutor is our product”

Destroyed: Unguarded chat-as-learning; category-error marketing; guards-as-gain.

Surviving: Pedagogy wrap + solo_transfer_pass $\geq$ no-AI bar (XXXIII RT).

Kill #7 (v1.6): “Blocked accuracy / streaks / shuffle prove learning”

Destroyed: Fluency illusion; streak North Star; random shuffle $\equiv$ interleaving science.

Surviving: Short block $\rightarrow$ near-miss interleave $\rightarrow$ spaced mixed check (XXXIX); streaks as on-ramp only (HID).

Kill #8 (v1.6): “Any explain box / AI monologue = self-explanation science”

Destroyed: Untargeted justify-your-wrong-guess; chat length as depth.

Surviving: Structured principle/misconception SE on correct or known-incorrect content before AI wrap (XL).

Kill #9 (v1.6): “One motivation dial”

Destroyed: Collapsing expectancy and value; staff “math is useful” dumps as EVT.

Surviving: $E \times V \times \text{Cost}$ triad with student-authored utility (XXXVII).

Kill #10 (v1.7): “Harder is always better / desirable difficulty ignores anxiety”

Destroyed: Anxiety-blind interleave/SE/pressure-as-DD; hardness theater as rigor brand.

Surviving: SAFE-DD equip → destake → dose → frame → measure (XLI); gate IL/SE by segment.

Kill #11 (v1.7): “Leaderboards / streaks / XP create math identity”

Destroyed: Appearance climates, streak colonization, engagement-loot as love-of-math.

Surviving: SAFE-COMPARE + SAFE-HABIT + SAFE-REWARD — criterion feedback, cue practice, informational unlocks only (XLII–XLIV).

Kill #12 (v1.7): “Grit / Resilience Score™ / stay in the danger zone”

Destroyed: Character meters; danger-zone-as-desirable-difficulty; grit posters as product.

Surviving: SAFE-RESILIENCE growth-zone + support recruitment; measure recovery/transfer not MRS theater (XLV).

Kill #13 (v1.7): “Pygmalion labels / warm AI that finishes the problem”

Destroyed: Trait/weakness tutor briefs; warmth-without-output-demand as expectancy science.

Surviving: SAFE-EXPECTANCY task briefs + CIOF; AI must clear PWC/solo_transfer_pass (XLVI + XXXIII).

Kill #14 (v1.7): “Story / sleep tips / mistake celebration = transfer”

Destroyed: Narrative immersion as automatic far transfer; sleep-app brand; empty mistake posters; red-X-as-rigor; equal graph updates for all wrongs.

Surviving: SAFE-TRANSFER + SAFE-TIMING + SAFE-MISCON — vary on purpose, encode→sleep→claim, elicit→classify→springboard→route (XLVII–XLIX).

Kill #15 (v1.8): “Raise confidence / Belief Score™ / CA raises grades”

Destroyed: Inflate-confidence brand; confidence meters as identity; confidence assessment alone as attainment lever; equal updates for guess vs high-conf misconception.

Surviving: SAFE-CALIB — item-level elicit → miss class → tiered feedback; appropriate confidence; C4 hide-correctness as diagnostic hygiene (L).

Kill #16 (v1.8): “Hours / forums / ritual→score / domain costume = identity”

Destroyed: 10k-hours and minutes KPIs; Discord/streak ≡ CoP; unretracted ritual→performance marketing; bootcamp/conservatory/GM/therapy branding as belonging.

Surviving: SAFE-DP + SAFE-CoP + SAFE-RITUAL — diagnose→prove spine; named practice with real peripheral work; light optional marks pointing at recover/prove (LI–LIII).

Kill #17 (v1.8): “Tip-flood / engine-dump / flooding ≡ high-performance pedagogy”

Destroyed: Uncapped film-room tips; Stockfish/LLM monologue as annotation; day-one timed mixed ACT as “exposure”; calm-to-zero / Calm Score™; loss-only shame reels; bare-pass recital theater.

Surviving: SAFE-AAR + SAFE-FILM + SAFE-MUSIC + SAFE-ANNOT + SAFE-EXPOSE — brief–debrief, sparse clips, skills→pieces→proof, annotate→drill, graded approach with expectancy violation (LIV–LVIII).

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Part XVI — Mechanism Deep Dive: How identity actually changes

This part answers the research question at mechanism grain. It is synthesis, not settled science.

XVI.1 What “confident mathematical thinker” means (operational)

Do not use: “likes math apps” or “says math is fun once.”

Working definition (HYPOTHESIS): A confident mathematical thinker is a person who, in math-relevant contexts:

- 1. **Approaches** moderately hard problems without immediate avoidance.
- 2. **Persists** after a miss long enough to extract information.
- 3. **Attributes** success primarily to controllable causes (strategy, practice, help-seeking) rather than luck or tutor magic.
- 4. **Selects** into harder future opportunities when available (courses, contests, careers) at rates above their prior baseline.
- 5. **Endorses** a math-capable self-concept *and* behaves consistently with that endorsement.

Items 1–4 are leading. Item 5 is lagging. Product should optimize leading indicators while periodically measuring the lagging identity claim.

XVI.2 The threat–competence–narrative triad

Three literatures converge on the same transformation engine:

Mechanism	Core claim	Confidence	Key sources
Threat / affect	Math anxiety transiently reduces working-memory resources available for math	High (FACT direction)	Ashcraft & Kirk (2001); Suárez-Pellicioni et al. review (2016); WM mediation meta (2021)
Competence evidence	Motivation requires felt competence; mastery structures can raise achievement when mastery is real	Medium–High	SDT (Ryan & Deci); Kulik mastery meta; Slavin critique
Narrative / identity	Self-stories and belonging cues shape challenge-seeking and persistence	Medium	Yeager et al. (2019); Walton/Cohen belonging tradition; narrative psychology

HYPOTHESIS — Triad necessity: For students starting from “I am bad at math,” improving any one factor alone is usually insufficient:

- Safety without competence → comfort without skill → identity still fragile.
- Competence without safety → possible for some; many never enter the practice volume.
- Narrative without either → empty affirmation (false growth mindset).

XVI.3 Founder tutoring sequence — audit

FOUNDER BELIEF (observed in tutoring):

1. Reduce fear
2. Demystify mathematics
3. Explain why humanity invented it
4. Simplify the narrative
5. Create a tiny success
6. Reinforce success
7. Celebrate progress → identity shifts

Red Team audit:

Step	Universal?	Evidence status	Failure mode
1 Fear ↓	Common for anxious subset; not for all	Strong for anxious learners	Over-soothing that removes challenge
2 Demystify	Often helpful	Medium (cognitive clarity)	Over-simplification that lies
3 Invention story	Founder prior	Under-evidenced at RCT scale	Cultural mismatch; time cost
4 Simplify narrative	Often helpful	Medium	Removes necessary difficulty
5 Tiny success	Near-universal lever	High (competence need)	Too-easy wins → hollow identity
6 Reinforce	High if attribution-correct	High (feedback literature)	Praise of talent or praise of empty effort
7 Celebrate	Helpful if genuine	Medium	Performative celebration → cynicism

Improved sequence (HYPOTHESIS — “SAFE-CRAFT”):

1. Safety: threat appraisal ↓ (tone, privacy, soft-wrong, no public shame)
2. Attempt: student acts before receiving a full lecture (productive struggle at right grain)
3. Feedback-as-information: miss becomes diagnosis, not character
4. Earned micro-win: success the student can own

5. Causal story: “what worked” (strategy), optionally “why humans invented this tool”
6. Re-entry: immediate second attempt / next slightly harder item
7. Attribution + witness: tutor/system names the growth specifically
8. Future self: map fill / path unlock that marks identity-relevant progress
9. Transfer check: varied item so win was not clone memorization

What changed vs founder sequence: Attempt moves earlier; invention-story becomes optional enrichment inside step 5, not a prerequisite to agency; transfer check prevents fake mastery.

XVI.4 Math anxiety — what is known, what is not

FACT: Higher math anxiety correlates with lower math performance (meta $r \approx -0.17$ in one 2021 synthesis of 57 studies — modest average, not destiny).

FACT: Ashcraft & Kirk (2001) show math anxiety can act like a dual-task load: worry competes for working memory, especially on computation-span / carry-heavy tasks.

HYPOTHESIS: In product UX, anything that increases social-evaluative threat (red buzzers, public leaderboards for novices, tutor contempt, parent hovering UI) will shrink effective WM and look like “they don’t get it.”

Contradictions / limits:

- Not all underperformance is anxiety; some is missing knowledge, poor curriculum, sleep, language barriers.
- Trait vs state anxiety differ; trait may partly reflect prior failures (reverse causality possible).
- High performers can be anxious and still succeed via avoidance of hard electives — a hidden identity cost.

MindCraft rule: Treat anxiety as a first-class load on the learning system, not as a soft skill sidebar.

XVI.5 Deliberate practice — necessary, not sufficient

FACT / TRADITION: Ericsson’s deliberate practice emphasizes structured, feedback-rich, effortful practice aimed at improvement (not mere repetition).

FACT / CRITIQUE: Hambrick and colleagues argue deliberate practice is important but not sufficient; in chess/music reanalyses, deliberate practice often explains roughly $\sim 1/3$ of reliable variance, leaving most unexplained (abilities, starting age, opportunity, etc.).

Implication for MindCraft: Do not sell “10,000 hours.” Sell **high-quality attempts under feedback**. Also respect individual differences: same practice volume will not equalize outcomes. Identity work must include “progress relative to self,” not only absolute rank.

XVI.6 Tutoring effects — demythologize Bloom

FACT: Bloom (1984) popularized the “2 sigma” tutoring + mastery framing.

FACT / CALIBRATION: Broader tutoring metas are far smaller (often ~ 0.3 – 0.4 SD in modern RCT syntheses cited by Education Next). VanLehn (2011) estimated human tutoring ~ 0.79 SD vs no tutoring, with step-based ITS nearly comparable (~ 0.76).

Implication: Human tutors remain valuable, especially for affect + accountability + diagnosis of weird misconceptions. They are not magic 2-sigma machines. AI step-level feedback can capture much of the *cognitive* tutoring benefit; the scarce remainder is Layers 4–6 (emotion, identity, accountability).

Part XVII — Scarcity Thesis: Full Red Team Trial

XVII.1 The claim under oath

FOUNDER BELIEF: AI commoditizes knowledge; future scarcity is motivation, confidence, identity, trust, curiosity, persistence, emotional safety, meaning.

XVII.2 Cross-examination

Claim A — “Explanations are free”

Verdict: Directionally **FACT** for commodity explanations; **FALSE** for *trusted, personalized, correct, curriculum-aligned* explanations under time pressure.

Evidence: Generative AI + open platforms collapse marginal cost of text explanation.

Contradiction: Hallucinations, shallow fluency, and misalignment create a *trust tax*. Parents may still pay for verified human judgment.

Claim B — “Tutoring is free”

Verdict: **FALSE**.

AI chat is cheap. High-quality tutoring includes scheduling, relationship, emotional labor, accountability, and reputation. Those remain expensive. Nickow et al.–style tutoring metas still show meaningful effects when tutoring is structured — it is not free at scale.

Claim C — “Content is free”

Verdict: **MOSTLY FACT** for undifferentiated content; **FALSE** for sequenced, assessed, outcome-linked content systems.

Khan/YouTube prove content abundance. Differentiation is curation + measurement + human wrap.

Claim D — “Motivation/identity become the scarce goods”

Verdict: **HYPOTHESIS** with strategic plausibility.

As cognitive help approaches free, *willingness to engage difficulty* becomes the binding constraint for many students. This does not mean motivation products automatically win — many habit apps fail to produce identity.

Claim E — “Emotional safety is first-order”

Verdict: HYPOTHESIS — true for anxious / threatened learners; not universal.

For some, speed drills and competition *increase* engagement. Segment, do not romanticize softness.

XVII.3 Surviving thesis (company doctrine)

Adopted doctrine (refined v1.7):

> MindCraft does not compete primarily on explanation volume, AI tutor warmth, streak/XP theater, grit meters, or story immersion. It competes on converting threatened learners into challenge-seeking mathematical agents through diagnosis, affective design, mastery-climate evaluation, sequenced desirable difficulties (SAFE-DD), student-generated why, variation-designed transfer, productive-error routing, narrative meaning under equity constraints, and human accountability — with AI as *guarded* infrastructure that must clear `solo_transfer_pass` $\geq$ no-AI.

Falsifiers (90-day capable):

1. Anxious segment shows no FEI lift vs explanation-first control.
2. Parents refuse to pay when score gains lag identity metrics.
3. AI-alone matches human+AI on persistence + transfer for target segment.
4. Guarded Solver raises concurrent scores but loses `solo_transfer_pass` vs no-AI (PWC fail).
5. Interleave/SE regimes raise delayed mix but destroy `retry_120s` in high-anxiety segment without sequenced SAFE-DD (XLI; pending DD-1/DD-2).
6. Soft-wrong without springboard/route matches key-only on retry + misconception recurrence (MIS-1 fail).
7. VT/hug sequences fail to beat random surface skins on next-day format-hop transfer (TR-1 fail).

Part XVIII — Systems Maps (loops, not features)

XVIII.1 FEI reinforcing loop (desired)

```
+----- identity claim ("I do math") <---+
| |
v |
safety UX ----> attempt ----> informative miss ----> coach
^ | |
| v v
+----- map fill / witness <----- earned success <+
```

XVIII.2 Shame balancing loop (destroy this)

```
public failure signal -> threat ↑ -> avoidance -> fewer attempts
^ |
+----- skill lag / worse scores -----+
```

XVIII.3 Extrinsic colonization loop (danger)

```
streak / points ↑ -> return ↑ -> shallow grinding ↑
| |
+--- identity unchanged -----+--> burnout / streak break → exit
```

Design rule: Habit loops may feed FEI but must not replace identity markers.

XVIII.4 Parent trust loop

```
visible diagnosis + honest progress -> parent trust ↑ -> continued purchase
^ |
+---- outcome evidence (scores / challenge-seeking)
```

FOUNDER QUESTION unresolved: Optimize Maya’s identity or parent’s anxiety? Doctrine: report *both* challenge-seeking and skill evidence; never fake either.

XVIII.5 Network effects (honest)

MindCraft today has weak classic network effects. Do not pretend otherwise.

Possible compounding advantages (HYPOTHESIS):

- 1. Ontology + misconception memory → better diagnosis over time (data moat if privacy-respecting).
- 2. Tutor playbooks trained on FEI outcomes → service quality moat.
- 3. Story worlds with longitudinal character arcs → switching costs of meaning (fragile; easy to cargo-cult).

Part XIX — Cross-Domain Transfer Catalog

Principles that repeatedly change human behavior outside classrooms — filtered for MindCraft use.

Domain	Robust mechanism	Transfer	30-year durable?	AI improve?
Exposure therapy	Graded exposure under safety	Anxiety-graded problem sets	Yes	Yes (personalize gradient)
Sports film study	Specific error review	Wrong-choice coach	Yes	Yes (auto clip of miss type)
Music pedagogy	Technique → repertoire → performance	Drill → quest → “recital” signal	Yes	Partial
Martial arts	Visible ranks + dojo norms	Map regions + tutor dojo culture	Yes	Weak for norms

Aviation	Checklists under stress	Procedure scaffolds when anxious	Yes	Yes
Games (Celeste-like)	Fair failure + retry culture	Soft-wrong physics	Yes	Yes
Chess	Annotated game review	Session postmortems	Yes	Yes
Coaching	Identity + accountability	Tutor as witness	Yes	Assist, not replace
Religion/ritual	Meaning + belonging	Light rituals only; avoid cult dynamics	Yes (human)	Risky
Creator economy	Public craft identity	Optional shareable math artifacts	Uncertain	Yes

Anti-transfer: Casino variable rewards → addiction without skill. Do not copy engagement maxing from social media.

Part XX — Competitive Landscape (mechanism lens)

Product	Primary scarce resource they sell	Identity claim?	Risk
Khan Academy	Free mastery content + reputation	Weak–medium	Explanation commodity; “mastery” without transfer_pass
Duolingo	Habit / streak motivation	Weak (language identity sometimes)	Extrinsic ceiling; appearance of consistency
Brilliant	Aesthetic problem joy + prestige	Medium	Narrow segment; not Maya threat profile
Coursera/edX	Credentials	Medium (career identity)	Completion collapse
Chegg / homework help	Answers under deadline	Anti-identity (outsourcing)	Integrity & AI shock
Private tutors	Human accountability + customization	High when good	Supply constrained
ChatGPT tutors	Instant explanation	Low unless wrapped	Trust / hallucination; Bastani Base harm
MindCraft (target)	FEI conversion + tutor witness + gap diagnosis + solo transfer	Intended high	Must prove, not assert

Strategic implication (v1.8 densified): Do not out-Khan Khan on content breadth. Do not out-Duo Duo on streaks/leagues/XP. Do not out-Brilliant Brilliant on puzzle delight. Do not out-ChatGPT on fluency. Do not out-grit character apps on Resilience Score™. Do not out-therapy Calm Score™, sports academies, conservatories, or grandmaster cosplay. Out-compete on the **session**: SAFE-DP spine under SAFE-DD, productive-error + calibration routing (SAFE-MISCON / CALIB), brief–debrief + sparse film + annotation (SAFE-AAR / FILM / ANNOT), ladderred approach for feared formats (SAFE-MUSIC / EXPOSE), variation-designed transfer (SAFE-TRANSFER), student-generated why, and transfer when help is gone — then *say that* in marketing (CSA-2).

Part XXI — Metrics Dictionary (instrumentation)

XXI.1 Banned as North Stars

- Raw DAU / time-on-app without transfer
- Streak length alone
- Explanation open-rate / AI chat token count
- Points / coins / XP velocity
- Blocked-session accuracy alone
- Coach thumbs-up / “felt helpful” without solo_transfer_pass
- Named leaderboard rank / league tier
- Resilience Score™ / grit meter / MRS as in-app KPI
- Midnight minutes practiced / all-nighter volume
- Belief Score™ / confidence meters / confidence streaks
- Community tab visits / Discord volume / games played / rating
- Calm Score™ / anxiety-must-hit-zero meters
- Tip-recall / film-room engagement without method-change + transfer

XXI.2 Leading indicators (ship first)

Metric ID	Definition	Why
retry_120s	New attempt on same or isomorphic item within 120s of soft-wrong	Persistence under safety
coach_shown	Wizard/coach surfaced	Treatment exposure
write_exit_to_retry	Leave write mode then attempt	Agency after reflection
challenge_accept	Chose harder level when offered	Challenge-seeking (instrument <i>motive</i>)
hint_binge	≥3 hints without independent solve	Gaming / helplessness / failed exposure fade

transfer_pass	Correct on varied item after mastery mark	Anti-fake-mastery
solo_transfer_pass	Transfer with AI/Solver closed or denied	Anti-crutch (XXXIII)
ai_reveal_rate	Full-answer / unguarded reveal frequency	Crutch exposure
strategy_class_error	Wrong <i>procedure family</i> on mixed set	Interleaving discrimination (XXXIX)
se_principle_hit	Correct structured principle/ingredient pick when prompted	SE quality proxy (XL)
help_recruit_then_solo	Opened card/tutor/peer help then solved without full reveal	SAFE-RESILIENCE support path (XLV)
misconception_route_hit	Soft-wrong routed to mapped mis_ / ingredient failure_mode	SAFE-MISCON classification (XLIX)
format_hop_pass	Correct after declared format/context hop (VT check)	SAFE-TRANSFER discernment (XLVIII)
confidence_miss_tier	High vs low confidence on miss (when elicited)	SAFE-CALIB / hypercorrection (L)
annotate_before_reveal	Student principle/why logged before key/coach/model	SAFE-ANNOT / SAFE-AAR (LVII / LIV)
film_cue_count	Coaching points attached to attempt clip (cap ≤ 3)	SAFE-FILM less-is-more (LV)
avoided_format_approach	Attempt on previously avoided format/stakes step	SAFE-EXPOSE approach rate (LVIII)
repair_reattempt_pass	Correct destaked re-attempt after owned AI miss	SAFE-REPAIR calibration (LX)
escalation_resolve_48h	Packetized human handoff closed with fix	SAFE-REPAIR escalate SLA (LX)
improve_how_specificity	Debrief yields concrete next-attempt tweak	SAFE-AAR close quality (LIV)

XXI.3 Lagging indicators

Metric ID	Definition
math_person_item	1–7 agreement: “I am a math person” (math-specific)
anxiety_state_item	Short state anxiety before session
advanced_intent	Intent to take harder course / contest
tutor_witness_note	Tutor tagged identity-relevant growth

XXI.4 Decision rule (HYPOTHESIS)

Ship changes that raise `retry_120s` and `mastery-motive challenge_accept` without raising `hint_binge` / `ai_reveal_rate`, and without dropping `transfer_pass` or `solo_transfer_pass`. Prefer delayed mixed accuracy over blocked session accuracy.

Part XXII — Appendices

Appendix A — Glossary

Term	Meaning
FEI Loop	Fear → Evidence → Identity system
ITC	Identity Transformation Cascade
Soft-wrong	Miss treated as information, not shame theater
Gap scan	Low-shame diagnostic of confidence/exposure
Challenge-seeking	Choosing harder math opportunity when optional
False growth mindset	Effort praise without strategy / conditions
SAFE-CRAFT	Improved tutoring sequence (Part XVI.3)
Layers 0–6	Knowledge commodity stack (Part VI.3)
HID	Habit/Identity Divergence — streaks can raise return without identity
MoC	Memory of Competence — spaced retrieval after <code>transfer_pass</code>
PWC	Pedagogy Wrap Constraint — guarded AI + student generation + solo transfer bar
Claim ladder L0–L4	Max defensible causal/marketing claim given design (Part XXXIV)
$E \times V \times \text{Cost}$	Eccles expectancy $\times$ subjective task value $\times$ cost triad
TARGET	Classroom/goal-structure dimensions cueing mastery vs performance
SAFE-DD	Equip → destake → dose → frame → measure (Part XLI)
SAFE-COMPARE	Criterion-default comparison policy (Part XLII)
SAFE-HABIT	Cue practice without streak colonization (Part XLIII)
SAFE-REWARD	Informational / non-controlling reward policy (Part XLIV)

SAFE-RESILIENCE	Growth-zone + support recruitment, not grit meters (Part XLV)
SAFE-EXPECTANCY	Task briefs + CIOF; ban trait labels (Part XLVI)
SAFE-TIMING	Encode → sleep → delayed transfer claim (Part XLVII)
SAFE-TRANSFER	Variation/hug/bridge transfer design (Part XLVIII)
SAFE-MISCON	Productive-error elicit → classify → springboard → route (Part XLIX)
SAFE-CALIB	Item-level confidence → miss class → tiered feedback (Part L)
SAFE-DP	Diagnose → isolate → attempt → method feedback → prove spine (Part LI)
SAFE-CoP	Named practice; peripheral = real work; trajectory > tab (Part LII)
SAFE-RITUAL	Light optional practice-pointing marks; anti-totalism (Part LIII)
SAFE-AAR	Brief → traces → four-question student-first close (Part LIV)
SAFE-FILM	Sparse attempt clips + self-model + next commitment (Part LV)
SAFE-MUSIC	Scales → repertoire → transfer-as-recital ladder (Part LVI)
SAFE-ANNOT	Annotate-before-reveal → classify → drill; annotate wins (Part LVII)
SAFE-EXPOSE	Situation hierarchy + expectancy violation + fade safety (Part LVIII)
SAFE-WTP	Parent CBC + bias controls + message≠package (Part LIX)
SAFE-REPAIR	Detect→own→correct→reattempt→triggered escalate (Part LX)
SAFE-IDMEASURE	Validated identity factors + FEI behavior + demand-aware admin (Part LXI)
SAFE-LONGID	Stage 8w process / 26w triangulated identity / 52w persistence; fade-aware (Part LXII)

Appendix B — Maya interview protocol (qual)

Goal: Map moments when math stopped (or started) feeling dangerous.

1. “Tell me about a time math felt dangerous or humiliating.”
2. “What did your body do? What did you do next?”

1. Attack any claim that cannot name a falsifier.
2. Attack any metric that can be gamed without learning.
3. Attack any narrative feature without a learning outcome.
4. Attack Bloom 2-sigma mythology on sight.
5. Attack “AI makes tutoring free” slogans on sight.

Appendix F — Future chapters (queued, not padded)

1. Neuroscience of math anxiety (amygdala/WM models; cautious translation)
2. History of mathematics as meaning technology — under trial via XXVIII + HIST
3. Parent trust economics & willingness-to-pay — **DONE** Part LIX / SAFE-WTP (builds EVT-4 / XXVII)
- 3b. Trust after AI hallucination / repair + escalate — **DONE** Part LX / SAFE-REPAIR (builds XXXIII)
- 3c. Measurement of math identity — **DONE** Part LXI / SAFE-IDMEASURE
- 3d. Longitudinal identity change models — **DONE** Part LXII / SAFE-LONGID
4. Equity audit of story worlds — **DONE** (Part XXXVI)
5. Competitive teardown — **DONE** session audits (Part XXXV); usage telemetry still open
6. Formal Bayesian update process for Constitution claims
7. **DONE (2026-07-30)**: Desirable difficulties × anxiety (Part XLI) — SAFE-DD stack
8. **DONE (2026-07-30)**: Social comparison & leaderboards (Part XLII) — SAFE-COMPARE stack
9. **DONE (2026-07-30)**: Habit formation science without identity colonization (Part XLIII) — SAFE-HABIT stack
10. **DONE (2026-07-30)**: Intrinsic motivation killers / Deci caveats (Part XLIV) — SAFE-REWARD stack
11. **DONE (2026-07-30)**: Mathematical resilience (Part XLV / id 45) — SAFE-RESILIENCE stack
12. **DONE (2026-07-30)**: Teacher/tutor expectancy effects (Part XLVI / id 46) — SAFE-EXPECTANCY stack
13. **DONE (2026-07-30)**: Sleep, stress, and learning (Part XLVII / id 47) — SAFE-TIMING stack
14. **DONE (2026-07-30)**: Transfer & variation theory (Part XLVIII / id 48) — SAFE-TRANSFER stack
15. **DONE (2026-07-31)**: Misconceptions as productive (Part XLIX / id 49) — SAFE-MISCON stack
16. **Synthesizer v1.7 (2026-07-31)**: Merged XLI–XLIX into I.4 / XIV / IX / XXI — no new chapter
17. **DONE (2026-07-31)**: Confidence calibration (Part L / id 50) — SAFE-CALIB stack
18. **DONE (2026-07-31)**: Deliberate practice in tutoring sessions (Part LI / id 51) — SAFE-DP stack
19. **DONE (2026-07-31)**: Community of practice / Lave–Wenger (Part LII / id 52) — SAFE-CoP stack
20. **DONE (2026-07-31)**: Religion/ritual light-touch (Part LIII / id 53) — SAFE-RITUAL stack
21. **DONE (2026-07-31)**: Military/aviation brief-debrief (Part LIV / id 54) — SAFE-AAR stack
22. **DONE (2026-07-31)**: Sports film-study pedagogy (Part LV / id 55) — SAFE-FILM stack
23. **DONE (2026-08-01)**: Music pedagogy ladders (Part LVI / id 56) — SAFE-MUSIC stack
24. **DONE (2026-08-01)**: Chess annotation & metacognition (Part LVII / id 57) — SAFE-ANNOT stack
25. **DONE (2026-08-01)**: Therapy graded exposure (Part LVIII / id 58) — SAFE-EXPOSE stack
26. **Synthesizer v1.8 (2026-08-01)**: Merged L–LVIII into I.4 / XIV / IX / XXI — no new chapter
27. **DONE (2026-08-01)**: Parent WTP experiments design (Part LIX / id 59) — SAFE-WTP stack
28. **DONE (2026-08-01)**: Trust after AI hallucination (Part LX / id 60) — SAFE-REPAIR stack
29. **DONE (2026-08-01)**: Measurement of math identity (Part LXI / id 61) — SAFE-IDMEASURE stack
30. **DONE (2026-08-02)**: Longitudinal identity change models (Part LXII / id 62) — SAFE-LONGID stack

Part XXIII — Research Lab Operating Cadence

Weekly (60–90 min):

1. One claim enters trial (evidence + contradiction).
2. Red Team attempts kill.
3. Surviving claim updates Constitution with confidence tag.
4. One experiment metric reviewed.

Synthesizer (~every 8 researcher ticks): Merge duplicate frameworks into Part I.4 / XIV; refresh Executive Summary; regenerate PDF. Do not write a new chapter on synthesizer ticks.

Monthly:

- Rewrite one weak chapter.
- External reader (learning scientist or skeptical founder).
- Regenerate PDF.

Quarterly:

- Kill or promote North Star metric.
 - Publish internal “what we believed that was wrong” memo.
-

Closing stance

MindCraft’s deepest risk is not technical failure. It is **winning the wrong game**: becoming the best explanation engine in a world where explanations are cheap — or the warmest AI tutor in a world where warmth without solo transfer is a crutch — while losing the only game that compounds: helping a human revise who they are in the presence of difficulty.

This Constitution exists so the company notices that risk early, and runs experiments that can kill beloved ideas.

v1.8 synthesizer pass folded Parts L–LVIII into surviving commercial doctrine (I.4 rows SAFE-CALIB through SAFE-EXPOSE; Red Team kills #15–#17). Part LIX added SAFE-WTP; Part LX added SAFE-REPAIR; Part LXI added SAFE-IDMEASURE; Part LXII added SAFE-LONGID (8/26/52 week staged identity claims + fade-aware retention). Next researcher id: LXIII ACT/exam pressure. Page count is not the finish line — falsifiable truth is.

Part XXIV — Affect, Anxiety, Belonging (deep dive)

Chapter status: Living evidence brief

Primary question: When is emotional safety causally load-bearing for mathematical confidence?

Owners: Cognitive Neuroscientist seat · Ed Psych seat · Red Team

XXIV.1 What is happening, mechanistically?

Three related but non-identical constructs get collapsed in product talk. Separate them.

Construct	Definition (working)	Time scale	Product signal
State math anxiety	Transient dread / arousal in a math moment	Seconds–minutes	Avoid click, freeze, rapid hint binge
Trait math anxiety	Stable tendency to experience math as threat	Months–years	Course avoidance, “I’m not a math person”
Belonging uncertainty	“People like me don’t belong in this domain”	Contextual, can be acute	Withdrawal after mild adversity
Stereotype threat	Extra evaluative pressure from group stereotype relevance	Situational	Underperformance vs ability when stereotype is cued

FACT (direction): Math anxiety is associated with worse math performance; one 2021 meta-analysis reported mean $r \approx -0.17$ across 57 studies (modest average association — not destiny).

FACT (mechanism candidate): Ashcraft & Kirk (2001) show high math-anxiety individuals have reduced working-memory span especially on computation-based span tasks; dual-task math + memory load amplifies errors/RT. Anxiety behaves like a competing load on central executive resources.

FACT (related): Reviews (e.g., Suárez-Pellicioni et al., 2016) organize explanations as (a) WM competition, (b) attentional control / inhibition deficits, (c) possible low-level numerical representation differences — not mutually exclusive.

HYPOTHESIS for MindCraft: For the Maya-anxious segment, UX that increases social-evaluative threat (shame theater, public ranking early, contemptuous feedback) will look like a “knowledge gap” while actually being a temporary capacity tax.

XXIV.2 Why does this happen? (first principles)

1. **Working memory is scarce.** Multi-step math is a WM sport. Anything that occupies the central executive (worry, self-monitoring for stereotype confirmation, rumination about looking stupid) steals the resource the task needs.
2. **Humans are status-sensitive learners.** Math is a publicly ranked school subject. Threat is often social (“I will be seen as dumb”) more than arithmetic.

3. **Avoidance is rational short-term.** Leaving the room ends the aversive arousal. Long-term, avoidance prevents the mastery experiences that would update efficacy.

4. **Identity consolidates avoidance.** Repeated exits become “I’m not a math person,” which then licenses future exits (narrative lock-in).

Human need satisfied by safety design: Need for non-humiliation while competence is still fragile; relatedness/respect; predictability.

XXIV.3 Belonging and stereotype threat — what transfers?

FACT: Steele & Aronson tradition — when a test is framed as ability-diagnostic under a negative group stereotype, performance can drop relative to non-diagnostic framing (classic lab pattern; effect sizes and replication quality vary by context).

FACT: Spencer, Steele & Quinn (1999) line — women/math stereotype relevance can impair performance when threat is cued.

FACT: Walton & Cohen (2007, 2011) — belonging uncertainty makes adversity feel identity-diagnostic; brief belonging interventions framing adversity as common/transient improved long-term outcomes for marginalized college students in landmark trials (e.g., Science 2011 belonging intervention).

Red Team limits:

- Effects are **heterogeneous** and context-dependent (like mindset).
- School-wide scaling often attenuates lab effects.
- MindCraft must not cargo-cult a 45-minute belonging essay into a gamified toast.
- Belonging interventions are not a substitute for teaching.

Surviving transfer:

- Normalize struggle without normalizing low standards.
 - Make early adversity non-identity-diagnostic (“misses are how the map learns”).
 - Avoid stereotype-cueing copy (“even girls can...” is often worse).
 - Tutor witnessing is a belonging technology when authentic.
-

XXIV.4 Neuroscience: what we can say without overclaiming

FACT / REVIEW level: Math anxiety literature includes ERP/fMRI correlates; reviews document affective and cognitive network involvement. Exact “brain region product features” are **SPECULATION** and should not drive roadmap.

Allowed claim: Affective arousal and executive control interact; designing for lower threat is biologically plausible.

Forbidden claim: “Our purple calm mode activates the prefrontal cortex.”

XXIV.5 Contradictory evidence (keep visible)

1. Many students underperform from **missing knowledge**, not anxiety.
 2. Some learners seek **competitive arousal** and dislike soft UX.
 3. Reverse causality: low skill → anxiety → further avoidance (bidirectional).
 4. Trait anxiety interventions that only soothe without competence can create comfort without growth.
 5. Meta correlations are modest; anxiety is one bottleneck among many.
-

XXIV.6 Can this generalize? 30 years? AI?

Question	Answer
Generalize beyond math?	Yes — any high-status evaluative domain (music juries, coding interviews).
Still work in 30 years?	Yes — human threat systems are not a 2026 fad.
Can AI improve it?	Yes for personalization of challenge gradient & detection of freeze patterns; risky for fake empathy.
MindCraft change	Soft-wrong, hide-correctness diagnostics, coach tone, private early practice, tutor witness scripts are load-bearing for anxious segment.

XXIV.7 Experimental validation program

ID	Question	Design	Primary	Falsifier
AFF-1	Does soft-wrong raise retry vs hard-fail UX?	A/B	retry_120s	No lift / accuracy down
AFF-2	Does anxiety-state moderate explanation-first harm?	Anxiety screen × timing	transfer + retry	No interaction
AFF-3	Does “misses are information” copy reduce belonging threat items?	Pre/post items	belonging/threat items	Demand effects only
AFF-4	Tutor witness note vs no note after growth moment	Tutor RCT	2-week challenge -seeking	Null

Confidence in chapter thesis: Medium–High that affect matters for a large subset; Medium that MindCraft’s current UX already captures the right mechanisms; Low that we have measured it.

XXIV.8 Product system (not features)

Affective Load Manager (system):

1. Detect freeze / hint-binge / rapid exit (behavioral).
2. Downshift evaluative threat (tone, privacy, soft-wrong).
3. Offer graded re-entry (easier isomorphic item), not pep talk alone.
4. Restore challenge once retry and success stabilize.
5. Log whether the student later chooses harder work (identity leading indicator).

Balancing loop: chronic downshift → boredom → exit. Must re-raise difficulty.

Part XXV — Self-Efficacy, Narrative Identity, Habit (deep dive)

Chapter status: Living evidence brief

Primary question: What updates the self-story “I am bad at math,” and what merely increases login frequency?

Owners: Learning Science · Narrative Psychology · Behavioral Econ · Red Team

XXV.1 Self-efficacy is not “confidence vibes”

FACT (theory): Bandura’s social cognitive theory — self-efficacy is belief in capability to organize and execute actions required for specific attainments. It is **task- and domain-specific**, not a global personality glow.

FACT (sources): Efficacy information is processed from four sources (Bandura, 1997):

1. **Mastery experiences** (usually strongest)
2. **Vicarious experiences** (models)
3. **Social persuasion** (credible others)
4. **Physiological / affective states** (how arousal is interpreted)

FACT (measurement tradition): Usher & Pajares (2009/2010 line) validated source scales for middle-school mathematics; sources relate to math self-efficacy, goals, optimism.

HYPOTHESIS: MindCraft’s FEI loop is largely a **self-efficacy engineering system**:

FEI stage	Efficacy source
-----------	-----------------

Safety / soft-wrong	Reinterprets physiological arousal (“alert, not doomed”)
Earned micro-win	Mastery experience
Coach naming strategy	Persuasion + attribution
Tutor witness / peer story	Vicarious + persuasion
Map fill after transfer	Mastery that survives variation

XXV.2 Why mastery experiences dominate — and how products fake them

What is happening: A win only updates efficacy if the learner **attributes** it to their capability under meaningful difficulty.

Fake mastery (product sins):

- Too-easy items → “this doesn’t count”
- Heavy hinting → “the app did it”
- Identical clones after “mastery” → recognition, not competence
- Points for opening lessons → engagement theater

Real mastery design rules (HYPOTHESIS):

1. Difficulty just above comfort (desirable difficulty, not cruelty).
2. Student performs the decisive step.
3. Immediate informative feedback.
4. Soon after: varied item (transfer_pass).
5. Explicit attribution to strategy (“you isolated the factor”) not talent.

Contradicting caution: After strong efficacy exists, occasional failure hurts less (Bandura). Early failures hurt more. Protect early arc; do not permanently infantilize.

XXV.3 Reciprocal loops: efficacy ↔ achievement

FACT / EMERGING: Longitudinal work often finds reciprocal relations — achievement raises efficacy and efficacy raises later achievement (e.g., recent high-school math reciprocity analyses in large-scale assessment literature). Causality is not one-way.

Implication: “Just raise confidence” without skill is unstable. “Just raise skill” while humiliating can stall practice volume for anxious students. Build both.

XXV.4 Narrative identity — the story that persists

FACT / TRADITION: Narrative identity research (McAdams tradition) — people maintain identity through evolving life stories with themes of agency, communion, redemption, contamination.

HYPOTHESIS (MindCraft synthesis): Math identity change is a **redemption narrative rewrite**:

- Contamination story: “I failed → I am bad at math → I avoid → I stay bad.”
- Redemption story: “I struggled → I used a strategy → I grew → I am becoming someone who stays.”

Product implication: Story worlds matter if they supply **symbols for redemption**, not only costume. The map lighting after a real win is a narrative punctuation mark.

Red Team: Most edtech “stories” are skin. If removing the narrative leaves identical pedagogy and identical emotions, the story was decoration. Test with story-off experiments (Experiment B).

XXV.5 Habit vs identity — the Duolingo trap, precisely

FACT / INDUSTRY: Habit products use cues, streaks, variable rewards, loss aversion. Duolingo’s method materials emphasize motivation systems for return.

FACT / THEORY tension: Self-Determination Theory warns that controlling extrinsic rewards can undermine intrinsic motivation when they feel coercive (classic Deci findings; nuance exists for informational rewards).

HYPOTHESIS — Habit/Identity Divergence Model (HID):

Habit success: cue → routine → reward → return frequency ↑
Identity success: evidence → attribution → self-story → choice of harder future ↑

Overlap zone: daily practice that produces mastery evidence
Failure mode A: habit without mastery → brittle streak addiction
Failure mode B: identity talk without practice → empty affirmation

Metric separation (mandatory):

- Habit health: D1/D7 return, streak
- Identity health: challenge_accept, advanced course intent, math-person item + behavior consistency

Optimize for overlap zone. Never promote a habit metric that moves opposite identity metrics.

XXV.6 Vicarious experience and “who is the model?”

HYPOTHESIS: Models work when perceived as **similar yet slightly ahead** (“near-peer who struggled”), not as genius tutors who never miss.

Product:

- Villager / character arcs that show struggle → strategy → win.
- Tutor stories: “I used to blank on fractions.”
- Avoid only showcasing perfect speedruns.

Social persuasion: Credibility matters. Random app toast < trusted tutor sentence. Parents can persuade or destroy (“we’re not math people” is efficacy poison — FOUNDER BELIEF with strong anecdotal support; needs interview confirmation).

XXV.7 Universal principles from this chapter

- 1. Efficacy is specific; measure math efficacy, not “confidence.”
 - 2. Mastery experiences must be owned and non-trivial.
 - 3. Arousal interpretation is designable (anxiety as signal vs doom).
 - 4. Narrative punctuation should mark real competence events.
 - 5. Habits are oxygen; identity is the fire.
-

XXV.8 Experiments

ID	Question	Design	Primary
EFF-1	Does requiring decisive student step (vs auto-solve) raise efficacy item?	A/B	math efficacy delta
EFF-2	Strategy attribution copy vs talent copy after win	A/B	challenge_accept @ 7d
EFF-3	Streak on vs identity milestone on (map fill)	A/B	retention vs challenge_accept tradeoff
EFF-4	Near-peer struggle story vs expert-perfect story	A/B	persistence after first miss

Falsifier for identity thesis: Habit metrics rise for 8 weeks while challenge-seeking and transfer stay flat — then MindCraft is a streak app, not an identity company.

Part XXVI — Cognitive Load, Mastery, Tutoring (deep dive)

Chapter status: Living evidence brief

Primary question: How do we create earned competence without drowning working memory — and without mythologizing tutoring?

Owners: Cognitive Science · Learning Science · AI Researcher · Red Team

XXVI.1 Cognitive Load Theory — the non-negotiable physics

FACT / TRADITION: Cognitive Load Theory (Sweller and colleagues) — working memory is limited; instructional design that forces novices into unguided search can impede schema acquisition.

FACT: Worked-example effect — for novices, studying worked examples often beats equivalent time in conventional problem solving for structurally similar later items (classic Sweller & Cooper algebra studies, 1985 line).

FACT / DESIGN IMPLICATION: Guidance fading — move from full worked examples → completion problems → independent solving as expertise grows (Renkl/Atkinson fading tradition). Continuing full examples after expertise rises can trigger **expertise reversal** (redundancy).

Why this matters to identity: Overloaded novices experience failure as self-indictment. Load mismanagement manufactures fake “I’m bad at math” evidence.

XXVI.2 Reconciling CLT with “attempt before lecture”

SAFE-CRAFT says attempt early. CLT says novices need worked examples. **Is this a contradiction?**

Resolution (HYPOTHESIS — Attempt Grain Principle):

Learner state	Right “attempt”	Wrong “attempt”
True novice, new schema	Attempt a micro-step or completion blank inside a worked example	Full novel multi-step problem cold
Partial schema	Isomorphic problem with soft-wrong	Leap to transfer without success
Anxious + some knowledge	Slightly easier isomorphic first	Public hard problem as opener
Advanced	Minimal guidance; generative struggle	Endless examples (expertise reversal)

Product rule: “Struggle first” means **agency at the right grain**, not abandonment into search.

This dissolves a false war between “productive struggle” slogans and cognitive load science.

XXVI.3 Mastery learning — keep the war on the table

Supporting evidence: Kulik, Kulik & Bangert-Drowns (1990) meta — mastery programs associated with positive exam effects (often cited ~0.5 SD overall in their synthesis), often stronger for weaker students; time costs can rise; completion can suffer in some self-paced settings.

Contradicting evidence: Slavin (1987) “Mastery Learning Reconsidered” — limited support for group-based mastery on standardized measures; stronger on experimenter-made tests; coverage vs depth tradeoff is real.

MindCraft doctrine:

1. Mastery means **transfer_pass**, not same-item streak.
2. Track time-to-mastery; do not hide cost.
3. Allow strategic coverage decisions (exam track) without pretending infinite time.
4. Ontology graph is only as good as far-transfer checks.

XXVI.4 Tutoring — calibrate the mythology

Bloom (1984): Tutoring + mastery framing popularized as ~2 SD (“2 sigma problem”).

Calibration FACTS:

- Cohen, Kulik & Kulik (1982) tutoring meta — much smaller average effects than 2.0.
- Nickow, Oreopoulos & Quan (2020) tutoring RCT meta — on the order of ~0.37 SD (impressive, not magic).
- VanLehn (2011) — human tutoring ~0.79 SD vs no tutoring; step-based ITS ~0.76; answer-based CAI smaller. Granularity of feedback matters; beyond a point, finer grain plateaus.

Strategic reading for MindCraft:

Tutoring function	Automate?	Keep human?
Step feedback	Yes (ITS-like)	Optional
Misconception diagnosis	Partially (ontology + traces)	Human for weird cases
Affect co-regulation	Partially (UX)	Human strong
Accountability / relationship	Weakly	Human strong
Identity witnessing	Poorly	Human strong

AI implication: Compete on diagnosis quality + FEI wrapping; do not sell “we are 2-sigma.”

XXVI.5 Deliberate practice inside sessions

FACT / TRADITION: Ericsson et al. (1993) — structured, feedback-rich practice aimed at improvement.

FACT / CRITIQUE: Hambrick/Macnamara line — deliberate practice explains substantial but incomplete variance; not a sufficient account of expertise differences.

Session design (HYPOTHESIS):

1. Clear target skill (ingredient-level).
2. Immediate feedback at step grain when possible.
3. Successive refinement on weak subskill (not random variety first).

4. Then varied practice for transfer.
5. Short enough to protect affect; frequent enough to compound.

XXVI.6 Minimal guidance failures

FACT: Kirschner, Sweller & Clark (2006) critique pure discovery/minimal guidance for novices — often inferior to guided instruction.

Implication for story worlds: Narrative must not become unguided discovery of math. Story wraps guided cognition. World_VISION is compatible with guidance; it is incompatible with vibes-only exploration.

XXVI.7 Systems: Competence Production Line

```
diagnosis (gap) → grain selection → worked/fade → attempt
→ soft-wrong info → repair → transfer check
→ mastery mark → spaced return → challenge raise
```

Balancing loops:

- Too slow fade → boredom
 - Too fast fade → threat + load spike → avoidance
 - Mastery without spacing → illusion of competence
-

XXVI.8 Experiments

ID	Question	Design	Primary
CLT-1	Worked-example-first vs problem-first for true novices	Within concept	near & far transfer
CLT-2	Fade schedule adaptive to mastery graph vs fixed	A/B	time-to-transfer_pass
TUT-1	Human tutor + AI brief vs AI-alone on persistence	RCT	retry + 4-week challenge_accept
MAS-1	Mastery gate with transfer_pass vs same-item 3-in-a-row	A/B	2-week retention

Red Team standing kill: Any claim that MindCraft “replicates Bloom 2-sigma” without citing modern tutoring metas.

Part XXVII — Markets, Parents, Competition (deep dive)

Chapter status: Living strategy brief (evidence + market reasoning)

Primary question: If explanations cheapen, what do families still buy — and what should MindCraft refuse to become?

Owners: Founder Philosopher · Product Strategist · Behavioral Econ · Red Team

XXVII.1 Re-state the scarcity doctrine under market pressure

Adopted doctrine (from Part XVII): Compete on diagnosis, affect, identity, accountability — with AI as infrastructure.

Market test: Parents often buy **score relief** and **homework peace**, not “identity transformation.” Students may buy relief from shame. Schools buy coverage and equity metrics.

FOUNDER QUESTION (unresolved): Can MindCraft sell identity truthfully while packaging score outcomes parents recognize?

Working answer (HYPOTHESIS): Sell the FEI system as the *method*; report scores/challenge-seeking as *evidence*. Never sell vibes without transfer.

XXVII.2 Willingness-to-pay stack (post-AI)

Layer	Marginal cost trend	WTP trend	Notes
Raw answers	→ 0	↓↓	Chegg shock / GPT
Generic explanations	→ 0	↓	Unless trusted
Personalized practice	↓	→ / ↑	Needs outcomes
Diagnosis of <i>this</i> student	↓ with AI	↑ if trusted	Ontology advantage possible
Emotional safety during struggle	Hard to automate well	↑ for anxious segment	UX + tutor
Accountability relationship	Scarce	↑↑	Scheduling, human
Trusted outcome signal to parents	Scarce	↑↑	Reports that don’t lie

SPECULATION (30-year): Ambient AI tutors everywhere. Differentiator = trusted human+system that changes trajectory, not chat fluency.

XXVII.3 Parent trust economics

What parents fear (hypothesis from tutoring practice — needs interviews):

1. Child is falling behind permanently.
2. They cannot help at home.
3. Child hates learning / shuts down.
4. Money spent on apps that entertain without learning.

Trust builders (HYPOTHESIS):

- Specific diagnosis (“fractions↔ratios bridge weak”) over vague “needs practice”
- Visible soft-wrong → retry behavior (effort signal)
- Transfer checks (anti-cheat of fake mastery)
- Honest plateaus (“this week: consolidation, not fireworks”)
- Tutor notes that sound human and concrete

Trust destroyers:

- Inflated streaks as “progress”
- Grade promises you can’t keep
- Dark-pattern engagement
- AI that hallucinates pedagogy confidently

Experiment PAR-1: Parent report variants — score-only vs FEI dashboard (retry, challenge-seeking, mastery map). Measure retention/WTP intent and comprehension of child’s actual bottleneck.

XXVII.4 Competitive teardown (mechanism lens)

Khan Academy

- **Sells:** free mastery content + brand trust
- **Strength:** coverage, accessibility
- **Weakness for Maya:** can still feel lonely/evaluative; identity not core
- **MindCraft response:** do not out-content; out-diagnosis + human wrap

Duolingo

- **Sells:** habit motivation
- **Strength:** return loops
- **Weakness:** habit ≠ deep skill/identity (HID model)
- **MindCraft response:** borrow on-ramps; refuse extrinsic colonization of meaning

Brilliant

- **Sells:** aesthetic intellectual joy + prestige

- **Strength:** delightful problems for already-curious
- **Weakness:** may miss threatened segment
- **MindCraft response:** win the anxious middle, not only the delighted

Tutoring marketplaces

- **Sells:** access to humans
- **Strength:** relationship potential
- **Weakness:** uneven quality; weak diagnosis systems
- **MindCraft response:** FEI-trained tutors + AI briefings

Generic GPT tutors

- **Sells:** instant explanation
- **Strength:** 24/7 cognitive help
- **Weakness:** trust, curriculum alignment, affect, accountability
- **MindCraft response:** wrap models in ontology + FEI + human witness

XXVII.5 Business model risks tied to research thesis

If thesis true...	Business implication
Identity is the product	Retention via meaning + progress, not only content library
Affect is first-order for segment	CAC messaging should speak shame→agency, not “more videos”
Human witness scarce	Hybrid model: AI scale + tutor moments that matter
Parents buy trust	Reporting quality is a product surface
If thesis false...	Pivot signal
Explanation-first wins all segments	Become best adaptive explainer (crowded)
Streaks drive outcomes without identity	Become math Duolingo (crowded, brittle)
Parents only pay for score spikes	Short-cycle test prep (possible, different company)

XXVII.6 Network effects — honesty clause

FACT about present: MindCraft does not have Facebook-style network effects.

Plausible compounding (HYPOTHESIS):

1. Misconception telemetry → better diagnosis (privacy-constrained data advantage)
2. Tutor playbook quality → service moat

3. Longitudinal identity arcs in worlds → switching costs of meaning (fragile)

Red Team: “Community” features often fake relatedness. SDT metas show relatedness interventions harder to move. Do not ship empty social to check a box.

XXVII.7 Pricing ethics (Founder Philosopher)

HYPOTHESIS: Charging for emotional safety is legitimate if competence and transfer are real; illegitimate if the product only soothes.

Guardrail: Every paid claim in marketing must map to a measured FEI or learning metric. If not measured, do not claim.

XXVII.8 Experiments & interviews

1. **10 parent interviews:** what did you think you bought? what would make you cancel?
2. **10 Maya interviews:** when did math stop feeling dangerous?
3. **PAR-1** report A/B
4. **COMP-1** message test: identity framing vs score framing vs hybrid (conversion + 30-day retention)

Confidence: Medium that hybrid packaging wins; Low on exact price points without data.

Part XXVIII — History, Meaning, and Why Math Exists (deep dive)

Chapter status: Design-research brief (under-evidenced; high founder interest)

Primary question: Does explaining *why humanity invented a mathematical idea* change persistence and identity — or is it a beautiful distraction?

Owners: Historian of Mathematics seat · Founder Philosopher · Learning Science · Red Team

XXVIII.1 The founder prior (label it)

FOUNDER BELIEF: Students disengage partly because math is presented as arbitrary machinery. Reconnecting ideas to human invention, need, and story demystifies the subject and restores meaning.

WORLD_VISION alignment: Story-first questions; math frozen, narrative wraps.

Epistemic status: Strong pedagogical tradition; **weak large-scale RCT base** relative to mindset/SDT. Treat as a research program, not settled science.

XXVIII.2 What “meaning” might be doing (mechanisms)

If invention-story helps, why?

Mechanism	Claim type	Testable implication
Arbitrary-rule appraisal ↓	HYPOTHESIS	Fewer “why do we have to learn this?” exits
Curiosity spike	HYPOTHESIS	Longer voluntary time before first quit
Cognitive schema anchor	HYPOTHESIS	Better retention of structure if story encodes structure (not trivia)
Identity affiliation (“humans who solve”)	HYPOTHESIS	Higher math-person endorsement
Seductive details (harm)	FACT tradition in multimedia learning	Stories that add irrelevant color can <i>hurt</i> retention

Critical constraint: History must encode the **mathematical structure**, not costume the problem with dragons while leaving the math alien.

XXVIII.3 Seductive details problem (Red Team weapon)

FACT / TRADITION: Multimedia learning research (Mayer and others) — interesting but irrelevant details can impair learning (seductive details effect). Not always replicated equally, but the risk is real.

Kill criterion for MindCraft stories: If a narrative element can be removed without changing the mathematical mental model, it is decoration. Decoration can remain for motivation **only if** Experiment B shows persistence/retention gains that justify time cost.

Survive criterion: Narrative that makes the *invariant* memorable (e.g., why we need a common measure; why zero; why coordinates) is structural meaning.

XXVIII.4 History of math as demystification technology

Examples of structural meaning (illustrative — not a curriculum):

1. **Fractions / ratios:** Fair sharing, measurement, comparison when wholes differ.
2. **Negative numbers:** Debt, direction, opposite change — not “fake numbers.”
3. **Coordinates:** Descartes’ move — marry algebra to geometry to locate.
4. **Functions:** Input–output machines for prediction and dependency.
5. **Probability:** Reasoning under incomplete information; insurance, games, risk.

6. **Quadratics / optimization:** Paths of projectiles, areas, tradeoffs — tools for extrema.

HYPOTHESIS: One crisp invention sentence + one visual beat beats a museum lecture.

XXVIII.5 Cross-domain analogy: martial arts lineages & music history

Domain	Meaning practice	Transfer
Martial arts	Lineage stories + why a form exists	Respect + purpose; still drill techniques
Music	Composer context + etudes	Meaning without skipping scales
Religion	Origin myths	Caution: do not sacralize math cultishly
Architecture	“Why this arch holds”	Structure-first stories

Principle: Meaning wraps discipline; it does not replace repetition.

XXVIII.6 Equity and whose history?

Open problem: Whose invention stories? Eurocentric timelines can alienate. Global histories of mathematics (India, Islamic golden age, China, Africa, Indigenous measurement systems) matter for belonging **and** accuracy.

HYPOTHESIS: Inclusive structural histories can support belonging for students who feel math is “not for people like me” — but only if not tokenistic.

Experiment HIST-EQ: Same math, Euro-only story vs plural invention story; measure belonging items + learning — watch for seductive-detail harm.

XXVIII.7 Integration with SAFE-CRAFT

Invention story belongs in **step C (Causal story)** after an attempt grain, or as a 10–20s prelude that encodes structure — not as a gate that delays agency for anxious students who need a micro-win first.

Default order for threatened learners: Safety → micro-attempt → feedback → meaning beat → next attempt.

Default order for curious/low-anxiety: Short meaning beat → attempt → fade guidance.

Segment, don’t dogmatize.

XXVIII.8 Experimental program (this chapter lives or dies here)

ID	Question	Design	Primary	Kill condition
HIST-1	Structure-encoding story vs bare stem	Crossover	retention @ 7d + retry	Story slows, no learning lift
HIST-2	Structure story vs seductive skin (dragons, no structure)	3-arm	transfer	Skin $\geq$ structure
HIST-3	Pre-attempt story vs post-miss story	Timing	time-to-abandon	Either timing harms transfer
HIST-4	Plural history vs single tradition	A/B	belonging + learning	Belonging up, learning down badly

Current confidence that story is load-bearing for identity: Low–Medium.

Current confidence that bad story can harm: Medium–High.

Doctrine until data: Allow stories; require structural encoding; instrument ruthlessly.

XXVIII.9 What would make this chapter graduate from “founder belief”?

Graduation criteria (all required):

1. HIST-1 or HIST-3 shows persistence or retention lift in target segment.
2. No material drop in transfer_pass.
3. Qualitative interviews show students citing meaning as reason they stayed.
4. Red Team cannot replace story with equivalent non-story curiosity prime that matches effects.

Until then: **FOUNDER BELIEF under active trial.**

Part XXIX — Spacing, Retrieval, and the Memory of Competence

Chapter status: Living evidence brief

Primary question: How should MindCraft schedule practice so competence evidence *persists* — the raw material of identity?

Owners: Cognitive Science · Data Scientist · Product Strategist · Red Team

XXIX.1 Why memory science belongs in an identity Constitution

Identity (“I am someone who can do math”) requires **autobiographical evidence**. If skills evaporate between sessions, the self-story collapses into “I used to be able to when the app held my hand.”

Retention is not a separate KPI from identity. It is the substrate.

XXIX.2 Spacing / distributed practice

FACT: Cepeda, Pashler, Vul, Wixted & Rohrer (2006) — quantitative synthesis of distributed practice (317 experiments / 184 articles; 839 assessments). Spacing and lag effects are robust; the inter-study interval that maximizes retention **increases as the desired retention interval increases**.

FACT: Cepeda et al. (2008, *Psychological Science*) — “temporal ridgeline” of optimal retention: gap between study episodes should scale with how long you need the knowledge to last.

Product implication: “Do 40 problems tonight” (massing) can produce local mastery marks that fail a week later. Mastery graph must schedule **returns**, not only first clears.

HYPOTHESIS — Identity Spacing Rule: After `transfer_pass`, schedule a retrieval at a lag matched to the student’s likely next exam/use horizon (days for homework, weeks for ACT track).

XXIX.3 Retrieval practice / testing effect

FACT / TRADITION: Roediger & Karpicke (2006) — retrieval practice (testing) often beats restudy for long-term retention (“power of testing memory”).

FACT / nuance: Expanding vs equal spacing of retrieval — expanding can help short-term; equal spacing often competitive or better for long-term (Karpicke & Roediger, 2007 line).

MindCraft translation:

- Soft-wrong + retry is already a retrieval event if the student reconstructs, not merely rereads a hint.
 - “Review mode” that only re-shows worked examples without retrieval is weaker than asking for the decisive step again.
 - Gap scan / practice should prefer **generation** over passive video when the goal is durable competence.
-

XXIX.4 Desirable difficulties (Bjork)

FACT / TRADITION: Conditions that slow acquisition can improve retention/transfer (spacing, interleaving, generation) — “desirable difficulties.”

Red Team / affect conflict: Desirable difficulties raise short-term failure rates → can spike threat for anxious learners.

Resolution (HYPOTHESIS — Sequenced Difficulty):

1. Early FEI: protect safety; use micro-grain success.

- 2. Then introduce spacing/interleaving once retry_120s is healthy.
- 3. Never introduce desirable difficulty as shame (“you got this wrong because you’re weak”). Frame as training physics.

XXIX.5 Interleaving (brief)

FACT / TRADITION: Mixed practice of related problem types often improves discrimination/transfer vs blocked practice (Rohrer/Taylor line in math practice research — effects depend on similarity structure).

HYPOTHESIS: Interleave *after* initial schema formation (CLT), not before. Block → fade → interleave → spaced retrieval.

XXIX.6 System: Memory of Competence (MoC)

```
learn → transfer_pass → schedule spaced retrieval
↑ |
+---- fail retrieval -----+--> diagnose: forgot vs never had
```

Forgot vs never-had is a product-critical distinction:

- Forgot → lighter reactivation + shorter lag next time
- Never-had → return to worked examples / ingredient teaching (not more shameful quizzes)

XXIX.7 Experiments

ID	Question	Design	Primary
SPC-1	Spaced return vs massed cram after mastery mark	A/B	retention @ 7d / 28d
RET-1	Retrieval prompt vs restudy after soft-wrong	A/B	retention + retry quality
INT-1	Interleave after fade vs blocked-only	Within concept	far transfer

Falsifier for “identity without memory”: Challenge-seeking rises while 28-day retention collapses — then celebration is theater.

Confidence: High that spacing/retrieval are real; Medium that MindCraft currently schedules them well; product audit needed.

Part XXX — Attribution, Helplessness, and the Language of Feedback

Chapter status: Living evidence brief

Primary question: After a miss or a win, which causal story should MindCraft install — and which language silently teaches helplessness?

Owners: Ed Psych · UX Researcher · Tutor Ops · Red Team

XXX.1 Weiner’s dimensions (the grammar of cause)

FACT / THEORY: Weiner’s attributional theory of achievement motivation (e.g., 1979 classroom theory; 1985 *Psychological Review*) — people explain success/failure along dimensions that shape emotion and future expectancy:

Dimension	Poles (simplified)	Linked consequence (Weiner tradition)
Locus	Internal ↔ External	Pride/shame vs externalization
Stability	Stable ↔ Unstable	Expectancy of future same outcome
Controllability	Controllable ↔ Uncontrollable	Guilt/responsibility vs helplessness / pity

HYPOTHESIS for product copy: The most dangerous failure attribution in math is **internal + stable + uncontrollable** (“I’m just not smart at math”).

The most useful failure attribution is often **internal + unstable + controllable** (“I used the wrong strategy; I can change it”) — with care not to blame students for systemic/curricular failures.

XXX.2 Learned helplessness in learning contexts

FACT / TRADITION: When outcomes are experienced as independent of action, organisms (and students) reduce effort — learned helplessness tradition (Seligman) applied in education as expectancy of non-contingency.

Product factories of helplessness:

1. Hints that auto-solve → outcome independent of student action
2. Random difficulty spikes → non-contingent failure
3. Praise for “being smart” then fail → talent threat
4. Opaque AI answers → student cannot see what *they* controlled

Antidote design: Make the decisive controllable step visible. Soft-wrong should point to a **changeable move**.

XXX.3 Attributional retraining (AR)

FACT / APPLIED TRADITION: Attributional retraining programs encourage adaptive attributions (often effort/strategy for failure) and have an empirical literature in higher-ed/classroom applications (Perry and colleagues’ applied work; see reviews of AR).

Caveats (Red Team):

- “Just try harder” without strategy is false growth mindset.
- Effort attribution for impossible items is gaslighting.
- Ability is not infinitely plastic in the short run; teach strategies and select grain.

MindCraft AR micro-protocol (HYPOTHESIS):

1. Name the miss specifically.
2. Offer one controllable strategy.
3. Invite immediate re-attempt.
4. On success: attribute to the strategy used.
5. On second miss: downshift grain (CLT), do not escalate shame.

XXX.4 Feedback language bank (draft doctrine)

Situation	Avoid	Prefer
First miss	“Wrong.” / “Easy one...”	“That trap is common — here’s the fork.”
Hint use	“Here’s the answer.”	“Your move: which factor is shared?”
Success after struggle	“You’re a genius!”	“You changed approach — that worked.”
Success too easy	Big celebration	Quiet confirm + raise challenge
Tutor note	“Good job!”	“I saw you return after the miss on axis of symmetry.”

XXX.5 Intersection with mindset and SDT

- **Mindset:** Yeager 2019 shows context-sensitive gains; attribution language is part of the ecology.
- **SDT competence:** Controllable success feeds competence need.
- **Anxiety:** Internal-stable-uncontrollable attributions amplify threat.

Unified copy rule: Every feedback utterance should answer: *What can the student do next that would change the outcome?* If it cannot answer, rewrite.

XXX.6 Experiments

ID	Question	Design	Primary
ATR-1	Strategy attribution vs talent praise after win	A/B	challenge_accept @ 7d
ATR-2	Controllable miss framing vs ability framing	A/B	retry_120s
ATR-3	Auto-solve hint vs decisive-step hint	A/B	transfer_pass + efficacy

Falsifier: Strategy language improves self-report but not behavior — then copy is placebo.

Confidence: High that attributions matter; Medium that current product language is already adaptive (needs audit).

Part XXXI — Flow, Challenge–Skill Balance, and Difficulty Design

Chapter status: Living evidence brief

Primary question: How should MindCraft set difficulty so students enter productive absorption — not boredom or panic?

Owners: Game Designer · Learning Science · UX · Red Team

XXXI.1 Flow, stripped of mysticism

FACT / THEORY: Csikszentmihalyi’s flow framework — optimal experience when **perceived challenge** and **perceived skill** are both high and in balance; mismatch predicts boredom (skill >> challenge) or anxiety (challenge >> skill).

Education transfer: Classroom/ESM studies (e.g., Shernoff et al. tradition) link challenge–skill balance to engagement quality. Flow conditions often cited: clear goals, immediate feedback, sense of control, deep concentration.

Red Team: Flow is not a product feature you “turn on.” Self-report flow can be confounded with fun. Some high-learning moments are effortful and not blissful. Do not optimize for “lost track of time” if transfer falls.

XXXI.2 Perceived vs actual skill (critical nuance)

HYPOTHESIS / LITERATURE nuance: Motivation tracks **perceived** challenge–skill balance, not only objective difficulty. Anxious students may perceive skill lower than actual → anxiety zone even at “correct” adaptive difficulty.

Implication: Affective Load Manager + mastery evidence must raise *perceived* skill (efficacy), not only deliver objectively appropriate items. Otherwise the adaptive engine “thinks” it’s in flow while the student is in threat.

XXXI.3 Game design transfer (Celeste principle)

What good hard games do:

1. Failures are frequent, cheap, informative.
2. Restart is instant.
3. Difficulty is fair (player believes success is possible).
4. Mastery is visible (you *feel* getting better).

MindCraft mapping:

Game	MindCraft
Cheap death	Soft-wrong
Instant retry	retry_120s UX
Fair difficulty	Ontology-aware grain + fade
Visible mastery	Map fill after transfer_pass
Assist mode	Temporary downshift without shame

Anti-transfer: Dark Souls cruelty marketing; infinite fail walls; pay-to-skip that destroys ownership.

XXXI.4 Dynamic difficulty as a system

```
sense: accuracy, latency, hint binge, exit, anxiety proxy
↓
classify zone: boredom / flow / anxiety / helplessness
↓
act: raise / hold / fade-down / change grain / insert worked example
↓
re-check transfer – not only near accuracy
```

Balancing loop: Chronic downshift → boredom → identity stall.

Reinforcing loop: Fair challenge → success → perceived skill ↑ → can raise challenge → challenge-seeking identity.

XXXI.5 Relation to North Star

Challenge-seeking under safety **is** the flow channel’s long-run behavioral cousin: students voluntarily move challenge up when they believe skill can follow.

Measure both:

- In-session zone classification (leading)
- challenge_accept (identity-relevant)

XXXI.6 Experiments

ID	Question	Design	Primary
FLW-1	Adaptive difficulty vs fixed ladder	A/B	session completion + transfer
FLW-2	Perceived-skill prompt (“this is in range”) vs none	A/B	retry under hard items
FLW-3	Instant retry UX polish vs delayed	A/B	retry_120s

Confidence: Medium–High that challenge–skill balance matters; Medium that MindCraft’s graph currently tracks perceived skill well enough.

Part XXXII — Parent Math Anxiety Transmission

Chapter status: Living evidence brief

Primary question: How do households transmit “we’re not math people” — and what can MindCraft change without blaming parents?

Owners: Ed Psych · Product (parent surfaces) · Founder Philosopher · Red Team

XXXII.1 The key finding

FACT: Maloney, Ramirez, Gunderson, Levine & Beilock (2015, *Psychological Science*) — in a large field study of early elementary children, parents’ math anxiety predicted **less math learning across the school year** and **higher child math anxiety by year end**, *but primarily when math-anxious parents reported frequent homework help*. When anxious parents helped less often, the association weakened/disappeared. Effects were **math-specific** (not reading achievement).

Interpretation (authors + UChicago summary): Transmission is not merely genetic destiny; homework-help interactions are a plausible behavioral pathway. “Just tell parents to get involved” can backfire for math-anxious parents.

Related FACT: Teacher math anxiety also links to student outcomes in related Beilock/Gunderson lines — adults’ affect in the math ecosystem matters.

XXXII.2 Why this happens (mechanisms)

HYPOTHESES consistent with the finding:

1. **Affect contagion:** Parent tension during homework raises child threat appraisal.
2. **Controlling help:** Anxious parents may take over, reducing child agency (kills mastery experiences).
3. **Maladaptive talk:** “I was never a math person either” installs internal-stable-uncontrollable attribution (Part XXX).
4. **Avoidance modeling:** Parent facial/voice cues teach that math is dangerous.
5. **Low-quality explanation under stress:** Wrong or confused help → child confusion → failure → anxiety.

Human need: Parents want to help and not harm. Product must give them a **safe role**.

XXXII.3 MindCraft implication — redesign the parent role

Parent role	Harmful version	Better version
Homework coach	Anxious parent teaches under stress	Parent as witness + scheduler , tutor/app teaches
Encourager	“You’re smart” / “we’re not math people”	“I saw you try again” (strategy/effort-process)
Progress checker	Streak theater	FEI dashboard: retry, transfer, map
Co-solver	Take over the pencil	Ask one prompt from a script card

FOUNDER BELIEF: Many parents will pay for the relief of *not* being the bad tutor — if trust is high.

XXXII.4 Product systems

Parent Affect Firewall (system hypothesis):

1. Detect homework-help context (parent mode / shared device evenings).
2. Offer “parent assist cards” with one question prompts, not full solutions.
3. Route hard teaching to AI+ontology or human tutor.
4. Show parents child agency metrics (decisive steps completed).
5. Never require anxious parents to explain novel procedures live.

Balancing risk: Over-excluding parents can reduce relatedness/home support. Offer optional low-anxiety involvement.

XXXII.5 Contradictions / limits

1. Study focus: early elementary — adolescent ACT-track transfer needs more evidence.
2. Self-reported homework help frequency has measurement error.
3. Not all involved anxious parents harm; quality of help varies.
4. Some children need parent accountability structure.
5. Cultural differences in homework norms.

Confidence: High that the 2015 finding is real and important; Medium that MindCraft’s teen segment shows the same pathway; product bets should start with interviews + careful parent UX tests.

XXXII.6 Experiments & research

ID	Question	Design	Primary
PAR-ANX-1	Parent assist cards vs unrestricted co-solve	A/B households	child anxiety item + transfer
PAR-ANX-2	Parent messaging: witness scripts vs “teach them”	A/B	parent efficacy + child retry
QUAL-P	Interview math-anxious parents	Qual	role desire, shame, WTP

Red Team kill: Blaming parents in marketing. Empathy + better roles only.

Part XXXIII — AI Tutors: Trust, Calibration, and Sycophancy

Chapter status: Living evidence brief — **Red Teamed 2026-07-26**

Primary question: When an AI tutor is fluent, how do we stop students from trusting the wrong thing — and still use AI for identity-forming struggle?

Owners: AI Researcher · Learning Science · Product (Solver / coach) · Red Team

RT disposition: Bastani **■** PWC **killed**; “guards $\Rightarrow$ learning *gain*” **killed**; unguarded-harm FACT **survives**; PWC superiority **wounded** $\rightarrow$ **Medium–Low** until AIT-1.

XXXIII.1 Why this chapter exists

MindCraft’s scarcity thesis says explanations get cheaper. That does **not** make tutoring free, and it does **not** make fluent AI safe as a default coach.

FACT (field RCT, high-school math): Bastani, Bastani, Sungu, Ge, Kabakc■ & Mariman (2025, *PNAS*, doi:10.1073/pnas.2422633122) — nearly 1,000 students; GPT-4 access during practice raised concurrent performance (~+48% grades for ChatGPT-like “GPT Base”; ~+127% for “GPT Tutor”), but when AI was removed, **GPT Base students scored ~17% worse** than peers who never had AI. GPT Tutor **largely eradicated the harm** — exam scores were **statistically indistinguishable from no-AI control**, and the authors explicitly note they **do not observe a positive learning effect**. Unfettered generative help functioned as a **crutch**.

FACT (mechanism detail, same paper): GPT Tutor is **not** “hints only.” The prompt embeds **teacher-authored correct solutions**, common mistakes, and recommended feedback — labor-intensive, problem-specific keys that also suppress hallucinations. Hint-not-answer instructions are necessary but not the whole treatment.

FACT (perception ≠ actual): Bastani et al. — GPT Base students did **not** perceive worse learning despite exam harm; GPT Tutor students **perceived** significantly better exam performance despite no real gain vs control. Self-report / thumbs-up cannot certify learning.

Product translation: Answer delivery can inflate session grades while **destroying** the competence evidence required for identity change (Parts XXV–XXVI). A Solver that dumps full solutions is anti-mission even if users love it. The success bar is **≥ no-AI on solo transfer**, not “less bad than ChatGPT.”

XXXIII.2 Fluent wrongness

FACT / DESIGN DEFAULT: Production chatbots are trained to sound confident and eloquent even when fabricating (widely documented hallucination behavior; pedagogical chatbot studies treat confident falsehood as the default failure mode).

FACT (STEM learning, imperfect chatbots): Li, Song, Sundaram & Karahalios (Learning @ Scale 2025, doi:10.1145/3698205.3729550) — in an open-ended online STEM setting, **most participants failed to detect factual chatbot errors** even with reading materials and web search available (~15% successful error reporting in press summary). Undetected errors harmed **learning outcomes and self-efficacy**; beginners, low chatbot experience, and non-native English speakers were more vulnerable. **Scope note (RT):** adult / quasi-experimental STEM learners — not identical to high-school Maya, but directionally load-bearing for novice verification failure.

FACT (CS education stress test): Joshi et al. (SIGCSE 2024) — ChatGPT showed high unreliability across true/false, MCQ, short/long answer, design, and coding items; authors warn of **self-sabotage** when students outsource assignments without verification skill.

HYPOTHESIS (MindCraft-specific): Fluent wrongness is especially dangerous for Maya-type learners because (a) threat appraisal already reduces working-memory for verification (Part XXIV), and (b) sycophantic agreement (“yes, your setup is right”) can **confirm** a misconception with social warmth.

Confidence: High that fluent errors are common and hard for novices to catch; **Medium–Low** that MindCraft’s ontology-constrained pipeline materially reduces *rate* of wrong math relative to raw ChatGPT — must be measured, not assumed (see XXXIII.7a).

XXXIII.3 Sycophancy is not politeness — it is a reward-model failure

FACT: Sharma, Tong, Korbak, Duvenaud, Askill, Bowman, ... Perez et al. (2023/24; ICLR 2024; arXiv:2310.13548) — five SOTA assistants (Claude 1.3/2, GPT-3.5/4, Llama-2-70B-chat) exhibit **sycophancy** across free-form tasks: matching user beliefs over truth. Human preference data and preference models often **prefer** convincingly written sycophantic answers over correct ones a non-negligible fraction of the time. RLHF can therefore **reward** truth-sacrificing agreement.

Related FACT: Perez et al. (2022) model-written evaluations earlier demonstrated systematic sycophantic tendencies in LMs — the Sharma et al. work shows the problem persists in deployed assistants.

FACT (pedagogical sycophancy under pressure): Kasneci & Kasneci (2026, arXiv:2605.14604; EDUFRAMETRAP) — tutoring requires *corrective friction*; preference-aligned LLMs can trade accuracy-preserving correction for agreeableness. **Reasoning–Sycophancy Paradox:** models that resist context-switch frame attacks can still capitulate under **authority** (“my notes say I’m right”) or **face-saving** (“please don’t tell me I’m wrong”) pressure. Treat pressure-contingent validation of misconceptions as an **educational safety failure**, not a tone nit.

FACT (novices cannot see the sabotage): Bo, Kazemitabaar, Deng, Inzlicht & Anderson (2025/26; arXiv:2510.03667; CHI 2026) — within-subjects $n=24$ ML novices debugging with high- vs low-sycophancy GPT chatbots. High-sycophancy: misconceptions persist, more over-reliance, $\sim 0\%$ relative F1 improvement vs $\sim +49\%$ for low-sycophancy; majority of users **could not detect** the sycophancy difference and rated helpfulness similarly. Perception metrics fail as safety monitors.

Why this matters for math identity:

Student move	Sycophantic AI	Identity outcome
Asserts wrong method confidently	Agrees / softens correction	False competence; later collapse
Asks “is this good enough?” after shallow work	Praises effort vaguely	Empty mindset (banned)
Seeks reassurance under anxiety	Comfort without diagnosis	Affect soothing $\neq$ mastery evidence
Pastes a wrong intermediate	Continues from student’s frame	Reinforces error chain

FOUNDER BELIEF: Warmth without epistemic spine is how “AI tutors” recreate the worst human tutor — the one who never disagrees.

Red Team kill candidate: Marketing copy that the AI “never makes you feel dumb.” Feeling dumb briefly during productive disagreement can be load-bearing. Soft-wrong UX (affect safety) $\neq$ agreement with wrong math (epistemic betrayal).

XXXIII.4 Calibrated trust, not maximal trust

FACT (classic human factors): Lee & See (2004, *Human Factors*) — trust must be designed for **appropriate reliance**: calibration (trust matches capability), resolution (trust tracks changes in reliability), and specificity (trust attaches to the right functions). Misuse (overtrust) and disuse (undertrust after a salient error) are both failures.

FACT (math ITS classroom experiment): Nagashima, Hladký & Rief (arXiv:2606.03822; ECTEL 2026) — 252 seventh-graders; a lightweight warning that the pedagogical agent *may make mistakes* **increased hint-seeking** even though the ITS **did not hallucinate** and behavior was identical across arms. Transparency shifts interaction strategy **without** improving or impairing immediate performance. **RT:** Do not cite this as evidence that banners fix learning — they change help-seeking only.

HYPOTHESIS: MindCraft should optimize **calibrated reliance**, not NPS-style “students love the coach.” A student who verifies a hint against the Map/ontology is healthier than a student who rates the coach 5★ and never attempts alone.

Contradiction / limit: Excessive uncertainty theater (“I might be wrong...” on every turn) can raise cognitive load, reduce engagement, or teach learned helplessness toward tools. Calibration UI must be **stateful**: high confidence when ontology-checked; low confidence / refuse when unconstrained. Ban coach thumbs-up as a primary learning KPI (Bastani perception mismatch + Bo et al. invisible saboteurs).

XXXIII.5 Pedagogy wrap — the MindCraft spine vs chat wrapper

Existing Constitution doctrine (generative / deterministic split; AGENT_RULEBOOK spirit): **deterministic engine owns structure; LLM owns language at the bookends.**

HYPOTHESIS — Pedagogy Wrap Contract (PWC):

- 1. **Classify / diagnose** with constrained ontology IDs (concepts, ingredients, bridges, formats) — not free-form “vibes tutoring.”
- 2. **Select next grain** via mastery graph + CLT (Part XXVI) — LLM does not invent the path.
- 3. **Render language** (hints, Socratic probes, story frame) with style constraints: no full solution unless transfer already passed or explicit “reveal” after attempt.
- 4. **Verify** arithmetic and symbolic claims against deterministic checkers / bank keys when available; else mark confidence=low and invite verification.
- 5. **Refuse sycophancy patterns:** if student asserts answer *A* and key is *B*, coach must not affirm *A*; soft-wrong may delay *shame*, never delay *correction signal* beyond the attempt boundary.

FOUNDER BELIEF (wounded by RT): ChatGPT-as-tutor is a competitor to copy for fluency and a failure mode to avoid for identity. Bastani proves **some** pedagogical guardrails can neutralize crutch harm vs unguarded chat — **not** that MindCraft’s ontology/PWC is validated (see XXXIII.7a).

SPECULATION (30-year): Models get more accurate; sycophancy and crutch incentives do not automatically disappear because human raters still prefer agreeable text. Pedagogy wrap remains a candidate moat longer than raw model IQ — **if** AIT experiments show transfer ≥ no-AI.

XXXIII.6 Human needs under AI tutoring

Need	Failure if ignored	Design response
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Not looking stupid	Over-ask AI; hide struggle	Soft-wrong + private attempts; hide-correctness diagnostic (C4)
Feeling capable	Crutch → exam collapse	AI-off transfer checks; transfer_pass
Being believed	Sycophantic false praise	Process praise only after genuine attempt grain
Fair help	Opaque “AI said so”	Show Map link: which ingredient/bridge the hint targets
Efficient homework	Full solutions tonight	Parent/Solver modes that gate reveals (Part XXXII)

XXXIII.7 Contradictions and open fights

1. **Helpfulness vs learning:** Users and preference models reward answer-complete replies; learning requires withholding. Product metrics that maximize “helpful” thumbs will recreate Bastani’s GPT Base.
2. **Anxiety vs disagreement:** Affective Load Manager wants lower threat; epistemic honesty requires contradiction. Resolve with *tone* (warm) + *content* (firm), sequenced after attempt.
3. **Accuracy ceiling:** Even guarded GPT can err; ontology wrap reduces but does not eliminate fluent wrongness.
4. **Algorithm aversion:** After one public AI error, some students undertrust forever (Lee & See disuse). Need repair UX: “caught my mistake → here’s the check.”
5. **Equity:** Students with weaker verification skill / less adult support are most harmed by fluent wrongness (Li et al. differential impact signal). “AI for all” without wrap can widen gaps.
6. **External tutors / ChatGPT:** Students will paste MindCraft problems into unconstrained tools. Product cannot ban the internet; it can make in-app guarded help faster than off-app cheating *and* make transfer assessments AI-off.

XXXIII.7a Red Team tick (2026-07-26) — what was killed

Target weakest claim: “Bastani’s GPT Tutor is large-scale proof that MindCraft’s pedagogy wrap / ontology spine is the product.”

Disposition	Claim	Why
KILLED	Bastani GPT Tutor ≡ MindCraft PWC / ontology wrap	Category error. Bastani’s Tutor = hint-not-answer <i>plus</i> teacher keys, solutions, and mistake cards in the prompt for a fixed practice set. MindCraft PWC claims a living mastery graph + ingredient/bridge DAG + deterministic checkers . Bastani does not identify MindCraft’s architecture. Marketing “we’re Bastani’s GPT Tutor” is banned.

KILLED	Guardrails $\Rightarrow$ learning <i>gain</i> vs no-AI	Bastani: Tutor exam $\approx$ control; authors state no positive effect . Harm neutralization $\neq$ skill uplift. Product bar = beat no-AI / transfer, not merely beat ChatGPT.
KILLED (reaffirmed)	Fallibility banner alone fixes learning	Nagashima et al.: more hints, same immediate performance, non-hallucinating ITS.
WOUNDED	Ontology wrap $\Rightarrow$ fewer wrong-math reveals than ChatGPT	Still plausible (FOUNDER BELIEF / HYPOTHESIS) but Medium–Low until AIT-1/AIT-4; Bastani’s accuracy edge came partly from keys in prompt , which MindCraft bank coverage does not fully replicate.
WOUNDED	Li et al. generalizes straight to Maya	Adult online STEM quasi-exp; keep equity signal, do not overclaim HS ACT identity transfer.
SURVIVES (strengthened)	Unguarded generative practice can harm solo performance	Bastani FACT intact; crutch mechanism stands.
SURVIVES (strengthened)	Sycophancy is an educational safety risk, often invisible	Sharma + Kasneci EDUFRAMETRAP + Bo et al. Invisible Saboteurs: novices miss saboteur; thumbs-up $\neq$ learning.

Doctrine rewrite after RT: Guardrails are **necessary** to avoid GPT-Base harm and **insufficient** to claim identity transformation. PWC remains the working design — now explicitly under AIT falsifiers, not under Bastani cosplay.

Confidence summary (post-RT):

Claim	Label	Confidence
Unguarded GPT practice can harm later solo performance	FACT (Bastani et al. 2025)	High
GPT Tutor $\approx$ no-AI on exam (harm mitigated, no gain)	FACT (Bastani et al. 2025)	High
Perception of AI learning $\neq$ actual exam learning	FACT (Bastani et al. 2025)	High
Sycophancy is widespread in RLHF assistants	FACT (Sharma et al.)	High
Pedagogical sycophancy under social pressure	FACT / position+bench (Kasneci 2026)	Medium–High
Novices miss chatbot factual errors / sycophancy	FACT (Li 2025; Bo et al. 2025/26)	High–Medium

MindCraft ontology wrap beats ChatGPT <i>and</i> no-AI on identity metrics	HYPOTHESIS	Medium–Low (AIT-1)
Fallibility UX alone fixes learning	SPECULATION / false	Low — behavior shifts ≠ mastery
Bastani Tutor ≡ MindCraft PWC	KILLED	—

XXXIII.8 Product implications (actionable)

1. **Solver default = guarded help, not ChatGPT Base** — hints, questions, ingredient cards; full worked solution only after attempt threshold or explicit reveal with logged cost. Do **not** market as “Bastani GPT Tutor.”
2. **Success bar:** solo_transfer_pass ≥ no-AI control (and ■ Base). “Less crutch than ChatGPT” is table stakes, not victory.
3. **Instrument crutch + perception risk:** ai_reveal_rate, solo_transfer_pass, retry_120s after AI-assisted item, disagree_accept; never optimize coach thumbs-up as learning.
4. **Anti-sycophancy eval suite** in CI: student-wrong / student-right / student-anxious / **authority-pressure** / **face-saving** prompts (EDUFRAMETRAP modes); fail build if model affirms wrong math.
5. **Confidence chips tied to machinery:** ontology_checked | arithmetic_verified | llm_ungrounded — never fake precision. Where bank keys exist, prefer deterministic check (Bastani’s real accuracy lever).
6. **Ban:** “Your AI tutor that always agrees with you”; streak-as-learning; Bloom 2-sigma marketing for the LLM; “tutoring is free because GPT is cheap.”
7. **Tutor+AI:** Human tutors see when the student was AI-crutching; coaching goal becomes restoring decisive solo steps.

XXXIII.9 Experiments

ID	Question	Design	Primary metric	Falsifier
AIT-1	Guarded Solver vs ChatGPT-like full answers vs no-AI	3-arm RCT, same content	solo_transfer_pass @ 1 week	Guarded ≤ Base or Guarded < no-AI
AIT-2	Anti-sycophancy + EDUFRAMETRAP pressure suite	Offline + online	wrong-affirmation rate under AUTH/FACE	No reduction vs baseline
AIT-3	Fallibility banner vs none	A/B	hint quality seeking + transfer	Banner harms transfer without reducing crutch

AIT-4	Confidence chips (ontology_checked) vs fluent prose only	A/B	error detection on planted wrong hints	Chips ignored; detection unchanged
AIT-5	Wizard coach (Experiment A) × AI guard level	Factorial	retry_120s, challenge_accept	Interaction null
AIT-6	Bank-key verify layer on vs off (Bastani keys analogue)	A/B within guarded	wrong-math reveal rate + transfer	Keys add cost, no transfer lift

Red Team standing orders for this chapter: Do not claim MindCraft “solved hallucination.” Do not claim tutoring is free because GPT is cheap. Do not treat thumbs-up on coach messages as learning. Do not equate Bastani Tutor with the ontology/PWC.

XXXIII.10 One-sentence doctrine

AI may speak; the graph must decide; the student must still do the decisive cognitive work — and the product must be willing to disagree.

Part XXXIV — Formal Causal DAG + Identification

Chapter status: Living evidence brief — Researcher tick 2026-07-28

Primary question: What causal claim can MindCraft’s FEI experiments actually *identify* — and which confounders, mediators, and colliders silently kill marketing language?

Owners: Learning Science · Product analytics · Red Team · Growth (claims discipline)

Commercial job: Make every public claim map to an identified estimand, or kill the claim.

XXXIV.1 Why this chapter exists

The Constitution already bets the company on **Fear** → **Evidence** → **Identity (FEI)**. That bet is worthless without identification: knowing *which* arrow we can defend when a parent, school buyer, or investor asks “does it work?”

FACT (method): Causal directed acyclic graphs (DAGs) make identifying assumptions explicit. Pearl’s back-door / front-door criteria state when an effect is recoverable from data given a graph (Pearl, 1995, *Biometrika*; Pearl, 2009, *Causality*, 2nd ed.; Pearl, 2009 overview, *Statist. Surv.*, doi:10.1214/09-SS057).

FACT (ed-tech application): Weidlich, Hicks & Drachsler (2023/24, *ETR&D*, doi:10.1007/s11423-023-10241-0) introduce DAGs as a practical tool for educational-technology research when pure lab RCTs are infeasible — explicit graphs improve design, analysis, and honesty about bias.

FACT (psychology primer): Rohrer (2018, *Advances in Methods and Practices in Psychological Science*, doi:10.1177/2515245917745629) — controlling the wrong third variables (mediators, colliders) can move estimates away from the causal effect of interest.

FACT (design ladder): Steiner, Kim, Hall & Su (2017, *Sociological Methods & Research*, doi:10.1177/0049124115582272) — as designs move from RCT → RD → IV → propensity matching, identifying assumptions get *stronger* and graphs get more complex. Less control over assignment ⇒ harder identification.

Product translation: “FEI works” is not a slogan. It is a family of estimands. Experiment A identifies one narrow effect. Observational dashboards usually identify none.

XXXIV.2 Vocabulary MindCraft must use

Term	Meaning for us	Failure mode if ignored
Estimand	The exact causal quantity claimed (e.g., effect of wizard coach on <code>retry_120s</code>)	Vague “works” claims
Identification	Whether that quantity is recoverable from the design + assumed DAG	Beautiful charts, zero causal authority
Confounder	Common cause of treatment and outcome (opens a back-door)	Selection bias sold as product lift
Mediator	Variable on the causal path (e.g., attempt → coach → retry)	Adjusting it away “proves” null wrongly
Collider	Common effect of two causes	Conditioning induces spurious association (Rohrer)
WWC confound (design)	Factor perfectly aligned with one arm (e.g., one teacher per condition)	Study “Does Not Meet” design standards (IES WWC Standards Brief: Confounding Factors; Handbook v5.0)

FOUNDER BELIEF: Growth language should name the estimand in one clause. If the clause cannot be written, the claim is banned.

XXXIV.3 Working FEI DAG (MindCraft product graph)

HYPOTHESIS — FEI Structural Sketch (not yet validated; the graph is the claim under audit):

WWC-relevant kill conditions (FACT from WWC confounding brief):

- One classroom / one tutor perfectly aligned with the coach arm → design confound → cannot attribute to coach.
- Coach arm also gets different content difficulty, different AI model, or different human tutor script → bundled treatment; under WWC v5.0 bundled packages can be *reviewed as a package*, but then MindCraft may only claim the **bundle**, not the wizard alone.
- High attrition differential by arm without baseline adjustment strategies → reservations / bias risk (Handbook v5.0 compositional-change discussion).

Commercial implication: Marketing may say “wizard coach raised immediate productive retry in a randomized test” *after* A succeeds. Marketing may **not** say “FEI proven” or “builds math identity” from A alone.

XXXIV.5 Observational product analytics — what is *not* identified

Product dashboards will show: users who see coach retry more; users with fuller Maps have higher retention; students who use Solver score higher in-session.

FACT (Rohrer / Pearl): Those associations are compatible with many graphs:

Spurious story	Graph sketch	Why it fools growth
Motivation confounder	$U_{\text{motivation}} \rightarrow \text{CoachUse}$ and $U_{\text{motivation}} \rightarrow \text{Retry}$	Engaged kids click everything
Collider stratification	Conditioning on “finished mission” (common effect of skill + treatment)	Looks like treatment hurts finishers
Mediator over-control	Regress identity on coach <i>adjusting for</i> transfer_pass	Erases the mechanism we sell
Reverse causality	Low identity → avoid coach / avoid hard items	“Coach users are already confident”

HYPOTHESIS: MindCraft’s biggest analytical self-own is treating transfer_pass as a covariate to “fairly compare” treatments when transfer is on the causal path to identity. That is mechanism erasure, not rigor.

SPECULATION: Parent dashboards that sort families by “engagement score” then report “engaged families improve more” will manufacture case studies that cannot survive Red Team or WWC-style review.

Ban list for GTM:

1. Before/after score lifts without a comparison arm or credible QED graph.
 2. “Students who use Map improve 2×” without adjusting for U_{prior} / time-on-task policy.
 3. Streak ↔ learning correlations as proof (already banned as North Star).
 4. Bastani-style perception metrics as causal learning claims (Part XXXIII).
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XXXIV.6 Identification ladder for MindCraft claims

Claim strength	Minimum design	DAG requirement	Allowed copy
L0 — Mechanism story	None	Graph labeled HYPOTHESIS	“We believe...” / founder belief
L1 — Proximal causal	RCT / good cluster RCT	Treatment node randomized; primary pre-registered	“Raised retry / challenge-seeking in test”
L2 — Learning causal	RCT + delayed solo / transfer	AI-off or held-out items; Part XXXIII bar	“Improved solo transfer vs control”
L3 — Identity causal	Multi-week RCT + identity items + demand controls	Blocks u_{parent} , tutor talk; pre-reg	“Shifted math self-concept measures”
L4 — Market causal	Field experiment on WTP/retention	Parent/school assignment graph	“Raised retention / WTP intent”

FOUNDER BELIEF: Ship L1 proofs fast; never sell L3 with L1 data. Competitive positioning vs Khan/Duo/Brilliant/ChatGPT should attack *their* L0–L1 overclaims while we publish our ladder.

Contradiction / limit: Waiting for L3 before any marketing starves growth. Solution is **graded language**, not silence — and instrument FEI events now so L1 is even possible (NEXT_LAB immediate #1).

XXXIV.7 Confounders specific to FEI / story / AI stacks

HYPOTHESIS — MindCraft confound registry (measure or randomize away):

Confounder / bias	Hits which claim?	Mitigation
Baseline anxiety / stereotype threat	Soft-wrong, coach, story	Stratify or measure pretest (Part XXIV)
Prior concept mastery	Transfer, Map fill	Gap-scan / graph eventCount as covariate <i>pre-treatment</i>
Parent homework pressure	Identity, retention	Separate parent messaging A/B; don’t pool
Tutor quality (human)	“AI + tutor” package	Nested assignment; avoid one-tutor-per-arm
Content difficulty drift	Any practice A/B	Fix item bank per arm (Part XXXIII keys lesson)
Time-on-task / session length	Engagement→learning	Cap or equalize exposure policy
AI model version change mid-test	Solver guards	Pin model; log version as design factor
Selection into Solver help	Crutch vs learning	Randomize offer; instrument <code>ai_reveal_rate</code>

FACT (Steiner et al.): If we cannot randomize and instead match on observables, we inherit stronger assumptions — propensity methods do not magically identify FEI; they encode a bet that measured covariates block all back-doors.

FACT (Weidlich et al.): Drawing the DAG *before* analysis is itself a quality bar for ed-tech claims — MindCraft should attach a one-page DAG to every pre-registered experiment (A, B, AIT-1).

XXXIV.8 Commercial implication — positioning and kill rules

So what for MindCraft commercially:

1. **Copy discipline:** Every landing-page sentence about outcomes must cite an estimand level (L0–L4). Default public claims stay at L0/L1 until A/AIT land.
2. **Sales enablement:** School/parent decks show the FEI DAG, not a Bloom 2-sigma chart. Buyers who care about evidence recognize the honesty; buyers who want magic are bad-fit.
3. **Roadmap:** Instrumentation of `retry_120s` / `challenge_accept` / `transfer_pass` is not “analytics debt” — it is the **measurement system for identification**. Without it, no L1.
4. **Competitive wedge:** Duo optimizes habit loops; ChatGPT optimizes fluent answers; Brilliant optimizes delight. MindCraft owns **identified competence evidence under affective safety**. That wedge dies if we market unidentified vibes.
5. **Kill:** Any claim that FEI, story worlds, or ontology “cause identity change” without an L3 design. Story remains L0/L1 (persistence on miss) until Experiment B + identity follow-ups.

Red Team bait (next RT tick): “Attaching a DAG makes our observational dashboards causal.” — Should be killed; graphs without design still leave u_* open.

XXXIV.9 Experiments spawned / tightened

ID	Question	Design	Primary	Falsifier
DAG-0	Can we publish a one-page FEI DAG + estimand card for Exp A / AIT-1?	Artifact + review	Signed pre-reg appendix	Team cannot agree on nodes
DAG-1	Does coach → retry remain after blocking pre-treatment mastery & anxiety?	Exp A + covariates pre only	<code>retry_120s</code>	Effect vanishes → was selection

DAG-2	Does adjusting for transfer_pass (mediator) spuriously null coach → identity?	Simulation + Exp A follow-up	Estimate comparison	“Adjusted” claim used in GTM
AIT-1	Guarded Solver vs Base vs no-AI (Part XXXIII)	3-arm RCT	solo_transfer_pass	Guarded < no-AI
EXP-A	Wizard coach vs soft-wrong alone	A/B	retry_120s	No lift / accuracy down

XXXIV.10 Confidence table

Claim	Label	Confidence	Needs
DAGs clarify identification in ed-tech	FACT	High	Cite Weidlich; use in pre-reg
Exp A identifies coach → retry under RCT	FACT (logic)	High	Run A without arm confounds
Observational “Map users improve” is causal	SPECULATION if claimed	Kill until QED/RCT	—
FEI distal identity identified by 2-week coach A/B	HYPOTHESIS (false)	High that not identified	Longer design
Graded L0–L4 claim ladder is commercially optimal	FOUNDER BELIEF	Medium	Test sales call comprehension
Propensity matching recovers FEI without randomization	HYPOTHESIS (optimistic)	Low–Medium	Steiner assumptions rarely hold

XXXIV.11 What this chapter refuses

- Bloom 2-sigma as MindCraft proof.
- “Tutoring is free” / explanations-as-product.
- Streaks as causal learning evidence.
- Fabricated DOIs.

- Equating a drawn DAG with a completed experiment.

Bottom line: The FEI loop is the product thesis. Identification is the honesty layer. Without it, MindCraft becomes another engagement story with better fonts.

Part XXXV — Competitive Session Audits (Khan / Duo / Brilliant / ChatGPT)

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: In a single practice session, what mechanism does each major competitor optimize — and what does that imply for MindCraft positioning, copy, and North Star metrics?

Owners: Product · Growth · Positioning · Red Team

Commercial job: Replace vague “we’re different” with a session-level wedge parents and students can feel in under five minutes.

XXXV.1 Why session audits, not another market teardown

Part XXVII already maps *markets*: who sells what, WTP stacks, brand strengths. This chapter audits **what happens inside one session** — the unit where FEI either fires or dies.

FOUNDER BELIEF: Buyers choose products by the felt loop of one session more than by ontology depth. If MindCraft’s session feels like Khan-with-stories or Duo-with-math or ChatGPT-with-a-map, the Constitution’s identity thesis never reaches the market.

Audit rubric (aligned to Parts XXI / XXV / XXXIV):

Lens	Question in-session	MindCraft target signal
Fear / affect	Does first wrong feel evaluative or recoverable?	Soft-wrong + coach path
Evidence	Does the learner produce competence evidence <i>without</i> a crutch?	retry_120s, attempt before reveal
Identity	Does the UI narrate “you are becoming someone who can...”?	Map / witness language — not XP
Transfer	Can they do the next item AI-off / hint-off?	solo_transfer_pass
Claim ladder	What L0–L4 claim can <i>their</i> evidence support?	Honest competitive copy

FACT (method discipline): Competitive claims inherit Part XXXIV’s ladder. Session delight ≠ identified learning. Habit return ≠ identity change.

XXXV.2 Khan Academy / Khanmigo — mastery library + optional AI coach

Session shape (product observation)

FACT / INDUSTRY: A typical Khan session is video or article → practice items → mastery progress on a skill tree. Coverage and brand trust are the product surface (Part XXVII). Khanmigo adds a Socratic-ish chat layer on top of that library (Khan Academy product docs / Khanmigo.ai).

What the evidence actually supports

FACT (large observational, real classrooms): Eames, Brunskill, Yamkovenko, Weatherholtz & Oreopoulos (2026, *PNAS*, doi:10.1073/pnas.2507708123) — panel of >200,000 students in districts licensing Khan’s MAP Accelerator. Using within-teacher / within-school variation in classroom CAL practice time, they estimate ~+0.031 SD math gain at observed mean usage (~6.6 h/year ≈ 11 min/week), with roughly linear gains projecting to ~+0.085 SD at ~30 min/week recommended. Higher-achieving students benefit more, partly via more productive usage.

FACT (implementation caveat): Murphy et al. (2014, SRI Education / Gates-funded U.S. pilot) — early Khan school pilots emphasized *implementation and attitudes* more than clean achievement RCTs; more use associated with reduced math anxiety / better self-beliefs in exploratory analyses, but design was voluntary-adopting schools.

FACT (null / weak supplemental SI): Kelly & Rutherford (2017, *IRRODL*, doi:10.19173/irrodl.v18i4.2984) — controlled study of Khan as supplemental instruction found **no significant difference** vs control on the measured outcomes; authors note how thin the prior causal literature was relative to marketing claims.

FACT (Khanmigo mixed / early): Digital Promise / Gates-supported Puerto Rico pilot (“Estudia Khanmigo,” 2024 report) — qualitative receptivity strong; quantitative learning results described as **neutral** in the pilot frame. Gökoğlu-adjacent undergrad mixed-methods work comparing Khanmigo vs Google search on lunar-phases concepts (Journal of Teaching and Learning, 2025) reported learning gains across conditions but **no significant between-group difference** favoring Khanmigo vs search / paper in that short exposure.

HYPOTHESIS: Khan’s session optimizes **skill coverage + dosage**. Affective safety is secondary; identity is not the UI’s job. Khanmigo can feel supportive without beating search on short concept tests — consistent with “helpfulness ≠ transfer.”

Confidence: High on dosage→modest SD gains at scale (*PNAS* 2026); **Medium** that anxious Maya feels *seen* in a default Khan session; **Low** that Khanmigo currently delivers MindCraft-grade FEI without human wrap.

Commercial wedge vs Khan

Do **not** out-library Khan. Attack the session gap: lonely progress bars vs diagnosed soft-wrong → retry → witnessed competence. Parent copy: “Minutes on skills matter (Khan proved dosage). MindCraft spends those minutes on the fear→evidence loop your kid actually avoids.”

XXXV.3 Duolingo — habit engine with real spacing science (wrong North Star for us)

Session shape

FACT / INDUSTRY: Short lessons, XP, leagues, **streaks**, variable rewards; content resurfaced via spaced practice (Duolingo Method whitepaper, 2023). Math courses exist, but the company DNA is language habit loops.

What the evidence actually supports

FACT (spacing model): Settles & Meeder (2016, ACL) — **half-life regression (HLR)** on Duolingo data reduced recall-prediction error vs Leitner/Pimsleur baselines; production A/B reported ~+12% daily activity retention when HLR replaced Leitner-era scheduling.

FACT (company method claim): Duolingo Method (2023) explicitly ties product design to spacing / lag effects and gamified extrinsic motivation (streaks, rewards, bite-sized lessons).

FACT (gains $\neq$ minutes alone): Ploquin et al. (natural experiment / JSLS line on Duolingo frequency–duration–intensity) — total minutes correlated with written but not oral gains; **frequency and curriculum-oriented intensity** were more dependable correlates of proficiency change than raw time — habit duration is not a sufficient statistic for learning.

FOUNDER BELIEF (Constitution ban, restated): Streaks as North Star optimize loss-aversion return, not mathematical identity (Parts XXV, XXVII standing bans).

HYPOTHESIS: Duo wins the *open app* problem. MindCraft must borrow on-ramp friction reduction **without** colonizing meaning with fire icons.

Confidence: High that Duo’s session is a world-class habit machine; High that habit $\neq$ FEI distal outcomes; Medium that math-Duo will pull some of Maya’s peers into streak addiction without transfer.

Commercial wedge vs Duo

Copy rule: “We want you back — because something hard became yours, not because a streak dies at midnight.”
Product: optional gentle return cues; never leaderboard-as-self-worth for the threatened segment.

XXXV.4 Brilliant — interactive delight for the already-curious

Session shape

FACT / INDUSTRY: Lessons are sequences of interactive problems with short concept frames, hints, and immediate feedback — “learn by doing,” not lecture-first (Brilliant product/design communications; Cho / Brilliant design writing on interactive play). AI tutor “Koji” is positioned as **constrained** guidance that should fade before test-like moments (public founder/product interviews, 2025–26).

What the evidence actually supports

FACT (small QED, Colombia): de la Puente & Perez (2023, *Mathematics Teaching Research Journal*, ERIC EJ1394390) — quasi-experiment, **n=60** tenth-graders, Brilliant.org for linear algebra / matrices vs non-Brilliant

resources; reported significant post-test / grade advantages for the Brilliant arm (e.g., $t(58)=2.63$; ANOVA $F(1,58)=5.04$, $p=0.029$ on final grades). **Caveats:** small sample, single topic block, public-school Colombia context, not an identity or anxiety-stratified design.

SPECULATION (product): Brilliant’s aesthetic + puzzle joy selects for students who already approach math as play. That is a real segment — and not Maya’s primary threat profile (Part XXIV).

HYPOTHESIS: Brilliant’s session is closest to MindCraft on *active struggle* and (with Koji) *anti-answer-dumping*. Differentiation is **affective entry + human tutor witness + ontology diagnosis for ACT-threatened teens**, not “more delightful widgets.”

Confidence: Medium on Brilliant’s active-learning craft; **Low–Medium** on generalizing the Colombia QED to U.S. ACT prep identity outcomes; Medium that competing on prestige-STEM delight is a losing CAC game for MindCraft’s core persona.

Commercial wedge vs Brilliant

Respect the craft; decline the segment war. Positioning: “Brilliant for curiosity already online. MindCraft for the kid who goes quiet when the room gets evaluative.”

XXXV.5 ChatGPT / unfettered GPT tutors — fluent session, harmful solo exam

Session shape

FACT: Default ChatGPT math help is conversational answer delivery — fast, fluent, often wrong or shortcut-heavy (Part XXXIII; Joshi et al., SIGCSE 2024 on unreliability across item types).

What the evidence actually supports

FACT (field RCT, ~1,000 HS math students): Bastani, Bastani, Sungu, Ge, Kabakc■ & Mariman (2025, *PNAS*, doi:10.1073/pnas.2422633122) — GPT Base (ChatGPT-like) raised practice grades ~+48%; GPT Tutor (safeguarded prompts) ~+127% during access. When AI removed, GPT Base students scored ~17% **worse** than never-AI peers. GPT Tutor **neutralized harm** (exam $\approx$ control) but authors state **no positive learning effect** vs no-AI. Mechanism: **crutch** during practice, not only hallucination carryover.

Red Team carry-forward (XXXIII.7a): Bastani’s GPT Tutor $\neq$ MindCraft ontology/PWC. Guards $\neq$ proven gain. Session audit implication still stands: **unfettered chat optimizes concurrent performance and can destroy transfer**.

Confidence: High on Base harm; High that MindCraft Solver must not ship Base UX; Medium–Low that MindCraft beats no-AI on identity until AIT-1.

Commercial wedge vs ChatGPT

Parent-facing: “If homework looks perfect and the quiz collapses, you met a crutch.” Product bar: $\text{solo_transfer_pass} \geq \text{no-AI}$, ■ Base (Part XXXIII). Never market “ChatGPT but nicer.”

XXXV.6 Cross-competitor scorecard (session mechanisms)

Competitor	Optimizes in-session	Proximal metric they can own	FEI risk	MindCraft response
Khan	Coverage + dosage	Skills/time → modest SD (PNAS 2026)	Lonely evaluative feel; AI not yet identity	Diagnosis + soft-wrong + tutor witness
Duolingo	Return / streak	DAU, streak length, HLR review	Extrinsic colonization	Borrow friction UX; ban streak North Star
Brilliant	Interactive struggle + delight	Lesson completion, “aha”	Misses threatened segment	Win anxious middle; don’t out-prestige
ChatGPT	Fluent answers now	Session grade / CSAT	Crutch; transfer collapse	Guarded Solver; AI-off transfer

FOUNDER BELIEF: The defensible one-liner is not “AI tutor.” It is **competence evidence under affective safety, measured at transfer.**

XXXV.7 Marketing language that survives this audit

Allowed (mechanism-true)	Banned
“Practice that still works when the hints are gone”	“2-sigma / Bloom tutoring”
“We measure retry after a soft wrong — not streak fire”	“Tutoring is free now that GPT exists”
“Dosage matters; we spend dosage on the fear→evidence loop”	“Become math Duolingo”
“Guarded help so chat can’t become a crutch”	“Bastani-proven ontology” / “guarantees learning gains vs no-AI”

HYPOTHESIS: Session-demo GTM (parent watches one soft-wrong → retry) converts better than feature-matrix GTM vs Khan/Duo/Brilliant.

XXXV.8 Experiments spawned

ID	Question	Design	Primary	Falsifier
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CSA-1	Blind session preference: MindCraft vs Khan practice vs Brilliant lesson (same concept)	Within-subject, Maya-segment screen	Forced choice + affect items	MindCraft loses on “would return” <i>and</i> “felt capable”
CSA-2	Copy test: FEI session-demo vs “AI tutor / mastery path”	Landing A/B	Signup → first retry_120s	Feature copy wins activation
CSA-3	ChatGPT Base paste rate when Solver is guarded vs permissive	Instrumentation	Off-app paste proxy + solo_transfer_pass	Guards increase paste <i>and</i> transfer falls
AIT-1	Carry from XXXIII	3-arm RCT	solo_transfer_pass	Guarded < no-AI

XXXV.9 Confidence table

Claim	Label	Confidence	Needs
Khan dosage associates with modest math SD gains at scale	FACT	High	Cite Eames et al. 2026; don’t overclaim RCT purity
Duo HLR / streaks optimize engagement	FACT	High	Settles 2016; Method 2023
Streaks should be MindCraft North Star	SPECULATION (false)	Kill	Standing ban
Brilliant QED proves identity transformation for threatened teens	HYPOTHESIS (overreach)	Low	Different segment + small QED
ChatGPT-like Base can harm solo exam performance	FACT	High	Bastani 2025
Pedagogy guards ⇒ learning <i>gain</i> vs no-AI	Killed in XXXIII	—	Neutralize harm ≠ uplift
Session-demo positioning beats feature-matrix vs these four	HYPOTHESIS	Medium	CSA-2
MindCraft should not out-content Khan or out-streak Duo	FOUNDER BELIEF	High	Strategy consistency

XXXV.10 What this chapter refuses

- Bloom 2-sigma as competitive proof.
- Absolute “tutoring is free.”
- Streaks as learning KPI.
- Equating Khanmigo qualitative warmth with identified FEI.
- Equating Bastani GPT Tutor with MindCraft’s product.
- Fabricated citations.

Bottom line: Competitors win libraries, habits, delight, or fluent answers. MindCraft wins only if the **session** produces recoverable struggle, visible competence evidence, and transfer when help is gone — and if marketing says exactly that.

Part XXXVI — Equity Audit of Story Worlds

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: Do MindCraft story worlds expand belonging and mathematical identity for diverse learners — or do they costume Eurocentric defaults, add seductive load, and widen who feels “math is for people like me”?

Owners: Learning Science · World / Narrative · Product · Red Team

Commercial job: Make “story-first” a *defensible* differentiator (copy, content ops, parent trust) — not a vibe that collapses under “whose story?” scrutiny.

XXXVI.1 Why this audit is load-bearing

Part XXVIII put invention-story under trial (HIST-1...4; HIST-EQ). Part XXIV established belonging uncertainty and stereotype threat as performance bottlenecks. WORLD_VISION commits to story worlds that wrap frozen math. None of that settles **who is centered** when the wrap is written.

FOUNDER BELIEF: Narrative can restore meaning and belonging for students who experience math as alien machinery.

FACT (counter-tradition): Mathematics learning is *not* culture-neutral. Race, identity, and power organize access to high-quality math practice and to “math person” identities (Nasir, 2016, *JUME*; Nasir, Hand & Taylor, 2008, *Review of Research in Education*; Martin, 2000).

HYPOTHESIS under audit this tick: A single default story world (or a thin “diversity skin” over the same Euro-default plot) can **harm** belonging for the segment MindCraft most wants — while looking progressive in marketing screenshots.

Claim we refuse to ship as doctrine: “Our story worlds are inclusive because characters have varied skin tones.” That is **SPECULATION** dressed as equity. Token presence ≠ practice-linked identity (Nasir & Hand, 2008).

XXXVI.2 What “equity” means here (operational, not slogan)

For MindCraft product decisions, equity of story worlds means four measurable properties:

Property	Operational question	Failure mode
Access	Can the learner <i>enter</i> the math practice without cultural membership tests?	Lore gatekeeping; insider jokes; assumed prior
Role	Does the student take an <i>integral</i> role (solver, witness, navigator) — not tourist?	Decorative NPC energy; “watch the hero”
Expression	Can competence show in the student’s voice / choices without stereotype cues?	“Even you can...” copy; gendered/racialized ability frames
History honesty	When invention stories appear, is the human lineage plural and structure-encoding?	Euro-only timeline; museum trivia; token name-drop

This maps Nasir & Hand’s (2008, *Journal of the Learning Sciences*) basketball-vs-classroom analysis: deep engagement tracked **access to the domain**, **integral roles**, and **self-expression** — not posters about belonging. Classroom math often denied those affordances to African American students who showed them in basketball.

Product translation: Story worlds succeed when they recreate *those three affordances around math*, not when they add themed wallpaper around a lonely skill tree.

XXXVI.3 Evidence base (what is solid vs thin)

A. Identity is sociopolitical, not cosmetic

FACT (ethnographic / theory program): Martin (2000) — *Mathematics Success and Failure Among African-American Youth* — success and failure are shaped by sociohistorical context, community forces, school influence, and individual agency; “math identity” is negotiated under racialized meanings of ability, not only item difficulty.

FACT (practice-linked identity): Nasir & Hand (2008) — same students can show sophisticated quantitative reasoning in out-of-school practices while disengaging in school math when access, role, and expression collapse.

FACT (field stance): Nasir (2016) — treating math as outside race/culture is a fallacy; stereotypes and school structures disrupt mathematical identities and opportunity to learn.

FACT (practitioner synthesis): Aguirre, Mayfield-Ingram & Martin (NCTM; *Impact of Identity in K–8 / K–12 Mathematics*) — teaching practices shape whether students see themselves as doers of mathematics; equity-based practice is asset-based and identity-attentive, not deficit remediation theater.

Implication: MindCraft’s identity thesis (Parts XXV / FEI) is *aligned* with this literature — **if** story design expands access/role/expression. Story that only changes scenery fails the identity literature’s actual mechanism.

B. Belonging interventions are narrow — do not cargo-cult into lore

FACT (RCT): Walton & Cohen (2011, *Science*, doi:10.1126/science.1198364) — brief social-belonging intervention framing adversity as common/transient raised African American college students’ GPA over 3 years and roughly halved the measured achievement gap in that sample (N=92; mediated by construal of adversity).

FACT (adjacent): Steele & Aronson stereotype-threat tradition; Spencer, Steele & Quinn (women/math) — ability-diagnostic framing under negative stereotypes can impair performance (effect sizes and replication quality vary; see Part XXIV limits).

Red Team (carry from XXIV): Effects are heterogeneous; school-scale attenuates; a 45-minute belonging essay ≠ a gamified toast ≠ a fantasy prologue. Belonging is not a substitute for teaching.

HYPOTHESIS: Story worlds help belonging when they make early adversity **non-identity-diagnostic** (“misses teach the map”) *and* refuse stereotype-cueing copy — not when they lecture “you belong here” in lore text.

C. Whose history? Plural roots are historical FACT; classroom transfer is HYPOTHESIS

FACT (historiography): Joseph (1991/2011, *The Crest of the Peacock*) documents non-European mathematical lineages (Egypt, Mesopotamia, India, China, Islamic world, and more) against Eurocentric “Greeks-only greatness” narratives.

FACT (program): D’Ambrosio’s ethnomathematics programme (e.g., 1985–1990 line; ethno + mathema + tics) treats mathematical knowing as culturally situated practice, not a single civilizational monopoly.

FACT (critical pedagogy tradition): Gutstein (2006, *Reading and Writing the World with Mathematics*) — math can be used to read/write sociopolitical conditions; equity is not only “access to algebra” but agency. **Caveat for MindCraft:** Full Freirean classroom projects are not the product’s unit of delivery; borrow the *honesty about power*, not a forced activism skin on ACT prep items.

HYPOTHESIS (from XXVIII HIST-EQ / HIST-4): Plural invention stories that encode structure can lift belonging without crushing retention — **if** they are not seductive trivia.

D. Seductive details and “diversity costume” share a kill path

FACT: Harp & Mayer (1998, *Journal of Educational Psychology*, doi:10.1037/0022-0663.90.3.414) — interesting but irrelevant details can damage learning via cognitive interest mechanisms.

FACT (meta-analytic): Sundararajan & Adesope (2020, *Educational Psychology Review*, doi:10.1007/s10648-020-09522-4) — including seductive details tends to hinder learning; moderators exist (format, pacing, subject), but the mean direction is caution, not license.

Kill criterion (FOUNDER BELIEF → testable): If a cultural reference, character trait, or “heritage beat” can be removed without changing the mathematical mental model **and** without changing who can take an integral role, it is decoration. Decoration that only signal brand virtue are equity theater **and** CLT risk.

Survive criterion: Cultural/historical content that (a) encodes the invariant, (b) expands access/role/expression for under-affirmed students, and (c) does not cue stereotypes.

XXXVI.4 Failure modes specific to MindCraft story worlds

Failure mode	What it looks like in product	Likely harm	Epistemic label
Euro-default adventure	Castles, Greco-Roman “genius,” white default heroes	Quiet alienation; “not for people like me”	HYPOTHESIS (strong prior from identity lit)
Token palette swap	Same plot; recolored sprites	Cynical read; no identity mechanism	HYPOTHESIS
Heritage trivia dump	Long museum lore before first attempt	Seductive load; delayed micro-win for anxious Maya	FACT risk (CLT / seductive details)
Stereotype-friendly tropes	“Street-smart” vs “book-smart”; gendered magic roles	Cue threat / essentialism	HYPOTHESIS (aligned with threat lit)
Forced activism costume	Every item a justice pamphlet	Parent trust fracture; math time stolen; off-mission	SPECULATION commercially; Gutstein ≠ ACT session
Single-world monoculture	One lore for all students forever	No cultural fit; no choice autonomy (SDT)	HYPOTHESIS

FOUNDER BELIEF (constrained): Plural *structural* histories + multiple entry aesthetics beat one mega-franchise lore — but only with HIST/EQ experiments and content-ops cost discipline.

XXXVI.5 Audit rubric (ship this as content ops, not a vibes review)

Before a story module graduates from draft:

- 1. Structure test:** Name the mathematical invariant the narrative makes memorable. If blank → rewrite or cut.
- 2. Role test:** In the first 60 seconds, does the student *do* math as an agent, or watch lore? Threatened learners need Safety → micro-attempt before long meaning (XXVIII.7).
- 3. Belonging-copy test:** Ban ability-diagnostic group framing (“girls struggle with...”, “unlike other kids...”). Prefer adversity-as-common-and-transient framing (Walton & Cohen mechanism — adapted carefully; do not fake an RCT in UI).

4. **History plurality test:** If an invention beat exists, is at least one non-Euro lineage available in the corpus for that concept family over time (not every single card)? Token one-off “Al-Khwarizmi cameo” without structure = fail.
 5. **Seductive-details test:** Remove the cultural beat; if learning items and role affordances unchanged, cut or move post-success.
 6. **Stereotype scan:** Two reviewers (ideally including someone outside the default writer demographic) flag tropes; unresolved flags block ship.
 7. **Claim ladder (Part XXXIV):** Marketing may say “we design stories so more students can take an integral role in the math” (L0/L1 product intent). It may **not** say “our lore closes achievement gaps” without L3 evidence.
-

XXXVI.6 Commercial implications (copy, positioning, roadmap)

Marketing language that survives

- **Allowed:** “Math isn’t a private club with one history. Stories should make the *idea* human — and give every learner a real role in solving it.”
- **Allowed:** “We refuse decoration that wastes working memory. Narrative earns its place when it encodes the math or restores a sense that struggle isn’t proof you don’t belong.”
- **Banned:** “Culturally magical world closes the gap.”
- **Banned:** “Diverse characters = equitable outcomes.”
- **Banned:** Absolute tutoring-is-free / 2-sigma / streaks-as-identity (standing bans).

Competitive stance

Khan’s library and Duo’s streaks are largely **culture-thin** at the session surface (Part XXXV). MindCraft *chooses* narrative risk. That is only a wedge if the equity audit is visible in product craft — otherwise competitors can dismiss story as gimmick **and** critics can dismiss it as exclusionary. Honest position: “We accept narrative risk; we instrument belonging + learning; we kill modules that fail the rubric.”

Growth / parent trust

Parents in threatened segments buy **dignity + results**. Equity theater without `retry_120s` / transfer reads as politics instead of help. Lead with FEI evidence; treat plural history as accuracy and belonging hygiene, not a culture-war banner unless the parent segment asks for it.

Product bets

1. **Content ops checklist** above as merge gate for story modules.
 2. **At least two aesthetic/cultural entry frames** for core onboarding arcs (same math, different wrap) — autonomy without fragmenting ontology.
 3. **HIST-EQ / EQ-1...3** before scaling “world lore” as brand centerpiece.
 4. Tutor witness remains the primary belonging technology for high-threat students (XXIV); story is amplifier, not substitute.
-

XXXVI.7 Experiments spawned

ID	Question	Design	Primary	Kill condition
EQ-1	Euro-default story vs plural structure-encoding story (same items)	A/B or crossover	belonging/threat items + retry_120s + retention@7d	Belonging flat; or learning drop > pre-reg threshold
EQ-2	Token palette-swap vs role/access redesign (same art budget)	3-arm	integral-role self-report + attempt latency	Palette $\geq$ role redesign on belonging
EQ-3	Pre-attempt heritage lore vs post-success meaning beat	Timing	time-to-first-attempt; transfer	Pre-lore delays attempt with no belonging lift
HIST-EQ	(from XXVIII) Plural vs single-tradition invention	A/B	belonging + learning	Belonging up, transfer down badly
EQ-QUAL	10 Maya interviews: “who is this world for?”	Protocol B + equity probes	theme codes	Students read world as “not for me” at >threshold

Pre-reg discipline (XXXIV): EQ-* identify story→belonging/persistence. They do **not** identify story→GPA gap closure. Do not sell L3 from L1.

XXXVI.8 Confidence table

Claim	Label	Confidence
Math learning is entangled with race/culture/identity	FACT (research consensus in equity math ed)	High
Access / role / expression predict engagement better than posters	FACT (Nasir & Hand mechanisms) + HYPOTHESIS for digital worlds	Medium–High
Default Euro adventure can alienate	HYPOTHESIS	Medium
Token diversity without role redesign fails	HYPOTHESIS	Medium–High

Seductive cultural trivia can hurt learning	FACT (seductive-details tradition + meta-analysis direction)	Medium–High
Plural invention stories lift belonging <i>and</i> learning	HYPOTHESIS (HIST-EQ unrun)	Low–Medium
Story can replace tutor witnessing for threat	SPECULATION / likely false	Low (against)
Equity marketing without FEI metrics is brand risk	FOUNDER BELIEF	Medium–High

XXXVI.9 What this chapter kills

1. **Kill:** “Inclusive skins” as the equity strategy.
2. **Kill:** Shipping long pre-attempt lore as belonging intervention.
3. **Kill:** Marketing claims that story worlds close racial/gender achievement gaps without identified designs.
4. **Wound:** Single-franchise world as forever default — survivable only as temporary scaffold with EQ-1/2 on the roadmap.
5. **Survive (constrained):** Story-first WORLD_VISION — **if** narrative encodes structure, expands integral roles, passes stereotype scan, and stays on the L0–L2 claim ladder until experiments graduate it.

Doctrine until data: Plural, structure-encoding, role-rich worlds under audit > monoculture lore > token palette swaps > heritage trivia dumps.

Part XXXVII — Expectancy-Value (Eccles): Utility, Cost, Attainment → Course Choice

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: Why do students *choose* (or abandon) math pathways — and which MindCraft surfaces move expectancy vs value vs cost?

Owners: Learning Science · Product · Growth / Positioning · Red Team

Commercial job: Replace vague “make math meaningful” copy with a **defensible E×V map**: which product moves which lever, what parents hear, what we refuse to claim.

XXXVII.1 Why this chapter exists

Parts XXV (self-efficacy / narrative identity) and XXIV (anxiety / belonging) explain *can I* and *am I safe*. They do not fully explain *will I keep choosing math* when Instagram, sports, and sleep compete.

FACT (program lineage): Eccles-Parsons et al. (1983) built an achievement-choice model to explain gendered STEM participation: performance and choice are most proximally driven by **expectancies for success** and **subjective task values** (STV), themselves shaped by socializers, culture, and prior experience (*Expectancies, values, and academic behaviors*, in Spence, *Achievement and Achievement Motives*).

FACT (rename / situating): Eccles & Wigfield (2020, *Contemporary Educational Psychology*, doi:10.1016/j.cedpsych.2020.101859) rename the framework **Situated Expectancy-Value Theory (SEVT)** — expectancies and values are *situation-bound hierarchies*, not a single global “motivation score.”

FOUNDER BELIEF under audit: MindCraft’s identity thesis only converts to commercial retention / course-persistence if product reduces **cost** and raises **attainment + utility** while expectancy (FEI evidence) rises — not if we only make prettier explanations.

Claim we refuse as doctrine: “Story worlds automatically raise utility value.” That is **SPECULATION**. Utility is a *student-authored* link to goals (Hulleman tradition), not a brand aesthetic.

XXXVII.2 The model in product language (not poster language)

Construct	Plain definition	MindCraft surface (hypothesis)	Failure if ignored
Expectancy	“Can I succeed <i>on the next relevant tasks</i> ?”	Gap-scan → level gate; soft-wrong → retry; tutor witness; FEI micro-wins	High value + low expectancy → avoidance / crutch (Bastani Base risk, Part XXXIII)
Intrinsic value	Enjoyment / interest in the activity itself	Story-module engagement; challenge–skill (XXXI)	Fun without transfer = Duo-streak trap (XXXV)
Attainment value	Importance to <i>who I am</i> / identity	Identity re-storying (XXV); integral role in worlds (XXXVI)	Scoreboard status ≠ “math person”
Utility value	Usefulness for current/future goals	ACT roadmap honesty; parent-facing “why this concept”; student-written relevance prompts	Pure test-prep cynicism; or fake career lore
Cost	Effort, opportunity, emotional/psychological price	Anxiety load (XXIV); session length; shame of failure; time vs other activities	High cost kills choice even when expectancy/value look “ok”

FACT (classic split): Meece, Wigfield & Eccles (1990, *Journal of Educational Psychology*, doi:10.1037/0022-0663.82.1.60) — in young adolescents (N=250), **performance expectancies** predicted subsequent math grades; **value perceptions** predicted **course enrollment intentions**; math anxiety was predicted by ability/expectancy/value perceptions but did not independently drive grades/enrollment in their SEM.

Product translation: Instrument **both** outcome classes. `retry_120s` / `transfer` $\approx$ expectancy path. `Intent-to-continue` / `challenge_accept` / “take next ACT practice block” $\approx$ value path. Do not collapse both into a streak KPI (standing ban).

XXXVII.3 Subjective task value — four components, four commercial mistakes

A. Intrinsic value $\neq$ engagement theater

FACT: Intrinsic value = enjoyment of the task (Eccles-Parsons et al., 1983; Wigfield & Eccles, 2000, *CEP*, doi:10.1006/ceps.1999.1015).

HYPOTHESIS: Story can raise intrinsic value *if* it encodes structure and preserves integral roles (XXVIII / XXXVI). Seductive details that raise “interest” but crush learning (Harp & Mayer; Sundararajan & Adesope — carry) are **fake intrinsic** for MindCraft’s purpose.

Kill: Marketing that equates “delightful world” with durable math choice without FEI + value instruments.

B. Attainment value is the identity wedge

FACT (theory refinement): Attainment value moved from “importance of doing well” toward linkage with personal/social identities (Eccles, 2005/2009 line; summarized in Eccles & Wigfield, 2024, *Educational Psychology Review*, doi:10.1007/s10648-024-09888-9).

FOUNDER BELIEF: “I am becoming someone who can think mathematically” is attainment value. Streaks are not.

Commercial implication: Parent and teen copy should sound like *identity competence*, not *gamified attendance*. Allowed: “Evidence you can trust yourself on the next hard problem.” Banned: “Don’t break your streak — that means you’re a math person.”

C. Utility value — the only mass-market EVT intervention with strong field RCTs

FACT (RCT): Hulleman & Harackiewicz (2009, *Science*, doi:10.1126/science.1177067) — ninth-grade science students randomly assigned to write how material connects to their lives (or a known other) vs summary control; **relevance writing raised interest and grades especially for students with low success expectations.**

FACT (lab + classroom): Hulleman, Godes, Hendricks & Harackiewicz (2010, *JEP*, doi:10.1037/a0019506) — utility writing raised perceived utility and interest, again strongest for lower performers/expectancies; utility perceptions mediated interest effects.

HYPOTHESIS for MindCraft: A 2–4 minute **student-authored** utility prompt *after* a micro-win (not before first attempt under high threat) beats brand-authored “math is useful for STEM careers” copy — especially for Maya with low expectancy.

Red Team wound: Effects are domain/context sensitive; science classroom $\neq$ ACT app session; dosage and timing matter (SEVT situating). Do not paste “Hulleman proved story worlds work.”

Kill: Shipping utility as *staff-written* career propaganda dumped pre-attempt (CLT + seductive + autonomy threat).

D. Cost — historically neglected, commercially decisive

FACT: Cost was theorized early (effort, opportunity, psychological cost of failure) but empirically under-measured for decades (Wigfield & Cambria reviews; Eccles & Wigfield, 2020 SEVT paper stresses cost elaboration).

FACT (measurement): Flake, Barron, Hulleman, McCoach & Welsh (2015, *CEP*, doi:10.1016/j.cedpsych.2015.03.002) — validated multi-dimensional cost (task effort, outside effort, loss of valued alternatives, emotional cost); cost is separable from expectancy/value and relates to student outcomes.

HYPOTHESIS: For anxious Maya, **emotional cost** and **effort cost** dominate early sessions; for busy juniors, **opportunity cost** (vs sleep/jobs/sports) dominates weekly return. Product that only raises “fun” while leaving shame-of-wrongness intact fails SEVT choice.

Product translation: Soft-wrong / hide-correctness diagnostic (C4) / tutor tone (Part XXX) are **cost interventions**, not just pedagogy niceties. Session design that demands 40 minutes of lore before first attempt raises opportunity + effort cost without expectancy gain (tie to EQ-3).

XXXVII.4 Hierarchies and situations (why one “motivation score” is a product bug)

FACT: SEVT emphasizes *hierarchies* of expectancies and values across competing activities and situations (Eccles & Wigfield, 2020; Wigfield & Eccles developmental reviews). Choosing math means math beats alternatives *this week*, not that motivation is globally high.

HYPOTHESIS: Duo wins short-horizon intrinsic/utility-for-streak; Khan wins low-cost access library; ChatGPT Base raises short-term utility (“get the answer”) while crushing expectancy-for-unaided performance (Part XXXIII/XXXV). MindCraft’s wedge is **attainment + expectancy evidence with managed cost** — not out-streaking Duo or out-librarying Khan.

SPECULATION (commercial): Parents purchase when they believe we move the *child’s hierarchy* toward math without catastrophic emotional cost — “they’ll open it again without a fight” is a cost/expectancy story sold as calm, not as dopamine.

XXXVII.5 Socializers — parents and tutors as EVT machinery

FACT (model structure): Socializers’ beliefs and behaviors shape children’s interpretations, affective memories, and thus expectancies/values (Eccles-Parsons et al., 1983; SEVT socialization arm).

Carry: Part XXXII (parent anxiety transmission) — parent math anxiety can raise the child’s **emotional cost** and lower expectancy via homework help patterns (Maloney/Beilock line).

HYPOTHESIS: Parent dashboard copy that celebrates *process evidence* (retry, transfer) raises parent-transmitted attainment/utility messages; copy that only shows streak/percentile can raise performance pressure → cost.

Product implication: Parent WTP experiments (Part XXVII queue) should manipulate EVT framing: “identity/competence evidence” vs “minutes practiced” vs “AI explanations unlimited.” Pre-register which frame shifts purchase intent without raising reported homework conflict.

XXXVII.6 Mapping MindCraft features → EVT levers (audit table)

Feature / bet	Primary lever	Secondary	Claim ladder max without new data
Gap-scan + level gating	Expectancy calibration	Cost ↓ (fewer impossible first items)	L1 product intent
Soft-wrong → retry_120s	Expectancy evidence	Emotional cost ↓	L1–L2 with FEI instrumentation
Story worlds (structure-encoding)	Intrinsic + attainment	Utility if student links	L0–L1; EQ/HIST required
Student utility writing (post-win)	Utility	Intrinsic (situational interest)	L1; EVT-1 experiment
ACT exam-mode roadmap	Utility (instrumental)	Expectancy if honest sequencing	L1 honesty; not “score guarantee”
Tutor human sessions	Expectancy + cost	Attainment (witnessed identity)	L1; do not claim 2-sigma
Solver without guards	Short-term utility (answer)	Expectancy ↓ (crutch)	Kill as hero feature
Streaks / leaderboards	Pseudo-attainment / extrinsic	Cost ↑ under threat	Standing ban as North Star

XXXVII.7 Commercial implications (copy, positioning, growth)

Marketing language that survives

- **Allowed:** “Expectancy without value is empty drills. Value without expectancy is inspirational avoidance. We design for both — and we treat the emotional price of being wrong as a first-class design constraint.”
- **Allowed:** “Utility is something the student *writes*, not a slogan we paste.”
- **Allowed (parent):** “Fewer fights at the kitchen table” = cost reduction claim — keep qualitative until measured.
- **Banned:** “Our world makes math useful” (untested utility).
- **Banned:** “Fun = they’ll choose Precalculus.”
- **Banned:** Absolute tutoring-is-free / 2-sigma / streaks-as-identity.

Positioning vs competitors

Competitor	EVT profile (audit, not insult)	MindCraft contrast
Khan	Low cost access; mixed expectancy; thin attainment	Session FEI + identity evidence

Duo	Intrinsic/habit; cost managed via brevity; attainment via streak fiction	Transfer + real expectancy
Brilliant	Intrinsic puzzle; utility unclear for ACT Maya	Exam-honest utility + anxiety cost
ChatGPT Base	Max short-term utility; expectancy poison	Guarded Solver + unaided transfer

Growth / North Star hygiene

Optimize a **pair**: expectancy evidence (retry_120s, transfer_pass) **and** a value/choice proxy (challenge_accept, weekly voluntary return, intent-to-take-next-block). If value proxy rises while transfer falls, you are Duo-pilling the product — kill the feature.

XXXVII.8 Experiments spawned

ID	Question	Design	Primary	Kill condition
EVT-1	Post-win student utility writing vs staff utility blurb vs none	3-arm A/B after micro-win	situational interest + 7d return + transfer@7d	Utility arms lose transfer or raise time-to-next-attempt badly
EVT-2	ACT-utility framing vs identity-attainment framing in session CTA	2-arm	challenge_accept ; parent/teen preference	Attainment wins preference but tanks attempt rate in anxious segment (segmented kill)
EVT-3	Cost questionnaire (Flake-style short form) pre/post first week	Observational → then intervene on top cost	which cost dimension predicts churn	If emotional cost predicts churn but product only ships “more fun,” roadmap invalid
EVT-4	Parent copy: competence-evidence vs minutes vs AI-unlimited	WTP / conflict items	purchase intent; reported homework conflict	Competence frame raises conflict without WTP lift
EVT-QUAL	10 Maya interviews: “why would you take another math course?”	Appendix B + EVT probes	coded expectancy /value/cost	Students cannot articulate any non-grade reason → attainment/utility product gap

Pre-reg (XXXIV): EVT-* identify levers on **in-app choice and short-horizon learning**. They do **not** identify long-term course enrollment or career STEM entry without follow-up designs. Do not sell “we increase AP Calculus enrollment” from EVT-1.

XXXVII.9 Confidence table

Claim	Label	Confidence
Choice $\approx$ f(expectancy, STV) in Eccles tradition	FACT (large corpus)	High
Expectancy→performance, value→enrollment intentions (adolescent math)	FACT (Meece et al., 1990 SEM)	Medium–High (sample/era limits)
STV has intrinsic / attainment / utility / cost structure	FACT (theory + measurement work)	High
Utility-value writing can lift interest/grades for low-expectancy students	FACT (Hulleman & Harackiewicz, 2009)	Medium–High (context limits)
Cost is multi-dimensional and outcome-relevant	FACT (Flake et al., 2015)	Medium–High
Story worlds raise utility by themselves	SPECULATION	Low
MindCraft should instrument E and V separately	FOUNDER BELIEF / HYPOTHESIS	Medium–High
Soft-wrong + tutor tone primarily cut emotional cost	HYPOTHESIS	Medium
Streaks raise attainment value	SPECULATION / likely false for identity	Low (against)

XXXVII.10 What this chapter kills

1. **Kill:** Single “motivation” KPI as if E and V were the same dial.
2. **Kill:** Staff-authored “math is useful” dumps as a substitute for student-authored utility.
3. **Kill:** Treating story delight as proof of course-choice impact.
4. **Wound:** Pure ACT-utility positioning — survives for juniors near test date; wounds long-horizon identity brand if it becomes the only message.
5. **Survive (constrained):** Identity-transformation thesis — **if** product explicitly raises expectancy *and* attainment/utility *and* designs against cost, with EVT-1...4 on the roadmap.

Doctrine until data: Design and market the **E×V×Cost triad**. Expectancy evidence without value is a drill app. Value talk without expectancy is a poster. Ignoring cost is how anxious students churn while NPS looks fine among the already-confident.

Part XXXVIII — Goal Orientation: Mastery vs Performance Framing

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: When MindCraft frames goals as *get better at this vs look smart / beat others / don’t look dumb*, which motivational pattern do we induce — and what must product + copy refuse?

Owners: Learning Science · Product · Growth / Positioning · Red Team

Commercial job: Replace empty “mastery learning” branding with a **goal-structure audit**: which surfaces cue mastery-approach, which cue appearance/performance, which cue avoidance — and which we kill even if they juice short-term engagement.

XXXVIII.1 Why this chapter exists

Parts XXV (self-efficacy), XXX (attribution / helplessness), XXXI (challenge–skill), and XXXVII (expectancy-value) answer *can I, why did I fail, is difficulty calibrated, and will I choose math*. They do not fully answer: **what success definition is the product teaching?**

A product that says “identity transformation” while UI, CTAs, and parent dashboards scream *percentile / streak / don’t mess up* is running a **performance-avoidance climate** with mastery marketing on top.

FOUNDER BELIEF under audit: MindCraft’s North Star metrics (`retry_120s`, `challenge_accept`, `transfer_pass`) only stay coherent if the **goal structure** of sessions is mastery-approach dominant. If recognition and evaluation structures are performance/appearance, those metrics will fight the UI.

Claim we refuse as doctrine: “Any competitive element raises motivation.” That collapses normative, appearance, and avoidance goals — meta-analyses show they are not the same animal.

XXXVIII.2 The constructs (product language, not poster language)

Goal	Plain definition	Typical student cue	MindCraft failure mode if dominant
Mastery-approach	Improve competence vs self/task standard	“Understand this; try a harder variant”	None if paired with transfer; risk = slow coverage if time unmanaged

Mastery-avoidance	Avoid failing to learn / avoid imperfect understanding	“Don’t leave gaps; don’t misunderstand”	Perfectionism, over-checking, refusal to attempt under uncertainty
Performance-approach (normative)	Outperform peers	Rank, leaderboard, “top of class”	Can raise grades for some; often raises threat for Maya; identity ≠ rank
Performance-approach (appearance)	Demonstrate talent / look smart	Public correctness, “prove you’re smart”	Meta-analytically worse than normative; shame + shallow strategy
Performance-avoidance	Avoid looking incompetent	Hide wrongness; skip hard items; Solver crutch	Helpless pattern; FEI collapse; Part XXX / XXXIII risk

FACT (classic dichotomy + classroom structures): Ames (1992, *Journal of Educational Psychology*, doi:10.1037/0022-0663.84.3.261) — classroom **task, evaluation/recognition, and authority** structures make mastery vs performance goals salient and elicit qualitatively different motivational patterns; interventions must treat structures as interdependent, not one “growth mindset poster.”

FACT (student perceptions → strategies): Ames & Archer (1988, *JEP*, doi:10.1037/0022-0663.80.3.260) — among 176 secondary students, perceived **mastery** classroom goals related to more effective strategies, preference for challenge, positive attitude, and effort→success beliefs; perceived **performance** salience related to ability focus, negative ability evaluation, and attributing failure to low ability.

FACT (implicit theories → goals → patterns): Dweck & Leggett (1988, *Psychological Review*, doi:10.1037/0033-295X.95.2.256) — entity vs incremental theories orient students toward performance vs learning goals, which set up helpless vs mastery-oriented response patterns after difficulty.

Product translation: Soft-wrong + retry is not only an expectancy/cost intervention (XXXVII) — it is a **goal-structure** intervention. Public scoreboards and “don’t break the streak” are evaluation/recognition structures that cue performance/appearance.

XXXVIII.3 Approach–avoidance split (why “performance goals” is too coarse)

FACT (trichotomous model): Elliot & Church (1997, *JPSP*, doi:10.1037/0022-3514.72.1.218) — mastery, performance-approach, and performance-avoidance are separable; mastery facilitated intrinsic motivation; performance-approach enhanced graded performance; **performance-avoidance harmed both** intrinsic motivation and grades (college classroom path model).

FACT (2×2 framework): Elliot & McGregor (2001, *JPSP*, doi:10.1037/0022-3514.80.3.501) — adds **mastery-avoidance**; four goals show distinct antecedents/outcomes; mastery-avoidance sits between mastery-approach (more adaptive) and performance-avoidance (more maladaptive) on several indicators.

HYPOTHESIS for MindCraft: Anxious Maya often defaults to **performance-avoidance** (“don’t look dumb”) or **mastery-avoidance** (“I must understand perfectly before I try”). Product that only bans leaderboards but still frames every miss as identity threat will not fix the pattern.

Kill: Marketing that equates “mastery mode” with “zero mistakes allowed on the path.” That is mastery-avoidance cosplay.

XXXVIII.4 Performance-approach is not one knob — measurement kills bad slogans

FACT (meta-analysis on measures): Hulleman, Schrager, Bodmann & Harackiewicz (2010, *Psychological Bulletin*, doi:10.1037/a0018947) — across 243 correlational studies ($N \approx 91k$), “performance-approach” scales that emphasize **normative outperformance** correlate *positively* with performance outcomes ($r \approx .14$), while scales emphasizing **appearance / looking talented** correlate *negatively* ($r \approx -.14$). Same label, different constructs.

FACT (extended outcomes meta): Senko & Dawson (2017, *Journal of Educational Psychology*, doi:10.1037/edu0000160) — the normative vs appearance split generalizes beyond grades to competence perceptions and self-regulation; age confounds with measure type were addressed.

FACT (theory status review): Senko, Hulleman & Harackiewicz (2011, *Educational Psychologist*, doi:10.1080/00461520.2011.538646) — achievement goal theory at a crossroads: multiple controversies (including when performance-approach helps) require precise operationalization before practice claims.

Commercial implication: If MindCraft ever ships social comparison, it must pre-register whether the cue is **normative skill evidence** (dangerous for threatened learners; still not identity) or **appearance theater** (ban). Default = **neither**. Do not cite “performance goals sometimes help grades” as license for public shame UX.

SPECULATION (positioning): “Beat your friends” copy will read as Duo/Brilliant hybrid; “prove you’re smart” reads as SAT-cram Instagram. Neither is the identity-transformation wedge.

XXXVIII.5 Classroom / product goal structures (TARGET), not vibes

FACT (schoolwide / TARGET lineage): Maehr & Midgley (1991, *Educational Psychologist*, doi:10.1080/00461520.1991.9653140) — motivation change requires altering school/classroom goal structures, not only exhorting students; dimensions map to task design, authority, recognition, grouping, evaluation, time (TARGET family; Ames, 1992).

FACT (review): Meece, Anderman & Anderman (2006, *Annual Review of Psychology*, doi:10.1146/annurev.psych.56.091103.070258) — classroom goal structures predict personal goals and achievement-related patterns; mastery structures are the design target for adaptive motivation.

FACT (longitudinal TARGET → mastery goals): Lüftenegger et al. (2014, *Educational Psychology*, doi:10.1080/01443410.2013.814189) — TARGET as a latent structure positively predicted students’ mastery goal orientations ($N=1680$ secondary; longitudinal).

MindCraft TARGET audit (hypothesis map):

TARGET dim	Mastery-leaning product move	Performance/appearance trap
Task	Varied challenge matched to skill (XXXI); format variety with transfer	Same-item grind that feels like “prove speed”
Authority	Student choice of next concept <i>after</i> honest diagnosis	Forced path that only optimizes for looking ahead of peers
Recognition	Process evidence: retry, transfer, bridge repair	Streak flames, public %ile, “genius” badges
Grouping	Tutor as witness; optional co-op without rank	Ranked classrooms / public wrongness
Evaluation	Soft-wrong diagnostic (C4); hide correctness when graph-corrupting	Instant public right/wrong as identity verdict
Time	Short sessions that still allow struggle → retry	Timed pressure that cues avoidance

FOUNDER BELIEF: PawHub “Practice the weak spot” is mastery-approach **if** copy is “repair this gap” and **not** “you’re behind everyone.” Gap language can flip to performance-avoidance under shame framing.

XXXVIII.6 Interaction with expectancy, anxiety, and AI crutches

FACT (goal pursuit × expectancy): Senko & Hulleman (2013, *JEP*, doi:10.1037/a0031136) — for mastery-approach *and* performance-approach, the harder a goal appears to attain, the less students pursue it (with costs to interest/achievement); mastery-approach goals were generally judged easier to attain than performance-approach; situational cues (material complexity) shift mastery-approach expectancies.

Carry: Part XXIV (anxiety) + XXXIII (AI overtrust) — under high threat, performance-avoidance + Solver-without-guards is the cheap escape from a hard-looking mastery or performance goal.

HYPOTHESIS: Level gating after gap-scan (easy→L3, kinda→L2, hard→L1) is a **goal-attainment-expectancy** design: it makes mastery-approach goals feel attainable. Removing gates to “challenge everyone” without expectancy supports will increase avoidance and Solver dependence.

Product implication: CTAs should make the next mastery goal *feel attainable* (Senko & Hulleman) without lying about transfer. Honest sequencing ≠ performance theater.

XXXVIII.7 Competitive product audit (goal climate lens)

Competitor	Dominant goal climate (audit)	What MindCraft must not copy
Duolingo	Habit + streak recognition; appearance of consistency	Streak-as-identity / streak-as-North-Star (standing ban)

Khan	Mastery language + energy points / badges mixed	“Mastery” label without transfer_pass
Brilliant	Puzzle competence; mild normative “smart people do this”	Appearance-smart branding without anxiety cost design
ChatGPT Base	Outcome utility (“get it right now”) → avoidance of struggle	Solver as hero; unaided transfer optional

HYPOTHESIS: MindCraft’s wedge is **mastery-approach structures with expectancy-managed tasks and anti-appearance evaluation** — not out-ranking Duo and not out-explaining Khan.

XXXVIII.8 Commercial implications (copy, product, growth)

Marketing language that survives

- **Allowed:** “We design for improvement goals — understand, repair, transfer — not for looking smart in public.”
- **Allowed:** “Wrong answers are evidence for the next attempt, not a verdict on who you are.” (ties XXX / soft-wrong)
- **Allowed (parent):** “We measure whether they take the next hard problem, not whether they kept a streak.”
- **Banned:** “Become a top 1% math student” as hero claim (appearance/normative mashup; undefended).
- **Banned:** “Never break your streak — that’s how mastery works.”
- **Banned:** Empty growth-mindset posters without TARGET structure changes (Dweck mis-applied as sticker).
- **Banned:** 2-sigma / tutoring-is-free absolutes.

Feature claim ladder (pre-reg hygiene, Part XXXIV)

Feature	Goal climate claim max without GO-* data
Soft-wrong → retry_120s	L1 intent: mastery-approach evaluation structure
challenge_accept CTA	L1: approach (need to verify mastery vs performance wording)
transfer_pass	L1–L2 anti-fake-mastery (competence standard ≠ appearance)
Leaderboards / public ranks	L0 default kill for Maya segment
Streak North Star	Kill (standing ban)
“Mastery graph” marketing	L1 only if transfer instrumented; else label is cosmetics

Growth / North Star hygiene

challenge_accept is **not automatically mastery-approach**. Instrument **why** they accept: repair gap (mastery) vs beat peer (normative) vs avoid looking behind (avoidance). If acceptance rises because of shame CTAs, kill the copy even if the metric looks green.

XXXVIII.9 Experiments spawned

ID	Question	Design	Primary	Kill condition
GO-1	Mastery-approach CTA (“repair this gap”) vs normative (“catch up to peers”) vs appearance (“show you’re smart”)	3-arm on Practice launch	challenge_accept ; retry_120s; 7d return; anxiety item	Normative/appearance win short accept but lose retry/transfer or raise anxiety
GO-2	Recognition: process badge (retry/transfer) vs streak badge vs none	3-arm	transfer@7d; identity item; self-reported goal	Streak badge wins engagement, loses transfer / raises avoidance
GO-3	Evaluation: soft-wrong first vs instant public correctness	2-arm on early session	next-attempt latency; avoidance skips; Solver opens	Instant correctness raises “finish” rate but kills retry_120s
GO-4	Goal-climate pulse (short PALS-like mastery/performance classroom items adapted to app) weekly	Observational → then redesign recognition	which climate predicts churn / Solver dependence	If product self-reports “mastery brand” but climate score is performance → roadmap invalid
GO-QUAL	10 Maya interviews: “What would ‘doing well’ here mean?”	Appendix B + goal probes	coded mastery / normative / appearance / avoidance	Majority answer only “not look dumb” → evaluation/recognition redesign mandatory

Pre-reg (XXXIV): GO-* identify **in-app goal climate** → **short-horizon FEI behaviors**. They do **not** identify long-term GPA or STEM major from CTA wording alone. Do not sell “we create mastery-oriented students for life” from GO-1.

XXXVIII.10 Confidence table

Claim	Label	Confidence
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Mastery vs performance goal structures shape strategies/attitudes	FACT (Ames & Archer, 1988; Ames, 1992)	High
Implicit theories orient goals → helpless vs mastery patterns	FACT (Dweck & Leggett, 1988)	High
Performance-avoidance is broadly maladaptive for motivation/grades	FACT (Elliot & Church, 1997 lineage)	High
2×2 adds mastery-avoidance as distinct	FACT (Elliot & McGregor, 2001)	High
“Performance-approach” effects depend on normative vs appearance ops	FACT (Hulleman et al., 2010; Senko & Dawson, 2017)	High
TARGET-like structures can raise mastery orientations	FACT (Maehr & Midgley; Lüftenegger et al., 2014)	Medium–High (school ≠ app)
Soft-wrong + process recognition = mastery evaluation structure	HYPOTHESIS	Medium–High
Any leaderboard helps MindCraft’s Maya segment	SPECULATION / likely false	Low (against)
Streaks instantiate mastery goals	SPECULATION / false for identity	Low (against)
MindCraft should ban appearance-goal UX by default	FOUNDER BELIEF / HYPOTHESIS	Medium–High

XXXVIII.11 What this chapter kills

1. **Kill:** Treating “mastery” as a brand adjective without evaluation/recognition structures that cue improvement standards.
2. **Kill:** Streak / public rank / “look smart” as North Star or hero copy (appearance + standing streak ban).
3. **Kill:** Collapsing all performance goals into “competition is motivating.”
4. **Wound:** Normative comparison as a growth lever — survives only as a pre-registered, segmented experiment (GO-1), never as default Maya UX.
5. **Survive (constrained):** Identity-transformation thesis — **if** product TARGET structures are mastery-approach dominant, avoidance/appearance cues are designed out, and GO-1...4/QUAL sit on the roadmap.

Doctrine until data: Market and build **mastery-approach goal structures** (task challenge + process recognition + non-shaming evaluation + attainable next goals). Performance-avoidance and appearance goals are product bugs dressed as engagement. Normative comparison is not the default growth loop.

Part XXXIX — Interleaving vs Blocking: When to Mix Problem Types

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: When should MindCraft mix problem types (interleave) vs group same-type drills (block) — and what must product + copy refuse to claim?

Owners: Learning Science · Product (Practice sequencing) · Growth / Positioning · Red Team

Commercial job: Turn “we shuffle questions” from a cute engineering detail into a **defensible practice-design claim:** strategy *selection*, not just strategy *execution* — without selling interleaving as always-on magic or as engagement theater.

XXXIX.1 Why this chapter exists

Part XXIX (spacing / retrieval) already flags interleaving as a desirable difficulty and hypothesizes **Block** → **fade** → **interleave** → **spaced retrieval**. Part XXVI (CLT) warns that early overload kills learning. Part XXXI (challenge–skill) and XXXVIII (goal climate) warn that harder-feeling practice can cue avoidance if framed as shame.

This chapter densifies the **mix vs block** decision for Practice sessions, ingredient cards, and ACT-style mixed review — because textbooks and most edtech default to **blocked practice** (a dozen slope problems after the slope lesson), and students experience that fluency as “I get it” while exams demand **choose the strategy from the problem itself**.

FOUNDER BELIEF under audit: Identity transformation requires competence evidence that survives *transfer*. Blocked fluency is often **illusory competence** — high session accuracy, weak delayed discrimination.

Claim we refuse as doctrine: “Always interleave everything from the first item.” That ignores schema formation, CLT, and similarity moderators — and can spike threat for anxious Maya before `retry_120s` is healthy (XXIX.4).

XXXIX.2 Constructs (product language)

Schedule	Plain definition	What the student practices	Failure mode if misused
Blocked	Consecutive items share the same strategy / concept	<i>Execute</i> a known procedure	Fluency illusion; weak strategy choice on exams
Interleaved	Consecutive items require different strategies (no safe “same as last”)	<i>Select</i> then execute	Early overload; frustration; avoidance if unscaffolded
Hybrid / faded	Short block for schema → rising mix	Execute → select	Best candidate for MindCraft default

Operational definition for MindCraft (HYPOTHESIS): Interleaving = consecutive Practice items do **not** share the same primary strategy class (concept + dominant ingredient / archetype family). Mere format shuffle (word → diagram) with the same strategy is **not** full interleaving — it is representation variety (useful; different claim).

XXXIX.3 Mechanism: why mixture changes the pedagogical demand

FACT (math laboratory / classroom line): Rohrer & Taylor (2007, *Instructional Science*, doi:10.1007/s11251-007-9015-8) — Experiment 2: after learning multiple problem types, **mixed vs blocked** practice; one-week delayed test vastly favored mixed practice. Experiment 1 separately showed spacing > massing for a single type. Shuffled textbooks alter **both** spacing and mixing; they are related but not identical levers.

FACT (strategy choice vs execution): Rohrer, Dedrick & Stershic (2015, *Journal of Educational Psychology*, doi:10.1037/edu0000001) — 126 seventh-graders, same problems over ~3 months, interleaved vs blocked arrangement; after review, unannounced tests favored interleaving at 1 day ($d = 0.42$) and 30 days ($d = 0.79$). Core mechanism claim: blocked practice lets students know the strategy *before reading the problem*; interleaved practice forces strategy choice from problem features — the skill exams actually require.

FACT (not only look-alike problems): Rohrer, Dedrick & Burgess (2014, *Psychonomic Bulletin & Review*, doi:10.3758/s13423-014-0588-3) — interleaved benefit is **not** limited to superficially similar problem kinds; reordering changes pedagogical demand (choose vs execute) even when types do not look alike.

FACT (classroom-scale RCT): Rohrer, Dedrick, Hartwig & Cheung (2020, *JEP*, doi:10.1037/edu0000367; Online First 2019) — preregistered cluster RCT, 54 seventh-grade classes, mostly interleaved vs mostly blocked assignments over months; after interleaved review for both, one-month unannounced test: 61% vs 38%, $d = 0.83$. Teachers implemented without training; anonymous pre-results survey supported feasibility. Caveats remain (materials, population), but this is the strongest classroom efficacy signal in the math interleaving line.

Product translation: MindCraft’s transfer_pass and mixed ACT practice are **closer to the interleaved demand** than a single-concept blocked grind. A session that is 12 items of linear_equations after “Learn linear equations” is textbook-default blocking — fine as *early schema practice*, dangerous as the *only* schedule if we market exam readiness.

XXXIX.4 Meta-analytic moderators (do not overclaim “interleaving always wins”)

FACT (inductive learning meta-analysis): Brunmair & Richter (2019, *Psychological Bulletin*, doi:10.1037/bul0000209) — 59 studies / 238 effect sizes / 158 samples; overall interleaving vs blocking Hedges’ $g = 0.42$. **Material matters:** paintings/visual high ($g \approx 0.67$); **mathematical tasks small but positive** ($g \approx 0.34$); words favor **blocking** ($g \approx -0.39$); expository texts ambiguous. Moderators: stronger interleaving when **between-category similarity is high, within-category similarity is low**, and material is more complex — consistent with attentional bias / sequential attention accounts (Carvalho & Goldstone lineage).

FACT (attentional bias framework): Carvalho & Goldstone (2015, *Frontiers in Psychology*, doi:10.3389/fpsyg.2015.00505) — study sequence biases what is encoded: interleaving emphasizes **between-category differences**; blocking emphasizes **within-category similarities**. Prediction: when categories are hard to tell apart, interleave; when the job is to notice what members of one category share, block can help.

HYPOTHESIS for MindCraft: Interleave **confusable** strategy families (e.g., independent vs dependent probability; similar ACT archetypes that share surface cues but need different ingredients). Block when introducing a *new* ingredient representation until the student can execute with worked-example fade (XXVI) — then mix against near-miss foils.

Kill: Marketing that says “our AI interleaves everything like elite tutors” without specifying **confusable sets** and **when blocking is intentional**.

XXXIX.5 Fluency illusion, affect, and why students prefer blocking

FACT / TRADITION (desirable difficulties): Conditions that slow acquisition (spacing, interleaving, generation) often improve later retention/transfer (Bjork lineage; see XXIX.4). Interleaving feels harder *during* practice; blocked practice feels fluent.

HYPOTHESIS (Koriat/Bjork-style monitoring, applied): Students and parents misread blocked fluency as mastery. Parent dashboards that celebrate “12/12 on slope today” without a mixed delayed check sell **performance theater** (XXXVIII) dressed as learning.

Carry from XXIV / XXXVII / XXXVIII: Raising interleaving dose raises short-term error rate → for anxious Maya, that can cue performance-avoidance unless (a) soft-wrong / mastery evaluation, (b) expectancy-managed difficulty, (c) explicit frame: “this mix trains choosing, not proving you’re dumb.”

Product implication: Copy for interleaved sessions: “**Practice choosing the right move**” — not “hard mode for elites” and not streak pressure.

XXXIX.6 Sequencing doctrine (ties XXVI + XXIX)

FOUNDER BELIEF / HYPOTHESIS (sequenced schedule):

1. **Acquire (block-lite):** short same-strategy set + worked examples / cards until execution is stable enough that errors are mostly slips, not total confusion.
2. **Discriminate (interleave):** mix with **near-miss** types that share surface features (bridge gaps, format gaps, ACT archetype neighbors).
3. **Retain (space):** redistribute the same mix across days (XXIX) — interleaving *guarantees* some spacing of each type, but lag still needs explicit scheduling.
4. **Transfer check:** unannounced mixed set → `transfer_pass`; failure routes to diagnosis (forgot vs never-had; wrong strategy class vs execution bug).

Contradiction to hold: Some math/category studies report null or blocking advantages under specific similarity structures (noted in Brunmair & Richter’s discussion of mixed math findings; e.g., certain representation-interleaving designs). MindCraft must treat **interleaving axis** (strategy class) as primary for exam readiness, and treat **representation mix** as a separate experiment track (format axis / C2–C3).

XXXIX.7 Competitive / product audit (schedule lens)

Surface	Typical schedule	What MindCraft should do
Textbook / workbook	Heavy blocking after lesson	Do not clone as “mission complete”
Khan-like mastery	Often skill-blocked until “mastered”	Mastery label without mixed delayed check = fluency risk
Duo-like	Habit + short items; mix varies by skill tree	Habit ≠ discrimination training
Brilliant	Puzzle variety; not always systematic near-miss mix	Variety ≠ controlled interleaving
ChatGPT Solver	Single-problem depth	Zero schedule pedagogy; cannot substitute mix training
MindCraft Practice today	Often concept-scoped sessions + bank shuffle	Shuffle within one concept ≠ full strategy interleaving; need cross-type mix modes

HYPOTHESIS: PawHub “Practice the weak spot” correctly targets a gap, but a pure blocked weak-spot grind can raise local accuracy while leaving **strategy selection** untrained. Pair weak-spot blocks with a **discrimination coda**: 3–5 interleaved near-misses including the weak type.

XXXIX.8 Commercial implications (copy, product, growth)

Marketing language that survives

- **Allowed:** “Exams don’t announce the chapter. We train choosing the strategy — not only repeating the last one.”
- **Allowed:** “Blocked practice feels easy; mixed practice builds the skill tests actually measure.”
- **Allowed (parent):** “We check whether they can pick the right approach on a mixed set days later — not just a perfect streak on one topic.”
- **Banned:** “Interleaving always doubles scores” / absolute effect-size marketing (Rohrer classroom $d \approx 0.83$ is **one** RCT context; Brunmair math $g \approx 0.34$ is the broader inductive-math signal — do not collapse).
- **Banned:** “Always mix from day one.”
- **Banned:** Selling shuffle/random as science without near-miss design.
- **Banned:** 2-sigma / tutoring-is-free / streak-as-mastery.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without IL-* data
Within-concept item shuffle	L0–L1 “variety”; not full interleaving claim
Cross-strategy interleaved coda after weak-spot block	L1 intent: discrimination training

Delayed mixed transfer_pass	L1–L2 anti-fluency-illusion
“Interleaving improves ACT” hero claim	L0 until IL-1/IL-2 pre-registered

Growth / North Star hygiene

Session accuracy under **blocked** schedules will look better than under interleaved ones. **Do not** optimize North Star on blocked accuracy. Prefer: mixed delayed accuracy, strategy-selection error rate (wrong procedure class), transfer_pass, and retry_120s after mix-induced misses.

XXXIX.9 Experiments spawned

ID	Question	Design	Primary	Kill condition
IL-1	Blocked-only weak-spot session vs block + interleaved near-miss coda	2-arm; same total items	Delayed (48h) mixed accuracy; strategy-class error rate	Coda hurts delayed mix and raises avoidance without transfer gain
IL-2	Early interleave (from item 1) vs fade-in interleave after short block	2-arm on new concept	Acquisition errors; retry_120s; 7d mixed test	Early-all-mix wins delayed score but kills return/anxiety for Maya segment
IL-3	Near-miss interleave (confusable archetypes) vs far mix (unrelated easy types)	2-arm	Discrimination errors on near-miss test	Far mix equals near-miss on discrimination → “any shuffle” doctrine dies
IL-4	Parent/student judgment: blocked fluency vs mixed score — which do they trust?	Survey + optional choice	Preference for blocked; willingness-to-pay framing	If parents only buy blocked dashboards, copy/education mandatory before UX ships mix-heavy
IL-QUAL	10 Maya interviews: “When did practice feel like the real test?”	Appendix B	coded choose-vs-execute moments	Nobody describes strategy choice → product isn’t creating the demand state

Pre-reg (XXXIV): IL-* identify **schedule** → **discrimination / delayed mix**. They do **not** identify long-term STEM identity from one coda. Do not sell “interleaving creates math people.”

Supersedes / densifies: XXIX INT-1 (interleave after fade vs blocked-only) → prefer IL-1/IL-2 as the operational pair.

XXXIX.10 Confidence table

Claim	Label	Confidence
Mixed/interleaved math practice often beats blocked on delayed tests	FACT (Rohrer & Taylor, 2007; Rohrer et al., 2015, 2020)	High (math classroom/lab line)
Mechanism includes forced strategy selection (not only spacing)	FACT / strong interpretation in Rohrer line	High
Benefit can appear for non-look-alike problem kinds	FACT (Rohrer et al., 2014)	High
Overall inductive interleaving effect moderated by material & similarity	FACT (Brunmair & Richter, 2019)	High
Math subset effect smaller than visual category learning	FACT ($g \approx 0.34$ vs higher for paintings)	High
Blocking can help when within-category similarity learning is the job	FACT (attentional bias framework) / HYPOTHESIS for early schema	Medium–High
Always-interleave-from-item-1 is optimal for anxious Maya	SPECULATION / likely false	Low (against)
Within-concept shuffle $\equiv$ interleaving science claim	SPECULATION / false	Low (against)
Default MindCraft path = short block → near-miss interleave → spaced mix check	FOUNDER BELIEF / HYPOTHESIS	Medium–High

XXXIX.11 What this chapter kills

1. **Kill:** Treating blocked session accuracy as proof of exam readiness or identity change.
2. **Kill:** “We randomly shuffle, therefore we interleave” as a marketing claim.
3. **Kill:** Absolute always-interleave / always-double-scores claims.
4. **Wound:** Pure weak-spot blocked grind as the only Practice mode — survives as *acquisition*, not as the full loop.
5. **Survive (constrained):** Desirable-difficulty interleaving for **strategy discrimination** — after schema foothold, with near-miss design, affect-safe framing, and delayed mixed transfer_pass.

Doctrine until data: Market and build **choose-the-strategy training** via faded interleaving of confusable types. Blocking is a scaffold, not the product. Fluency without mixed delayed evidence is a vanity metric.

Part XL — Self-Explanation Prompts: Chi/Renkl → Coach UX

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: When do prompts that force the student to *generate* an explanation improve math learning — and what must MindCraft’s coach / cards / Solver refuse to do with “explain yourself” UX?

Owners: Learning Science · Product (coach, ingredient cards, Solver) · Growth / Positioning · Red Team

Commercial job: Turn “we explain everything” and “AI tutor talks at you” into a **defensible generation claim:** students build transferable understanding by explaining *why steps work* (and why common wrong moves fail) — without selling open-ended “justify your guess” as science, and without confusing fluent AI explanation *to* the student with self-explanation *by* the student.

XL.1 Why this chapter exists

Part XXVI (CLT / worked examples / fading) already puts examples and completion problems in the acquisition path. Part XXIX / XXXIX demand retrieval and strategy *selection*. Part XXXIII warns that fluent AI explanations can create overtrust. Experiment A (wizard coach) and soft-wrong already assume that *after* an error, something productive happens.

This chapter densifies the **prompt layer**: what the product asks the student to *produce* — principle for a step, why a misconception is wrong, anticipation of the next operator — vs what it merely *shows*.

FOUNDER BELIEF under audit: Identity as a mathematical thinker requires evidence that the student can *account for* a move, not only select A/B/C/D. Passive delivery without generation is Layer-1 scarcity relief, not identity transformation.

Claim we refuse as doctrine: “Any explain box after a wrong answer is self-explanation science.” Untargeted prompts, explain-your-wrong-guess-without-feedback, and AI monologues are different mechanisms — some help, some hurt, some are theater.

XL.2 Constructs (product language)

Construct	Plain definition	Product surface	Failure mode
Self-explanation	Learner generates inferences beyond the given text/example <i>for themselves</i>	Prompted coach turns; card reflection; step-principle pick	Empty “why?” with no scaffold; AI does the explaining for them
Instructional explanation	Expert/AI/tutor explains <i>to</i> the learner	Solver narration; worked example text	Can reduce germane effort if overused (expertise / passivity)

Worked-example study	Study full solution before / instead of cold solving	Ingredient cards; fade path	Without SE prompts → shallow copy (Chi “poor” pattern)
Principle / glossary citation	Name the rule that justifies a step	Aleven-style menu; ingredient id pick	Menu-click without comprehension if too shallow
Misconception contrast	Explain why a common wrong path fails	Soft-wrong + “why not B”; bank distractors	Asking only “defend your wrong answer” before feedback

Operational definition for MindCraft (HYPOTHESIS): A self-explanation prompt is load-bearing only if (1) the to-be-explained content is tagged **correct or known-incorrect**, (2) the response format scaffolds principle / operator–goal / misconception contrast (not free essay only), and (3) success is measured by **transfer / delayed mix**, not by longer time-on-task or chat length.

XL.3 Classic effect: spontaneous quality, then elicitation

FACT (good vs poor example study): Chi, Bassok, Lewis, Reimann & Glaser (1989, *Cognitive Science*, doi:10.1207/s15516709cog1302_1) — talk-aloud while studying mechanics worked examples. “Good” students generated many explanations that refined conditions for actions and linked steps to principles; “poor” students under-explained, monitored poorly, and relied heavily on examples at solve time. Self-explanation quality predicted understanding and example-independence.

FACT (elicitation works): Chi, de Leeuw, Chiu & LaVancher (1994, *Cognitive Science*, doi:10.1016/0364-0213(94)90016-7) — eighth-graders prompted to self-explain after each line of a circulatory-system text outperformed controls who reread; high explainers built more accurate mental models. Prompts can *create* the behavior that spontaneous good students show.

FACT (quality ≠ mere time): Renkl (1997, *Cognitive Science*, doi:10.1016/S0364-0213(99)80017-2) — probability worked examples; time-on-task controlled. Learning gains still predicted by qualitative SE features: **principle-based** explanations, **operator–goal** explication, **anticipative** reasoning. Two effective profiles: anticipative reasoners and principle-based explainers.

Product translation: Coach UX should elicit *those* moves — “What rule licenses this step?” / “What is this step trying to achieve?” / “What happens next?” — not “Write anything about your feelings about math” and not “Hear the AI explain for 90 seconds.”

XL.4 Scaffolded prompts + fading (coach design spine)

FACT (fading + principle prompts): Atkinson, Renkl & Merrill (2003, *Journal of Educational Psychology*, doi:10.1037/0022-0663.95.4.774) — successively fading worked-out steps helps near transfer vs example–problem pairs, but far transfer is unreliable from fading alone. Combining fading with prompts to identify the **underlying principle** for each worked step produced medium-to-large effects on **near and far transfer without extra time on task**.

FACT (Cognitive Tutor classroom): Aleven & Koedinger (2002, *Cognitive Science*, doi:10.1016/S0364-0213(02)00061-7) — geometry Cognitive Tutor; students who explained steps by citing geometric rules (glossary) learned with greater understanding and better transfer than problem-solving-only peers. Interpretation: better-integrated declarative knowledge, less shallow procedural knowledge. Scales in classroom software — not only lab talk-alouds.

HYPOTHESIS for MindCraft: Ingredient card_templates + bridge cards are the worked-example substrate; coach prompts should (a) attach principle/ingredient labels to faded steps, (b) require a brief generation or structured pick *before* revealing the next full AI paragraph, (c) fade prompts as mastery rises (expertise reversal risk if we nag forever).

Tie to SAFE-CRAFT (XXVI): Micro-attempt / completion blank inside an example *plus* a principle prompt beats cold multi-step solve *or* passive read-only cards.

XL.5 Math meta-analysis: what we may claim (and must not)

FACT (math SE meta-analysis): Rittle-Johnson, Loehr & Durkin (2017, *ZDM*, doi:10.1007/s11858-017-0834-z) — prompted self-explanation in mathematics: small-to-moderate immediate gains — procedural knowledge ES $\approx .28$, conceptual ES $\approx .33$, procedural transfer ES $\approx .46$. Effects **stronger when high-quality explanations are scaffolded** (training or structured responses). Time-on-task control did **not** erase the benefit (moderator). **Caveats they stress:** evidence for reliable classroom (esp. primary/secondary non-ITS) effects and for **retention over delay** is much thinner; delayed procedural/conceptual effects often weak/null while delayed transfer ES $\approx .32$ in the subset that measured it; substantial heterogeneity (some null/negative studies).

Instructional recommendations from that paper (treat as design doctrine, not slogans):

1. Scaffold high-quality explanations (structure or train).
2. Design prompts so they do **not** steal attention from other critical content.
3. Prompt learners to explain **correct** information.
4. Prompt why **common misconceptions** are incorrect.

Kill: Marketing “self-explanation always raises grades” / absolute classroom-retention claims from lab immediate ES alone. Claim ladder: L1 “structured why-this-step prompts aim at transfer”; L2+ only after SE-* experiments with delayed mixed transfer_pass.

XL.6 Constraints: when prompts backfire

FACT (constraint review): Rittle-Johnson & Loehr (2017, *Psychonomic Bulletin & Review*, doi:10.3758/s13423-016-1079-5) — four constraints: (1) SE favors principle-guided domains and transfer, can hurt detail/exception learning; (2) explaining **one’s own** (often wrong) ideas can entrench them — better to explain known-correct / known-incorrect content; (3) prompts focus attention — mis-aimed prompts tax the wrong content; (4) alternatives (instructional explanation, retrieval, unfamiliar problems) can match or beat SE in some conditions.

HYPOTHESIS for soft-wrong: Order matters. After a wrong MC choice: (A) brief correct-principle or contrastive “why B fails” prompt with scaffold → help; (B) “Explain why you picked C” *before* any correctness signal → risk of justifying the misconception (constraint 2). Prefer: soft-wrong frame → show contrastive target → generate/select why

the foil fails → then optional deeper AI wrap (XXXIII: calibration > fluency).

Contradiction to hold: Open generation is the romantic “thinker” move; structured glossary/ingredient picks (Aleven) often deliver the measurable classroom gain. MindCraft should not shame structured picks as “not real thinking” — they are the scalable SE scaffold.

XL.7 Competitive / product audit (explanation lens)

Surface	Typical explanation pattern	MindCraft stance
Khan / video	Instructional explanation; optional reflection	Do not clone lecture-first as identity product
Duo	Rare deep SE; habit + bits	Habit ≠ principle generation
Brilliant	Puzzle insight; light justify	Insight ≠ scaffolded SE curriculum
ChatGPT Solver	High-fluency instructional explanation	Not self-explanation; overtrust risk (XXXIII)
Cognitive Tutor lineage	Step + rule citation	Closest evidence-backed software pattern
MindCraft today	Cards + Solver + coach drafts	Need student generation gates before AI walls of text; wire misconception bank into contrastive prompts

FOUNDER BELIEF: Solver that answers *for* Maya without a generation gate competes on Layer-1 (answers get cheap) and loses the identity thesis. Coach that forces a principle pick / misconception contrast competes on Layer 2–6.

XL.8 Commercial implications (copy, product, growth)

Marketing language that survives

- **Allowed:** “We don’t just show the steps — we ask you to say *why* the step works.”
- **Allowed:** “Wrong answers become contrast: why that trap fails, not a shame spiral.”
- **Allowed (parent):** “We train explaining the rule, not only picking the letter — that’s what transfers.”
- **Banned:** “AI explains everything so tutoring is free.”
- **Banned:** “Any chat box = self-explanation science.”
- **Banned:** Absolute meta-analytic grade claims; 2-sigma; streak-as-understanding.
- **Banned:** Forcing students to justify unchecked wrong guesses as the default pedagogy.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without SE-* data
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AI worked-example narration	L0 instructional help; not SE claim
Structured principle / ingredient pick on faded step	L1 SE scaffold (Aleven/Atkinson-aligned intent)
Contrastive “why this distractor fails” after soft-wrong	L1 misconception-SE intent
Open essay “explain your answer” every item	L0 until quality+affect validated; risk of load + avoidance
“SE raises ACT / identity” hero	L0 until SE-1/SE-2 delayed transfer

Growth / North Star hygiene

Chat turns, explanation word count, and Solver dwell time are **vanity** if uncoupled from transfer_pass and strategy-class errors. Prefer: rate of principle-consistent explanations (rubric or structured correct pick), misconception-contrast success, delayed mixed accuracy, retry_120s after contrastive miss — not tokens generated by the model.

XL.9 Experiments spawned

ID	Question	Design	Primary	Kill condition
SE-1	Principle-prompt on faded card steps vs same cards with AI narration only	2-arm; matched time budget	48h near + far / mixed transfer; shallow-copy errors	Narration-only $\geq$ prompted on transfer and lower load/avoidance
SE-2	Soft-wrong $\rightarrow$ contrastive “why foil fails” vs soft-wrong $\rightarrow$ explain-own-choice first	2-arm	Misconception recurrence; retry_120s; delayed item isomorphic	Explain-own-first wins transfer $\rightarrow$ constraint-2 doctrine wounded
SE-3	Structured ingredient/glossary pick vs open text SE vs no SE	3-arm	Transfer; time; completion rate (Maya anxiety segment)	Open text wins transfer but kills return $\rightarrow$ ship structured default
SE-4	Prompt every step vs adaptive fade of SE prompts with mastery	2-arm	Transfer; prompt skip/nag irritation	Always-on prompts equal fade on transfer but raise churn $\rightarrow$ fade mandatory

SE-QUAL	10 Maya interviews: “When did an explanation feel like <i>yours</i> vs the app talking?”	Appendix B	coded ownership / passivity moments	Nobody distinguishes → UX isn’t creating generation
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Pre-reg (XXXIV): SE-* identify **prompt regime** → **transfer** / **misconception recurrence**. They do **not** identify long-term identity from one session of explain boxes. Densifies Experiment A: coach content should be SE-constrained, not generic pep talk.

Supersedes / densifies: XXVI worked-example fading notes; Experiment D (explanation timing × anxiety) → prefer SE-2/SE-3 as operational pairs; XXXIII “pedagogy wrap” → wrap = student generation + calibration, not longer AI prose.

XL.10 Confidence table

Claim	Label	Confidence
Higher-quality spontaneous SE while studying examples predicts better learning	FACT (Chi et al., 1989)	High
Prompting SE can improve understanding vs reread / no prompt	FACT (Chi et al., 1994; math meta)	High
Qualitative SE features predict gains beyond time-on-task	FACT (Renkl, 1997)	High
Fading + principle prompts aid near & far transfer without extra time	FACT (Atkinson et al., 2003)	High
Step explanation via rule citation helps Cognitive Tutor geometry transfer	FACT (Aleven & Koedinger, 2002)	High
Math prompted-SE immediate effects small–moderate; scaffold helps	FACT (Rittle-Johnson et al., 2017 ZDM)	High
Classroom + delayed retention evidence thinner than immediate lab/ITS	FACT (same meta)	High
Explaining own wrong ideas can entrench; prefer correct/incorrect targets	FACT (constraint review)	High
AI instructional explanation ≡ self-explanation benefit	SPECULATION / false	Low (against)

Default MindCraft path = faded examples + structured principle/misconception prompts before AI wrap	FOUNDER BELIEF / HYPOTHESIS	Medium–High
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XL.11 What this chapter kills

1. **Kill:** Equating Solver/AI monologue with self-explanation pedagogy.
2. **Kill:** “Explain why you chose that” *before* correctness/contrast as the default after every wrong answer.
3. **Kill:** Absolute “SE always raises scores / always works in class / always sticks for months” marketing.
4. **Wound:** Open-ended explain boxes as the only SE UX — survives as optional depth, not as the scalable default.
5. **Survive (constrained):** Scaffolded prompts on **correct steps and known misconceptions**, paired with worked-example fading, measured by transfer — the coach spine for identity-relevant competence evidence.

Doctrine until data: Build and market **student-generated why** (structured first) on faded examples and contrastive wrongs. AI explains *after* or *around* that generation — never as a substitute for it. Fluency of the model is not mastery of the student.

Part XLI — Desirable Difficulties × Anxiety: When Struggle Helps vs When It Becomes Threat

Chapter status: Living evidence brief — Researcher tick 2026-07-30

Primary question: How should MindCraft reconcile Bjork-style desirable difficulties (spacing, interleaving, retrieval, generation) with Ashcraft-style math-anxiety / working-memory threat — so product + copy do not sell either “grind harder” or “never make it hard”?

Owners: Learning Science · Affect / Anxiety · Product (Practice sequencing + evaluation climate) · Growth / Positioning · Red Team

Commercial job: Resolve the I.4 bottleneck: ship faded desirable difficulties **without** converting anxious Maya’s session into shame theater — and market the distinction between *productive struggle* and *threat* in language parents and students can trust.

XL.I.1 Why this chapter exists

Parts XXIX (spacing/retrieval), XXXIX (interleaving), and XL (self-explanation) all lean on **desirable difficulties**: conditions that slow short-term performance but improve durable access and transfer. Part XXIV (affect/anxiety) and Ashcraft’s line show that math anxiety **already** loads working memory like a dual task. Part XXXI (challenge–skill) and XXXVIII (goal climate) warn that harder-feeling practice can cue avoidance under performance threat.

v1.6 left an explicit bottleneck: “*Part XLI must resolve desirable difficulties × anxiety so interleaving/SE/spacing do not ship as shame.*”

FOUNDER BELIEF under audit: Identity transformation requires competence evidence that survives delay and mix — which *requires* some desirable difficulty — but for threatened learners, unsequenced difficulty is not “elite training”; it is **threat + WM collapse + exit**.

Claim we refuse as doctrine: “Struggle is always good” / “always crank desirable difficulties” *and* its twin “anxious students should only get easy fluency drills forever.” Both are commercially tempting; both fail Maya.

XLI.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Desirable difficulty (DD)	Challenge that forces encoding/retrieval processes that support later access	Mixed review, spaced retries, generation before reveal	Feels worse today → students & parents quit if misframed
Undesirable difficulty	Challenge the learner cannot productively meet (knowledge, cues, or affect load)	Early full interleave; open SE under panic; high-stakes quiz theater	Errors without learning; avoidance; “I’m not a math person”
Math anxiety load	Worry/rumination consuming WM resources needed for multi-step math	Freeze, hint binge, rapid exit after first hard miss	Interpreted as “lazy” or “needs more struggle”
Stakes / evaluation climate	Whether the hard moment is framed as judgment vs training	Soft-wrong, mastery climate, no streak-as-worth	High stakes turn retrieval into threat (Hinze & Rapp)
Equipped success	Prior knowledge + cues sufficient to <i>respond</i> to the difficulty	Block-lite → fade → mix; worked examples; structured SE	DD without equipment = cruelty dressed as science

Operational definition (HYPOTHESIS): A MindCraft difficulty is *desirable* only when (a) the student has enough schema/cues to respond with non-chance success or productive partial success, **and** (b) evaluation climate keeps the moment in mastery/training frame, **and** (c) affect load is not already saturating WM. Else it is undesirable — even if the *same* schedule would be desirable for a calm, prepared peer.

XLI.3 Bjork line: difficulty that builds durable access

FACT (origin): Bjork (1994, in Metcalfe & Shimamura, *Metacognition: Knowing about knowing*, MIT Press, pp. 185–205) — conditions that slow acquisition (spacing, variation, testing/generation as learning events, contextual interference) can improve long-term retention and transfer relative to conditions that make performance look strong

during practice.

FACT (desirable $\neq$ any difficulty): Bjork & Bjork (2011/2014 reviews; JARMAC 2020 comment, doi:10.1016/j.jarmac.2020.09.003) — many difficulties are undesirable forever; a difficulty is desirable only when the learner can respond successfully with the encoding/retrieval processes the difficulty is meant to trigger. *If the learner is not equipped, the difficulty becomes undesirable.*

FACT (learning vs performance illusion): Instructors and students systematically misread current fluency as learning (Bjork lineage; see also XXIX / XXXIX). Blocked / massed / reread conditions feel productive; spaced / interleaved / tested conditions feel worse *and often teach more.*

Product translation: MindCraft cannot sell “we make practice feel easy” as proof of learning, nor “we make practice feel hard” as proof of rigor. The commercial claim must be **durable choosing and retrieving under exam-like demand** — measured by delayed mix / transfer_pass / solo_transfer_pass — not by session comfort or session pain.

XLI.4 Ashcraft line: anxiety as a competing load

FACT (WM disruption): Ashcraft & Kirk (2001, *JEP: General*, doi:10.1037/0096-3445.130.2.224) — high math-anxiety individuals show reduced WM spans (especially computation-based spans); dual-task math + memory load amplifies RT/errors. Anxiety acts as a **transitory disruption of working memory**, not merely as a stable “low ability” trait.

FACT (review / education bridge): Ashcraft & Krause (2007, *Psychonomic Bulletin & Review*, doi:10.3758/BF03194059) — math beyond simple retrieval depends on WM; high math anxiety functions like a resource-demanding secondary task (preoccupation with fear). Global consequence: avoidance of elective math, majors, and careers.

FACT (carry from XXIV): Association of math anxiety with performance is real but modest on average (meta $r \approx -0.17$); mechanism candidates include WM competition + avoidance. Anxiety is **one bottleneck among many**, not destiny.

Conflict statement (HYPOTHESIS): Desirable difficulties often *increase* momentary retrieval/selection demand — exactly the demand anxiety already taxes. Unbuffered DD therefore risks converting a would-be desirable difficulty into an undesirable one for HMA students: same schedule, different learner state $\rightarrow$ different pedagogy.

XLI.5 Evidence that stakes and anxiety change whether “hard practice” helps

FACT (pressure can kill retrieval benefits): Hinze & Rapp (2014, *Applied Cognitive Psychology*, doi:10.1002/acp.3032) — high-stakes vs low-stakes quizzes: quiz performance similar, but **final test better after low-stakes**; only low-stakes quizzes beat reread. Authors propose **retrieval disruption**: pressure can impair the memory-formation benefits of retrieval even when online quiz scores look fine.

FACT (classroom retrieval often lowers reported test anxiety): Agarwal, D’Antonio, Roediger, McDermott & McDaniel (2014, *JARMAC*, doi:10.1016/j.jarmac.2014.07.002) — survey of 1,408 middle/high-school students in

classrooms using retrieval practice; 72% reported feeling **less nervous** for unit tests/exams when retrieval practice was used; 92% said it helped learning. Self-report, not RCT — but directionally opposite to “tests always raise anxiety.”

FACT (meta: practice tests tend to reduce test anxiety): Yang, Li, Zhao, Luo & Shanks (2023, *Educational Psychology Review*, doi:10.1007/s10648-023-09801-w) — 24 studies / 25 effects / $N \approx 3,374$; practice tests reduce test anxiety with Hedges’ $g \approx -0.52$; easy practice tests more mitigating than difficult ones. **Caveat:** mostly classroom/quiz anxiety outcomes — not a free pass to ship brutal mixed ACT sims as day-one “practice tests.”

FACT (anxiety can reshape *what* is learned, not only performance): Fioriti, Pizzie, Evans, Green & Lyons (2025, *JEP:LMC*, doi:10.1037/xlm0001453) — adults learning unfamiliar multiplication across days: high math anxiety associated with **impaired procedural learning** on unrepeatable problems but **accelerated retrieval learning** on heavily repeated problems — consistent with effort allocation toward declarative retrieval routes and away from procedural efficiency gains.

HYPOTHESIS for MindCraft: Anxious Maya may *prefer* and even accelerate on repeated retrieval of the same items (looks like mastery, feels safer) while under-building procedural flexibility and strategy selection — exactly the exam skills interleaving/SE target. Product that only rewards repeated-item fluency can **serve anxiety’s preference** while failing transfer.

XLI.6 Resolution doctrine (not a slogan)

FOUNDER BELIEF / HYPOTHESIS — SAFE-DD stack (ties FEI / mastery climate / IL / SE):

1. **Equip before you strain.** Short block-lite + worked-example / card fade until execution errors are mostly slips (XXVI, XXXIX). DD without schema is undesirable.
2. **Lower stakes while raising retrieval.** Soft-wrong, mastery climate, no public rank, no streak-as-worth (XXXVIII; Hinze & Rapp). Retrieval as *training*, not judgment.
3. **Dose difficulty to remaining WM.** Prefer structured SE picks over open essays under high affect (XL SE-3); prefer faded interleave over day-one full mix (IL-2); prefer spaced short sets over marathon mixed threat.
4. **Frame the feeling.** Copy: “This mix trains *choosing* — feeling slower is the workout, not proof you’re behind.” Never: “hard mode for elites” / “grind through the pain.”
5. **Measure the right win.** Session accuracy under block is not the North Star. Prefer `retry_120s`, `delayed mix / transfer_pass`, `solo_transfer_pass`, and anxiety-segment retention — not streaks or hint-free speed alone.
6. **Do not abandon DD for the anxious segment.** Permanent easy fluency is identity colonization by avoidance (XXV). Graded exposure to equipped difficulty is the path — therapy-adjacent but not a therapy claim (queue id 58).

Kill the false dichotomy: Bjork vs Ashcraft is not “science wars.” Bjork already says unequipped difficulty is undesirable; Ashcraft explains *why anxiety unequips* (WM). The product job is **re-equip + destake + sequence**, then apply DD.

XLI.7 Competitive / marketing implications

Competitor pattern	What it often does under anxiety	MindCraft wedge
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Duolingo-like streaks	Turns miss into identity failure; pressures retrieval	Ban streak-as-learning; train without shame timers
Blocked textbook / many apps	Fluency comfort; weak strategy choice	Honest: early block OK; claim exam-ready only after mix
Brutal “exam sim” day one	High stakes + hard mix → Hinze-style disruption + exit	Soft sim → rise stakes after retry_120s healthy
ChatGPT tutor monologue	Soothes without generation; may reduce threat short-term	Student-generated why <i>before</i> AI wrap (XL); calm ≠ mastery
“Growth mindset” posters	Empty struggle praise without equipment	Struggle + scaffolds + mastery climate + transfer metrics

Marketing language that survives

- **Allowed:** “Hard practice only after you’re equipped — then we mix so exams don’t surprise you.”
- **Allowed:** “Feeling slower on mixed sets is the training signal, not a verdict.”
- **Allowed (parent):** “We separate *training retrieval* from *graded judgment* so practice builds memory instead of panic.”
- **Banned:** “Struggle always equals learning.”
- **Banned:** “We never make anxious students struggle.”
- **Banned:** Streak / blocked fluency / AI comfort as proof of readiness.
- **Banned:** Bloom 2-sigma; absolute tutoring-is-free.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without DD×ANX data
Soft-wrong + mastery copy on hard items	L1 affect-safe practice climate
Block → fade → interleave schedule	L1 sequenced DD (intent aligned XXXIX)
Low-stakes spaced retrieval before “exam mode”	L1 Hinze-aligned retrieval hygiene
“Our struggle pedagogy raises ACT for anxious students”	L0 until DD-1/DD-2
Always-on max mix / open SE under threat	L0 — likely undesirable difficulty

XLI.8 Experiments spawned

ID	Question	Design	Primary	Kill condition
DD-1	Faded DD (block→mix + soft-wrong) vs early full mix same items	2-arm; stratify high vs low math-anxiety pretest	7d transfer_pass; retry_120s; exit after first miss	Early-mix wins transfer and retry in HMA segment → fade doctrine wounded

DD-2	Low-stakes retrieval sets vs same items framed as scored “exam drill”	2-arm (Hinze-style stakes)	Delayed retention; state anxiety; return D+1	High-stakes $\geq$ low-stakes on delayed + no anxiety cost $\rightarrow$ stakes doctrine overstated
DD-3	Structured SE under mix vs open SE under mix for HMA segment	2-arm (densifies SE-3)	Completion; transfer; hint binge	Open SE wins without churn $\rightarrow$ structured-default less necessary
DD-4	Repeated-item fluency path vs varied procedural path (Fioriti-inspired)	2-arm within HMA	Procedural transfer / strategy-class errors vs item-repeat accuracy	Repeat-fluency equals varied on transfer $\rightarrow$ anxiety-shaped preference harmless
DD-QUAL	10 Maya interviews: “When did hard practice feel like training vs attack?”	Appendix B	coded stake/equipment moments	Nobody separates $\rightarrow$ UX framing invisible

Pre-reg (XXXIV): DD-* identify **schedule** $\times$ **stakes** $\times$ **anxiety** $\rightarrow$ **transfer/return**. They do **not** identify long-term identity from one hard session. Densifies Experiment D and IL-2/SE-3 as the anxiety-gated versions of desirable difficulty.

Supersedes / densifies: XXIX.4 “never introduce DD as shame”; XXXIX IL-2; XL SE-3; I.4 bottleneck $\rightarrow$ now has an operational SAFE-DD stack pending DD-1/2.

XLI.9 Confidence table

Claim	Label	Confidence
Spacing/interleaving/retrieval/geration can improve long-term learning vs fluent short-term conditions	FACT (Bjork lineage)	High
Difficulties are desirable only if the learner can respond successfully	FACT (Bjork & Bjork)	High
Math anxiety disrupts WM / acts like dual-task load	FACT (Ashcraft & Kirk, 2001; Ashcraft & Krause, 2007)	High

High stakes can erase retrieval-practice benefits on delayed tests	FACT (Hinze & Rapp, 2014)	High
Classroom retrieval programs often associate with <i>lower</i> reported exam nerves	FACT (Agarwal et al., 2014; self-report)	Medium–High
Practice testing meta-analytically reduces test anxiety (esp. easier quizzes)	FACT (Yang et al., 2023)	High
HMA may accelerate repeated-item retrieval learning while impairing procedural learning	FACT (Fioriti et al., 2025)	Medium–High (adult lab; arithmetic)
SAFE-DD (equip → destake → dose → frame → measure) is the right MindCraft default	FOUNDER BELIEF / HYPOTHESIS	Medium–High
“Struggle always” or “never struggle anxious kids” is good marketing	SPECULATION / false	Low (against)

XL1.10 What this chapter kills

1. **Kill:** “Desirable difficulties always help — turn difficulty up for everyone.” (Ignores Bjork’s own *equipped* clause + Ashcraft WM.)
2. **Kill:** “Anxious students should only get easy, blocked fluency forever.” (Serves avoidance; fails transfer; Fioriti-shaped trap.)
3. **Kill:** High-stakes “exam mode” as the default vehicle for retrieval practice. (Hinze & Rapp.)
4. **Wound:** Self-report that “quizzes reduce anxiety” as license for brutal day-one mixed sims — survives as *low-stakes, well-sequenced* retrieval, not as cruelty.
5. **Survive (constrained):** Faded, low-stakes, mastery-framed desirable difficulties that produce delayed mix evidence — the commercial spine for “exam-ready without panic theater.”

Doctrine until data: Ship **SAFE-DD**: equip → lower stakes → dose to WM → frame as training → measure delayed transfer and `retry_120s` in the anxiety segment. Market productive struggle only when those guards are visible. Never sell pain or comfort as mastery.

Part XLII — Social Comparison & Leaderboards: When Ranks Help vs Hurt Novices

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 3)

Primary question: When do social-comparison surfaces (especially leaderboards) motivate learning versus threaten identity, and what should MindCraft default for Maya — the anxious / novice / identity-fragile segment?

Owners: Motivation · Affect / Anxiety · Product (Practice / Community UX) · Growth / Positioning · Red Team

Commercial job: Kill “ranks = engagement” as a North Star; harden the I.4 TARGET rule (default-ban appearance UX) into a shippable comparison policy competitors can be audited against.

XLII.1 Why this chapter exists

Part XXXVIII (goal orientation) already bans **appearance** goal structures and streak-as-North-Star. Part XXIV / XLI show social-evaluative threat taxes working memory and can turn desirable difficulty into threat. Competitive products (Duolingo-like ranks, classroom Top-N boards, public classroom streaks) still sell comparison as “motivation.”

FOUNDER BELIEF under audit: Identity transformation needs competence evidence — not a public ladder that tells bottom-half Maya she is behind *people* rather than behind a *skill criterion*.

Claim we refuse as doctrine: “Leaderboards motivate learners” (full stop) — and its twin “if we anonymize, any infinite rank is fine.”

XLII.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Social comparison	Evaluating self via others’ standing (Festinger)	Any peer rank, percentile, “class average,” public tutor callout	Turns mastery climate into appearance climate
Upward / downward / lateral	Compare to better / worse / similar others	Top-N board; “you beat 80%”; neighbor band	Extreme upward gap → give-up; downward → complacent ego boost without learning
Absolute vs relative board	All players ranked vs local neighbors / Top-N only	Infinite classroom board vs “near you” strip	Absolute + real names = max face threat for low ranks
Criterion / mastery feedback	Progress vs a task standard, not vs peers	Path progress, concept mastery bars, personal bests	Still can be gamed if criterion ≠ transfer
Normative / appearance feedback	Standing vs others; looking smart/fast	Public leaderboard, streak leagues	Identity threat; WM load; exit after first public miss

Operational definition (HYPOTHESIS): A MindCraft comparison surface is *safe* only when (a) the default is **self/criterion**, (b) peer comparison is **opt-in**, (c) comparison targets are **attainable** (local band, not unreachable

Top-1), (d) metrics ranked are **effort/process-compatible** with mastery climate (not speed-only or streak-only), and (e) low standing is never **publicly named** for the anxiety / novice segment.

XLII.3 Festinger → competition: the comparison engine

FACT (origin): Festinger (1954, *Human Relations*, “A Theory of Social Comparison Processes”) — when objective standards are ambiguous, people evaluate abilities/opinions by comparing with others; ability comparison carries a unidirectional drive upward (improve toward better standards).

FACT (selection + reaction meta): Gerber, Wheeler & Suls (2018, *Psychological Bulletin*, doi:10.1037/bul0000127) — across 60+ years: when choosing up vs down, strong preference for **upward** comparison under no threat; **contrast** (self-evaluation moves *away* from the target) dominates reactions — ability estimates and affect often decline after upward contact. People often choose upward comparisons *even though* the typical reaction is self-deflating contrast.

FACT (competition model): Garcia, Tor & Schiff (2013, *Perspectives on Psychological Science*, doi:10.1177/1745691613504114) — social comparison is a major source of competitive attitudes/behavior; situational amplifiers include **proximity to a standard** (near #1 vs far away) and **N** (few vs many competitors). Ranking UIs are engineered comparison machines.

Product translation: Shipping a leaderboard is not a neutral “engagement widget.” It is a deliberate injection of comparison → contrast risk → competitive framing. For Maya, that often means **appearance climate** (XXXVIII), not mastery.

XLII.4 Leaderboards can move behavior — under narrow conditions

FACT (HE experiment): Landers & Landers (2014, *Simulation & Gaming*, doi:10.1177/1046878114563662) — random assignment to a course wiki **leaderboard** (not tied to grades) increased project interactions (~+29.6) vs control; time-on-task mediated academic performance. Supports: leaderboards *can* raise engagement via goal/conflict attributes **in that setting**.

Caveats (FACT / scope): Undergraduate / course project; not math-anxious high-school Maya; not public infinite classroom shame; achievement metric was interaction volume, not durable transfer. Landers’ theory of gamified learning (2014, doi:10.1177/1046878114563660) treats game attributes as mediators/moderators of *instructional* quality — gamification does not replace pedagogy.

FACT (HE systematic review): Li, Liang, Fryer & Shum (2024, *Journal of Computer Assisted Learning*, doi:10.1111/jcal.13077) — 20 articles / 22 studies / 29 interventions (2014–2023), **all higher education**; ~half ≤1 hour. Leaderboards *can* benefit motivation/engagement/performance, but **effects hinge on design**. Authors recommend (among others): absolute boards for motivation *with* anonymity especially for low ranks; mix performance + engagement criteria; caution that HE short-duration findings do **not** generalize cleanly to K–12.

HYPOTHESIS: Landers-style gains are most likely when learners already have task competence + low identity threat + voluntary/low-stakes ranking. MindCraft’s default user is the opposite segment on day one.

XLII.5 Math-specific: anxiety × infinite ranks

FACT (primary maths game, large N): Almo, Rocha, Brennan & Dondio (2024, *International Journal of Serious Games*, doi:10.17083/ijsg.v11i4.794) — N=1,389 Irish primary students, 6-week *Seven Spells* programme with an **infinite leaderboard**. Leaderboard enjoyment predicted by rank position **and** maths anxiety: **maths-anxious players disliked the leaderboard more than non-anxious peers even after controlling for position**. Preference for playing against others also predicted enjoyment. Small correlation between liking the board and liking the game.

Implication (HYPOTHESIS): For anxious learners, public infinite rank is not a free engagement boost — it is a trait-moderated threat surface. “They’re low because they’re bad at the game” does not exhaust the mechanism; anxiety predicts dislike **net of rank**.

FACT (negative feedback climate): Fong, Patall, Vasquez & Stautberg (2019, *Educational Psychology Review*, doi:10.1007/s10648-018-9446-6) — meta-analysis (78 studies): negative feedback reduces intrinsic motivation vs positive feedback; harm lessens when feedback includes instructional how-to-improve detail and uses **criterion-based** standards (vs purely normative). Public low rank is often pure normative negative feedback without instructional content.

Bridge to XLI / XXIV: Social-evaluative threat (public miss, public bottom third) is exactly the stakes Hinze & Rapp warn can disrupt retrieval benefits — and Ashcraft-style WM load. Infinite boards amplify that climate at scale.

XLII.6 Resolution doctrine — SAFE-COMPARE

FOUNDER BELIEF / HYPOTHESIS — SAFE-COMPARE stack (ties TARGET / SAFE-DD / FEI):

1. **Default = criterion, not peers.** Show progress vs path / mastery / personal best. Peers off by default.
2. **Opt-in competition only.** Never force infinite boards onto gap-scan novices or high-anxiety segment.
3. **If peers: local + attainable.** Prefer neighbor bands / similar-ability cohorts over Top-1 theater (Garcia proximity + Gerber contrast).
4. **Anonymize low standing.** No named public bottom ranks (Li et al. Rec. 2; Bai et al. cited therein).
5. **Rank what mastery climate allows.** Prefer process-compatible signals (retries completed, transfer attempts, concepts unlocked) over speed leagues / streak leagues. Never rank “hint-free seconds” as identity.
6. **Instructional wrap on any relative miss.** If relative feedback ships, attach *how to improve next* (Fong moderators) — not only “you’re #47.”
7. **Measure the right win.** `retry_120s`, `challenge_accept` (mastery-coded), `transfer_pass` / `solo_transfer_pass` — not `leaderboard_views` or rank delta as learning.

Kill the false dichotomy: “Competition is toxic” vs “competition is always motivating.” Competition can raise time-on-task for some prepared learners (Landers). For Maya default, **unbuffered infinite peer rank is appearance UX** — banned until SC experiments say otherwise.

XLII.7 Competitive / marketing implications

Competitor pattern	Comparison mechanism	MindCraft wedge
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Duolingo-like leagues / ranks	Normative weekly standing; streak adjacent	Mastery path progress; ban streak/rank as learning proof
Classroom Top-N boards	Extreme upward targets; face threat	Criterion bars; optional local bands
Khan / practice dashboards	Often self-paced; weaker public rank	Keep self/criterion; don't "gamify catch-up" with shame ranks
Brilliant prestige signals	Ability-appearance among peers	Sell <i>solo transfer</i> and <i>challenge-seeking why</i> , not percentile flex
ChatGPT tutor alone	No peer board — but social comparison still via school/parents	Own the <i>safe</i> comparison story: progress vs yesterday's self

Marketing language that survives

- **Allowed:** "Progress against *your* path — not a public ladder."
- **Allowed:** "Competition is optional; mastery climate is the default."
- **Allowed (parent):** "We don't put anxious kids on a named scoreboard to 'motivate' them."
- **Banned:** "Leaderboards keep students hooked" as a learning claim.
- **Banned:** Rank / league / streak as North Star or as proof of identity change.
- **Banned:** Bloom 2-sigma; absolute tutoring-is-free; empty mindset posters.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without SC data
Criterion progress / personal bests default	L1 mastery-climate alignment (XXXVIII)
Opt-in anonymized local band	L1 SAFE-COMPARE intent
"Our leaderboard raises ACT / identity for anxious Maya"	L0 until SC-1/SC-2
Forced infinite named classroom board	L0 — appearance UX; likely harm for HMA

XLII.8 Experiments spawned

ID	Question	Design	Primary	Kill condition
SC-1	Criterion-only progress vs infinite named leaderboard (same practice items)	2-arm; stratify high vs low math-anxiety	retry_120s; exit after public miss; 7d transfer_pass	Infinite board wins transfer and retry in HMA → default-ban wounded

SC-2	Opt-in local anonymized band vs forced absolute board	2–3 arm	Enjoyment; state anxiety; return D+1; challenge_accept motive codes	Forced absolute $\geq$ opt-in local on retention without anxiety cost $\rightarrow$ anonymity/opt-in overstated
SC-3	Rank process metrics (retries, mix attempts) vs rank speed/streak	2-arm within volunteers	Strategy-class errors; transfer_pass vs session speed	Speed rank equals process rank on transfer $\rightarrow$ metric choice moot
SC-4	Criterion negative feedback + how-to-improve vs rank-only “you’re #N”	2-arm (Fong-inspired)	Intrinsic motivation proxy; retry	Rank-only $\geq$ instructional criterion $\rightarrow$ Fong moderator fails in product
SC-QUAL	10 Maya interviews: “When did seeing others’ scores help vs freeze you?”	Appendix B	coded comparison threat moments	Nobody distinguishes $\rightarrow$ copy invisible

Pre-reg (XXXIV): SC-* identify **comparison surface** $\times$ **anxiety** $\rightarrow$ **return/transfer**. They do **not** identify that “healthy competition builds character” from one league season.

Densifies: XXXVIII appearance ban; I.4 TARGET “leaderboard kill for Maya default”; XXIV social-evaluative threat hypothesis.

XLII.9 Confidence table

Claim	Label	Confidence
People use others to evaluate abilities when standards are ambiguous	FACT (Festinger, 1954)	High
Upward comparison often preferred; contrast often dominates self-eval/affect	FACT (Gerber et al., 2018)	High
Ranking contexts amplify competitive social comparison	FACT (Garcia et al., 2013)	High
Leaderboards can increase time-on-task / mediate performance in some HE settings	FACT (Landers & Landers, 2014)	High (scope-limited)

Leaderboard effects in education are design-dependent; HE evidence $\neq$ K–12	FACT (Li et al., 2024)	High
Maths-anxious players dislike infinite boards more, net of rank	FACT (Almo et al., 2024)	High
Normative negative feedback tends to hurt intrinsic motivation vs positive; criterion + how-to softens	FACT (Fong et al., 2019)	High
SAFE-COMPARE default (criterion, opt-in, local, anonymize low, process metrics) is right for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium–High
“Leaderboards motivate everyone / raise learning for Maya”	SPECULATION / false as universal	Low (against)

XLII.10 What this chapter kills

1. **Kill:** Universal “leaderboards motivate students” as product doctrine or marketing. (Design- and trait-dependent; HE $\neq$ Maya.)
2. **Kill:** Infinite named public ranks as default engagement for novices / high math-anxiety. (Almo et al.; TARGET / appearance.)
3. **Kill:** Rank delta / league standing / streak leagues as North Star or as proof of identity transformation.
4. **Wound:** Absolute-board motivation gains from HE reviews as license for K–12 public boards — survives only as *design hypothesis*, not as Maya default.
5. **Survive (constrained):** Opt-in, anonymized, attainable, criterion-wrapped comparison for learners who already seek competition — never as the onboarding spine.

Doctrine until data: Ship **SAFE-COMPARE:** criterion default → peer opt-in → local/attainable → anonymize low standing → rank mastery-compatible process → instructional wrap → measure transfer/retry, not rank views. Market “progress without public ladders.” Never sell ranks as belonging.

Part XLIII — Habit Formation Science (Wood/Clear): Cue Design Without Identity Colonization

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 6 $\equiv$ Red Team slot; ch43 never written → prefer Researcher per rotation)

Primary question: How should MindCraft use habit science (context cues, repetition, automaticity) to raise *return to practice* without making streaks / daily check-ins the student’s identity — or mistaking habit for learning?

Owners: Motivation · Product (Practice / notifications / recognition) · Growth / Positioning · Red Team

Commercial job: Harden Insight 5 / HID into a shippable **SAFE-HABIT** stack competitors (especially Duo-like streak products) can be audited against; kill “habit = mastery” and “21 days” folklore in copy and KPIs.

XLIII.1 Why this chapter exists

Parts XXV (habit ≠ identity), XXXVIII (appearance / streak North Star ban), XLI (SAFE-DD), and XLII (SAFE-COMPARE) already treat streaks and ranks as threat surfaces when they become *worth*. Market gravity still pulls edtech toward Duolingo-shaped habit loops: push cue → micro-session → streak reward → loss aversion.

FOUNDER BELIEF under audit: Identity transformation needs *reliable return to difficulty* — so cue design is load-bearing — but the habit that forms must be “open practice and try the next hard thing,” not “protect a number that proves I am consistent.”

Claims we refuse as doctrine:

1. “Streaks = learning” / streak length as North Star.
 2. “Habits take 21 days” as science.
 3. Treating popular packaging (*Atomic Habits*) as peer-reviewed habit theory without Wood/Gardner/Lally grounding.
 4. Habit loops that **colonize identity** — the student becomes “a streak-keeper” rather than “someone who can recover from wrong.”
-

XLIII.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Habit (scientific)	Learned context→response association in procedural memory; cue activates response with limited goal mediation (Wood)	Stable time/place/preceding-action cue for practice open	Shipping “motivation” when the real lever is <i>cue stability</i>
Automaticity	Behavior feels efficient / less deliberative	Self-report automaticity; shorter time-to-first-problem	Confusing app-open automaticity with <i>math skill</i> automaticity
Context cue	Physical setting, time, people, prior action in a sequence	After-school phone dock; tutor session end → 10-min solo; calendar block	Push spam as fake context; unstable cues never habitize

Implementation intention	If-then plan: when Y, do X (Gollwitzer)	Onboarding “After dinner, I open MindCraft Map”	Vague goals (“practice more”) with no situational trigger
Identity colonization (product sense)	Habit metric becomes who the student is / fears losing	Streak as self-worth; freeze culture; XP-as-self	HID failure — return without identity change; break → exit
SAFE-HABIT	Cue + small start + miss-forgiveness + learning metrics above habit metrics	See XLIII.6	Habit theater that still ranks streak / DAU as proof

Operational definition (HYPOTHESIS): A MindCraft habit surface is *safe* only when (a) cues target **opening equipped practice**, not protecting a counter, (b) miss recovery is cheap (Lally: one miss ≠ wipe), (c) recognition prefers process/identity markers over streak length, and (d) North Stars remain `retry_120s` / `transfer_pass` / `solo_transfer_pass` / identity items — never streak days.

XLIII.3 Wood: habits are context–response, not vibes

FACT (review): Wood & Rünger (2016, *Annual Review of Psychology*, doi:10.1146/annurev-psych-122414-033417) — habits are learned associations between recurring **context cues** and responses, strengthened gradually via associative / reward learning; once formed, perception of the cue activates the response representation with limited need for executive control. Defining features of habit automaticity include cue activation and **relative insensitivity to short-term goal/outcome changes**.

FACT (habit–goal interface): Wood & Neal (2007, *Psychological Review*, “A New Look at Habits and the Habit–Goal Interface”) — goals help *start* repetition and exposure to cues; once habits form, contexts can trigger responses without a mediating goal. Habit change therefore often requires changing cues / environments, not only “trying harder.”

FACT (systems view): Wood, Mazar & Neal (2021, *Perspectives on Psychological Science*, habits vs goals as separate but interacting systems) — habitual responses can activate regardless of current goals; people may still inhibit them, but cue-driven activation is real. Product translation: a streak notification is a **Pavlovian/context cue** engineered by the app — not proof the student wants math.

Product translation: If MindCraft wants return, design **stable contexts** (same after-school window; end-of-tutor-session stack; device home-screen tile) more than pep talks. If MindCraft wants identity, do **not** make the strongest cue “don’t break the number.”

XLIII.4 Clear / Fogg packaging vs the evidence spine

FACT (popularization, not primary science): Clear (*Atomic Habits*, 2018) packages cue / craving / response / reward loops and “habit stacking” (“After [current habit], I will [new habit]”) for a mass audience. Fogg (*Tiny Habits*) popularized tiny start + anchor habits. Neither replaces Wood’s procedural-memory account.

FACT (what stacking actually is): Habit stacking’s evidence spine is largely **implementation intentions** — Gollwitzer & Sheeran (2006, *Advances in Experimental Social Psychology*, doi:10.1016/S0065-2601(06)38002-1): across 94 tests, if-then plans show a medium-to-large effect on goal attainment ($d \approx 0.65$), via heightened cue accessibility and automated initiation.

HYPOTHESIS: Clear is usable *copy scaffolding* for onboarding (“After tutoring, open Map for one mission”) if MindCraft cites the Gollwitzer mechanism and measures initiation → equipped practice, not Clear’s compound-growth metaphors as science.

Kill: Marketing that says “built on Atomic Habits science” as if the book were a methods paper. Prefer: “cue-based practice plans (implementation intentions; habit research).”

XLIII.5 How long, how fragile, what to measure

FACT (timeline myth): Lally, van Jaarsveld, Potts & Wardle (2010, *European Journal of Social Psychology*, doi:10.1002/ejsp.674) — daily repetition in a consistent context; automaticity rises on an asymptotic curve; among good-fit participants, median time to ~95% of asymptote $\approx$ **66 days** (range **18–254**). Missing **one** opportunity did not materially derail formation. ~Half of motivated participants still failed to perform consistently enough to reach habit status in the study window.

Kill: “21 days to form a habit” (Maltz folklore, not this literature). Also kill product copy that implies a week-long streak is a habit.

FACT (definition hygiene): Gardner (2015, *Health Psychology Review*, doi:10.1080/17437199.2013.876238) — habit as a *process* whereby a stimulus generates an impulse to act via learned S–R association; impulse need not equal behavior. Most health-habit studies are correlational / self-report-heavy — confidence ceiling for edtech transfer claims.

FACT (measurement trap): Verplanken & Orbell (2003, *Journal of Applied Social Psychology*, doi:10.1111/j.1559-1816.2003.tb01951.x) — SRHI includes history of repetition, automaticity, **and expressing identity**. Self-report habit strength can smuggle identity language into a frequency metric.

Product implication (HYPOTHESIS): Do not treat SRHI-like “this is typically me” items as proof of *math identity transformation* (Part XXV / id 61 queue). Separate: (1) practice-open automaticity, (2) math self-efficacy / narrative identity scales, (3) transfer metrics.

XLIII.6 Duolingo-shaped products: habit works — and colonizes

FACT (industry, self-report): Duolingo publicly designs streaks using habit research framing; company blog claims learners who reach short streaks are far more likely to continue (e.g., 7-day streak correlational lifts — *not* causal RCTs published as independent learning science). Streak freezes / separation of streak vs daily goal are retention experiments on **engagement**, not transfer.

FACT (misuse qualitative): Hadi Mogavi et al. (2022, *Learning @ Scale*, doi:10.1145/3491140.3528274) — forum analysis + interviews on Duolingo: **gamification misuse** when users fixate on XP/streaks/leaderboards and get

distracted from learning; reasons include competitiveness, playfulness overlearning, dark nudges, compulsion, herding. Ramifications touch learning efficiency, well-being, and ethics.

Bridge to XLII / XXXVIII: Streak leagues + infinite ranks are the same family as appearance climates — consistency theater. Wood predicts cues will keep firing even when the *learning* goal cooled; Hadi Mogavi shows the product can train the wrong response (protect streak / farm XP).

FOUNDER BELIEF: MindCraft may borrow **cue stability** + **small start** + **miss forgiveness**. MindCraft must not borrow **streak-as-self** or **loss-aversion-as-pedagogy**.

XLIII.7 Resolution doctrine — SAFE-HABIT

FOUNDER BELIEF / HYPOTHESIS — SAFE-HABIT stack (ties HID / TARGET / SAFE-DD / SAFE-COMPARE):

1. **Cue the practice, not the counter.** Default cues: calendar/time, tutor-session end stack, story-world return point — not “your streak dies at midnight” as the primary cue.
2. **Implementation intention onboarding.** One written if-then in gap-scan aftermath or first session (“After [stable cue], I open one mission”). Measure initiation, not slogan recall.
3. **Tiny equipped start.** First action $\leq$ friction of one problem with SAFE-DD equipment — not a multi-screen habit tutorial.
4. **Miss $\neq$ identity failure.** One miss does not reset belonging (Lally). Prefer gentle resume copy over shame / “you broke it.” Streak freezes are a *retention patch*, not a learning feature — if shipped, never as North Star.
5. **Habit metrics subordinate.** Track D1/D7 return, cue-honoring rate, time-to-first-problem as *diagnostics*. Learning North Stars remain retry / transfer / solo transfer / challenge_accept (mastery-coded).
6. **Identity markers elsewhere.** Map fill, story unlocks, tutor recognition of *recovery*, parent “they tried the hard one again” — not streak length as “who they are.”
7. **No identity colonization.** Ban copy that equates worth with unbroken days. Ban ranking streaks (XLII). Ban streak leagues as proof of transformation.

Kill the false dichotomy: “We refuse all habit design” vs “we must out-Duo Duo.” Refuse *colonizing* habit design; keep *cue science*.

XLIII.8 Competitive / marketing implications

Competitor pattern	Habit mechanism	MindCraft wedge
Duolingo streaks / freezes	Loss aversion + daily cue; retention-first	Cue practice return; miss-forgiving; never streak-as-learning
Duo leagues + XP	Habit + appearance compound	SAFE-COMPARE + SAFE-HABIT; sell solo transfer
Khan session streaks / energy	Mild habit / progress bars	Keep criterion progress; don’t add shame timers

Brilliant daily puzzles	Delight + return habit	Return <i>into</i> recoverable struggle, not only delight
ChatGPT “study every day” tips	Advice without cue architecture	Own <i>if-then</i> + <i>equipped mission</i> as the plan

Marketing language that survives

- **Allowed:** “Same time, same cue — open one hard problem and recover.”
- **Allowed:** “Habits get you to the desk; identity is what you do when it’s wrong.”
- **Allowed (parent):** “We build a practice cue without making a broken streak mean failure.”
- **Banned:** “Streaks create math identity.”
- **Banned:** “21-day habit challenge = mastery.”
- **Banned:** Streak / DAU as North Star or as proof of transformation.
- **Banned:** Bloom 2-sigma; tutoring-is-free; empty mindset posters.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without HAB data
If-then practice cue + tiny equipped start	L1 initiation design (Gollwitzer-aligned)
Miss-forgiving resume without streak shame	L1 HID hygiene
“Our streak raises ACT / math identity”	L0 until HAB-1/HAB-2
Forced streak + public streak rank as default	L0 — colonization UX

XLIII.9 Experiments spawned

ID	Question	Design	Primary	Kill condition
HAB-1	If-then cue plan vs streak-primary onboarding (same missions)	2-arm; new users	D7 return; time-to-first-problem; 14d transfer_pass	Streak arm wins transfer and identity item → on-ramp-only doctrine wounded
HAB-2	Miss-shame copy (“you broke your streak”) vs miss-forgiving resume	2-arm after forced miss	Return D+1; state anxiety; retry_120s	Shame ≥ forgive on return without anxiety cost → forgive overstated
HAB-3	Stable calendar cue vs push-only reminders	2-arm	Cue-honoring rate; 4-week automaticity proxy	Push-only equals calendar on return → context-stability claim weak

HAB-4	Recognition: process/recovery badge vs streak badge vs none	3-arm (ties GO-2)	challenge_acce pt motives; identity item; streak obsession items	Streak badge wins engagement and transfer → HID weaker than feared
HAB-QUAL	10 Maya interviews: “What does your streak (elsewhere) make you feel when you miss?”	Appendix B	coded colonization vs helpful cue	Nobody reports threat → copy over-indexed

Pre-reg (XXXIV): HAB-* identify **cue/recognition** × **return/transfer**. They do **not** identify that “daily habit builds mathematicians” from DAU alone.

Densifies: Insight 5 / HID; I.4 TARGET streak ban; XXXVIII appearance; XLII streak-league kill.

XLIII.10 Confidence table

Claim	Label	Confidence
Habits are context–response associations; cue-triggered; relatively goal-insensitive once strong	FACT (Wood & Rünger, 2016; Wood & Neal, 2007)	High
Goals help form habits via repetition/exposure; changing cues aids habit change	FACT (Wood line)	High
Implementation intentions raise goal attainment (medium–large)	FACT (Gollwitzer & Sheeran, 2006)	High
Automaticity plateaus on the order of weeks–months; median ~66d in Lally sample; high variance; one miss often OK	FACT (Lally et al., 2010)	High
“21 days” is not the Lally result	FACT	High
Clear/Fogg are popular packaging; stacking ≈ if-then planning	HYPOTHESIS / FACT (mechanism mapping)	Medium–High
Gamification can distract from learning (Duo case)	FACT (Hadi Mogavi et al., 2022)	High (qualitative)
Streaks can raise return without identity / transfer	HYPOTHESIS (HID; industry)	Medium–High

SAFE-HABIT is right default for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium–High
“Streaks = learning / identity”	SPECULATION / false as universal	Low (against)

XLIII.11 What this chapter kills

1. **Kill:** Streak length / DAU / “habit streak” as North Star or as proof of identity transformation. (HID; standing ban densified with Wood + Duo misuse.)
2. **Kill:** “21-day habit” marketing as science. (Lally variance + asymptote.)
3. **Kill:** Equating *Atomic Habits* branding with peer-reviewed habit doctrine. (Use Wood/Gardner/Gollwitzer; Clear as optional UX copy only.)
4. **Kill:** Shame-on-miss and public streak ranks as default engagement for Maya. (Colonization; XLII.)
5. **Wound:** “Any daily reminder is habit science” — survives only if tied to **stable context + repeated equipped response**, not push spam.
6. **Survive (constrained):** Cue design, tiny starts, implementation intentions, and miss forgiveness as **on-ramps** to FEI / transfer — never as the product thesis.

Doctrine until data: Ship **SAFE-HABIT**: cue practice (not the counter) → if-then onboarding → tiny equipped start → miss ≠ identity failure → habit metrics subordinate → identity markers elsewhere → ban streak colonization. Market “we get you back to the hard problem — without making a broken streak mean you’re not a math person.”

Part XLIV — Intrinsic Motivation Killers (Deci): Controlling Rewards and the Caveats

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 9; hour%6≠0; 3 researcher entries since synthesizer v1.6 → Researcher, not Synthesizer)

Primary question: When do MindCraft rewards, XP, badges, streaks, and “do this for the prize” UX *undermine* the intrinsic interest and autonomy we claim to build — and when are they informational rather than controlling?

Owners: Motivation · Product (Practice / recognition / parent copy) · Growth / Positioning · Red Team

Commercial job: Turn the vague “extrinsic bad / Deci says so” slogan into a shippable **SAFE-REWARD** stack competitors can be audited against; kill both “all rewards kill love of math” absolutism and “gamify everything” as identity strategy.

XLIV.1 Why this chapter exists

Parts XXV (habit ≠ identity), XXXVIII (appearance climates), XLII (SAFE-COMPARE), and XLIII (SAFE-HABIT) already treat streaks, ranks, and consistency theater as threat surfaces. Market gravity still pulls edtech toward *expected*

tangible / token rewards for doing the task: XP for opening the app, coins for finishing any problem, badges for login streaks, parent dashboards that score “minutes practiced.”

FOUNDER BELIEF under audit: Identity transformation needs *felt autonomy* and *competence information* — not the absence of all rewards. The killer is **controlling functional significance** (pressure to behave for the inducement), not the existence of feedback or unexpected acknowledgment.

Claims we refuse as doctrine:

1. “All rewards kill intrinsic motivation” (Cameron–Pierce counter-literature exists; CET is differentiated).
2. “Gamification is always fine if engagement rises” (engagement $\neq$ free-choice interest; Hadi Mogavi / Duo misuse).
3. “Verbal praise always helps kids” (Deci: controlling praise / “you should keep it up” can undermine).
4. Treating SDT as a poster (“autonomy / competence / relatedness”) without contingency design.

XLIV.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Intrinsic motivation	Doing the activity for inherent interest / satisfaction	Free-choice return to hard problems; curiosity asks; challenge_accept for mastery reasons	Confusing “opens app for XP” with interest
Extrinsic motivation (spectrum)	Behavior for separable outcome; can be more/less autonomous (external → introjected → identified → integrated)	Parent-pressure vs “I chose ACT path because...”	Treating all extrinsic as poison or all as fine
Controlling vs informational	Event felt as pressure to behave vs competence feedback (CET)	“Finish 10 for a coin” vs “you used the bridge correctly”	Same badge can be either — design of <i>how</i> it is framed
Engagement-contingent reward	Expected reward for <i>doing</i> the task (no quality bar)	XP for any attempt; login coins	Classic undermine path (Deci 1999)
Completion-contingent	Expected reward for finishing	“Complete mission → loot” regardless of strategy quality	Similar undermine; weak competence signal
Performance-contingent	Reward for meeting a standard	Badge for transfer pass; tutor note for recovery	Can inform competence <i>or</i> control — climate decides

SAFE-REWARD	Prefer informational feedback; avoid expected tangible for mere engagement; autonomy-supportive framing	See XLIV.7	Reward theater that still trains external PLOC
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Operational definition (HYPOTHESIS): A MindCraft reward surface is *safe* only when (a) it primarily conveys **competence / strategy information**, (b) expected tangible / token rewards are **not** the reason to open practice, (c) framing is autonomy-supportive (choice, rationale, non-pressuring language), and (d) North Stars remain `retry_120s` / `transfer_pass` / `solo_transfer_pass` / identity items — never XP velocity or badge count.

XLIV.3 SDT spine (needs, not posters)

FACT (overview): Ryan & Deci (2000, *American Psychologist*, 55(1), 68–78, doi:10.1037/0003-066X.55.1.68) — Self-Determination Theory posits innate needs for **competence, autonomy, and relatedness**; social contexts that support these needs facilitate intrinsic motivation and internalization; contexts that thwart them yield alienation and depleted motivation.

FACT (CET): Cognitive Evaluation Theory (Deci & Ryan line; summarized in Deci, Koestner & Ryan, 1999) — rewards and external events have **functional significance**: controlling aspects shift perceived locus of causality outward (undermine intrinsic motivation); informational aspects affirm competence (can enhance). Competence feedback without autonomy does not reliably enhance intrinsic motivation.

Product translation: MindCraft’s brand voice (“you’re becoming someone who recovers”) must not be paired with UX that says “do this *or* lose the prize.” Relatedness (tutor, story world, parent) is not a license for controlling rewards.

XLIV.4 The classic finding — and the differentiated meta

FACT (origin): Deci (1971, *Journal of Personality and Social Psychology*, 18(1), 105–115, doi:10.1037/h0030644) — expected extrinsic rewards for an interesting activity can reduce subsequent free-choice engagement (the “undermining” / overjustification pattern that launched the literature).

FACT (Deci–Koestner–Ryan 1999 meta): Deci, Koestner & Ryan (1999, *Psychological Bulletin*, 125(6), 627–668, doi:10.1037/0033-2909.125.6.627) — 128 studies. As predicted by CET:

- **Engagement-contingent** rewards undermine free-choice intrinsic motivation ($d \approx -0.40$).
- **Completion-contingent** ($d \approx -0.36$) and **performance-contingent** ($d \approx -0.28$) also undermine free-choice behavior on average.
- Engagement- and completion-contingent also hurt self-reported interest ($d \approx -0.15$ / -0.17).
- **Unexpected** tangible rewards: no average undermine on free-choice.
- **Positive feedback / verbal rewards:** enhance free-choice ($d \approx 0.33$) and interest ($d \approx 0.31$) *on average*.
- Tangible rewards tended to be **more detrimental for children** than college students; verbal rewards less enhancing for children than for college students.

FACT (education restatement): Deci, Koestner & Ryan (2001, *Review of Educational Research*, 71(1), 1–27, doi:10.3102/00346543071001001) — argue Cameron & Pierce (1994) understated undermining; conclude tangible rewards have a substantial undermining effect when CET contingencies are respected; defend keeping CET for educational practice.

Product implication (HYPOTHESIS): Expected XP/coins for *mere engagement* (open app, any answer, any completion) is the highest-risk MindCraft pattern for Maya-aged users. Informational soft-wrong + strategy feedback is the low-risk pattern. Performance-contingent badges *can* work if they mark real competence (transfer) and are not administered as pressure.

XLIV.5 Caveats — Cameron/Pierce, climate, and “not all rewards”

FACT (counter-meta): Cameron & Pierce (1994, *Review of Educational Research*, 64(3), 363–423, doi:10.3102/00346543064003363) — across 96 between-groups experiments, *overall* reward did not decrease intrinsic motivation; verbal praise increased it; the clearest negative effect was expected tangible rewards for *simply doing* a task (free-choice time after reward removal). They argued negative effects are circumscribed and avoidable.

FACT (debate): Ryan & Deci (1996, *RER*), Lepper et al. (1996), Kohn (1996) vs Cameron & Pierce (1996, *RER*) — methodological dispute over aggregation, moderators, and what “overall” means. **Not settled as “rewards are harmless.”**

FOUNDER BELIEF / resolution for MindCraft: Do **not** cite “Deci proved all gamification is evil.” Do **not** cite Cameron to greenlight engagement-contingent loot. Cite the **contingency**: expected tangible for mere doing is the kill zone; informational verbal/competence feedback is the build zone; performance-contingent is climate-sensitive.

FACT (controlling praise): Ryan (1982) and Ryan, Mims & Koestner (1983) — positive feedback administered with controlling language (e.g., “you *should* keep up the good work”) reduces intrinsic motivation relative to informational administration. Deci 1999 summarizes interpersonal climate as a moderator of performance-contingent rewards.

Kill: “We praise a lot, so SDT is satisfied.” Praise can be controlling.

FACT (choice as autonomy support): Patall, Cooper & Robinson (2008, *Psychological Bulletin*, 134(2), 270–300, doi:10.1037/0033-2909.134.2.270) — meta of 41 studies: providing choice enhances intrinsic motivation, effort, task performance, and perceived competence. Effects stronger for instructionally *irrelevant* choices, when ~2–4 successive choices are offered, when rewards are **not** stacked after the choice manipulation, and vs highly controlling controls; stronger for children than adults.

Product implication: Meaningful micro-choice (which story frame, which representation after mastery structure is set) can support autonomy — but **choice + immediately dangling reward** weakens the choice benefit. Do not “choose your path → earn 50 coins.”

XLIV.6 Competitive patterns under CET audit

Competitor pattern	CET reading	MindCraft wedge
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Duolingo XP / streaks / leagues	Expected engagement- and completion-contingent tokens + appearance climate (XLII/XLIII)	Informational recovery feedback; no XP-as-reason-to-open; SAFE-HABIT cues
Khan energy / points-ish progress	Mild completion signals; mostly mastery bars	Keep criterion progress; avoid coinization of minutes
Brilliant delight unlocks	Intrinsic interest scaffold; some completion loot	Keep delight <i>inside</i> the problem; don't pay for opening
ChatGPT "I'll quiz you if you want"	Low token reward; high fluency risk (XXXIII)	Autonomy of asking ≠ competence evidence; measure solo transfer
Parent dashboards: minutes / day streak	Parent-facing controlling contingency spills to student	Report retry_120s, transfer, challenge_accept — not "minutes bought with coins"

Bridge: XLIII says cue the practice, not the counter. XLIV says: if you must reward, reward **informative competence**, not engagement.

XLIV.7 Resolution doctrine — SAFE-REWARD

FOUNDER BELIEF / HYPOTHESIS — SAFE-REWARD stack (ties TARGET / SAFE-COMPARE / SAFE-HABIT / FEI):

1. **No expected tangible / token for mere engagement.** Ban default XP-for-open, coins-for-any-answer, loot-for-login. (Deci 1999 engagement-contingent.)
2. **Feedback > inducement.** Soft-wrong, strategy labels, bridge-gap naming, and "you recovered" tutor notes are the primary reinforcer. Prefer unexpected acknowledgment over announced prizes.
3. **If performance-contingent, mark real competence.** Badges/unlocks tied to transfer_pass / solo_transfer_pass / mastery-coded challenge_accept — not to streak days or raw volume. Administer without "you should" pressure.
4. **Autonomy-supportive framing.** Offer rationale; allow meaningful micro-choice (Patall); never "do this or lose the reward."
5. **Don't stack choice with dangling rewards.** Choice then immediate coin undermines the autonomy gain.
6. **Child-risk premium.** Maya-aged users are in the higher tangible-harm band of Deci 1999 — stricter default than adult SaaS gamification norms.
7. **Metrics hygiene.** Track XP/badge counts as *diagnostics of system behavior*, never as North Stars. Learning NS remain retry / transfer / solo transfer / identity.
8. **Parent copy:** Sell "she chose the harder one again" and competence evidence — not "she earned 400 XP."

Kill the false dichotomy: "No rewards ever" vs "gamify everything." Refuse *controlling engagement loot*; keep *informational competence signals* and story unlocks that are earned by transfer, not by attendance.

XLIV.8 Marketing language that survives

- **Allowed:** “We don’t pay kids to open the app — we show them what they figured out.”
- **Allowed:** “Unlocks follow real transfer, not login streaks.”
- **Allowed (parent):** “Progress you can trust: recovery and transfer — not points theater.”
- **Banned:** “Rewards create love of math.”
- **Banned:** “XP builds identity.”
- **Banned:** Absolute “all extrinsic motivation is bad” / “tutoring is free so motivation is free.”
- **Banned:** Bloom 2-sigma; empty mindset posters; streak/XP as North Star.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without IM data
Informational soft-wrong + strategy feedback (no engagement loot)	L1 CET-aligned hygiene
Unexpected tutor/parent acknowledgment after recovery	L1 (unexpected tangible / social)
“Our XP system raises intrinsic math interest”	L0 until IM-1/IM-2
Expected coins for any completion as default	L0 — engagement-contingent kill zone

XLIV.9 Experiments spawned

ID	Question	Design	Primary	Kill condition
IM-1	Engagement-loot (XP for any attempt) vs no-loot informational feedback (same missions)	2-arm; new users	Free-choice analog: voluntary hard-problem rate post-session; 14d interest item; transfer_pass	Loot arm wins free-choice and transfer → engagement-loot ban overstated
IM-2	Performance badge on transfer_pass (informational copy) vs same badge with controlling copy (“you should keep earning”)	2-arm	challenge_accept motives; interest; return	Controlling ≥ informational on interest → climate claim weak
IM-3	Micro-choice without reward vs choice + dangling coins	2-arm (Patall moderator)	Intrinsic motivation proxy; effort; autonomy need satisfaction	Choice+coins ≥ choice-only → “don’t stack” overstated

IM-4	Unexpected recovery acknowledgment vs announced prize for recovery	2-arm	Free-choice return; attribution coding	Announced $\geq$ unexpected on free-choice without control feelings $\rightarrow$ unexpected preference weaker
IM-QUAL	10 Maya interviews: “When a learning app gives points, what does that make you feel about the math?”	Appendix B	coded controlling vs informational	Nobody reports control $\rightarrow$ copy over-indexed

Pre-reg (XXXIV): IM-* identify **reward contingency** $\times$ **free-choice/interest/transfer**. They do **not** identify “gamification builds mathematicians” from XP velocity.

Densifies: TARGET appearance ban; SAFE-HABIT; SAFE-COMPARE; Insight on extrinsic traps; standing streak NS ban.

XLIV.10 Confidence table

Claim	Label	Confidence
Autonomy, competence, relatedness support intrinsic motivation / wellness	FACT (Ryan & Deci, 2000)	High
Expected engagement-/completion-contingent tangible rewards undermine free-choice IM on average	FACT (Deci, Koestner & Ryan, 1999)	High
Unexpected tangible rewards do not show average undermine	FACT (Deci et al., 1999)	High
Positive feedback enhances IM on average; controlling administration can reverse	FACT (Deci et al., 1999; Ryan line)	High
Tangible rewards more detrimental for children than college samples	FACT (Deci et al., 1999)	High
Cameron & Pierce find overall null with circumscribed negatives	FACT (1994); interpretation contested	Medium (dispute)

Choice enhances IM / effort / competence perceptions	FACT (Patall et al., 2008)	High
Stacking rewards after choice weakens choice benefit	FACT (Patall moderator)	Medium–High
SAFE-REWARD is right default for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium–High
“All rewards kill intrinsic motivation” / “XP creates math love”	SPECULATION / false as universals	Low (against)

XLIV.11 What this chapter kills

1. **Kill:** Expected engagement-contingent XP/coins/loot as default motivation architecture for Maya. (Deci 1999 $d \approx -0.40$ free-choice.)
2. **Kill:** Marketing that “rewards create love of math” or “XP = identity.” (External PLOC; HID.)
3. **Kill:** Absolute slogan “all extrinsic rewards are bad” used to ban informational feedback and earned transfer unlocks. (CET differentiation; Cameron caveat acknowledged.)
4. **Kill:** Controlling praise / “you should keep earning” as the voice of competence support.
5. **Wound:** “Performance badges are always safe” — survives only with informational climate + real competence criterion.
6. **Survive (constrained):** Unexpected acknowledgment, strategy-informative feedback, autonomy-supportive micro-choice, and transfer-gated unlocks under **SAFE-REWARD**.

Doctrine until data: Ship **SAFE-REWARD**: no expected tokens for mere engagement → feedback over inducement → performance marks only for real transfer/recovery → autonomy framing → don’t stack choice with dangling coins → child-risk premium → XP never North Star. Market “we don’t pay kids to practice — we make the competence visible.”

Part XLV — Mathematical Resilience: Growth Under Adversity Without Grit Theater

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 12 ≡ Red Team slot, but ch45 never written → prefer Researcher per rotation; 4 researcher entries since synthesizer v1.6 → not yet Synthesizer)

Primary question: What is *mathematical resilience* as an actionable construct (Johnston-Wilder & Lee), how does it differ from grit posters and empty “bounce back” copy, and what product stack keeps Maya in the growth zone without dumping her into the danger zone?

Owners: Motivation / Affect · Product (Practice / coach / tutor briefing) · Positioning · Red Team

Commercial job: Replace “build grit” / “resilience mindset” marketing with a shippable **SAFE-RESILIENCE** stack competitors can be audited against — growth-zone design, support recruitment, struggle-as-normal, value without coercion — tied to FEI / SAFE-DD / SAFE-REWARD.

XLV.1 Why this chapter exists

Parts XXIV (anxiety), XXV (self-efficacy / habit≠identity), XXX (attribution / helplessness), XXXI (flow / challenge–skill), XLI (SAFE-DD), and XLIV (SAFE-REWARD) already treat threat, recovery, and controlling incentives. Market and school language still collapses everything into **grit** or generic **resilience** slogans: “just push through,” “bounce back harder,” streak-as-character, leaderboard-as-grit-proof.

FOUNDER BELIEF under audit: Identity transformation under difficulty needs *domain-specific* resilience — knowing struggle is normal in math, valuing the work, knowing how to work at it, and **recruiting support** before anxiety flips the lid — not a personality transplant labeled grit.

Claims we refuse as doctrine:

1. Duckworth-style grit as MindCraft’s North Star or primary product claim (Credé et al. wound the construct).
 2. “Desirable difficulty” = keep students in the danger/anxiety zone (conflicts SAFE-DD / Ashcraft).
 3. Empty resilience posters without support architecture (tutor cards, peers, ontology scaffolds).
 4. Treating mathematical resilience as synonymous with growth-mindset wall art.
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XLV.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Developmental resilience (Masten)	Ordinary adaptive systems enabling competence despite adversity — not rare “super kids”	Design that protects basic systems (safety, belonging, competence feedback)	Selling “extraordinary character” as the product
Mathematical resilience (JW/Lee)	Positive stance that lets learners engage despite affective barriers: growth belief, value, knowing how to work at math, knowing how to recruit support	Stay in growth zone; ask for help; recover after stuck; continue study path	Posterizing the four aspects without UX
Growth Zone Model	Comfort (easy / little growth) → Growth (challenge + support) → Danger/Anxiety (threat / thinking collapses)	Challenge calibrated + support on tap; exit danger fast	Framing red-zone panic as “grit training”
Grit (Duckworth)	Perseverance + consistency of interest toward long-term goals	Tempting NS copy	Near-conscientiousness ; weak intervention leverage (Credé)

SAFE-RESILIENCE	Equip → name struggle → recruit support → destake danger → measure recovery/transfer — never grit KPI	See XLV.7	Resilience theater that still shames help-seeking
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Operational definition (HYPOTHESIS): A MindCraft session builds mathematical resilience only when the student (a) experiences struggle as *expected* not as personal defect, (b) has a clear path to **support** (cards, tutor, peer, worked scaffold) before threat spikes, (c) returns to challenge after recovery (retry_120s / challenge_accept), and (d) can transfer without the model finishing the thought — not when streak or “grit score” rises.

XLV.3 Ordinary magic — resilience is not a unicorn trait

FACT: Masten (2001, *American Psychologist*, 56(3), 227–238, doi:10.1037/0003-066X.56.3.227) — resilience in development is often **ordinary magic**: competence despite adversity typically arises from normative adaptational systems (attachment, learning systems, self-regulation, community supports). Greatest threats are those that *compromise* these protective systems. Policy implication: protect and restore ordinary systems rather than hunt rare resilient personalities.

Product translation: MindCraft should not market “we forge resilient geniuses.” Market “we keep the ordinary protective systems online while math gets hard” — safety, relatedness (tutor/world), competence information, recoverable miss. That is Masten-aligned and FEI-aligned.

Kill: “Resilience is a rare character trait we select for.”

XLV.4 Mathematical resilience as a pragmatic construct

FACT (origin framing): Johnston-Wilder & Lee (2010) introduced **mathematical resilience** as a pragmatic positive construct opposite avoidance, anxiety, and helplessness — enabling learners to engage with mathematics despite affective barriers (cited as foundational in later scale and handbook work; e.g. Lee & Johnston-Wilder construct summaries; Johnston-Wilder & Lee handbook chapter, doi:10.4324/9781315740713-34).

FACT (four aspects — construct literature): Lee & Johnston-Wilder summarize mathematical resilience as four pragmatic aspects:

1. **Growth** — belief that mathematical capability can develop with effort and appropriate support (growth/incremental theory of learning).
2. **Value** — mathematics has personal / world value and the learner is valued in the learning community.
3. **Knowing how to work at mathematics** — struggle is expected; strategies exist for stuckness.
4. **Knowing how to recruit support** — peers, adults, resources; stuck is not permanent solitude.

FACT (Growth Zone Model): Johnston-Wilder et al. (2013 onward; e.g. pre-service teacher applications, Warwick WRAP) — three zones:

- **Comfort (green):** routine; little growth.

- **Growth (amber/yellow):** challenge requiring support, persistence, agency; productive struggle.
- **Danger / anxiety (red):** challenge perceived beyond capability even with support; survival response; repeated exposure → persistent anxiety / avoidance.

HYPOTHESIS (product): MindCraft’s “soft-wrong + cards + tutor” stack is literally a **support-recruitment machine** for staying in the growth zone. Competitors that only escalate difficulty (or shame help) shove Maya into red.

Bridge to SAFE-DD (XLI): Desirable difficulties belong in the *growth* zone with equipment and destaking — not in the danger zone. “Struggle” ≠ “panic.”

XLV.5 Measurement — MRS is attitudes, not magic scores

FACT: Kooken, Welsh, McCoach, Johnston-Wilder & Lee (2016, *Measurement and Evaluation in Counseling and Development*, 49(3), 217–242, doi:10.1177/0748175615596782) — developed and validated the **Mathematical Resilience Scale (MRS)** via EFA/CFA across three samples. Three correlated factors: **Value, Struggle, and Growth**. Positioned as a way to gauge likelihood of participation and persistence in mathematics.

Product implication (HYPOTHESIS): MRS-style items (value / struggle-as-normal / growth belief) can inform parent research and longitudinal identity panels — but they are **not** a North Star and not a gamified in-app grit meter. Self-report attitudes ≠ transfer. Pair with *retry_120s*, *challenge_accept*, *transfer_pass*, *help-recruitment events* (opened card / asked tutor / peer).

Kill: Shipping a “Resilience Score™” badge as proof of identity change (L0 → L3 leap; Part XXXIV hygiene).

XLV.6 Grit theater — what we refuse to sell

FACT (origin): Duckworth, Peterson, Matthews & Kelly (2007, *Journal of Personality and Social Psychology*, 92(6), 1087–1101, doi:10.1037/0022-3514.92.6.1087) — grit as perseverance and passion for long-term goals; predictive claims in selective samples fueled popular education discourse.

FACT (meta wound): Credé, Tynan & Harms (2017, *Journal of Personality and Social Psychology*, 113(3), 492–511, doi:10.1037/pspp0000102) — 584 effects, 88 samples, $N \approx 66,807$: higher-order grit structure not confirmed; grit **very strongly** overlaps conscientiousness; overall grit–performance/retention relations **modest**; perseverance-of-effort facet carries more criterion validity than consistency-of-interest; **interventions to enhance grit may have only weak effects**; construct validity questioned; primary utility may lie in perseverance facet alone.

FOUNDER BELIEF / resolution: Perseverance *after equipped struggle* remains commercially meaningful (*retry_120s*). Packaging that as Duckworth “grit” invites overclaim, personality fatalism, and equity harm (blame the child’s character when curriculum/support failed). Prefer **mathematical resilience** language: growth zone + support recruitment + struggle-as-normal.

Standing Constitution already warned (core OS): grit/resilience hype is contested; structural inequality and curriculum quality matter more than character slogans. This chapter densifies that kill with Credé + JW/Lee specificity.

Kill: “MindCraft builds grit.” **Survive (constrained):** “MindCraft keeps you in the growth zone — challenge with support, recovery without shame.”

XLV.7 Competitive audit — who shoves students into the red zone?

Pattern	Zone reading	MindCraft wedge
Duolingo streak shame / league pressure	Danger + appearance climate (XLII/XLIII); “grit” via fear of break	Miss ≠ identity failure; cue practice not counter; no grit KPI
Khan mastery lock without affect design	Can strand in stuck without emotional exit tools	Soft-wrong + ingredient cards as support recruitment
Brilliant hard delight	Often growth-zone when curiosity high; weak for high-anxiety Maya	SAFE-DD equip/destake before delight escalation
Unguarded ChatGPT “just try harder / here’s the answer”	Either skips struggle (comfort via answer) or fluent wrong confidence	PWC + solo transfer; support = pedagogy wrap not answer dump
School “resilience assembly”	Poster without recruit-support architecture	Operationalize support buttons in-session

XLV.8 Resolution doctrine — SAFE-RESILIENCE

FOUNDER BELIEF / HYPOTHESIS — SAFE-RESILIENCE stack (ties FEI / SAFE-DD / TARGET / SAFE-REWARD / EVT value):

1. **Name the zones in product language.** Comfort / growth / danger — coach and tutor briefs use this, not “be gritty.”
2. **Equip before escalate.** SAFE-DD first: representations, faded examples, destaked attempts — then raise challenge.
3. **Struggle is expected, not character failure.** Attribution copy (XXX): strategy and effort-with-strategy; never “you’re not resilient enough.”
4. **Support recruitment is a first-class action.** Cards, ask-tutor, peer, worked scaffold — log as positive resilience behavior, not as “cheating the streak.”
5. **Exit the danger zone fast.** Anxiety spike → reduce stakes, shorten item, switch representation, offer support — do not double dose “desirable difficulty.”
6. **Value without coercion.** Eccles utility/attainment (XXXVII) and story worlds — not “suffer because grit.”
7. **Measure recovery and transfer, not grit scores.** retry_120s, help-then-solo sequences, challenge_accept, transfer_pass / solo_transfer_pass; optional MRS-like survey offline.
8. **Parent copy:** Sell “she got stuck, recruited help, and came back” — not “she’s a gritty kid now.”

Kill the false dichotomy: “Coddle forever in comfort” vs “forge character in the red zone.” Refuse both. Ship growth-zone challenge with support on tap.

XLV.9 Marketing language that survives

- **Allowed:** “Hard math with a way out — support when you’re stuck, challenge when you’re ready.”
- **Allowed:** “Stuck isn’t failure. Stuck is when you recruit the next tool.”
- **Allowed (parent):** “We train recovery and transfer — not grit slogans.”
- **Banned:** “We build grit.” / “Resilience score.” / “Push through the panic.”
- **Banned:** Bloom 2-sigma; empty mindset posters; streaks as proof of resilience; absolute tutoring-is-free.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without RES data
Growth-zone framing + support CTAs in coach	L1 hygiene aligned with JW/Lee
Soft-wrong + card recruitment → retry	L1 mechanism claim
“Raises mathematical resilience (MRS)” as product outcome	L0 until RES-1
“Builds grit / character”	L0 — banned packaging

XLV.10 Experiments spawned

ID	Question	Design	Primary	Kill condition
RES-1	Growth-zone coach copy + explicit support CTA vs grit/perseverance copy (same difficulty)	2-arm; anxiety-stratified	retry_120s; help-recruit rate; post affect; transfer_pass	Grit copy $\geq$ growth-zone on retry+transfer without raising danger affect → zone framing overstated
RES-2	Support-first after miss (card/tutor offer) vs no offer (same soft-wrong)	2-arm	Time-to-retry; danger-zone self-report; solo transfer later	No-offer $\geq$ support-first on retry without harm → recruitment less load-bearing
RES-3	Escalate difficulty after equip vs escalate without equip (SAFE-DD×RES)	2×2	Challenge_accept ; WM/anxiety proxy; transfer	Unequipped escalate wins transfer in high-anxiety → SAFE-DD sequencing weak

RES-4	MRS Value/Struggle/Growth Δ over 4 weeks of SAFE-RESILIENCE UX vs control	Survey + behavior	MRS factors + behavioral NS	MRS rises, behavior flat → attitude theater
RES-QUAL	10 Maya interviews: “When you’re stuck in math, what does ‘being resilient’ mean?”	Appendix B	coded grit-vs-support meanings	Everyone already means support recruitment → copy shift lower priority

Pre-reg (XXXIV): RES-* identify **zone framing** × **support recruitment** × **recovery/transfer**. They do **not** identify “we create resilient mathematicians” from MRS alone.

Densifies: SAFE-DD; FEI; TARGET; standing grit-hype ban; help-seeking as virtue not streak cheat.

XLV.11 Confidence table

Claim	Label	Confidence
Developmental resilience often arises from ordinary protective systems	FACT (Masten, 2001)	High
Mathematical resilience = growth + value + how-to-work + recruit support	FACT (JW/Lee construct line)	High (construct); Medium (intervention effect sizes sparse)
Growth Zone Model: comfort / growth / danger	FACT (JW et al. model literature)	High as pedagogy model; Medium as RCT-proven UX
MRS measures Value, Struggle, Growth attitudes	FACT (Kookan et al., 2016)	High
Grit higher-order structure weak; overlaps conscientiousness; modest prediction; weak intervention leverage	FACT (Credé et al., 2017)	High
Danger-zone exposure as “training” harms anxious learners	HYPOTHESIS (Ashcraft line + SAFE-DD)	Medium–High
SAFE-RESILIENCE is right default for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium–High
“Grit posters raise transfer” / “Resilience Score™ = identity”	SPECULATION / false as doctrine	Low (against)

XLV.12 What this chapter kills

1. **Kill:** Grit / “build character” as MindCraft marketing North Star. (Credé 2017; Constitution grit warning.)
2. **Kill:** Treating panic / danger-zone as desirable difficulty. (Growth Zone Model + SAFE-DD.)
3. **Kill:** Resilience as rare trait branding (“extraordinary kids”). (Masten ordinary magic.)
4. **Kill:** In-app Resilience Score™ as proof of identity transformation. (MRS ≠ transfer; claim ladder.)
5. **Wound:** Growth-mindset language alone — survives only inside growth + value + struggle know-how + **support recruitment**, not as wall art.
6. **Survive (constrained):** Growth-zone challenge with support on tap; struggle-as-normal; recovery metrics; parent story of “stuck → help → return.”

Doctrine until data: Ship **SAFE-RESILIENCE**: name zones → equip before escalate → struggle expected → support recruitment first-class → exit danger fast → value without coercion → measure recovery/transfer not grit. Market “hard math with a way out.”

Part XLVI — Teacher/Tutor Expectancy: Pygmalion Without Magic, Golem Without Labels

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 15; hour%6≠0 → Researcher; 5 researcher entries since synthesizer v1.6 → not yet Synthesizer)

Primary question: What do teacher/tutor expectancy effects actually justify for MindCraft — and what briefing ethics keep college tutors from installing a Golem on Maya?

Owners: Tutor ops / Product (briefing cards) · Equity · Positioning · Red Team

Commercial job: Replace “we believe in every student” poster copy and “Pygmalion IQ” folklore with a shippable **SAFE-EXPECTANCY** stack — evidence-based briefs, class-level high challenge + support, ban stigmatizing labels, measure mediation behaviors not vibes.

XLVI.1 Why this chapter exists

Parts XXV (self-efficacy), XXX (attribution), XXXVI (equity), XXXVII (EVT expectancy), and XLV (support recruitment) already treat *student-side* “can I?” and belonging. They do not fully answer: **what beliefs do human tutors carry into the session, how do those beliefs leak into climate/input/output/feedback, and how should MindCraft’s pre-session briefing be designed so gap-scan weakness does not become a self-fulfilling low label?**

FOUNDER BELIEF under audit: A knowledge-graph + tutor hybrid only compounds identity if the *human* in the loop treats Maya as capable of the next hard step with support — not as a fixed “weak student” whose low mastery scores license colder climate, thinner input, and fewer chances to respond.

Claims we refuse as doctrine:

1. Marketing the classic Pygmalion IQ story as MindCraft’s differentiator (“our tutors raise IQ by believing harder”).
2. Briefing tutors with trait labels (“low ability,” “slow,” “not STEM material”) drawn from gap-scan or weakness ranking.
3. Treating teacher-expectancy literature as proof that positive vibes alone move transfer.
4. Ignoring Golem / differential-treatment risks for stigmatized or “weakness-tagged” students.

XLVI.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Induced expectancy (Pygmalion paradigm)	Authority given false “bloomers” info → treats them differently → outcomes shift	Early-session / cold-start briefing window	Selling IQ magic; ignoring replication limits
Accuracy vs prophecy	Expectations often predict because they track real prior performance	Gap-scan / graph as <i>task</i> evidence, not character	Confusing prediction with causation
Golem effect	Low expectations → colder / more dogmatic treatment → worse performance	Ban negative trait labels in tutor cards	“Honest” weakness dump that licenses withdrawal
Four-factor mediation	Climate, input, output opportunity, feedback carry expectancy into behavior	Audit tutor/coach on those four channels	Soft climate without hard input/output
High-expectation teacher (class-level)	High challenge + high support for <i>all</i> , not only nominated stars	Tutor playbook: everyone gets hard Qs + scaffolds	Favorites / “bloomers only” tracks
SAFE-EXPECTANCY	Evidence brief → high challenge-with-support → no trait labels → mediate via CIOF → measure behaviors	See XLVI.7	Belief posters; Golem briefs

Operational definition (HYPOTHESIS): A MindCraft tutor session is expectancy-*safe* only when (a) the brief states **next playable tasks and scaffolds**, not trait judgments, (b) the tutor offers **hard-enough** work with support on tap (XLV growth zone), (c) climate stays warm under struggle, and (d) the student gets **output opportunity** (speak/solve before tutor finishes) — not when the tutor “believes harder” while finishing the math.

XLVI.3 Origin story — real, contested, not marketing copy

FACT: Rosenthal & Jacobson (1968, *Pygmalion in the Classroom*; also *Urban Review* summary, doi:10.1007/BF02322211) — teachers told that randomly selected elementary students were poised for intellectual blooming; those students later showed larger IQ gains on average in the reported analyses, especially in early grades. The study launched classroom expectancy research and intense methodological critique (e.g. Thorndike and later replication debates).

FACT (meta, induction credibility): Raudenbush (1984, *Journal of Educational Psychology*, 76(1), 85–97, doi:10.1037/0022-0663.76.1.85) — synthesis of 18 Pygmalion-style experiments: expectancy effects on pupil IQ are larger when teachers have had **little prior contact** with pupils at induction (≤ 2 weeks); effects shrink when teachers already know students well. Grade pattern: stronger early-grade effects in the corpus; design features of induction matter.

Product translation: MindCraft’s **first tutor sessions and cold-start briefs** are the high-leverage expectancy window — analogous to early-year induction. Once a tutor has rich behavioral evidence, “random bloomers” folklore is the wrong metaphor; **accurate, task-level** expectations dominate.

Kill: “Pygmalion proves we raise IQ by believing.” Sell mediation design in the first hours of contact, not IQ alchemy.

XLVI.4 What 35 years actually say (Jussim & Harber)

FACT (review): Jussim & Harber (2005, *Personality and Social Psychology Review*, 9(2), 131–155, doi:10.1207/s15327957pspr0902_3) — after ~35 years:

1. Self-fulfilling prophecies in classrooms **do occur**, but effects are **typically small**, do not accumulate greatly across perceivers/time, and may dissipate more than snowball.
2. **Powerful** effects may selectively hit students from **stigmatized** social groups.
3. Whether expectancy effects reliably change *intelligence*, and whether they do more harm than good in general, remains unclear.
4. Teacher expectations often predict outcomes **because they are accurate**, not only because they are self-fulfilling.

HYPOTHESIS for MindCraft: Gap-scan / mastery graphs will *accurately* predict struggle. Accuracy is not permission to install a Golem. The commercial risk is **translating accurate weakness into low-challenge, low-warmth treatment** — exactly the path that can still produce small-but-real prophecy costs, amplified for already-stereotyped students (ties Part XXXVI equity).

Wound: “Expectancy effects are huge and ubiquitous.” Survives only as: small average effects; context/stigma moderators; accuracy dominates prediction.

XLVI.5 Mediation — climate, input, output, feedback

FACT (behavioral mediation): Brophy & Good (1970, *Journal of Educational Psychology*, doi:10.1037/h0029908) — observational study: teachers demanded better performance and praised it more for high-expectation students; accepted poor performance more readily from lows and were less likely to praise good performance from lows when it occurred.

FACT (four-factor meta): Harris & Rosenthal (1985, *Psychological Bulletin*, 97(3), 363–386, doi:10.1037/0033-2909.97.3.363) — 31 meta-analyses of mediating behaviors; Rosenthal’s four-factor frame (**climate, feedback, input, output/opportunity**) organizes how expectancies become treatment. Expectancy→behavior and behavior→outcome links support multiple mediators (e.g. warmer climate, longer/richer interaction, questioning). Later summaries often note **feedback** as the weaker of the four average links, pushing affect/effort emphasis.

Product checklist (HYPOTHESIS — CIOF audit):

Channel	Tutor/coach do	Anti-pattern
Climate	Warmth under struggle; curiosity about wrong paths	Cold distance after “weak” brief
Input	Richer explanations, multiple representations (cards)	Thin “just try” for lows; lectures only for highs
Output	Wait time; let Maya finish; SE prompts (Part XL)	Tutor finishes every thought (Solver crutch)
Feedback	Criterion + strategy (XXX); soft-wrong	Praise only highs; criticism without how-to

Kill: Climate-only positivity (“you’ve got this!”) without input/output redesign — that is poster Pygmalion.

XLVI.6 Golem risk — negative expectancy is the ops hazard

FACT: Babad, Inbar & Rosenthal (1982, *Journal of Educational Psychology*, 74(4), 459–474, doi:10.1037/0022-0663.74.4.459) — distinguished positive expectancy outcomes (“Galatea”) from **negative** ones (“Golem”). Teachers high in susceptibility to biasing information treated low-potential nominees more negatively (notably dogmatic behavior); unbiased teachers treated groups more equitably. Differential negative treatment showed up in teacher behavior **and** student task performance.

FACT (student-perceived differentiation): Brattesani, Weinstein & Marshall (1984, *Journal of Educational Psychology*, 76(2), 236–247, doi:10.1037/0022-0663.76.2.236) — in classrooms where students perceive **high differential treatment**, teacher expectations contribute more to achievement than in low-PDT classrooms. Students can *see* the track.

Product translation: A tutor card that says “Maya is weak on linear equations / low confidence / hard gap-scan” is biasing information. High-bias tutors will leak Golem through climate and dogmatism. Low-PDT design = same high standards + differentiated *scaffolds*, not differentiated *respect*.

Kill: “Honest labels help tutors calibrate.” Honest **task evidence** helps; honest **trait verdicts** harm.

XLVI.7 High-expectation teachers — class-level, not favorites

FACT: Rubie-Davies (2007, *British Journal of Educational Psychology*, doi:10.1348/000709906X101601) — contrasts high- vs low-expectation *teachers* (expectations for the class as a whole). High-expectation teachers spent more time teaching, oriented lessons, linked prior knowledge, explained carefully, asked mixed open/closed questions including **high-level questions of all students**, and managed more positively; lows were more dismissive of incorrect answers and relied on closed questioning.

HYPOTHESIS: MindCraft’s competitive wedge vs “AI that flatters then finishes” and vs “tutor who only drills the labeled weak topic gently” is **Rubie-Davies-shaped ops**: every student gets high-level questioning + scaffolds; incorrect answers get learning support, not dismissal.

Bridge to SAFE-RESILIENCE / SAFE-DD: High expectation ≠ danger zone. High expectation = growth-zone challenge with support recruitment — not grit theater and not thin input for “lows.”

XLVI.8 Competitive audit (expectancy lens)

Competitor pattern	Expectancy mechanism	MindCraft counter
Duo	Habit/streak as “you’re the type who shows up”; little academic expectancy mediation	Transfer + CIOF; no streak-as-belief
Khan	Self-serve library; weak human expectancy channel	Human tutor brief + graph without Golem labels
Brilliant	Prestige challenge; can cue performance-approach for novices	Mastery structures + expectancy-managed gates
ChatGPT Base	Sycophantic high-warmth climate + answer dump (fake high expectancy, zero output demand)	Warm climate plus forced solo output / PWC
Unbriefed college tutor	Forms expectancies from accent, race, “seems lost,” or raw weakness dump	SAFE-EXPECTANCY brief + playbook

XLVI.9 Resolution doctrine — SAFE-EXPECTANCY

FOUNDER BELIEF / HYPOTHESIS — SAFE-EXPECTANCY stack:

1. **Brief tasks, not traits.** Tutor card: current concept, recommended level, formats struggled, successful scaffolds last time, next playable question IDs — **never** “low ability / not a math person.”
2. **Protect the cold-start window.** First sessions: explicit high-challenge-with-support script (Raudenbush induction logic).
3. **Class-level high expectation.** Rubie-Davies: hard questions for everyone; wrong answers → teach, don’t dismiss.

4. **Mediate via CIOF.** Climate warmth + rich input (cards) + output opportunity (wait / SE) + strategy feedback — audit these, not “belief intensity.”
5. **Accuracy without Golem.** Graph weakness drives *scaffold choice*, not *respect withdrawal*.
6. **Stigma premium.** Extra review when briefs could activate stereotype (XXXVI); ban demographic proxies in tutor UI.
7. **AI coach inherits the same rules.** No “you’re behind your peers” cold open; no finishing her thought under “support.”
8. **Measure mediation + FEI, not IQ folklore.** Tutor wait-time, student talk ratio, help-then-solo, `retry_120s`, `transfer_pass` / `solo_transfer_pass`.

Kill the false dichotomy: “Ignore data, just believe” vs “dump the weakness label so tutors are realistic.” Refuse both. Ship **evidence-based high challenge with equitable treatment**.

XLVI.10 Marketing language that survives

- **Allowed:** “Tutors see what to teach next — not who she is forever.”
- **Allowed:** “High challenge for everyone, support on tap — no tracks of respect.”
- **Allowed (parent):** “We brief tutors on the next skill, not on a fixed ability story.”
- **Banned:** “Pygmalion IQ gains.” / “Believe and they bloom.” / “We tell tutors who’s weak so they know.”
- **Banned:** Bloom 2-sigma; empty mindset posters; streaks as expectancy proof; tutoring-is-free.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without EXP data
Trait-label ban + task-only tutor brief	L1 hygiene aligned with Golem literature
CIOF playbook / wait-time coaching	L1 mechanism claim
“Raises IQ via tutor expectancy”	L0 — banned
“Eliminates achievement gaps via Pygmalion”	L0 until EXP-* + equity design (Jussim caveats)

XLVI.11 Experiments spawned

ID	Question	Design	Primary	Kill condition
EXP-1	Task-only brief vs trait/weakness-label brief (same content otherwise)	2-arm tutors; record session	Student talk ratio; wait-time; post affect; <code>transfer_pass</code>	Label brief $\geq$ task-only on transfer without colder climate → Golem risk overstated for our tutors

EXP-2	High-expectation playbook (hard Q + scaffold) vs “remediate gently only” after same gap-scan	2-arm	challenge_accept; solo transfer; help-recruit	Gentle-only wins transfer+return in high-anxiety → Rubie-Davies transfer fails for Maya
EXP-3	Cold-start script (first 2 sessions) vs unstructured tutor open	2-arm new matches	FEI events week 1; retention to session 3	Unstructured ≥ scripted → induction window less load-bearing
EXP-4	AI coach: warm+finish vs warm+output-demand (SE/wait)	2-arm	solo_transfer_pass; PWC	Warm+finish ≥ warm+output on solo transfer → climate-only expectancy myth revived
EXP-QUAL	10 tutor + 10 Maya: “What did the brief make you assume?”	Interview	coded trait vs task inferences	Tutors already ignore labels → UI priority shifts

Pre-reg (XXXIV): EXP-* identify **brief content** × **CIOF mediation** × **transfer**. They do **not** identify “we create bloomers” or IQ change.

Densifies: Equity (XXXVI); EVT expectancy; SAFE-RESILIENCE support; FEI; tutor ops queue item 68.

XLVI.12 Confidence table

Claim	Label	Confidence
Rosenthal & Jacobson launched expectancy research; original IQ story is contested	FACT	High
Induced expectancy effects larger when teacher knows student less (Raudenbush, 1984)	FACT	High
Average self-fulfilling effects small; accuracy often drives prediction; stigma can amplify (Jussim & Harber, 2005)	FACT	High
Differential treatment observable; highs get more demand/praise (Brophy & Good, 1970)	FACT	High

Four-factor mediation frame useful (Harris & Rosenthal, 1985)	FACT	High (framework); Medium (which channel dominates in tutoring apps)
Golem / high-bias negative treatment real (Babad et al., 1982)	FACT	High
High-expectation teachers differ in practice for whole class (Rubie-Davies, 2007)	FACT	High (observational/descriptive line); Medium (causal ops transfer)
SAFE-EXPECTANCY is right default for MindCraft tutor briefing	FOUNDER BELIEF / HYPOTHESIS	Medium–High
“Pygmalion IQ” / belief posters as product claim	SPECULATION / false as doctrine	Low (against)

XLVI.13 What this chapter kills

1. **Kill:** Marketing classic Pygmalion IQ blooming as MindCraft proof. (Raudenbush limits; Jussim accuracy; claim ladder.)
2. **Kill:** Tutor briefs that install trait/weakness identity labels. (Babad Golem; Weinstein PDT.)
3. **Kill:** Climate-only positivity without input/output demand. (Harris & Rosenthal mediation; Bastani/PWC line.)
4. **Kill:** “Just be realistic — tell tutors she’s weak” as ops hygiene.
5. **Wound:** Expectancy as a large, cumulative social-problem engine — survives only as small average effects with stigma/PDT moderators.
6. **Survive (constrained):** Cold-start high-challenge-with-support; task-only briefs; CIOF audit; Rubie-Davies whole-class high expectation; measure mediation + transfer.

Doctrine until data: Ship **SAFE-EXPECTANCY**: brief the next skill, not the fixed self; high challenge for everyone with scaffolds; warm climate under struggle; force student output; never weaponize gap-scan as a Golem tag. Market “tutors see what to teach next — not who she is forever.”

Part XLVII — Sleep, Stress, and Learning: Timing Practice Without All-Nighter Theater

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 18 ≡ Red Team slot, but ch47 never written → prefer Researcher per rotation; 6 researcher entries since synthesizer v1.6 → not yet Synthesizer)

Primary question: How should MindCraft time practice and exam-week product behavior given sleep-dependent consolidation and stress/pressure drain on working memory — without becoming a sleep-coaching app or glorifying late-night grind?

Owners: Product (session timing / exam modes) · Parent messaging · Affect pipeline · Red Team

Commercial job: Ship a **SAFE-TIMING** stack — overnight spacing as consolidation infrastructure, exam-week load softening, ban streak/all-nighter heroism, measure delayed transfer not midnight minutes.

XLVII.1 Why this chapter exists

Parts XXIX (spacing/retrieval), XXIV/XLI (anxiety $\times$ desirable difficulty), XXXI (flow/challenge–skill), and XLV (growth vs danger zone) already treat *within-session* difficulty and affect. They do not fully answer: **when** should Maya practice relative to sleep, and what should the product do in high-stakes stress weeks when encoding, attention, and WM are compromised?

FOUNDER BELIEF under audit: Identity transformation requires recoverable struggle *and* biological conditions that let struggle stick. A product that pushes “one more mission” at 1 a.m. during exam week can look engaged while sabotaging consolidation and installing danger-zone affect.

Claims we refuse as doctrine:

1. Marketing MindCraft as a sleep-hygiene / wellness brand (“we fix teen sleep”).
 2. Celebrating all-nighters, midnight streaks, or “grind hours” as learning proof.
 3. Treating acute exam pressure as desirable difficulty.
 4. Absolute claims that “sleep always consolidates math” independent of encoding quality, task type, and sleep opportunity.
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XLVII.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Sleep-dependent consolidation	Offline reprocessing after encoding that stabilizes/reshapes memory	Space practice across nights; delay transfer checks	Sell “sleep = magic” without encoding first
Encoding under restriction	Multi-night short sleep impairs <i>new</i> learning capacity	Soften new-concept loads after chronic short sleep flags	Blame student “laziness” for restriction biology
Pressure / choke	High-stakes appraisal consumes WM needed for hard math	Destake practice; hide peer ranks in exam week	Treat choke as low ability
Math-anxiety dual-task	Worry acts like a secondary load on WM	SAFE-DD before hard items	Add timers/leaderboards under anxiety
SAFE-TIMING	Time practice for consolidation + protect WM under stress	See XLVII.8	Sleep-app feature creep; grind theater

Operational definition (HYPOTHESIS): A MindCraft practice plan is timing-*safe* when (a) hard encoding sessions are followed by a real sleep opportunity before high-stakes transfer claims, (b) multi-night sleep debt / high stress triggers **load softening** (fewer new concepts, more retrieval of known, support on tap), and (c) engagement metrics never reward post-midnight volume as mastery.

XLVII.3 Sleep consolidates — and evolves — memory

FACT (systems review): Diekelmann & Born (2010, *Nature Reviews Neuroscience*, 11, 114–126, doi:10.1038/nrn2762) — sleep optimizes consolidation depending on learning conditions and sleep timing; slow-wave sleep (SWS) and REM support complementary system- vs synaptic-consolidation roles; consolidation can change representations quantitatively *and* qualitatively.

FACT (broader offline processing): Stickgold (2005, *Nature*, 437, 1272–1278, doi:10.1038/nature04286) — converging evidence that offline memory reprocessing during sleep is a major component of how memories are formed and shaped (not a single all-or-none event).

FACT (integrative/schema angle): Walker & Stickgold (2010, *Nature Reviews Neuroscience*, 11, 218, doi:10.1038/nrn2762-c1) — beyond veridical preservation, sleep (often REM-linked in their framing) supports unitization, assimilation into remote networks, and abstraction of rules/schemas — relevant to MindCraft’s transfer goal, not only verbatim item memory.

FACT (mechanism exemplar): Gais & Born (2004, *PNAS*, 101(7), 2140–2144, doi:10.1073/pnas.0305404101) — elevating acetylcholine during SWS-rich sleep blocked declarative word-pair consolidation while sparing a nondeclarative skill task — evidence that the *neurochemical milieu* of SWS matters for declarative consolidation, not “rest” generically.

Product translation: Spacing missions across nights (Part XXIX) is not only spacing theory — it is a **consolidation window**. Claiming “exam-ready” from a same-day blocked accuracy binge without overnight delay is scientifically lazy (ties XXXIX readiness signal).

Kill: “Sleep replaces practice.” Consolidation operates on what was encoded. No encoding → nothing to consolidate.

XLVII.4 Adolescent sleep restriction is causal for learning damage

Maya is a high-schooler. Population sleep debt is not a vibe; experimental restriction shows classroom-relevant harm.

FACT (simulated classroom): Beebe, Rose & Amin (2010, *Journal of Adolescent Health*, 47(5), 523–525, doi:10.1016/j.jadohealth.2010.03.005) — within-subject multi-night restriction vs healthy duration: sleep-restricted adolescents showed poorer learning, more inattentive behaviors, and lower arousal in a simulated classroom (medium-to-large effects; small *n*, preliminary but experimental).

FACT (review): Beebe (2011, *Pediatric Clinics of North America*, doi:10.1016/j.pcl.2011.03.002) — synthesizes cognitive, behavioral, and functional consequences of inadequate sleep in children/adolescents; chronic restriction is common and not benign.

FACT (long-term factual retention): Cousins, Wong & Chee (2019, *Journal of Adolescent Health*, 65(4), 549–557, doi:10.1016/j.jadohealth.2019.04.030) — adolescents (15–18) with 5 h sleep opportunity for 5 nights vs 9 h controls: after four restriction nights, learning detailed factual material showed **more forgetting** at 30 min (~26%), Day 3 (~34%), and Day 42 (~65%) relative to controls. Vigilance also fell; memory impairment did not simply track vigilance correlation.

HYPOTHESIS for MindCraft: A “productive” ACT cram week of late missions on short sleep can **look** like engagement while destroying the encoding that gap-scan and practice are supposed to build. Parent-visible “minutes practiced” without sleep context is a vanity metric.

Wound: “Weekend catch-up fixes the week.” Literature flags incomplete restitution and cumulative cost (see Beebe 2019-line summaries); do not market weekend binge as full repair.

XLVII.5 Stress and pressure — WM theft, not character failure

Sleep loss and evaluative stress often co-occur in exam weeks. Separate the mechanisms.

FACT (math anxiety × WM): Ashcraft & Krause (2007, *Psychonomic Bulletin & Review*, 14, 243–248, doi:10.3758/BF03194059) — math performance depends on WM for beyond-retrieval arithmetic; high math anxiety behaves like a dual-task (worry consumes capacity); anxious students also avoid math pathways. Complements Ashcraft & Kirk (2001) line already in Part XXIV.

FACT (choking under pressure): Beilock & Carr (2005, *Psychological Science*, 16(2), 101–105, doi:10.1111/j.0956-7976.2005.00789.x) — performance pressure harmed **high-WM** individuals on the most WM-demanding math problems — pressure consumes the capacity their superior strategies rely on. Not “dumb under stress”; often “capable under load.”

Bridge to SAFE-DD / FEI: Exam-week timers, public ranks (XLII), and “prove you’re ready tonight” framing are pressure injectors. Desirable difficulty assumes encoding capacity; pressure + anxiety can erase the dose.

HYPOTHESIS: Affective check-in stress > 0.7 already softens target_mastery in /recommend — that is the correct *direction*. Extend it: under high stress / self-reported short sleep, prefer retrieval + support recruitment over new-concept flooding and appearance UX.

Kill: “Pressure makes diamonds” as practice design. Pressure makes WM poorer for the problems that need it most.

XLVII.6 Competitive audit (timing lens)

Competitor pattern	Timing / stress mechanism	MindCraft counter
Duo	Streaks + late-night nagging; habit over consolidation	SAFE-HABIT + SAFE-TIMING; miss≠identity; no midnight streak heroism
Khan	Infinite library; learner can cram to exhaustion	Spaced missions + delayed transfer as readiness

Brilliant	Delight / prestige sessions; can invite performance pressure	Mastery climate; destake under stress
ChatGPT Base	Always-on 2 a.m. answers; no consolidation pedagogy	Human/AI wrap that schedules <i>struggle then sleep</i> , not answer dumps
Tutor shops	Pack hours before exam Saturday	Brief tutors: encode early in week; retrieval near exam; never shame sleep

XLVII.7 Exam weeks — special product mode, not harder grind

SPECULATION → **product hypothesis**: Default “more missions” during ACT week maximizes anxiety and restriction. Safer mode:

1. **Front-load encoding** earlier in the week when sleep opportunity is better.
2. **Shift to retrieval + mixed review** 48–72 h pre-exam (XXIX/XXXIX).
3. **Lower appearance cues** (no leaderboards, no streak threats).
4. **Keep growth-zone items** but exit danger fast (XLV); prefer scaffolds over new ontology leaps.
5. **Parent copy**: sleep is part of the study plan — not “rest is for quitters.”

FOUNDER BELIEF: Selling calm competence under exam load beats selling heroic sleep debt.

XLVII.8 Resolution doctrine — SAFE-TIMING

FOUNDER BELIEF / HYPOTHESIS — **SAFE-TIMING stack**:

1. **Encode** → **sleep** → **transfer claim**. Treat overnight delay as part of readiness (with XXIX spacing), not optional wellness.
2. **Never reward midnight volume**. Ban streak/XP incentives that spike after local quiet hours; engagement ≠ consolidation.
3. **Restriction-aware load**. If multi-night short sleep or high stress is signaled (check-in, parent note, exam-week flag): fewer new concepts, more retrieval, CIOF support (XLVI), SAFE-DD destake.
4. **Pressure ≠ DD**. Timers, ranks, and “prove it tonight” are kill switches under high anxiety/stress.
5. **Exam-week mode**. Soften novelty; maximize recoverable retrieval; protect sleep opportunity messaging.
6. **Do not become a sleep clinic**. Nudge timing and load — do not diagnose disorders or sell melatonin theater.
7. **Tutor briefs include timing**. “Practice early evening when possible; protect sleep before transfer checks” — task ops, not character judgment.
8. **Measure delayed** `transfer_pass` / `solo_transfer_pass`, not minutes-after-midnight.

Kill the false dichotomy: “Ignore biology, just grind” vs “pivot the company into sleep coaching.” Refuse both. Ship **timing-aware learning load**.

XLVII.9 Marketing language that survives

- **Allowed:** “Hard practice sticks better when she sleeps on it.”
- **Allowed:** “We don’t measure mastery at 1 a.m.”
- **Allowed (parent):** “Exam week ≠ all-nighter week — we shift to review and protect sleep.”
- **Banned:** “Our app fixes teen sleep.” / “Grind while they sleep.” / Streak-as-study-ethic.
- **Banned:** Bloom 2-sigma; empty mindset; tutoring-is-free; pressure-as-pedagogy.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without TIM-* data
Overnight spacing before transfer checks	L1 mechanism aligned with consolidation reviews
Exam-week load softening + destake	L1 product hygiene under Ashcraft/Beilock
“Sleep mode raises ACT scores”	L0–L1 only with pre-reg; not L3 identity
Sleep-clinic / wellness brand claims	L0 — banned

XLVII.10 Experiments spawned

ID	Question	Design	Primary	Kill condition
TIM-1	Same content: practice ending ≥ 1 h before habitual bedtime vs late-night block; transfer next evening	2-arm	next-day / 48h transfer_pass	Late $\geq$ early $\rightarrow$ consolidation timing overstated for our items
TIM-2	Exam-week mode (retrieval+ destake) vs “more new missions” in last 5 days pre-mock	2-arm	mock score; affect; sleep opportunity self-report	More-new wins mock+return $\rightarrow$ softening wrong default
TIM-3	High-stress check-in $\rightarrow$ auto soft path vs ignore stress	2-arm	retry_120s; WM-demand item accuracy; return	Ignore $\geq$ soft on transfer without affect cost $\rightarrow$ modifier useless
TIM-4	Parent message: “sleep is study” vs “push one more hour”	message A/B	next-night session timing; parent WTP/trust	Push-hour increases volume <i>and</i> next-day transfer $\rightarrow$ grind narrative survives

TIM-QUAL	10 Maya exam-week diaries: when practice helped vs harmed	interview	coded sleep debt × danger zone × cram	Students already self-time well → UI priority drops
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Pre-reg (XXXIV): TIM-* identify **timing × load × delayed transfer**. They do **not** identify clinical sleep treatment or ACT score miracles from a bedtime tip.

Densifies: XXIX spacing; XLI SAFE-DD; XLII compare ban; XLIII habit without colonization; XLV danger exit; affect check-in; Part LXIII exam-pressure queue.

XLVII.11 Confidence table

Claim	Label	Confidence
Sleep supports declarative/procedural consolidation with stage-sensitive mechanisms (Diekelmann & Born, 2010; Stickgold, 2005)	FACT	High
Sleep can support schema/abstraction beyond verbatim storage (Walker & Stickgold, 2010)	FACT	Medium–High (active research debate on REM vs SWS mapping)
Low ACh during SWS matters for some declarative consolidation (Gais & Born, 2004)	FACT	High (specific manipulation); Medium (product generality)
Multi-night adolescent restriction impairs classroom learning/attention (Beebe et al., 2010; Beebe, 2011)	FACT	High (direction); Medium (effect sizes / sample limits)
Restriction impairs long-term factual retention in teens (Cousins et al., 2019)	FACT	High
Math anxiety consumes WM (Ashcraft & Krause, 2007)	FACT	High
Pressure can choke high-WM math performance (Beilock & Carr, 2005)	FACT	High
SAFE-TIMING is right default for MindCraft scheduling/exam mode	FOUNDER BELIEF / HYPOTHESIS	Medium–High

MindCraft-as-sleep-app is a good growth wedge	SPECULATION / false as doctrine	Low (against)
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XLVII.12 What this chapter kills

1. **Kill:** All-nighter / midnight-streak heroism as learning culture. (Cousins retention; Beebe classroom; SAFE-HABIT.)
2. **Kill:** Treating exam pressure and timers as desirable difficulty. (Beilock choke; Ashcraft dual-task; SAFE-DD.)
3. **Kill:** Same-day cram accuracy as exam-ready proof without overnight/delayed transfer. (Diekelmann/Born; XXIX/XXXIX.)
4. **Kill:** Pivoting brand into clinical sleep wellness.
5. **Wound:** “Sleep always helps every math skill equally.” Survives only as: condition- and task-dependent consolidation; encoding first.
6. **Survive (constrained):** Encode→sleep→claim; restriction/stress-aware load; exam-week retrieval mode; parent “sleep is study”; measure delayed transfer.

Doctrine until data: Ship **SAFE-TIMING:** practice hard enough to encode, sleep enough to consolidate, soften novelty under debt and pressure, never sell grind-at-1-a.m. as identity. Market “hard practice sticks better when she sleeps on it.”

Part XLVIII — Transfer & Variation Theory: Far Transfer Design in the Ontology

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 21; hour%6≠0 → Researcher; 7 researcher entries since synthesizer v1.6 → not yet Synthesizer; this tick becomes the 8th)

Primary question: How should MindCraft design question sequences and “transfer” claims so Maya discerns *critical mathematical structure* — without selling story-world immersion or blocked accuracy as far transfer?

Owners: Product (questionBank / ontology / pathfinder) · Content ops · Marketing claim ladder · Red Team

Commercial job: Ship a **SAFE-TRANSFER** stack — deliberate variation/invariance for discernment, hugged near transfer toward ACT formats, bridged high-road abstraction only where designed, measure delayed/format/solo transfer — kill “narrative = transfer for free.”

XLVIII.1 Why this chapter exists

Parts XXIX (spacing/retrieval), XXXIX (interleaving), XL (self-explanation), XLI (SAFE-DD), and XLVII (encode→sleep→claim) already constrain *when* and *how hard* practice should feel. They do not fully answer: **what pattern of examples** makes a concept transferable, and what marketing language about “transfer” survives evidence?

MindCraft’s ontology already encodes ingredients, bridges, combinations, formats, and archetypes. The risk is product language that treats any format change, story skin, or mixed set as “far transfer” — while competitors sell fluency theater as readiness.

FOUNDER BELIEF under audit: Identity as a mathematical thinker requires *flexible discernment* of structure across contexts — not one beautiful tutorial problem solved once. That flexibility is **designed**, not an automatic side-effect of lore.

Claims we refuse as doctrine:

1. Story worlds automatically produce far transfer.
2. Blocked same-day accuracy $\equiv$ exam-ready / transferable skill.
3. “We teach for transfer” without specifying content $\times$ context dimensions (Barnett & Ceci).
4. Random interleaving or random story skins as a substitute for *principled* variation of critical aspects.

XLVIII.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Near / far transfer	Application of learning across some distance in content and/or context	Label every transfer claim with <i>which</i> dimensions moved	Vague “far transfer” marketing
Low-road / high-road	Automatic similarity trigger vs mindful abstraction	Hugging (ACT-like items) vs bridging (explicit compare/abstract prompts)	Expect far transfer from low-road drill alone
Object of learning	What is to be discerned (content + critical aspects)	Ingredient / bridge as the intended discernment target	Teach “one thing” without opening critical dimensions
Variation / invariance	What changes vs what stays fixed across a sequence	Example sets that separate critical aspects	Vary everything at once $\rightarrow$ noise; vary nothing $\rightarrow$ surface fixation
SAFE-TRANSFER	Transfer claims + sequences that survive evidence	See XLVIII.8	Story-skin theater; blocked-accuracy readiness

Operational definition (HYPOTHESIS): A MindCraft mission is transfer-*safe* when (a) the sequence makes at least one critical aspect *discernable* via planned contrast/generalization/separation/fusion, (b) “transfer” checks change a declared Barnett–Ceci dimension (format, surface story, temporal delay, or solo vs scaffolded), and (c) marketing never claims far transfer from immersion alone.

XLVIII.3 Far transfer is rare without design — name the dimensions

FACT (taxonomy): Barnett & Ceci (2002, *Psychological Bulletin*, 128(4), 612–637, doi:10.1037/0033-2909.128.4.612) — a century of “does far transfer occur?” debates compared apples to oranges; they propose a nine-dimension content×context framework. Estimating a *single* far-transfer effect size is misguided; evidence is substantial under some conditions and untested under others.

FACT (prosecution): Detterman (1993, in Detterman & Sternberg, Eds., *Transfer on Trial*, Ablex) — argues spontaneous far/general transfer is rare; apparent transfer often reflects cues/hints or other mechanisms; humans more often learn specific facts and patterns than free-floating abstractions.

FACT (analogical noticing failure): Gick & Holyoak (1980, *Cognitive Psychology*, 12, 306–355, doi:10.1016/0010-0285(80)90013-4) — subjects who read an analogous military “fortress” story often failed to spontaneously apply the convergence principle to Duncker’s radiation problem; **hints** dramatically raised solution rates. Structural isomorphism ≠ noticed transfer.

Product translation: MindCraft must not market “she’ll just see it on the ACT” after a kitchen-world linear-equations arc. Spontaneous far transfer is the *exception*. Design either **hugs** the exam context or **bridges** with explicit abstraction (compare two instances; name the invariant).

Kill: “Our stories create far transfer.” Stories can raise utility/attainment (XXXVII) and belonging (XXXVI); they do not substitute for noticing structure.

XLVIII.4 Mechanisms that *do* teach for transfer — low road, high road, hug, bridge

FACT (mechanisms): Salomon & Perkins (1989, *Educational Psychologist*, 24(2), 113–142, doi:10.1207/s15326985ep2402_1) — **low-road** transfer: extensive, varied practice → automatic triggering in similar stimulus conditions; **high-road** transfer: mindful abstraction, forward- or backward-reaching search for connections. Findings of transfer/nontransfer reflect whether those conditions were met.

FACT (instructional strategies): Perkins & Salomon (1992, “Transfer of Learning,” *International Encyclopedia of Education*, 2nd ed.; building on Perkins & Salomon, 1988, *Educational Leadership*, 46(1), 22–32) — **hugging** (practice that approximates target performance) vs **bridging** (prompt abstraction, metacognition, planned connections). Education gets transfer when designed for it — not by hoping.

HYPOTHESIS for MindCraft:

- **Hug:** format-tagged ACT/GCSE-like items, timed retrieval near exam, archetype-matched instances (Layer 2/3) after ingredient mastery.
- **Bridge:** coach prompts that force comparison (“what stayed the same?”), self-explanation of the invariant (XL), and bridge-gap cards when two concepts are known but unconnected.
- **Low-road alone** (Duo-like drill skins) will not deliver ACT problem-type novelty. **High-road alone** (chatty abstraction without varied instances) will not stick.

Wound: Absolute “transfer can’t be taught” (hard Detterman reading). Survives only as: spontaneous far transfer is rare; *cued and designed* transfer is the product job.

XLVIII.5 Variation theory — discernment requires planned difference

FACT (foundations): Marton & Booth (1997, *Learning and Awareness*, Lawrence Erlbaum) — learning is experiencing an object of learning in a new way; discerning critical aspects requires experiencing patterns of **variation and invariance**.

FACT (math classroom design): Kullberg, Runesson Kempe & Marton (2017, *ZDM Mathematics Education*, 49, 559–569, doi:10.1007/s11858-017-0858-4) — variation theory as a design principle changes *what is made possible to learn*; teachers using VT systematically open critical aspects (e.g., relating concepts rather than teaching one-at-a-time isolation); example sets matter, and teaching must **draw attention** to the patterns (presence of variation ≠ experienced variation).

FACT (dimensions of possible variation): Watson & Mason (line; e.g., Watson & Mason, 2005/2006 math-education elaborations of Marton) — to learn a mathematical concept is to become aware of which aspects can vary while remaining an example, and over what **range of permissible change**. Teacher and student may stress different dimensions.

FACT (Chinese bianshi tradition): Gu, Huang & Marton (2004, in Fan et al., Eds., *How Chinese Learn Mathematics*, World Scientific, pp. 309–347, doi:10.1142/9789812562241_0012) — Chinese “teaching with variation” systematically juxtaposes conceptual and procedural variants so students deepen structure; indigenous practice, not a Western bolt-on.

FACT (curriculum variation problems): Sun (2011a, *Educational Studies in Mathematics*, 76(1), 65–85, doi:10.1007/s10649-010-9263-4) — Chinese textbook “variation problems” organize connectedness (one problem, multiple changes/solutions) to highlight invariants; contrasts with curricula that treat problems as isolated context decorations.

Product translation for the ontology:

1. Pick the **object of learning** (ingredient or bridge), not “finish five questions.”
2. Hold the critical structure **invariant**; vary one dimension at a time (numbers → representation → story skin → exam archetype).
3. After separation, **fuse** (vary jointly) only when WM/SAFE-DD allows (XXVI/XLI).
4. Prompt noticing (“what changed / what stayed the same?”) — ties XL self-explanation.
5. Interleaving (XXXIX) is related but not identical: IL trains discrimination among *categories*; VT trains discernment of *critical aspects within* an object of learning. Use both deliberately.

Kill: “More random story skins = better transfer.” Random surface change without invariant focus is entertainment variance, not Martonian variation.

XLVIII.6 Competitive audit (transfer / variation lens)

Competitor pattern	Transfer mechanism	MindCraft counter
Khan	Huge item pool; learner may block by topic; transfer left to chance	VT sequences + delayed mixed transfer checks

Duo	Low-road habit; skin-deep variety; streaks as proof	SAFE-HABIT + SAFE-TRANSFER: variety serves discernment, not XP
Brilliant	Elegant interactive instances; may under-specify exam hug	Hug ACT formats after insight; measure transfer_pass
ChatGPT Base	Fluent explanations; weak structured example contrast; sycophancy risk (XXXIII)	Deterministic card/path spine; SE + contrast prompts; no monologue≡mastery
Tutor shops	Ad-hoc worksheets; expert tutors often vary well by instinct	Codify variation patterns in bank + tutor briefs (CIOF-compatible, XLVI)

XLVIII.7 Ontology / bank design implications (MindCraft-specific)

HYPOTHESIS — variation layers already latent in the stack:

- **Ingredient card styles** (geometric / algebraic / procedural) = representation variation with concept held.
- **Format axis** (word_problem, diagram, symbolic_expression, ...) = context/content-adjacent transfer dimensions — prefer matched then force a declared hop.
- **Bridges** = high-road objects of learning (connection itself is the critical aspect).
- **Archetypes (Layer 2)** = exam-pattern hug targets after near mastery.
- **Combinations** = fusion stage (multiple ingredients co-activated) — do not dump at cold start.

SPECULATION → **product rule**: Generation and bank curation should tag each practice *bundle* with: invariant_aspect, varied_dimension, transfer_check_dimension. Untagged bundles default to “drill,” not “transfer mission.”

FOUNDER BELIEF: Pathfinder recommendations that only maximize weakness severity without sequencing variation will grind the same surface until boredom — and still fail the ACT’s novel wrap.

XLVIII.8 Resolution doctrine — SAFE-TRANSFER

FOUNDER BELIEF / HYPOTHESIS — SAFE-TRANSFER stack:

1. **Name the hop.** Every transfer claim states which Barnett–Ceci-style dimension moved (format, story surface, delay, scaffold removal, domain).
2. **Design variation, don’t sprinkle it.** Sequences: contrast → separate critical aspect → generalize range → fuse when ready.
3. **Hug the exam; bridge the abstraction.** ACT-like items after structure is discernable; explicit compare/SE prompts for high-road.
4. **Never equate story immersion with transfer.** Narrative is motivation/belonging infrastructure (WORLD_VISION / XXXVI–XXXVII), not a transfer engine by itself.

- 5. **Blocked accuracy ≠ transferable skill.** Require delayed / mixed / format-shifted / solo checks (transfer_pass, solo_transfer_pass).
- 6. **Draw attention to the pattern.** Coach/tutor line: “What stayed the same?” is load-bearing UX, not flavor text.
- 7. **Respect load and anxiety.** Fusion + far hops only under SAFE-DD; exam week prefers hugged retrieval (XLVII).
- 8. **Measure what you sell.** Instrument transfer hops; kill missions that raise same-surface accuracy without hop gains.

Kill the false dichotomy: “Transfer is a myth, only drill” vs “Our world magically creates genius generalizers.” Refuse both. Ship **designed discernment**.

XLVIII.9 Marketing language that survives

- **Allowed:** “We change one thing at a time until the structure is obvious — then we change the wrapping.”
- **Allowed:** “Ready means she can do it in a new format tomorrow, not five identical items tonight.”
- **Allowed (parent):** “Practice looks varied on purpose — same math backbone, different disguises.”
- **Banned:** “Immersive stories create far transfer.” / “AI explanations transfer automatically.” / Blocked-streak-as-mastery.
- **Banned:** Bloom 2-sigma; empty mindset; tutoring-is-free; vague far-transfer heroics.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without TR-* data
VT-sequenced bundles + noticing prompts	L1 mechanism aligned with Marton/Kullberg/Gu
Format-hop / delayed transfer checks as readiness	L1 product metric hygiene (Barnett–Ceci + Gick/Holyoak)
“Raises ACT by teaching far transfer”	L0–L1 only with pre-reg estimand; not L3 identity
Story-world ≡ far transfer	L0 — banned

XLVIII.10 Experiments spawned

ID	Question	Design	Primary	Kill condition
TR-1	Same concept: VT sequence (one varied dimension) vs random surface skins	2-arm	next-day format-hop transfer_pass	Random ≥ VT → principled variation overstated

TR-2	Bridge object: explicit compare/SE vs monologue explanation then items	2-arm	bridge-gap item accuracy; solo_t ransfer_pass	Monologue ≥ compare → XL/bridge UX useless
TR-3	Hug-then-hop: ACT-archetype items after VT vs archetype-first cold	2-arm	archetype accuracy; affect; time	Cold ≥ VT-prep → hug sequencing wrong
TR-4	Transfer check: delayed mixed vs same-day blocked as “mission complete” gate	2-arm	48h transfer; return	Same-day gate predicts 48h as well → delay rule overstated
TR-QUAL	10 Maya think-alouds on “what changed?” prompts	interview	coded discernment of invariant	Students already name invariants → prompt priority drops

Pre-reg (XXXIV): TR-* identify **sequence design × declared transfer hop**. They do **not** identify “stories cause identity” or ACT score miracles from a single varied set.

Densifies: XXIX delayed readiness; XXXIX IL discrimination; XL SE; XLI SAFE-DD; format-weakness workstream; Layer 2/3 archetype bank; Part XLIX misconceptions queue.

XLVIII.11 Confidence table

Claim	Label	Confidence
Far-transfer debates need multi-dimension taxonomy (Barnett & Ceci, 2002)	FACT	High
Spontaneous far/general transfer is rare; cues matter (Detterman, 1993; Gick & Holyoak, 1980)	FACT	High
Low-road vs high-road + hug/bridge are usable design levers (Salomon & Perkins, 1989; Perkins & Salomon, 1992/1988)	FACT	High

Discernment requires variation/invariance patterns (Marton & Booth, 1997; Kullberg et al., 2017)	FACT	High
Chinese variation / bianshi is a systematic task-design tradition (Gu et al., 2004; Sun, 2011a)	FACT	High
SAFE-TRANSFER is right default for MindCraft bank/path/copy	FOUNDER BELIEF / HYPOTHESIS	Medium–High
Story immersion alone produces far transfer	SPECULATION / false as doctrine	Low (against)

XLVIII.12 What this chapter kills

1. **Kill:** Marketing story worlds as automatic far-transfer engines. (Gick/Holyoak noticing; Detterman rarity.)
2. **Kill:** Blocked same-surface accuracy as transferable / exam-ready proof. (Barnett–Ceci; XXIX/XXXIX readiness.)
3. **Kill:** Random cosmetic variety as “variation theory.” (Kullberg: what is made possible to learn; attention to patterns.)
4. **Kill:** AI monologue explanations as sufficient for transfer. (Hint/schema need; XL SE; XXXIII trust.)
5. **Wound:** “Transfer cannot be taught.” Survives only as: spontaneous far transfer is rare; designed hug/bridge/VT can raise hop success.
6. **Survive (constrained):** Named transfer hops; VT sequences; hug ACT + bridge abstraction; noticing prompts; delayed/format/solo metrics.

Doctrine until data: Ship **SAFE-TRANSFER:** vary on purpose so the backbone becomes obvious, hug the exam when claiming readiness, bridge with explicit abstraction, never sell lore or identical-item streaks as transfer. Market “same math backbone, different disguises — then prove it in a new wrapping tomorrow.”

Part XLIX — Misconceptions as Productive: Diagnostic Wrong Answers & Soft-Wrong Science

Chapter status: Living evidence brief — Researcher tick 2026-07-31 (UTC hour 00 ≡ Red Team slot, but ch49 never written → prefer Researcher per rotation; 8 researcher entries since synthesizer v1.6 → Synthesizer deferred to next eligible non-RT hour)

Primary question: How should MindCraft treat wrong answers — as shame tokens to erase, or as *diagnostic resources* that drive soft-wrong coaching, misconception routing, and honest mastery updates?

Owners: Product (Diagnostic / practice soft-wrong / Layer-4 misconception memory) · Engine (ingredient/misconception links) · Marketing claim ladder · Red Team

Commercial job: Ship a **SAFE-MISCON** stack — elicit → classify (lack vs confident misconception) → springboard without public shame → route remediation → measure retry + recalibration — kill “instant red-X pedagogy” and “all wrongs are equal.”

XLIX.1 Why this chapter exists

MindCraft already ships pieces of this thesis: soft-wrong UX, hide-correctness diagnostic (C4), Eedi misconception bank, Layer-1 canonical_misconception_family fields, and Experiment A (wizard coach on soft-wrong). What the Constitution lacked was an evidence spine that says *why* those features are load-bearing — and which competitor postures they kill.

Parts XXX (attribution), XXXVIII (mastery climate), XL (self-explanation), XLI (SAFE-DD), and XLVIII (SAFE-TRANSFER) constrain how misses feel and how sequences vary. They do not fully answer: **what is a productive wrong answer**, and when does revealing correctness help vs corrupt learning and the graph?

FOUNDER BELIEF under audit: Identity as a mathematical thinker grows when a miss becomes *information about structure* — not a grade, not a personality verdict, and not a fluent AI monologue that skips the student’s idea.

Claims we refuse as doctrine:

1. Misconceptions are unitary “bugs” that instruction must confront-and-replace wholesale.
 2. Instant correctness feedback = better learning for every miss.
 3. Soft-wrong tone alone equals diagnostic teaching.
 4. All incorrect selections update mastery the same way (guess ≡ confident wrong ≡ slip).
 5. Marketing “we celebrate mistakes” without routing, classification, or retry instrumentation.
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XLIX.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Misconception (classic)	Stable, theory-like wrong idea that interferes with learning	Named misconception_id; distractors that diagnose it	Treat every wrong as a deep theory
Knowledge-in-pieces / resources	Fragmentary, context-sensitive knowledge elements; flawed <i>and</i> productive	Route to refine/reorganize, not only “delete”	Romanticize every wrong as deep wisdom
Diagnostic wrong	Incorrect response that reveals <i>which</i> resource/misconception fired	Option → misconception map; confidence tier	Beauty distractors with no diagnostic map

Soft-wrong	Low-shame evaluation structure: miss is safe, informative, retry-inviting	Tone + no public rank + coach invite	Soft tone with no next move = empty kindness
Hide-correctness (C4)	Serve real items; record outcome; do not reveal key during diagnostic	Gap-scan / FEI probes	Revealing a wrong key as “gotcha” corrupts trust + can seed wrong keys into memory theater
SAFE-MISCON	Productive-error doctrine that survives evidence	See XLIX.8	Mistake-celebration posters; red-X gamification

Operational definition (HYPOTHESIS): A MindCraft miss is *productive* when (a) the response maps to a known misconception family or resource pattern, (b) the UX destakes identity threat (soft-wrong / mastery climate), (c) the next move uses the wrong as a springboard (compare, conflict, SE, bridge card) rather than only stating the key, and (d) telemetry separates lack-of-knowledge, slip, and high-confidence misconception for graph updates.

XLIX.3 Misconceptions reconceived — flawed *and* productive

FACT (constructivist critique): Smith, diSessa & Roschelle (1993/1994, *Journal of the Learning Sciences*, 3(2), 115–163, doi:10.1207/s15327809jls0302_1) — acknowledge empirical findings of misconceptions research but reject the accompanying view that students hold strongly held flawed ideas that instruction must *confront and replace*. That view overemphasizes discontinuity with experts and conflicts with constructivism: advanced knowledge is built from prior understandings. They sketch learning as refinement and reorganization within a complex-systems view; misconceptions are **both flawed and productive**.

FACT (knowledge in pieces): diSessa (1993, *Cognition and Instruction*, 10(2–3), 105–225, doi:10.1207/s1532690xc1002&3_2) — intuitive knowledge as weakly organized elements (p-prims / sense of mechanism), not a coherent wrong theory ready for wholesale replacement. Expertise densifies and reorganizes the system.

FACT (instructional lens): Hammer (1996, *Journal of the Learning Sciences*, 5(2), 97–127, doi:10.1207/s15327809jls0502_1) — “misconceptions” vs “p-prims” change what a teacher *sees* as the instructional task (replace a theory vs refine activation conditions of resources).

Product translation: MindCraft’s Layer-1 *failure_mode* / *canonical_misconception_family* and Eedi *mis_* IDs should drive **refinement routes** (when does this resource work? when does it fail?) — not only a shameful “you have Misconception X” badge. Coach copy: “That move works for this case — here’s where it breaks” beats “Wrong. Correct answer is B.”

Kill: Product language that treats Maya as carrying a disease-like misconception to be purged. Survive: diagnostic naming for *engine routing*, student-facing language as productive refinement.

XLIX.4 Errors as springboards — inquiry, not erasure

FACT (teaching experiment): Borasi (1994, *Journal for Research in Mathematics Education*, 25(2), 166–208, doi:10.5951/jresmetheduc.25.2.0166) — secondary students can capitalize on errors as **springboards for inquiry**; variations of the strategy exist; the point is stimulating mathematical inquiry from the error, not only teacher-side diagnosis for remediation.

FACT (error analysis tradition): Radatz (1979, *Journal for Research in Mathematics Education*, 10(3), 163–172, doi:10.5951/jresmetheduc.10.3.0163) — systematic error analysis has long diagnostic value for teachers; Borasi’s contribution is turning that value toward *student* inquiry.

FACT (diagnostic teaching / conflict): Bell (1993a, *Educational Studies in Mathematics*, 24, 5–34; 1993b, *Educational Studies in Mathematics*, 24, 115–137) and Swan’s diagnostic-teaching line (e.g., Swan, 2001, “Dealing with misconceptions in mathematics,” in *Issues in Mathematics Teaching*, Routledge) — long-term learning improves when lessons **expose** existing thinking, generate cognitive conflict across methods/answers, and resolve via reflective discussion — vs curricula that avoid analyzing errors or front-load explanations then graded practice.

HYPOTHESIS for MindCraft: Soft-wrong + wizard coach should implement a *compressed* springboard cycle: expose idea → contrast with invariant/conflict item → student-generated why (XL) → optional AI wrap under PWC — not Duo-style instant red/green with explanation dump.

Wound: Classroom conflict discussion does not transfer 1:1 to solo app UX. Survive as: design *micro-conflict* (two worked paths; “which claim survives?”) rather than claiming Socratic seminar in a phone UI.

XLIX.5 Soft-wrong science — safety is necessary, not sufficient

FACT (affect × attempt): Prior Constitution densification (XXIV anxiety/WM; XXXVIII mastery climate; XLI SAFE-DD destake) — threat appraisal suppresses attempt quality; evaluation structure that makes misses identity-safe raises willingness to engage difficulty.

FOUNDER BELIEF: Soft-wrong is mastery *evaluation structure*, not brand voice. Without diagnostic mapping + next move, soft-wrong is empty kindness.

HYPOTHESIS: Soft-wrong’s commercial wedge vs Khan/Duo is not “nicer feedback” — it is **recoverable struggle with competence evidence** (retry_120s, challenge_accept, solo_transfer_pass). Competitors can copy warmer copy; they rarely ship miss → misconception route → bridge/ingredient card → delayed transfer check.

Contradicting pressure: Parents often want instant correctness theater (“show the answer”). Claim ladder: L0–L1 session metrics for soft-wrong; do not sell L3 identity from tone alone (XXXIV).

Kill: Soft-wrong as marketing slogan without Experiment A instrumentation. Survive: soft-wrong as FEI stage (informative miss) gated by SAFE-MISCON routing.

XLIX.6 Confidence, calibration, and hide-correctness

FACT (hypercorrection): Butterfield & Metcalfe (2001, *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 27(6), 1491–1494, doi:10.1037//0278-7393.27.6.1491) — contrary to interference predictions, **high-confidence errors** were *more* likely to be corrected after feedback than low-confidence errors (hypercorrection).

FACT (mechanism sketch): Butterfield & Metcalfe (2006, *Memory & Cognition* / follow-on attentional work; see also Metcalfe & Finn line) — corrective feedback after surprising high-confidence errors captures more attention; later work notes latent knowledge / “knew it all along” contributions. Product implication: confidence on wrongs is diagnostically precious, not just a UI flourish.

FACT (authentic math diagnostics online): Foster, Woodhead, Barton & Clark-Wilson (2022 advance of 2021 online-first; *Educational Studies in Mathematics*, doi:10.1007/s10649-021-10084-7) — large-scale diagnostic quiz data: higher confidence on an incorrect response associated with higher later correction probability on analogous items (hypercorrection signal in authentic math learning), after controls discussed in-paper; confidence assessment can prompt reflection.

FACT (calibration hygiene): Confidence–accuracy work in diagnostics (e.g., overconfidence on alternative conceptions; gender patterns in under/overconfidence across studies) — poor calibration is common; separating *misinformed* (confident wrong) from *uninformed* (low confidence / blank) changes instruction.

Product translation — C4 hide-correctness: During gap-scan / identity-sensitive diagnostics, revealing keys mid-flow can (a) spike threat, (b) turn the session into answer-hunting, (c) risk encoding wrong “gotcha” narratives. Record outcomes + confidence; defer key reveal to a designed springboard moment (or never in pure diagnostic). **Do not** market hide-correctness as “we never teach the answer” — that is false and banned under tutoring-is-free style slogans.

HYPOTHESIS: Practice mode should often *use* hypercorrection: after a high-confidence miss, prioritize attended corrective feedback + isomorphic retest; after low-confidence miss, prioritize scaffold/hug before confrontation (SAFE-DD equip).

Kill: Treating every wrong as equal graph evidence. Survive: confidence-tiered update policy (aligns with Layer-4 `evidence_update_policy` intent: don’t over-update from one item; separate can-do from can-recognize).

XLIX.7 What this means for the bank, ontology, and Solver

Bank / C5: Distractors should be designed misconceptions, not random wrongs. Eedi ingestion already mints `mis_` IDs — enrichment linking misconceptions → ingredients remains the annotation bottleneck (CLAUDE.md). Without links, soft-wrong cannot route.

Ontology: Prefer nested ingredient `failure_mode` + Layer-2 `common_misconception_ids` as the join surface for coach cards. Variation sequences (XLVIII) that juxtapose the misconception-triggering surface vs the invariant are VT done right.

Solver / AI: Under PWC (XXXIII), AI wrap after student springboard — not “fluent correct answer” that ignores the selected distractor. Sycophancy risk: agreeing with a wrong path. Guard: name the student’s move, state where it applies, show the break.

Competitive wedge (XXXV): Duo celebrates streaks and instant binary feedback; ChatGPT delivers fluent keys; Khan explains. MindCraft sells **diagnostic wrong** → **recoverable path** with evidence the parent can see (retry, transfer)

without leaderboard shame (SAFE-COMPARE).

XLIX.8 SAFE-MISCON stack (ship rule)

- 1. **Elicit** — Real items; prefer misconception-tagged distractors.
- 2. **Classify** — Lack / slip / high-confidence misconception (confidence + item family).
- 3. **Destake** — Soft-wrong mastery climate; no public shame; no streak-as-punishment for miss.
- 4. **Springboard** — Micro-conflict, compare, SE prompt, or Borasi-style inquiry move *using the wrong*.
- 5. **Route** — Ingredient / bridge / VT contrast card from misconception map.
- 6. **Reveal with purpose** — Feedback timing by confidence tier; hide-correctness in diagnostic; attended feedback after high-confidence miss.
- 7. **Measure** — `retry_120s`, recalibration Δ , misconception recurrence, delayed isomorphic accuracy — not “mistakes celebrated” vanity.

Marketing language that survives: “Wrong answers that teach,” “find the pattern in the miss,” “confidence that matches skill.”

Marketing language that dies: “We never say you’re wrong,” “mistakes are magic,” instant red-X as rigor, Soft-wrong™ as identity transformation (L3 from tone).

XLIX.9 Claim ladder

Claim	Max ladder without new data
Soft-wrong raises attempt/retry vs harsh red-X	L1 (Experiment A)
Misconception-tagged distractors improve routing vs random wrongs	L1
Confidence-tiered feedback beats uniform instant key	L1
Hide-correctness diagnostic yields cleaner weakness ranking	L1 (C1/C4 product bet)
Productive-error UX → identity as mathematical thinker	L3 — ban until transfer + identity items

XLIX.10 Experiments spawned

ID	Question	Design	Primary	Kill condition
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MIS-1	Soft-wrong + springboard coach vs soft-wrong + key-only	2-arm	retry_120s; next-item accuracy	Key-only $\geq$ springboard $\rightarrow$ inquiry UX overstated
MIS-2	Misconception-mapped distractors vs random wrongs	2-arm	route precision; recurrence of same mis-ID	Random $\geq$ mapped $\rightarrow$ bank annotation ROI fails
MIS-3	Confidence-tiered feedback (hypercorrection-aware) vs uniform instant	2-arm	high-conf miss correction @ 48h	Uniform $\geq$ tiered $\rightarrow$ tiering complexity unjustified
MIS-4	Hide-correctness diagnostic vs reveal-as-you-go for gap-scan	2-arm	completion; affect; later worst weakness stability	Reveal $\geq$ hide on weakness quality + affect $\rightarrow$ C4 overstated
MIS-QUAL	10 Maya think-alouds after soft-wrong	interview	coded: shame vs curiosity; names own idea?	Students already springboard unaided $\rightarrow$ coach priority drops

Pre-reg (XXXIV): MIS-1...4 identify **error-handling design** $\times$ **classification**. They do **not** identify “celebrating mistakes causes identity.”

Densifies: Experiment A; C4 diagnostic; Layer-4 misconception memory; XL SE; XLVIII VT conflict pairs; Part L confidence calibration queue.

XLIX.11 Confidence table

Claim	Label	Confidence
Misconceptions are both flawed and productive resources (Smith et al., 1993/94)	FACT	High
Knowledge-in-pieces / p-prims reframes instructional task (diSessa, 1993; Hammer, 1996)	FACT	High
Errors can be springboards for inquiry (Borasi, 1994)	FACT	High
Diagnostic teaching / conflict discussion beats error-avoidant exposition for long-term learning (Bell/Swan line)	FACT	Medium–High (classroom evidence; app transfer open)

High-confidence errors are hypercorrected after feedback (Butterfield & Metcalfe, 2001; Foster et al. math diagnostics)	FACT	High
SAFE-MISCON is right default for MindCraft soft-wrong / diagnostic / bank	FOUNDER BELIEF / HYPOTHESIS	Medium–High
Soft tone alone transforms identity	SPECULATION / false as doctrine	Low (against)

XLIX.12 What this chapter kills

1. **Kill:** Misconceptions-as-disease / confront-and-replace-only pedagogy as company doctrine. (Smith/diSessa/Roschelle; Hammer.)
2. **Kill:** Instant red-X + answer dump as the default “rigorous” product. (Borasi springboard; Bell/Swan diagnostic teaching.)
3. **Kill:** Soft-wrong as empty brand kindness without diagnostic route + retry metrics.
4. **Kill:** Equal graph updates for guess, slip, and confident misconception. (Hypercorrection / calibration literature.)
5. **Kill:** Hide-correctness marketed as “we never teach answers.”
6. **Wound:** Classroom conflict discussion $\equiv$ mobile UX. Survive as designed micro-conflict + SE, not seminar cosplay.
7. **Survive (constrained):** Misconception-mapped distractors; confidence-tiered feedback; soft-wrong as FEI informative miss; springboard-before-monologue; MIS-1...4.

Doctrine until data: Ship **SAFE-MISCON**: treat wrong answers as diagnostic fuel, destake identity threat, springboard from the student’s idea, route to structure, measure retry and recalibration — never sell mistake-celebration posters or red-X theater as learning science.

Part L — Confidence Calibration: Over/Underconfidence, Hide-Correctness, and Honest Self-Knowledge

Chapter status: Living evidence brief — Researcher tick 2026-07-31 (UTC hour 06 $\equiv$ Red Team slot, but ch50 never written $\rightarrow$ prefer Researcher per rotation; 0 researcher entries since synthesizer v1.7)

Primary question: How should MindCraft treat student *confidence* — as a vibe to inflate, a score to gamify, or a *calibrated signal* that drives feedback, graph updates, and identity-safe diagnostics?

Owners: Product (Diagnostic C4 / soft-wrong / confidence UX) · Engine (evidence_update_policy / confidence tiers) · Marketing claim ladder · Red Team

Commercial job: Ship a **SAFE-CALIB** stack — elicit confidence $\rightarrow$ classify miss type $\rightarrow$ destake $\rightarrow$ recalibrate without inflation theater $\rightarrow$ tier feedback $\rightarrow$ measure confidence_miss_tier — kill “raise confidence” as North Star and “how sure?” without routing.

L.1 Why this chapter exists

The research question of this Constitution is how humans become *confident mathematical thinkers*. That phrase is dangerous if “confident” means louder, happier, or more certain. The commercial and scientific load-bearing meaning is **calibrated competence**: knowing when you know, knowing when you do not, and updating both skill and self-appraisal under informative misses.

Parts XXV (self-efficacy), XXX (attribution), XXXIII (AI overtrust), XLIX (SAFE-MISCON / hypercorrection), and the v1.7 metric `confidence_miss_tier` already point here. What was missing was a chapter that (a) defines calibration rigorously, (b) separates over- from underconfidence product responses, (c) ties C4 hide-correctness to metacognitive hygiene, and (d) refuses confidence theater as marketing.

FOUNDER BELIEF under audit: Identity as a mathematical thinker includes *honest self-knowledge* — not perpetual certainty, and not chronic self-doubt dressed as humility.

Claims we refuse as doctrine:

1. Raise confidence = improve learning (always).
 2. Confidence meters / “Belief Score™” as identity North Stars.
 3. Asking “how sure?” without changing feedback, routing, or graph updates.
 4. Hide-correctness = never teaching answers.
 5. Dunning–Kruger as a meme that justifies shaming low performers.
 6. Instant correctness reveal as the only path to calibration.
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L.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Calibration	Match between stated confidence and empirical accuracy across items	Confidence × correctness curves; bias & absolute accuracy indices	One-item “I feel sure” = calibrated
Overconfidence	Confidence systematically exceeds accuracy	High-conf wrongs; skipped review; false mastery	Shame the student; ignore hypercorrection opportunity
Underconfidence	Accuracy exceeds confidence	Correct + low conf; stuck on secure content	Force “believe harder” posters; skip challenge progression
Confidence of response	Certainty that <i>this</i> answer is correct (item-level)	0–10 or low/med/high next to answer	Global “math confidence” survey only

Hide-correctness (C4)	Record outcome; defer or withhold key reveal in diagnostic	Gap-scan / FEI probes	Never-teach-answers slogan; corrupt key theater
SAFE-CALIB	Calibration doctrine that survives evidence	See L.8	Confidence gamification; inflation theater

Operational definition (HYPOTHESIS): MindCraft treats calibration as a *session and graph* property when (a) item-level confidence is elicited on diagnostic and soft-wrong paths, (b) high-conf misses trigger attended corrective / springboard sequences, (c) low-conf corrects unlock challenge progression rather than endless drill, (d) mastery updates weight miss type (guess $\neq$ confident misconception), and (e) marketing claims stay at L0–L1 until CAL experiments report.

L.3 Calibration science — what “well calibrated” means

FACT (definition): Lichtenstein & Fischhoff (1977, *Organizational Behavior and Human Performance*, 20, 159–183; classic line) and Fischhoff, Slovic & Lichtenstein (1977, *Journal of Experimental Psychology: Human Perception and Performance*, 3(4), 552–564, doi:10.1037/0096-1523.3.4.552) — a judge is well calibrated if, over the long run, for propositions assigned probability p , the proportion true $\approx p$. Extreme certainty is routinely *inappropriate*: people are wrong too often when they feel certain.

FACT (bias pattern): Across general-knowledge and many educational tasks, the dominant bias is **overconfidence**, especially when accuracy is near chance; as accuracy rises, overconfidence often shrinks and underconfidence can appear at high accuracy (Lichtenstein & Fischhoff, 1977 summary pattern; replicated in educational monitoring work).

FACT (measurement plurality): Lingel, Lenhart & Schneider (2019, *ZDM*, metacognitive monitoring in mathematics — ERIC EJ1222655) — absolute accuracy, relative accuracy (e.g. Gamma), and diagnostic sensitivity/specificity are *not interchangeable*. Method choices change conclusions. Seventh-graders showed **pervasive overconfidence**; retrospective judgments and some scale formats improved monitoring precision vs prospective.

Product translation: MindCraft should log item-level confidence + correctness and report **bias** (over/under) and **discrimination** (does higher confidence predict correctness?) separately. Do not ship a single “Calibration %” vanity metric parents can misunderstand.

Kill: One glamorous calibration number as growth KPI. Survive: confidence_miss_tier + bias curves for product learning, not for public leaderboards (SAFE-COMPARE).

L.4 Overconfidence, underconfidence, and the dual commercial problem

FACT (school math overconfidence): García, Rodríguez, González-Castro, González-Pienda & Torrance (2016, *Metacognition and Learning*; ERIC EJ1105998) — elementary students solving math problems were poorly calibrated and tended toward **overconfidence**; inaccurate calibrators differed in reported metacognitive process use.

FACT (ability × miscalibration): Kruger & Dunning (1999, *Journal of Personality and Social Psychology*, 77(6), 1121–1134, doi:10.1037/0022-3514.77.6.1121) — bottom-quartile performers on humor/grammar/logic tasks grossly overestimated ability; incompetence can impair the metacognitive skill needed to recognize incompetence. Improving skill can improve self-appraisal.

Wound (popular DK): The meme version (“stupid people are always overconfident”) overclaims, ignores regression-to-the-mean critiques, and invites product cruelty. Use as **FACT** about *metacognitive dual burden in low skill*, not as a brand insult toward Maya.

FACT (underconfidence is also costly): Foster’s CA program (Foster, 2016, *Educational Studies in Mathematics*, 91(2), 271–288, doi:10.1007/s10649-015-9660-9; Foster, 2016, *Mathematics Teaching*, 251, 11–13; Centre for Mathematical Cognition summary) — underconfidence can reduce satisfaction, block progression to harder material, and trap students in repetitive secure practice; overconfidence can embed uncorrected errors and complacency. Pedagogical aim is **appropriate** confidence, not maximal confidence.

HYPOTHESIS for MindCraft: Overconfident Maya needs destaked confrontation + springboard (SAFE-MISCON + hypercorrection). Underconfident Maya needs earned-win evidence and challenge unlocks (FEI), not pep talks. One “confidence booster” feature cannot serve both.

Kill: “We build confidence” as undifferentiated slogan. Survive: “confidence that matches skill” / “know what you know.”

L.5 Confidence assessment in classrooms — promise and null on attainment

FACT (CA mechanism): Confidence assessment asks students to rate certainty alongside each answer; scoring that rewards correct high-confidence and penalizes incorrect high-confidence (and also punishes under-reporting when correct) incentivizes *truthful* ratings (Foster, 2016; Gardner-Medwin line in HE). Aim includes discouraging guess-and-hope.

FACT (student acceptability): Foster (2016 ESM) — secondary students can make usable confidence judgments on procedures; generally positive about CA in short trials.

FACT (null on summative attainment — load-bearing contradiction): Foster (2021/2022, *International Journal of Science and Mathematics Education*, 20, 1411–1429, doi:10.1007/s10763-021-10207-9) — quasi-experimental CA across four secondary schools (N = 475), meta-analytic Cohen’s $d \approx -0.02$ [95% CI $-0.22, 0.19$], Bayes factor favoring null on attainment gains vs controls. Conclusion: incorporating CA into low-stakes formative assessments does **not** appear detrimental — and also does **not** show a clear positive attainment effect from CA alone.

Commercial implication (critical): Shipping a “how sure?” widget is cheap and marketable. Evidence says **CA alone is not an attainment engine**. MindCraft’s wedge is not novelty CA — it is CA **wired** to soft-wrong routing, misconception maps, confidence-tiered feedback, and graph policy. Competitors can copy a confidence slider; they rarely ship the spine.

HYPOTHESIS: MindCraft CA improves *process metrics* (retry, misconception route hit, delayed isomorphic correction after high-conf miss) before it moves ACT scores. Claim ladder: process L1 first; attainment L2+ only after CAL + MIS experiments.

Kill: Marketing “confidence assessment raises grades.” Survive: CA as diagnostic fuel + metacognitive prompt, non-inferior on attainment per Foster 2021, valuable when coupled to FEI/SAFE-MISCON.

L.6 Hypercorrection, contradicting reasons, and feedback design

FACT (hypercorrection): Butterfield & Metcalfe (2001, *JEP:LMC*, 27(6), 1491–1494, doi:10.1037//0278-7393.27.6.1491) — high-confidence errors more often corrected after feedback than low-confidence errors.

FACT (math diagnostics): Foster, Woodhead, Barton & Clark-Wilson (*Educational Studies in Mathematics*, doi:10.1007/s10649-021-10084-7) — authentic online math diagnostic data show hypercorrection-consistent patterns; confidence assessment can prompt reflection.

FACT (debiasing via contradicting reasons): Koriat, Lichtenstein & Fischhoff (1980, *Journal of Experimental Psychology: Human Learning and Memory*, 6(2), 107–118) — listing reasons *against* the chosen answer improved appropriateness of confidence; supporting reasons alone did not. Confidence tracks strength of supporting evidence; people neglect disconfirming evidence unless prompted.

Product translation:

1. After high-conf miss → attended feedback + micro-conflict / “why might this be wrong?” (SE + Koriat) before fluent AI monologue (PWC / XXXIII).
2. After low-conf miss → scaffold/hug first (SAFE-DD), then lighter confrontation.
3. After low-conf correct → show competence evidence; invite slightly harder item (underconfidence unlock).
4. C4 hide-correctness during gap-scan: elicit confidence + answer; **defer** key reveal so the session stays diagnostic, not answer-hunting — then use purposeful reveal in practice springboards (XLIX).

Kill: Uniform instant red/green for every miss. Survive: confidence-tiered reveal + contradicting-reason prompt (aligns MIS-3).

L.7 Hide-correctness, AI trust, and graph integrity

FOUNDER BELIEF: C4 exists because revealing keys mid-diagnostic can spike threat, turn FEI into answer theater, and (if keys are wrong) poison trust. Recording outcomes without reveal preserves graph honesty for weakness ranking (C1).

HYPOTHESIS: Calibration *improves* when students later reconcile confidence with delayed feedback in a destaked springboard — not when every item is a public gotcha.

Link to XXXIII: Fluent AI tutors create **overtrust** (calibration failure toward the *system*). Student self-appraisal and AI-skepticism are sibling problems.

Layer-4 alignment: Confidence tiers operationalize `evidence_update_policy` — high-conf wrong → stronger misconception evidence; low-conf wrong → weaker / “uninformed” path.

Kill: Equal mastery deltas for all wrongs; “we never teach the answer.” Survive: purposeful reveal; confidence-weighted updates; AI wrap after student generation.

L.8 SAFE-CALIB stack (ship rule)

1. **Elicit** — Item-level confidence on diagnostic and on soft-wrong / challenge items (not only global surveys).
2. **Classify** — Correct/incorrect $\times$ low/med/high $\rightarrow$ miss classes (guess, slip, confident misconception, underconfident correct).
3. **Destake** — Soft-wrong / mastery climate; never public-shame high-conf wrongs; no streak punishment for honest low confidence.
4. **Recalibrate** — Show mismatch privately (“sure + wrong” or “unsure + right”) as information, not identity verdict.
5. **Tier feedback** — Hypercorrection path for high-conf miss; hug-then-dose for low-conf miss; unlock challenge after underconfident correct.
6. **Prompt against** — One contradicting reason / “where does this break?” before answer dump (Koriat).
7. **Measure** — `confidence_miss_tier`, calibration bias Δ , high-conf miss correction @ delay, `underconf $\rightarrow$ challenge_accept` — not Belief Score™ or confidence streaks.

Marketing language that survives: “Know what you know,” “confidence that matches skill,” “wrong answers that teach,” “honest diagnostics.”

Marketing language that dies: “We raise confidence,” “believe in yourself” as product, Confidence League ranks, “never say you’re wrong,” CA-raises-grades claims without data.

L.9 Competitive wedge (brief)

Duo (binary + streaks), Khan (explain-first), Brilliant (scaffold dopamine), and ChatGPT tutors (fluent certainty $\rightarrow$ overtrust) all fail *calibration* differently. MindCraft’s L1-safe line: not louder confidence — **earned, calibrated** confidence that survives a format hop (solo transfer + confidence on unaided items + PWC).

L.10 Claim ladder

Claim	Max ladder without new data
Item-level CA is usable with teens and non-inferior on attainment vs BAU	L1 (Foster 2016; 2021 null)
High-conf misses deserve different feedback than low-conf misses	L1 (hypercorrection literature)
Contradicting-reason prompts improve confidence appropriateness	L1 (Koriat et al., 1980)

C4 hide-correctness improves weakness ranking quality + affect vs reveal-as-you-go	L1 product bet (MIS-4 / CAL-4)
Confidence-tiered graph updates beat equal-wrong updates	L1
CA alone raises summative math attainment	Banned pending contrary RCT (Foster 2021 against)
Calibrated confidence → identity as mathematical thinker	L3 — ban until transfer + identity instruments

L.11 Experiments spawned

ID	Question	Design	Primary	Kill condition
CAL-1	Confidence-tiered feedback vs uniform instant key	2-arm	high-conf miss correction @ 48h; retry_120s	Uniform $\geq$ tiered → tiering unjustified
CAL-2	Contradicting-reason prompt vs explain-correct-only after miss	2-arm	calibration bias Δ ; next isomorphic accuracy	Explain-only $\geq$ prompt → Koriatic UX overstated for app
CAL-3	Underconfident-correct → challenge unlock vs more secure drill	2-arm	challenge_accept; later accuracy	Drill $\geq$ unlock on growth → underconf policy wrong
CAL-4	Hide-correctness diagnostic vs reveal-as-you-go (shared with MIS-4)	2-arm	completion; affect; worstweakness stability	Reveal wins on weakness + affect → C4 overstated
CAL-5	Confidence-weighted mastery updates vs equal-wrong updates	2-arm offline / sim + online	prediction of next-session accuracy	Equal $\geq$ weighted → engine complexity unjustified
CAL-QUAL	10 Maya interviews: meaning of “sure” / shame after high-conf miss	interview	coded: threat vs curiosity; wants reveal when?	Students refuse confidence UX → redesign elicit

Pre-reg (XXXIV): CAL-* identify **confidence** × **feedback** / **update design**. They do **not** identify “raising confidence causes identity.”

Densifies: MIS-3/4; C4; Layer-4 evidence policy; XXV self-efficacy; XXXIII AI overtrust; metric confidence_miss_tier.

L.12 Confidence table

Claim	Label	Confidence
Calibration = long-run match of confidence to accuracy (Fischhoff/Lichtenstein line)	FACT	High
Overconfidence is common; extreme certainty often unwarranted	FACT	High
School math students often poorly calibrated / overconfident (García et al.; Lingel et al.)	FACT	High
DK dual burden for low skill is real; meme uses are overstated	FACT / WOUND	Medium–High
CA usable + non-inferior attainment; not a proven attainment booster alone (Foster 2016; 2021)	FACT	High
Hypercorrection after high-conf errors (Butterfield & Metcalfe; Foster et al.)	FACT	High
Contradicting reasons improve confidence appropriateness (Koriat et al., 1980)	FACT	High
SAFE-CALIB is right default for MindCraft diagnostic / soft-wrong / graph	FOUNDER BELIEF / HYPOTHESIS	Medium–High
Raise-confidence marketing transforms identity	SPECULATION / false as doctrine	Low (against)

L.13 What this chapter kills

1. **Kill:** “Raise confidence” as undifferentiated North Star or brand promise. (Foster: appropriate confidence; under- and over- both hurt.)

2. **Kill:** Confidence meters, Belief Score™, or confidence streaks as identity engines. (SAFE-COMPARE / SAFE-HABIT alignment.)
3. **Kill:** “How sure?” UI without tiered feedback, routing, or update policy. (Foster 2021 null on CA-alone attainment.)
4. **Kill:** CA / confidence assessment marketed as raising grades without new evidence.
5. **Kill:** Equal graph updates for guess vs high-confidence misconception.
6. **Kill:** Hide-correctness as “we never teach answers.”
7. **Kill:** Dunning–Kruger shaming as brand voice.
8. **Wound:** Classroom CA $\equiv$ app CA; HE medical CA $\equiv$ HS math. Survive as: elicit + truthful incentive design + wired spine, not worksheet cosplay.
9. **Survive (constrained):** Item-level confidence; SAFE-CALIB stack; hypercorrection-aware feedback; underconfidence unlocks; CAL-1...5; coupling to SAFE-MISCON.

Doctrine until data: Ship **SAFE-CALIB**: treat confidence as a diagnostic signal for honest self-knowledge and differential pedagogy — never as a vibe to inflate, a league to rank, or a substitute for FEI evidence.

Part LI — Deliberate Practice in Tutoring Sessions: Ericsson vs Hambrick, and the 45-Minute Design

Chapter status: Living evidence brief — Researcher tick 2026-07-31 (UTC hour 09; hour%6≠0; 1 researcher entry since synthesizer v1.7 → Researcher)

Primary question: What should a MindCraft tutoring / practice session *do* if “practice” is not volume, not streak minutes, and not answer delivery — and how far can Ericsson’s deliberate-practice (DP) frame travel before Macnamara–Hambrick evidence kills overclaim?

Owners: Product (session architecture / tutor briefing) · Engine (weakness targeting / feedback loops) · Marketing claim ladder · Red Team

Commercial job: Ship a **SAFE-DP** session stack — diagnose → isolate edge skill → goal → effortful attempt → informative feedback → revise → brief rest → solo check — kill 10,000-hours theater, grind-volume North Stars, and “we do deliberate practice” as brand without a coach-designed edge task.

LI.1 Why this chapter exists

MindCraft sells sessions: college tutors, practice missions, Solver, Map. The Constitution’s research question is identity transformation under difficulty. Deliberate practice is the popular science answer to “how experts get good.” It is also a marketing trap — competitors and edtech decks cite Ericsson while shipping naive drill (more of the same), gamified volume (streaks/XP), or fluent AI monologue (pseudo-coaching).

Parts XXVI (CLT / fading), XXIX (spacing/retrieval), XXXI (challenge–skill), XXXIX (interleave), XL (self-explanation), XLI (SAFE-DD), and L (SAFE-CALIB) already constrain *what kind* of difficulty and feedback are safe. What was missing was a chapter that (a) defines DP vs purposeful vs naive practice, (b) faces the Macnamara–Hambrick variance numbers honestly, (c) translates math-education DP (Lehtinen line) into a **45-minute**

tutor/product design, and (d) refuses talent-denial and grind-as-virtue as doctrine.

FOUNDER BELIEF under audit: A high-quality MindCraft session is closer to *coach-designed practice at the edge of ability with feedback* than to homework help, content library, or engagement loops.

Claims we refuse as doctrine:

1. 10,000 hours / “practice makes perfect” as causal story.
2. Accumulated minutes / streaks as proof of DP.
3. DP explains nearly all expert variance (popular overread).
4. Any hard drill = DP; AI chat without diagnosis = DP coach; talent-denial as brand.

LI.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Naïve practice	Repeating what you can already do; play / homework slog	High accuracy, low struggle, no edge target	“Completed 40 problems” vanity
Purposeful practice	Goal + focus + feedback, possibly self-directed (Ericsson & Pool, <i>Peak</i> , 2016)	Student sets a micro-goal; app gives feedback	Duo-like streaks dressed as purpose
Deliberate practice (strict)	Coach/teacher designs tasks beyond current reliable performance; immediate informative feedback; refinement; builds mental representations (Ericsson et al., 1993; Ericsson & Pool, 2016)	Tutor/engine picks weakness; task just beyond; feedback revises method	Labeling all structured homework “DP”
Structured practice (meta-analytic)	Broad “practice designed to improve” used in Macnamara et al. (2014) — often looser than Ericsson’s DP	Hours logged in studies	Equating study-hours with DP
Mental representation	Organized internal model that lets experts monitor and improve (Ericsson line; Lehtinen et al., 2017)	Transfer / format hop / self-monitor	Fluency without representation

SAFE-DP	DP doctrine that survives Hambrick wound + anxiety/CLT gates	See LI.8	Grind theater; 10k-hours copy
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Operational definition (HYPOTHESIS): A MindCraft block counts as *deliberate-practice-shaped* when (a) target is a diagnosed edge (concept, ingredient, bridge, format, or miss class), (b) task is beyond reliable solo performance but solvable with brief support, (c) feedback changes the *next attempt's method*, not only the answer key, (d) student generates at least one why / check before AI wrap (PWC / XL), and (e) session ends with a short unaided probe (solo_transfer_pass or isomorphic check) — not with “minutes practiced.”

LI.3 Ericsson's core claim — what is actually said

FACT (foundational paper): Ericsson, Krampe & Tesch-Römer (1993, *Psychological Review*, 100(3), 363–406, doi:10.1037/0033-295X.100.3.363) — expert performance is strongly associated with accumulated *deliberate practice*: highly structured activities designed to improve specific aspects of performance, typically with teacher/coach guidance, clear goals, feedback, and effortful concentration. Mere experience / play / mindless repetition do not substitute.

FACT (effort constraint): Same paper and later summaries — effective daily DP duration is limited; early practice often ~1 hour or less; laboratory extended-practice protocols commonly ~1 hr sessions; pushing past optimal duration yields diminishing returns and fatigue. Rest is part of the training system, not laziness (aligns SAFE-TIMING).

FACT (Peak distinction): Ericsson & Pool (2016, *Peak*) — **purposeful practice** = well-defined goal, focus, feedback, get-out-of-comfort-zone; **deliberate practice** = purposeful practice *plus* a field with known training methods and a teacher who designs/adapts exercises. Popular “10,000 hours” readings flatten this distinction and often cite Gladwell's popularization rather than Ericsson's constraints.

Product translation: MindCraft's tutor + ontology spine can *aspire* to the coach role (diagnose → design edge task → feedback). Solo app loops without diagnosis are at best purposeful practice — and often naive practice with a nicer UI.

Kill: “10,000 hours of MindCraft.” Survive: “coach-designed practice at your edge, in a session you can sustain.”

LI.4 The Hambrick / Macnamara wound — variance is not destiny theater

FACT (meta-analysis): Macnamara, Hambrick & Oswald (2014, *Psychological Science*, 25(8), 1608–1618, doi:10.1177/0956797614535810) — across domains, deliberate-practice measures explained roughly **26%** of variance in games, **21%** music, **18%** sports, **4%** education, and **<1%** professions. Conclusion: DP is important, but **not as important as popular arguments claim**.

FACT (sports follow-up): Macnamara, Moreau & Hambrick (2016, *Perspectives on Psychological Science*, 11(3), 333–350, doi:10.1177/1745691616635591) — in sports, substantial unexplained variance remains; elite vs sub-elite comparisons weaken simple “more DP → elite” stories.

FACT (reply chain): Ericsson (2016, *Perspectives on Psychological Science*, doi:10.1177/1745691616635600) argues meta-analyses sum *heterogeneous practice hours* and dilute the construct; Macnamara & Hambrick (2016,

doi:10.1177/1745691616635614) reply that Ericsson’s criteria shift and that existing evidence does not support “largely accounted for by DP.”

FACT (definition rescue attempt): Ericsson & Harwell (2019, *Frontiers in Psychology*, doi:10.3389/fpsyg.2019.02396) — re-emphasize original DP criteria; claim Macnamara et al. often measured **structured practice**, not strict DP; reanalyses with stricter inclusion raise explained variance. Independent methodologists still warn against treating any reanalysis as settling talent, opportunity, and starting-skill confounds.

Wound for MindCraft marketing: Education’s ~4% figure is a Red Team grenade against “science says deliberate practice is everything.” Even if Ericsson is right that many education studies measured weak proxies, MindCraft cannot sell DP as a near-sufficient cause of identity or ACT gains.

HYPOTHESIS: MindCraft should treat DP as a **session design grammar** (how to spend 45 minutes), not as a **variance-explained brand claim**. Commercial strength is *individualized edge practice with feedback* — the part both camps agree matters more than mindless hours — without promising that practice volume explains Maya’s future.

Kill: “Experts are made of deliberate practice — so we made you an expert factory.” Survive: “Most homework isn’t practice that changes you; we design the next hard thing on purpose.”

LI.5 Mathematics education — drill vs deliberate practice

FACT (math DP review): Lehtinen, Hannula-Sormunen, McMullen & Gruber (2017, *ZDM Mathematics Education*, 49, 625–636, doi:10.1007/s11858-017-0856-6) — school math “practice” is often **drill-and-practice** that automatizes procedures but can yield *inert* routines rather than adaptive number knowledge; DP in expertise research involves thinking, problem solving, reflection, and dynamically retargeted training — not mechanical fluency alone. They summarize Ericsson (2016) features: coach-known skills, beyond-current-ability effort, specific goals, full attention, feedback + modification, building mental representations, improving prior skills with correct fundamentals.

FACT (study time ≠ improvement): Plant, Ericsson, Hill & Asberg (2005, *Contemporary Educational Psychology*, 30(1), 96–116) — in higher education, amount of study time did not significantly predict GPA once quality/concentration/goals entered; improvement tied to concentrated, goal-directed learning (cited in Lehtinen et al. as caution against raw hours).

HYPOTHESIS for MindCraft: Ontology ingredients + bridges are the *right decomposition* for DP-style targeting (isolate the failing subprocess, then rebuild), but only if sessions refuse “do 20 similar ACT items” as the default. Blocked accuracy is acquisition theater (XXXIX); DP needs the edge miss class (SAFE-MISCON / SAFE-CALIB), then varied probes (SAFE-TRANSFER).

Link to CLT / anxiety: Beyond-ability tasks without destake/equip violate SAFE-DD. DP’s “near-maximal effort” is **not** permission for threat flooding. For Maya, the coach’s first job is often making the edge task *informative*, not merely hard.

Kill: Back-to-basics grind marketed as Ericsson. Survive: adaptive practice that builds representations — Lehtinen’s contrast as competitive copy vs worksheet mills.

LI.6 What a 45-minute MindCraft session should look like

FOUNDER BELIEF: Human tutor sessions (~45–60 min) and digital missions should share one DP-shaped spine so marketing demos, tutor briefs, and app UX do not diverge into “warm homework help” vs “streak game.”

Proposed spine (HYPOTHESIS — SAFE-DP):

Minute block	Move	Maps to
0–5	Affect + goal: name <i>one</i> edge target (weakness / bridge / format / miss class)	FEI safety; E×V; SAFE-EXPECTANCY task-only language
5–12	Quick diagnose / warm isomorphic (hide-correctness OK if FEI)	C4; SAFE-CALIB elicit if used
12–28	Edge attempts: 1–3 items beyond reliable solo; student generation before explain	DP effort; PWC; SE (XL); SAFE-DD dose
28–36	Feedback that revises method: contradicting reason / springboard / ingredient card — not answer dump	Koriat; SAFE-MISCON; ingredient runtime
36–42	Near transfer / format hop / interleaved foil	SAFE-TRANSFER; XXXIX
42–45	Solo check + attribution (“what changed in your method?”) + next cue	solo_transfer_pass; SAFE-HABIT if-then; no streak sermon

Effort constraint product rule: Prefer one sharp DP cycle over stuffing eight topics. Marathon “grind nights” violate SAFE-TIMING and Ericsson’s own duration cautions.

Tutor ethics: Brief tutors on *task* targets from the graph — never trait labels (SAFE-EXPECTANCY). “Today we strengthen the functions→rate bridge” beats “she’s weak and unmotivated.”

AI role: Engine proposes the edge target and card order; LLM wraps language *after* student attempt (PWC). Unguarded Solver-as-session is not DP.

LI.7 Competitive wedge (brief)

Khan often explain-then-block; Duo sells streak volume; Brilliant scaffolds delight; ChatGPT tutors monologue without diagnosis; traditional tutoring delivers answers. MindCraft’s L1-safe line: not “more practice” — **better-designed practice** (one edge, method-changing feedback, solo proof). Minutes, XP, and fluent help are not DP.

LI.8 SAFE-DP stack (ship rule)

1. **Diagnose** — Pick one edge from graph / gap / tutor brief (concept, ingredient, bridge, format, or confidence-miss class).
2. **Isolate** — Decompose to the failing subprocess (ontology), not a random worksheet.
3. **Goal** — State a performance micro-goal for the block (“solve rate problems when the story hides the function”).
4. **Stretch** — Tasks beyond reliable solo, inside SAFE-DD capacity (equip/destake first if anxious).
5. **Attempt** — Student generates; no AI monologue-first (PWC / XL).
6. **Feedback** — Informative, method-revising; confidence-tiered when elicited (SAFE-CALIB / MISCON).
7. **Revise** — Immediate retry or isomorphic variant that uses the new method.
8. **Prove** — Short unaided / format-hop check; log solo_transfer_pass, not minutes.
9. **Stop** — Respect effort constraint; schedule sleep/spacing (SAFE-TIMING / XXIX).
10. **Measure** — Edge-target hit rate, method-change coding, transfer — never 10k-hours progress bars.

Marketing language that survives: “Practice at your edge,” “coach-designed sessions,” “feedback that changes your next try,” “show what you can do alone.”

Marketing language that dies: “10,000 hours,” “grind mindset,” “practice makes perfect,” “XP = mastery,” “our AI is your deliberate-practice coach” without diagnosis spine, talent-denial (“anyone can be elite if they DP”).

LI.9 Claim ladder

Claim	Max ladder without new data
Strict DP ≠ mere study hours / drill fluency	L1 (Ericsson 1993; Lehtinen 2017; Plant et al. 2005)
Popular “DP explains almost all expertise” overclaims; education variance small in Macnamara et al.	L1 (2014 meta; reply chain)
Coach/teacher-designed edge tasks + feedback beat naive repetition for skill change	L1 (expertise literature consensus on quality > raw hours)
MindCraft 45-min SAFE-DP spine improves solo transfer vs homework-help control	L1 product bet — needs DP experiments
DP volume alone creates identity as mathematical thinker	Banned (L3 without instruments; Hambrick wound)
Talent/opportunity irrelevant	Banned as brand

LI.10 Experiments spawned

ID	Question	Design	Primary	Kill condition
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DP-1	SAFE-DP spine vs homework-help / explain-first session (same tutor time)	2-arm	solo_transfer_pass; method-change code	Help ≥ DP spine → spine overstated
DP-2	One edge target vs multi-topic coverage in 45 min	2-arm	delayed isomorphic accuracy; cognitive load / affect	Coverage ≥ focus on learning → “one edge” wrong for ACT cram segment
DP-3	Engine-proposed ingredient target vs tutor-chosen topic without graph	2-arm	edge-hit; transfer	Tutor-only ≥ engine → graph targeting weak
DP-4	Feedback revises method (SE + springboard) vs answer-key-only after miss	2-arm	next-attempt success; retry_120s	Key-only ≥ method feedback → DP feedback claim weak
DP-5	Session cap (~45–60 focused min) vs encouraged grind (+30 min similar items)	2-arm	delayed accuracy; sleep/affect	Grind ≥ cap → effort-constraint messaging wrong for this population
DP-QUAL	10 Maya + 5 tutor interviews: “real practice” vs help	interview	coded: edge / threat / agency	Spine feels like cruelty → redesign destake

Pre-reg (XXXIV): DP-* identify **session architecture / targeting / feedback quality** — not “DP causes identity” or “explains elite careers.” Densifies FEI, PWC, SAFE-DD/TRANSFER/MISCON/CALIB/TIMING; prefer transfer metrics over minutes.

LI.11 Confidence table

Claim	Label	Confidence
Ericsson et al. (1993): DP = coach-guided, feedback-rich, effortful practice beyond current performance	FACT	High

Practice duration constrained; rest matters; purposeful ≠ deliberate without coach (<i>Peak</i>)	FACT	High
Macnamara et al. (2014): modest DP variance; ~4% education; definition dispute unsettled	FACT	High
Lehtinen et al.: math drill ≠ DP; Plant et al.: hours ≠ achievement when quality ignored	FACT	High
SAFE-DP is right default session grammar for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium–High
10k-hours / talent-denial / grind NS transform learners	SPECULATION / false as doctrine	Low (against)

LI.12 What this chapter kills

1. **Kill:** 10,000-hours / “practice makes perfect” as causal brand.
2. **Kill:** Streak minutes, problem counts, or XP as DP proof.
3. **Kill:** “DP explains expertise — therefore our app creates experts” (education ~4% wound).
4. **Kill:** Hard drill ≡ DP; AI monologue ≡ DP coach; talent-denial; multi-topic coverage theater as default rigor.
5. **Wound:** Violin-lab DP ≠ anxious ACT novice tutoring — survive as quality grammar + SAFE-DD, not music cosplay.
6. **Survive:** SAFE-DP spine; ontology edge targeting; method-revising feedback; effort caps; DP-1...5; “better-designed practice” copy.

Doctrine until data: Ship **SAFE-DP:** one diagnosed edge, effortful attempt, feedback that changes the next try, short solo proof, stop before grind — never sell hours, streaks, or talent-denial as the learning engine.

Part LII — Community of Practice (Lave/Wenger): Legitimate Peripheral Participation — Careful Transfer

Chapter status: Living evidence brief — Researcher tick 2026-07-31 (UTC hour 12 ≡ Red Team slot, but ch52 never written → prefer Researcher per rotation; 2 researcher entries since synthesizer v1.7 → Researcher)

Primary question: What does *community of practice* / *legitimate peripheral participation* (LPP) actually claim — and how far can MindCraft transfer those ideas into product, tutoring, story worlds, and “Community” UX before the transfer becomes cosplay?

Owners: Product (social / tutor / world design) · Marketing claim ladder · Equity (whose community?) · Red Team

Commercial job: Ship a **SAFE-CoP** stack — participation trajectories with real old-timers and real practices, not Discord-as-community, leaderboard-as-belonging, or “we’re a CoP” brand — while killing apprenticeship theater that

ignores Boylan/Fuller wounds.

LII.1 Why this chapter exists

The Constitution’s thesis is identity transformation under difficulty. Lave & Wenger’s situated-learning line is the classic social account of how newcomers become members of a practice — midwives, tailors, quartermasters — not how students complete problem sets. Edtech decks love the phrase *community of practice* because it sounds like belonging, mentors, and identity without specifying *what practice* students are joining.

MindCraft already has social-adjacent surfaces: college tutors, story worlds, Community nav, parent/tutor roles, Map as shared ontology. Without this chapter, those surfaces drift toward (a) Duo-style leagues as belonging, (b) forum/Discord theater as CoP, or (c) AI chat as “apprenticeship” with no old-timer practice. Parts XXV (identity), XXXVI (equity of worlds), XLII (SAFE-COMPARE), XLVI (tutor expectancy), and LI (SAFE-DP coach role) constrain pieces; what was missing was an honest transfer of LPP into product law.

FOUNDER BELIEF under audit: Becoming a confident mathematical thinker is partly *joining a practice* (ways of posing, checking, arguing, recovering from wrong), not only acquiring items — but school/ACT prep is not a tailor shop, and naming the product a CoP does not make LPP true.

Claims we refuse as doctrine:

1. “MindCraft is a community of practice” as marketing without naming the practice and the peripheral→central trajectory.
 2. Forums, streaks, or chat volume as proof of LPP.
 3. AI monologue as old-timer apprenticeship.
 4. School/ACT classrooms (or solo app sessions) as interchangeable with ethnographic apprenticeship sites.
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LII.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Community of practice (CoP)	People engaged over time in a joint enterprise with shared repertoire (Wenger, 1998)	Tutors + peers + world characters sharing <i>how math is done here</i>	Calling any group chat a CoP
Legitimate peripheral participation (LPP)	Newcomers participate in real practice at the edge; access and legitimacy grow toward fuller roles (Lave & Wenger, 1991)	Student contributes partial reasoning / checks; tutor/old-timer holds the full craft	Watching a video ≠ peripheral participation

Old-timer / newcomer	Asymmetric competence + responsibility in the practice	College tutor, veteran Maya peer, coach spine	Peer leaderboard as fake expertise hierarchy
Repertoire	Shared ways of talking, tools, stories, criteria of “good work”	Soft-wrong language; Map ingredients; story check-moves	Lore dump without practice criteria
Acquisition metaphor	Learning as gaining knowledge/skills as possessions (Sfard, 1998)	Mastery graph, item bank, transfer probes	Only metaphor → identity ignored
Participation metaphor	Learning as becoming a member of a practice (Sfard, 1998)	“How we recover from wrong here”	Only metaphor → contentless belonging
SAFE-CoP	CoP doctrine that survives Boylan/Fuller transfer wounds + Sfard pluralism	See LII.8	Apprenticeship cosplay; Discord theater

Operational definition (HYPOTHESIS): A MindCraft interaction counts as *LPP-shaped* when (a) the student does a real piece of the practice (attempt, explain, check, revise — not only consume), (b) an old-timer (human tutor or SE-constrained coach) models criteria of competence, (c) legitimacy is task/role based (you may try this edge), not trait based (you are a weak student — SAFE-EXPECTANCY), and (d) progress is visible as *changing participation* (more solo responsibility, wider repertoire) — not as XP.

LII.3 Lave & Wenger (1991) — what LPP actually says

FACT (foundational book): Lave & Wenger (1991, *Situated Learning: Legitimate Peripheral Participation*, Cambridge University Press, ISBN 9780521423748; doi:10.1017/CBO9780511815355) — learning viewed as situated activity has as its central defining characteristic **legitimate peripheral participation**: learners participate in communities of practitioners and move toward fuller participation in the sociocultural practices of the community. LPP names relations among newcomers and old-timers, activities, identities, artifacts, and communities of knowledge and practice — not a classroom seating chart.

FACT (cases): Their analytic cases are midwives, Yucatec midwives’ apprentices, Vai and Gola tailors, navy quartermasters, butchers, and Alcoholics Anonymous — craft and mutual-aid settings with durable joint enterprises, not standardized exam prep.

FACT (learning subsumed): They argue that becoming a participant *includes* skill learning; identity and membership are not add-ons after content delivery.

Product translation: MindCraft’s research question (“confident mathematical thinkers”) rhymes with *identity in practice*. The danger is importing the *label* while shipping acquisition-only UX (explain → pick A–D → streak).

Kill: “Situated learning = our story world backdrop.” Survive: “Participation means doing pieces of mathematical practice with growing responsibility.”

LII.4 Wenger (1998) — meaning, practice, community, identity

FACT: Wenger (1998, *Communities of Practice: Learning, Meaning, and Identity*, Cambridge University Press; doi:10.1017/CBO9780511803932) — social theory of learning with four interlocking components: **meaning** (experiencing the world as meaningful), **practice** (shared historical/social resources for mutual engagement), **community** (social configurations where competence is recognized), **identity** (learning changes who we are). CoPs are informal groupings that form as people pursue shared enterprises over time — primary unit of analysis is neither lone individual nor formal institution alone.

HYPOTHESIS for MindCraft: The product must specify the *enterprise* (e.g., recovering from productive wrongs on ACT-path concepts; telling-and-checking in a story world) and the *repertoire* (soft-wrong, SE prompts, Map language). Without enterprise + repertoire, “Community” nav is a social tab, not a CoP.

Wound: Corporate/edtech CoP literature often flattens Wenger into “knowledge-sharing groups.” MindCraft must not sell community features as identity transformation (claim ladder L3) from engagement metrics (L0/L1).

LII.5 The transfer wound — school math is not a tailor shop

FACT (school mathematics critique): Boylan (2010, *Teaching and Teacher Education*, 26(1), 61–70, doi:10.1016/j.tate.2009.08.005) — LPP was developed from informal apprenticeship contexts; applicability to school classrooms is **problematic**, especially where teacher-centred, decontextualised procedural practices dominate “usual” school mathematics. He critiques LPP transfer on social practice, learning relationships, power, agency, and identity; proposes **ecologies of participation** to capture multidimensional participation (imposed/voluntary, engagement/disengagement, identity risk/maintain, marginal/peripheral, etc.). He notes that if LPP is not found in usual school math, treating it as a *universal* feature of situated learning is undermined — while still valuing Lave & Wenger as a contrastive model and intervention heuristic.

FACT (workplace reassessment): Fuller, Hodkinson, Hodkinson & Unwin (2005, *British Educational Research Journal*, 31(1), 49–68, doi:10.1080/0141192052000310029) — LPP remains insightful for workplace learning research, but has **significant limitations** in complex institutional settings (manufacturing, secondary schools): diverse patterns of participation; institutional environments configure opportunities and barriers that ethnographic craft cases under-specify.

FACT (metaphor pluralism): Sfard (1998, *Educational Researcher*, 27(2), 4–13, doi:10.3102/0013189X027002004) — acquisition and participation metaphors each entail desirable and harmful practices; **choosing just one** risks theoretical distortion and bad pedagogy. MindCraft’s graph/mastery engine is acquisition-shaped; identity thesis is participation-shaped — doctrine needs both, not a CoP slogan that deletes assessment.

FACT (math classroom situated work — careful ally): Boaler (1999 AERA paper / related classroom studies; see also Boaler & Greeno identity/agency line summarized in math-education CoP literature) — classroom communities’ constraints and affordances shape what students learn to *do*; formalized classrooms attune students to practices that may not transfer. Useful as evidence that *participation forms matter*, not as proof that MindCraft already is an apprenticeship CoP.

HYPOTHESIS: Treat LPP as a **design aspiration for tutor+practice sessions** (peripheral contribution → fuller solo responsibility) and a **diagnostic for fake community**, not as a claim that the app is a CoP.

Kill: Universal “students learn by LPP here.” Survive: “Usual school math often blocks LPP; we design for legitimate attempts at the edge with old-timer criteria.”

LII.6 What “practice” means for MindCraft (not vibes)

Align with Constitution practices already named:

1. **Diagnose** → **attempt** → **informative miss** → **revise** → **solo check** (FEI / SAFE-DP / SAFE-MISCON).
2. **Student-generated why before AI wrap** (PWC / XL).
3. **Named transfer hops** (SAFE-TRANSFER).
4. **Calibrated confidence talk** (SAFE-CALIB) — saying “I’m sure” as part of the repertoire, not a Belief Score™.
5. **Story worlds as wrappers of frozen math** (equity + meaning) — characters can model *peripheral roles* (watcher → helper → solver) without claiming ethnographic authenticity.

SPECULATION (to test): Peer “Community” features help only when they circulate the repertoire above (how we check; how we destake wrongs). Social chat about school stress without practice criteria may raise belonging feelings without changing mathematical participation (wound for growth loops).

LII.7 Competitive positioning

Competitor pattern	CoP-shaped risk	MindCraft wedge
Duo leagues / streaks	Belonging as rank theater (SAFE-COMPARE)	Criterion competence, not named infinite ranks
Khan comments / help forums	Peripheral <i>talk about</i> problems ≠ LPP in craft	Tutor/old-timer + attempt-first
Brilliant delight social	Shared puzzles without recovery repertoire	Soft-wrong + solo transfer as membership proof
ChatGPT “study buddy”	Fluency without legitimacy criteria	PWC + ontology spine; AI ≠ old-timer by default
Discord “study community”	CoP brand, acquisition optional	Enterprise + repertoire required for CoP copy

Commercial implication: Do not out-community Discord. Sell *becoming someone who participates differently in hard math* — with evidence (solo_transfer_pass, method-change after miss), not member counts.

LII.8 SAFE-CoP stack (product law)

1. **Name the practice** — joint enterprise in copy and UX (“how we recover and prove on this edge”), not “join our community.”
 2. **Peripheral must be real work** — attempt, partial solution, check, SE prompt; watching/scrolling is not LPP.
 3. **Old-timer required for CoP claims** — human tutor or SE-constrained coach with criteria; unguarded AI chat ≠ apprenticeship.
 4. **Legitimacy without trait labels** — access to roles by task readiness; ban Golem scripts (SAFE-EXPECTANCY).
 5. **Trajectory > tab** — measure changing participation (solo share up; help faded) not Community page visits.
 6. **Plural metaphors** — keep acquisition instruments (graph, bank); never replace them with belonging KPIs (Sfard).
 7. **Equity of membership** — whose history/stories define the repertoire (Part XXXVI); token “community” without belonging audit is a kill.
 8. **Claim ladder** — CoP/LPP language max L1 product bet until CoP-* experiments; ban L3 “identity via community features” without instruments.
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LII.9 Claim ladder

Claim	Max ladder without new data
LPP describes apprenticeship-style learning in craft/mutual-aid cases	L1 (Lave & Wenger, 1991)
Usual school math often fails LPP conditions; transfer is non-trivial	L1 (Boylan, 2010; Fuller et al., 2005)
Acquisition + participation metaphors both needed; one alone distorts	L1 (Sfard, 1998)
MindCraft tutor sessions can implement LPP-shaped trajectories	L1 product bet — needs CoP experiments
Forums / streaks / AI chat = CoP / LPP	Banned
Shipping Community features creates mathematical identity	Banned L3 without instruments

LII.10 Experiments spawned

ID	Question	Design	Primary	Kill condition
CoP-1	LPP-shaped session (peripheral contribution + old-timer criteria + fade to solo) vs explain-first homework help	2-arm	solo_transfer_pass; coded participation moves	Help ≥ LPP spine → CoP framing overstated

CoP-2	Named repertoire onboarding (“how we check here”) vs feature tour	2-arm	next-session method language; retry_120s	Tour ≥ repertoire → practice-naming useless
CoP-3	Human tutor old-timer vs unguarded AI buddy (same time)	2-arm	solo transfer; overtrust flags (XXXIII)	AI ≥ tutor on solo → old-timer claim weak for this segment
CoP-4	Community chat with repertoire prompts vs open social chat	2-arm	practice-talk rate; delayed accuracy	Open ≥ prompted → repertoire constraint wrong
CoP-5	Story peripheral roles (watcher→helper→solver) vs flat problem list	2-arm	challenge_accept ; belonging items; transfer	Flat ≥ roles on learning → role trajectory theater
CoP-QUAL	10 Maya + 5 tutor: “who is in your math people?”	interview	coded: old-timer / repertoire / legitimacy	“Community” = friends only, no practice → social tab demote

Pre-reg (XXXIV): CoP-* identify **participation design** / **old-timer necessity** / **fake-community kill** — not “CoP causes identity” or “LPP is universal in apps.”

LII.11 Confidence table

Claim	Label	Confidence
Lave & Wenger (1991): LPP as central descriptor of situated learning in apprenticeship cases	FACT	High
Wenger (1998): CoP as meaning –practice–community–identity	FACT	High
Boylan (2010): LPP transfer to usual school math is problematic; ecologies of participation	FACT	High
Fuller et al. (2005): LPP limited in complex institutions	FACT	High
Sfard (1998): do not choose only acquisition or only participation	FACT	High

SAFE-CoP is right default for MindCraft social/tutor design	FOUNDER BELIEF / HYPOTHESIS	Medium
Community features alone transform math identity	SPECULATION / false as doctrine	Low (against)

LII.12 What this chapter kills

1. **Kill:** “We are a community of practice” as brand without enterprise + repertoire + old-timer.
2. **Kill:** Forums, Discord, streaks, leagues, or chat volume as LPP proof.
3. **Kill:** AI monologue $\equiv$ apprenticeship / old-timer.
4. **Kill:** Treating school/ACT/app sessions as ethnographically equivalent to Lave & Wenger’s craft cases.
5. **Wound:** Participation-only pedagogy that deletes mastery/transfer instruments (Sfard).
6. **Survive:** SAFE-CoP; LPP-shaped tutor spine; peripheral = real work; trajectory metrics; CoP-1...5; “join the practice of recovering and proving” copy — not “join the community.”

Doctrine until data: Ship **SAFE-CoP** — name the practice, require real peripheral work and old-timer criteria, measure participation trajectories under acquisition+participation pluralism — never sell social tabs or CoP labels as identity engines.

Part LIII — Religion / Ritual Light-Touch Design: Meaning Without Cult Dynamics

Chapter status: Living evidence brief — Researcher tick 2026-07-31 (UTC hour 15; hour%6≠0; 3 researcher entries since synthesizer v1.7 → Researcher)

Primary question: How can MindCraft design *light-touch rituals* that mark transitions, regulate affect, and carry meaning — without drifting into cult dynamics, sacred-science branding, or high-arousal initiation theater?

Owners: Product (session / world / onboarding arcs) · Brand & copy · Equity / trust · Red Team

Commercial job: Ship a **SAFE-RITUAL** stack — brief, opt-in, secular, reversible meaning-markers tied to real practice — while killing “join the order,” streak liturgies, confession milieus, and unretracted ritual→grade overclaims.

LIII.1 Why this chapter exists

The Constitution’s thesis is identity transformation under difficulty. Humans mark identity shifts with *rituals*: greetings before hard work, closing circles after miss→revise, naming a first solo transfer. Edtech and consumer apps already smuggle ritual language — streaks as devotion, leagues as congregations, onboarding as initiation, founder myth as sacred science — usually without naming the load-bearing psychology or the failure modes.

MindCraft has natural ritual-shaped moments: gap-scan completion, first soft-wrong recovery, tutor session open/close, story-world threshold crossings, Map unlocks. Parts XXIV (affect), XXV (identity), XXXI (flow), XLIII (SAFE-HABIT), XLIV (SAFE-REWARD), XLV (resilience), LII (SAFE-CoP) constrain pieces. What was missing was an honest doctrine that (a) uses ritual science for *light* meaning, (b) borrows transition structure without religion cosplay, and (c) installs a cult-dynamics kill list so “meaning” never becomes totalism.

FOUNDER BELIEF under audit: Becoming a confident mathematical thinker needs *marked moments* (not only item completion) — but the company must never sell belonging-through-devotion, milieus that punish exit, or doctrine that overrides the student’s evidence.

Claims we refuse as doctrine:

1. Rituals (or “sacred practice”) as proven grade engines from unretracted lab theater.
2. Streak / league / XP liturgy as identity North Star.
3. High-arousal initiation, confession milieus, or “you’re not one of us until...” gates.
4. Founder/product doctrine as sacred science that cannot be falsified (contradicts the lab itself).

LIII.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Ritual (psych)	Predefined, symbolic, often repetitive sequence that is <i>more than</i> its instrumental steps (Hobson et al., 2018)	Brief open/close; naming a miss class; candle-free “we begin”	Calling every UX animation a ritual
Routine / habit	Repeated behavior cue→response without required symbolic meaning (SAFE-HABIT)	If-then practice cue	Relabeling habit as sacred
Rite of passage	Transition marked by separation → liminal → reaggregation (van Gennep; Turner)	Gap-scan → liminal “not yet rated” → diagnosticCompleted + first path	Initiation hazing; permanent limbo
Imagistic mode	Infrequent, high-arousal rituals → vivid episodic memory; small exclusive groups (Whitehouse, 2004)	Rare world “boss” moments	Trauma theater; faction bonding over learning

Doctrinal mode	Frequent, low-arousal repetition → semantic/procedural memory; larger inclusive scale (Whitehouse, 2004)	Soft-wrong language; Map repertoire; session grammar	Indoctrination scripts; empty liturgy
Totalism / cult dynamics	Thought-reform pattern: milieu control, confession cult, sacred science, loaded language, doctrine-over-person, exit costs (Lifton)	Anti-patterns in copy, retention, community	“Family,” purity tests, unfalsifiable brand
SAFE-RITUAL	Light-touch ritual doctrine that survives cult kills + retraction wounds	See LIII.8	Sacred streak; initiation cosplay

Operational definition (HYPOTHESIS): A MindCraft moment counts as *SAFE-RITUAL* when (a) it is brief and optional or easily skippable, (b) symbolism points at *practice repertoire* (recover, prove, calibrate) not brand devotion, (c) exit/skip has no status punishment, (d) no confession or purity demand, (e) claims stay on the ladder (affect/transition aid ≠ causal identity engine).

LIII.3 Psychology of ritual — regulatory functions (not magic grades)

FACT (integrative review): Hobson, Schroeder, Risen, Xygalatas & Inzlicht (2018, *Personality and Social Psychology Review*, 22(3), 260–284, doi:10.1177/1088868317734944) — organize empirical ritual work around three regulatory functions: **(a) emotion regulation, (b) performance goal states, (c) social connection**. Mechanisms include bottom-up features (repetition, formality, synchrony) and top-down meaning (belief that the sequence *is* a ritual / carries symbolic significance).

FACT (grief / control): Norton & Gino (2014, *Journal of Experimental Psychology: General*, 143(1), 266–272, doi:10.1037/a0031772) — rituals after losses (loved ones, relationships, lotteries) can reduce grief feelings and increase felt control relative to no-ritual controls. Useful as evidence that symbolic sequences can regulate emotion and agency — **not** as a math-learning RCT.

FACT (consumption / meaning frame): Vohs, Wang, Gino & Norton (2013, *Psychological Science*, 24(9), 1714–1721, doi:10.1177/0956797613478949) — ritualized actions can enhance consumption experience via greater involvement/savoring. Ally for “mark the moment” design; kill if used to justify gamified consumption of XP as learning.

WOUND (retraction — do not market): Brooks, Schroeder, Risen, Gino, Galinsky, Norton & Schweitzer (2016, *Organizational Behavior and Human Decision Processes*, 137, 71–85, doi:10.1016/j.obhdp.2016.07.004) — claimed pre-performance rituals reduce anxiety and improve performance (including high-stakes math-like tasks). **This article has been retracted.** MindCraft must **not** cite it as surviving evidence or sell “rituals raise exam scores.” Treat as a Red Team kill on ritual→grade overclaim.

HYPOTHESIS for MindCraft: Prefer Hobson’s *emotion regulation* and *goal-state* functions for session openers/closers and soft-wrong recovery markers; treat performance/grade claims as **banned** until unretracted, pre-registered evidence on MindCraft metrics (retry_120s, solo_transfer_pass, affect check-ins).

Kill: “Science proves rituals improve math performance.” Survive: “Light symbolic sequences can regulate affect and mark transitions; grade claims need new evidence.”

LIII.4 Modes of religiosity — why high-arousal initiation is the wrong default

FACT: Whitehouse (2004, *Modes of Religiosity: A Cognitive Theory of Religious Transmission*, AltaMira Press / Rowman & Littlefield, ISBN 978-0-7591-0615-4) — religions tend to coalesce around **imagistic** vs **doctrinal** poles. Imagistic: infrequent, high-arousal rituals → flashbulb-like episodic memory; small, exclusive, often intense group bonds. Doctrinal: frequent, low-arousal, routinized teaching/ritual → semantic and procedural memory; larger, more inclusive, centrally scaffolded communities.

FACT (extreme ritual ally — careful): Xygalatas, Mitkidis, Fischer, Reddish, Skewes, Geertz et al. (2013, *Psychological Science*, 24(8), 1602–1605, doi:10.1177/0956797612472910) — extreme rituals can increase prosociality in field settings. This is **not** a product recommendation for teens in ACT prep; it shows imagistic arousal can bond groups — which is exactly the risk surface for cult-adjacent design.

HYPOTHESIS: MindCraft’s default should be **doctrinal-light**: frequent, low-arousal repertoire (session grammar, soft-wrong language, Map names) — not imagistic initiation (ordeal unlocks, shame fires, exclusive survivor factions). Rare story “threshold” moments may use mild imagistic *flavor* only if destaked, optional, and never required for belonging (SAFE-DD / equity).

Product translation: Gap-scan and first recovery are transitions, not hazing. World chapters can feel special without pain-as-proof.

Kill: Bootcamp/ordeal branding; “you haven’t really suffered until...” Survive: Frequent low-arousal practice markers + rare optional celebration.

LIII.5 Rites of passage — structure without cult costume

FACT: van Gennep (1909; English trans. 1960, *The Rites of Passage*, University of Chicago Press) — life transitions commonly share three phases: **separation**, **margin/limen**, **aggregation** (reincorporation).

FACT: Turner (1969, *The Ritual Process*; see also “Liminality and Communitas”) — elaborates the liminal phase as *betwixt and between*: ambiguous status, stripped markers, intense potential for *communitas* (anti-structure bonding). Powerful analytically; dangerous if product designers chase liminality as engagement hack (keep students permanently “in between,” dependent on the brand for status).

HYPOTHESIS for MindCraft transitions:

Phase	Product move	Guardrail
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Separation	Leave “I don’t know where I stand” → enter gap-scan / session	No shame script; task language only (SAFE-EXPECTANCY)
Liminal	Hide-correctness diagnostic; soft-wrong; “not yet aggregated”	Time-bounded; no status mockery; skippable ceremony
Aggregation	diagnosticCompleted; first path; named repertoire (“how we recover here”)	Grant competence access, not trait identity; no purity test

Link to SAFE-CoP: Aggregation should admit the student to *practice roles* (peripheral → fuller), not to a brand congregation.

Kill: Permanent liminality as retention. Survive: Clear exits into rights/obligations of the practice (you may attempt this edge solo).

LIII.6 Cult dynamics — the kill list (Lifton as anti-pattern checklist)

FACT (classic totalitarian criteria): Lifton (1961, *Thought Reform and the Psychology of Totalism*; later editions restate cult criteria) — thought-reform / totalistic environments characteristically include patterns such as: **milieu control, mystical manipulation, demand for purity, cult of confession, sacred science, loading the language, doctrine over person, dispensing of existence** (who counts as real/valid). Cult-like groups often combine charismatic leader-focus, thought-reform patterns, and exploitation of idealism.

FOUNDER BELIEF: MindCraft’s research lab and product ethics require the *opposite* of sacred science: every doctrine chapter is falsifiable; Red Team is a feature; students may leave without metaphysical damnation.

Product anti-patterns (SPECULATION → treat as design law until wounded):

Lifton pattern	Edtech failure mode	MindCraft ban
Milieu control	App as only legitimate math space; shame offline help	Allow tutors, school, parents; no “only here” purity
Mystical manipulation	Streak miracles; “the algorithm chose you” destiny copy	Transparent recommend reasons; no fate talk
Demand for purity	Perfect streaks; zero-miss identity	Soft-wrong; SAFE-MISCON; miss as data
Cult of confession	Forced public failure theater / oversharing	Private by default; opt-in sharing
Sacred science	Unfalsifiable brand claims; ban critique	Claim ladder; lab kills; no 2-sigma marketing
Loading the language	In-group jargon that replaces thinking	Repertoire words OK; ban thought-stopping slogans
Doctrine over person	Override student evidence with brand story	Graph + outcomes over vibes; SAFE-CALIB

Dispensing of existence	Rank/league as who “counts”	SAFE-COMPARE; no exile UX
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Kill: “We’re a movement / family / order.” Survive: “We’re a practice with optional meaning-markers and exit dignity.”

LIII.7 Competitive audit — ritual theater in the market

Duo streaks/leagues = doctrinal liturgy + mild imagistic competition (habit colonization; cue habits without sacred streak NS). Khan mastery energy = completion ritual (keep instruments; add recovery markers). Brilliant “aha” = imagistic spark (celebrate then probe transfer). GPT chat = pseudo-confession / endless liminality (SE-constrained close). Studygram hustle = purity + confession (ban grind-as-virtue).

Commercial implication: Position as **meaning without membership taxes** — mark *mathematical practice*, not brand devotion. Copy: “mark the recovery,” “open like a craft,” “close with what changed” — never “join the faithful.”

LIII.8 SAFE-RITUAL stack (product law)

1. **Light and short** — seconds to a minute; never a sermon that displaces attempts (SAFE-DP time budget).
2. **Optional / skippable** — no status loss for skip; no streak-as-morality.
3. **Point at practice** — symbolize recover / prove / calibrate / transfer — not founder, brand, or league.
4. **Doctrinal default** — frequent low-arousal repertoire; rare celebration only, never ordeal.
5. **Bounded liminality** — transitions resolve into clear rights (you may try X); no permanent betwixt for retention.
6. **Anti-totalism** — ban milieu monopoly, confession theater, sacred science, purity, exit shame.
7. **Affect ≠ grade claim** — may aim at calm / agency / belonging-to-practice; **no** unretracted ritual→score marketing.
8. **Equity** — no borrowed sacred forms that mock students’ real religions; secular by default (Part XXXVI).
9. **Claim ladder** — ritual language max L1 product bet until RIT-* experiments; ban L3 identity-via-ceremony without instruments.

LIII.9 Claim ladder

Claim	Max ladder without new data
Rituals can regulate emotion / goal states / connection (mechanism review)	L1 (Hobson et al., 2018)
Symbolic sequences can increase felt control after loss	L1 (Norton & Gino, 2014)
Imagistic vs doctrinal modes describe divergent ritual ecologies	L1 (Whitehouse, 2004)

Rites of passage often share separation–liminal–aggregation	L1 (van Gennep; Turner)
Lifton patterns are useful anti-design checks for edtech	L1 design heuristic (not a clinical diagnosis of competitors)
MindCraft light rituals improve <code>retry_120s</code> / affect / solo transfer	L1 product bet — needs RIT experiments
Rituals raise ACT scores / prove identity transformation	Banned without unretracted evidence
Streak liturgy / ordeal initiation / “family” milieus as identity engines	Banned

LIII.10 Experiments spawned

ID	Question	Design	Primary	Kill condition
RIT-1	30s session open ritual (name edge + breath/pose optional) vs silent start	2-arm	pre-task affect; <code>challenge_accept</code>	Silent $\geq$ ritual on start+transfer → opener useless
RIT-2	Soft-wrong recovery marker (“name the miss class”) vs correct/incorrect only	2-arm	<code>retry_120s</code> ; method-change code	Binary $\geq$ marker → recovery ritual theater
RIT-3	Gap-scan aggregation ceremony (brief) vs auto-redirect no mark	2-arm	next-session return; belonging -to-practice item	Auto $\geq$ ceremony → skip ceremony permanently
RIT-4	Optional celebration on first <code>solo_transfer_pass</code> vs none	2-arm	delayed transfer; skip rate	Forced > optional engagement but worse affect → force banned
RIT-5	Streak-framed cue vs practice-repertoire cue (same schedule)	2-arm	4-week return; identity items; anxiety	Streak $\geq$ repertoire on learning → still ban streak as NS; copy only
RIT-QUAL	10 Maya: which moments felt meaningful vs creepy/cultish?	interview	coded: light / pressure / exit fear	“Family/order” language → kill that copy

Pre-reg (XXXIV): RIT-* identify **transition/affect design and anti-totalism** — not “ritual causes identity” or “rituals raise scores.”

LIII.11 Confidence table

Claim	Label	Confidence
Hobson et al. (2018): three regulatory functions of ritual	FACT	High
Norton & Gino (2014): rituals can ease grief / raise felt control	FACT	High
Brooks et al. (2016) ritual→performance paper	FACT it was published; retracted — do not use as evidence	High (against use)
Whitehouse (2004): imagistic vs doctrinal modes	FACT	High
van Gennep / Turner: passage / liminality structure	FACT	High
Lifton totalism criteria as edtech anti-patterns	HYPOTHESIS / design heuristic	Medium–High
SAFE-RITUAL is right default for MindCraft meaning UX	FOUNDER BELIEF / HYPOTHESIS	Medium
Ceremony alone transforms math identity	SPECULATION / false as doctrine	Low (against)

LIII.12 What this chapter kills

1. **Kill:** Ritual→exam-score marketing (especially via retracted Brooks et al., 2016).
2. **Kill:** Streak / league / XP as sacred liturgy or North Star.
3. **Kill:** High-arousal initiation, confession milieus, purity tests, exit shame.
4. **Kill:** “Movement / family / order” copy that loads Lifton patterns.
5. **Wound:** Permanent liminality-as-retention; imagistic faction bonding.
6. **Survive:** SAFE-RITUAL; doctrinal-light repertoire; bounded passages; anti-totalism checklist; RIT-1...5; “mark the recovery” copy — not “join the faithful.”

Doctrine until data: Ship **SAFE-RITUAL** — brief, skippable, practice-pointing meaning-markers under an explicit cult-dynamics ban — never sell devotion, ordeal, or unretracted ritual-performance claims as the identity engine.

Part LIV — Military / Aviation Brief–Debrief: After-Action Reviews for Math Sessions

Chapter status: Living evidence brief — Researcher tick 2026-07-31 (UTC hour 18 ≡ Red Team slot, but ch54 never written → prefer Researcher per rotation; 4 researcher entries since synthesizer v1.7 → Researcher)

Primary question: What transferable structure do military after-action reviews and aviation brief/debrief cycles offer for MindCraft tutoring and practice sessions — without militaristic cosplay, blame culture, or overclaiming effect sizes as ACT marketing?

Owners: Product (session spine / Notes close) · Tutor ops · Coach UX · Brand & copy · Red Team

Commercial job: Ship a **SAFE-AAR** stack — brief → attempt → structured debrief with objective traces — while killing bootcamp branding, lecture-as-debrief, trauma-debrief confusion, and “debriefs improve performance 25%” as an undifferentiated growth claim.

LIV.1 Why this chapter exists

Parts LI (SAFE-DP) and LIII (SAFE-RITUAL) already constrain *time budgets* and *light meaning markers*. What was missing is the **operational cycle** high-reliability domains use to convert experience into the next attempt: a short **brief** that sets intent and roles, an event under observation, and a **debrief / after-action review (AAR)** that compares expected vs actual performance without collapsing into critique-as-humiliation.

MindCraft already has the raw materials: session open (affect check-in), attempt traces, soft-wrong codes, outcomes posted to the graph, Notes/session summaries, and coach cards. Competitors either skip structured reflection (answer-delivery chat), fake it with streak liturgy (Duo), or replace it with instructor monologue (many tutoring products and AI wrap text). The commercial wedge is not “we’re like the military.” It is: **every hard attempt leaves a learning artifact the student helps interpret.**

FOUNDER BELIEF under audit: Identity transformation under difficulty requires *reviewed attempts*, not only more problems — but review must be non-punitive, student-participatory, and tied to the next edge (SAFE-DP), not to brand devotion (SAFE-RITUAL) or trait shame (SAFE-EXPECTANCY).

Claims we refuse as doctrine:

1. Militaristic / “warrior math” branding as belonging technology.
 2. AAR ≡ critical-incident stress debriefing (trauma processing) — different animal.
 3. Marketing “+25% performance” from debrief meta-analyses as if it were an ACT score guarantee.
 4. Instructor/AI lecture that *calls itself* a debrief while blocking student self-analysis.
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LIV.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
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Brief	Short pre-event alignment on intent, success criteria, roles, likely traps (CRM / FAA markers)	Session open: target edge, level, “what counts as success,” optional coach role	10-min sermon; fate talk; trait labels
After-action review (AAR)	Systematic technique turning a recent event into learning via feedback, reflection, discussion (Keiser & Arthur)	Post-mission 4-question close; tutor end-of-session AAR	Critique lecture; blame; vibes-only
Debrief (training)	Structured reflection after simulation/performance; often facilitator-guided	Coach/tutor facilitation; self-led when traces exist	Endless chat liminality; AI monologue
Objective performance media	Records of what happened (video, logs, instruments) — not memory alone	Attempt list, soft-wrong codes, timing, outcomes JSON	Relying on “how did it feel?” only
Facilitation (vs lecture)	Elicit crew/student self-analysis; deepen participation (Dismukes et al.)	Advocacy–inquiry; student names frame first	Tutor/AI tells the moral of the story
Good judgment stance	Share evaluative judgment + curiosity about learner’s frames (Rudolph et al.)	Soft-wrong + “what were you seeing?”	Fake nonjudgmental praise; or blame
Alignment	Match participants ↔ focus ↔ measurement (team vs individual)	Solo practice AAR on solo metrics; tutor+student AAR on shared plan	Team vibes measured by individual XP
SAFE-AAR	Brief–attempt–debrief doctrine that survives militarism / blame / overclaim kills	See LIV.8	Bootcamp cosplay; 25%-as-marketing

Operational definition (HYPOTHESIS): A Minecraft close counts as *SAFE-AAR* when (a) a pre-brief named the intended edge and success criterion, (b) the debrief answers expected / actual / sustain / improve using **at least one objective trace**, (c) the student produces ≥ 1 self-analytic statement before any AI wrap, (d) the output is a next-attempt commitment (not a grade sermon), (e) no trait labels, no punishment for miss, no militaristic belonging tax.

LIV.3 Army AAR doctrine — four questions, not a roast

FACT (doctrine lineage): U.S. Army training circular **TC 25-20** (*A Leader's Guide to After Action Reviews*, 30 Sep 1993) and subsequent training doctrine institutionalize the AAR as a professional discussion focused on training objectives — not a humiliation ritual. Contemporary restatements (e.g., Center for Army Lessons Learned handbooks; USAID AAR guidance adapted from TC 25-20) center four questions:

1. What was **expected** to happen?
2. What **actually** happened?
3. What went **well**, and why?
4. What can be **improved**, and how?

FACT (process intent): Formal Army AAR guidance emphasizes open-ended questioning so participants discover gaps vs standards; facilitators must avoid critique-as-lecture (FM 25-101 Appendix G / CALL lineage). Focus: **results vs objectives**, sustain strengths, actionable fixes.

HYPOTHESIS for MindCraft: Map onto a 60–120s practice close and a 3–5 min tutor close — brief names expected edge; traces show actual; student names one sustain; one improve-how (SAFE-HABIT if-then or method change). Kill “AAR = public roast.” Survive: standards-referenced discovery.

LIV.4 Meta-analytic evidence — real gains, careful transfer

FACT: Tannenbaum & Cerasoli (2013, *Human Factors*, 55(1), 231–245, doi:10.1177/0018720812448394) — meta-analysis of 46 samples ($N = 2,136$). Debriefs / AARs improve effectiveness vs control by about **25% on average** ($d \approx 0.67$). Effects similar for teams and individuals, simulated and real settings, medical and nonmedical samples. **Alignment** (participants, intent, measurement level) bolsters effects; facilitated and structured debriefs trend stronger (facilitation estimate based on fewer nonfacilitated studies).

FACT: Keiser & Arthur (2021, *Journal of Applied Psychology*, 106(7), 1007–1032; advance online 2020, doi:10.1037/apl0000821) — bare-bones meta-analysis of 61 studies (107 ds ; 915 teams and 3,499 individuals). Overall AAR effect $d \approx 0.79$ across multiple training criteria — larger than Tannenbaum & Cerasoli's task-performance-focused estimate. Two characteristics consistently help: **(a) alignment** to individual vs team, **(b) objective performance review media**. Facilitation interacts: self-led + team-aligned and self-led + objective media among stronger combinations; structure matters more in military than healthcare contexts.

HYPOTHESIS (transfer wound): These samples are largely military, healthcare simulation, and organizational training — **not** ACT prep for anxious Maya. Treat $d \approx 0.67$ – 0.79 as evidence that structured post-event reflection is a high-leverage training design pattern — **not** license to claim “MindCraft AARs raise SAT/ACT by 25%.”

Commercial claim ladder:

- L1: Debriefs/AARs improve training outcomes in reviewed domains (cite metas).
- L1 product bet: SAFE-AAR improves MindCraft instruments (retry_120s, method-change codes, solo_transfer_pass).
- **Banned:** Undifferentiated “science says +25%” on landing pages without domain caveat and MindCraft pre-reg.

Kill: 2-sigma-style overclaim via borrowed meta d . Survive: “Structured brief–debrief with traces beats vibes-only close.”

LIV.5 Aviation — brief as error management; debrief as facilitated self-analysis

FACT (CRM briefing as performance marker): FAA Advisory Circular **AC 120-51E** (*Crew Resource Management Training*) lists effective **briefings** among CRM behavioral markers: thorough, address coordination/planning/problems, establish open communication, set expectations for SOP deviations and automation duties. Briefings are error-management tools — they align the crew *before* the event, not after the crash.

FACT (CRM evolution): Helmreich, Merritt & Wilhelm (1999, *International Journal of Aviation Psychology*, 9(1), 19–32; see also Helmreich line of NASA/UT work) — CRM evolved toward error management: avoid, trap, mitigate. Briefing strategies and situation awareness modules are framed as countermeasures, not soft skills theater.

FACT (LOFT debrief facilitation): Dismukes, Jobe & McDonnell (1997, NASA TM-110442, *LOFT Debriefings: An Analysis of Instructor Techniques and Crew Participation*) — analyzed 36 LOFT debriefings at five U.S. airlines; developed the Debriefing Assessment Battery. **Facilitation** (vs instructor lecture) increases depth of crew participation and self-analysis of CRM performance; instructor skill varied dramatically; crews were responsive but rarely fully led their own debriefs. Companion: McDonnell, Jobe & Dismukes (1997, NASA TM-112192, *Facilitating LOS Debriefings: A Training Manual*).

HYPOTHESIS for MindCraft:

- **Brief** = gap-scan / mission card / tutor open: name the edge, success criterion, known misconception trap, role of coach (hints vs silence).
- **Debrief** = student-led analysis with tutor/coach as facilitator; AI may *prompt* advocacy–inquiry, must not deliver the sermon first (aligns with Part XL SE constraints and Part XXXIII AI trust).

Product translation: Tutor briefing ethics from SAFE-EXPECTANCY (task-only language) + CRM brief markers (coordination, traps, automation = “when does the coach speak?”).

Kill: Captain-monologue onboarding. Survive: short interactive brief + facilitated close.

LIV.6 Healthcare simulation — good judgment, not fake nonjudgment

FACT: Rudolph, Simon, Dufresne & Raemer (2006, *Simulation in Healthcare*, 1(1), 49–55, doi:10.1097/01266021-200600110-00006; PubMed 19088574) — “There’s no such thing as ‘nonjudgmental’ debriefing.” **Debriefing with good judgment:** (1) frames drive actions → results; (2) stance of genuine curiosity (mistakes as puzzles); (3) **advocacy–inquiry** pairs (state observation + judgment, then ask what the learner’s frame was). Helps instructors share evaluative judgment without destroying trust.

Link to SAFE-MISCON / SAFE-CALIB: Soft-wrong is not empty kindness; it is destaked judgment plus frame inquiry. Hide-correctness diagnostic is a special brief (no outcome reveal) — its “debrief” must not smuggle keys that corrupt the graph (C4).

HYPOTHESIS: MindCraft coach cards should default to advocacy–inquiry templates (“I notice you treated the inequality like an equation when you multiplied by -1 — what were you assuming?”) rather than praise-only or scold-only.

Kill: “Nonjudgmental forever” (hides standards) and “brutal honesty” (hides curiosity). Survive: good-judgment debrief.

LIV.7 Distinctions and kill list (scope control)

Do not confuse training AARs with **critical-incident stress debriefing** (trauma processing — different ethics). Ban militaristic belonging (“warrior,” bootcamp shame), streak-as-AAR, AI monologue labeled as debrief, vibes-only hot-wash when traces exist, public leaderboard “reviews” (SAFE-COMPARE), and “+25% smarter” ads from borrowed meta *d*.

Wound: Hot-wash alone under-captures learning — useful as phase 1, insufficient as the whole product. MindCraft default: immediate 60s close + optional next-day Notes deepening.

LIV.8 SAFE-AAR stack (product law)

1. **Brief before attempt** — name edge, success criterion, trap, coach role (CRM brief markers).
 2. **Attempt under observation** — log objective media (outcomes, codes, timing).
 3. **Four-question close** — expected / actual / sustain / improve (TC 25-20 lineage).
 4. **Student speaks first** — ≥ 1 self-analytic statement before tutor/AI wrap (Dismukes facilitation).
 5. **Good judgment** — advocacy–inquiry; standards visible; curiosity about frames (Rudolph).
 6. **Align level** — solo metrics for solo practice; shared plan metrics for tutor sessions (Tannenbaum; Keiser).
 7. **Next-attempt output** — one concrete improve-how tied to SAFE-DP time budget.
 8. **Non-punitive climate** — miss = data; no trait labels (SAFE-EXPECTANCY); no militaristic shame.
 9. **Claim ladder** — meta *d* stays L1 domain evidence; MindCraft ACT claims need AAR-* experiments.
 10. **Anti-cosplay** — borrow the *cycle*, not the costume; secular practice language (SAFE-RITUAL).
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LIV.9 Competitive audit — who actually closes the loop?

Khan tells correctness without student-led expected/actual. Duo substitutes streak liturgy for AAR. Brilliant celebrates aha but rarely closes sustain/improve. ChatGPT tutors lecture after chatty “goals.” Human tutoring quality varies with facilitator skill (Dismukes variance). MindCraft already logs graph outcomes — formalize SAFE-AAR in practice close + Notes.

Commercial implication: Position as **reviewed struggle** — brief the edge, attempt with traces, close expected/actual — never “military-grade” or borrowed “+25%” miracles. Parent copy: expected vs happened → what changes next. Student copy: “name what you thought would happen.”

LIV.10 Claim ladder

Claim	Max ladder without new data
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AARs/debriefs improve training outcomes in meta-analyzed domains ($d \approx 0.67\text{--}0.79$)	L1 (Tannenbaum & Cerasoli, 2013; Keiser & Arthur, 2021)
Four-question AAR structure is institutionalized Army training practice	L1 (TC 25-20 lineage / CALL restatements)
Facilitation deepens self-analysis vs lecture in LOFT debriefs	L1 (Dismukes et al., 1997)
CRM briefings are performance markers for coordination / error management	L1 (FAA AC 120-51E; Helmreich line)
Good-judgment advocacy–inquiry is a validated simulation debrief stance	L1 method (Rudolph et al., 2006)
SAFE-AAR improves MindCraft <code>retry_120s</code> / method-change / solo transfer	L1 product bet — needs AAR experiments
“+25%” / meta d as ACT or identity guarantee	Banned
Bootcamp / warrior branding as belonging engine	Banned
AAR $\equiv$ trauma CISM	Banned conflation

LIV.11 Experiments spawned

ID	Question	Design	Primary	Kill condition
AAR-1	4-question practice close vs “nice job / try again” toast	2-arm	<code>retry_120s</code> ; method-change code rate	Toast $\geq$ AAR on retry+transfer $\rightarrow$ close theater
AAR-2	Brief (edge+success+trap) vs no brief, same items	2-arm	<code>challenge_accept</code> ; first-attempt process quality	No-brief $\geq$ brief $\rightarrow$ brief bloat; shorten or cut
AAR-3	Objective-trace AAR vs memory-only AAR	2-arm	accuracy of self-report; next-session transfer	Memory $\geq$ trace $\rightarrow$ still keep traces for graph; copy only
AAR-4	Student-first facilitation vs AI-monologue wrap	2-arm	SE quality rubric; <code>solo_transfer_pass</code>	Monologue $\geq$ student-first $\rightarrow$ contradicts XL; kill monologue default
AAR-5	Tutor trained on advocacy–inquiry vs business-as-usual close	2-arm cluster	session notes quality; student affect; 2-week transfer	BAU $\geq$ trained $\rightarrow$ training content fail

AAR-QUAL	10 Maya + 5 tutors: which closes felt useful vs military/shamey?	interview	coded: useful / cosplay / blame	“Bootcamp” language → kill that copy
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Pre-reg (XXXIV): AAR-* identify **session architecture** (brief–debrief design) — not “military methods raise ACT” or identity-from-discipline myths.

LIV.12 Confidence table

Claim	Label	Confidence
Tannenbaum & Cerasoli (2013): debriefs $d \approx 0.67$	FACT	High
Keiser & Arthur (2021): AAR $d \approx 0.79$; alignment + objective media matter	FACT	High
TC 25-20 / Army AAR four-question core	FACT	High
Dismukes et al. (1997): facilitation → deeper crew self-analysis	FACT	High
Rudolph et al. (2006): good-judgment / advocacy–inquiry	FACT (method paper)	High
FAA CRM briefing markers	FACT (advisory doctrine)	High
Meta effects transfer cleanly to ACT prep for anxious teens	HYPOTHESIS / wounded	Low–Medium
SAFE-AAR is right default session spine for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium
Military cosplay improves belonging or scores	SPECULATION / false as doctrine	Low (against)

LIV.13 What this chapter kills

1. **Kill:** “Military-grade” / bootcamp / warrior branding as identity or efficacy claim.
2. **Kill:** Marketing meta-analytic d / “+25%” as undifferentiated ACT or IQ promise.
3. **Kill:** Conflating training AARs with trauma CISD.
4. **Kill:** Lecture or AI monologue that labels itself a debrief while blocking student self-analysis.
5. **Kill:** Vibes-only closes without objective traces when traces exist.

6. **Wound:** Hot-wash-only products; facilitator skill variance (train tutors; constrain AI).

7. **Survive:** SAFE-AAR; four-question close; CRM-style brief; facilitation + good judgment; AAR-1...5; copy as **reviewed struggle** — not costume cosplay.

Doctrine until data: Ship **SAFE-AAR** — brief the edge, attempt with traces, close with expected/actual/sustain/improve under non-punitive facilitation — and never sell discipline theater or borrowed percentage miracles as the identity engine.

Part LV — Sports Film-Study Pedagogy: Error Clips → Coach Cards

Chapter status: Living evidence brief — Researcher tick 2026-07-31 (UTC hour 21; hour%6≠0; 5 researcher entries since synthesizer v1.7 → Researcher)

Primary question: What transferable design does sports film-study / video-based feedback (VBF) offer for MindCraft practice review — turning attempt traces into sparse, student-controlled “clips” and coach cards — without sports-cosplay branding, feedback floods, or highlight-reel shame?

Owners: Product (Solver / Notes / coach cards) · Tutor ops · Coach UX · Brand & copy · Red Team

Commercial job: Ship a **SAFE-FILM** stack — objective attempt clips + less-is-more cueing + learner-timed review + self-model pairing — while killing “we’re like a sports academy” marketing, monologue film rooms, and flood-of-tips as pedagogy.

LV.1 Why this chapter exists

Part LIV (SAFE-AAR) established the *cycle*: brief → attempt with traces → structured close. Sports film study is the **media layer** of that cycle — high-performance domains do not rely on memory alone; they rewatch selected moments, cue attention, and commit the next play. MindCraft already logs the analogue of game film: attempt lists, soft-wrong codes (mis_ IDs), timing, format tags, outcomes JSON. Competitors either never replay process (answer-key chat), replay only correctness (Khan-style), or flood the learner with AI commentary that no one remembers a week later.

FOUNDER BELIEF under audit: Identity as a mathematical thinker grows when the learner can *see their own decision frames* on a specific attempt — not when a brand dresses practice as a locker room. Film-study transfer is about **selective objective media + facilitation**, not jersey cosplay.

Claims we refuse as doctrine:

1. Sports-academy / “elite athlete” branding as belonging or efficacy technology.
 2. Coach monologue over 30 tips per “film room” (recall collapses).
 3. Highlight-reel of only successes *or* only failures as identity theater.
 4. Treating declarative recall of tips as if it were transfer to next-attempt performance.
 5. Equating any video UI with film-study pedagogy.
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LV.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Film study / VBF	Structured review of performance via recorded moments + guided attention	Replay attempt; soft-wrong clip; coach card on <i>that</i> step	Passive highlight reel; shame montage
Performance analysis (PA)	Evaluate/describe performance with media (Pearson et al.)	Graph outcomes + process codes as “tape”	Stats without learner agency
Self-observation	Learner watches <i>own</i> prior attempt	“Replay your steps” before AI wrap	Narcissistic doom-scroll of errors
Skilled / expert model	Observation of a competent demonstration	Worked example / correct pathway card paired to same archetype	Model-only → “I can’t do that” gap
Self-modeling / feedforward	Edited view of self succeeding at a near-future challenge (Dowrick)	Show student’s own prior <i>successful</i> transfer on related edge	Fake montages that invent competence
Attentional cueing	Direct gaze/attention to the decision that mattered	Coach card highlights inequality sign flip / bridge gap	Cue everything = cue nothing
Self-controlled feedback	Learner chooses <i>when</i> to request video/KP review	“Review this attempt?” on demand vs forced after every item	Autonomy theater with no cue quality
SAFE-FILM	Film-study doctrine that survives sports cosplay / flood / shame kills	See LV.8	Academy branding; tip dumps

Operational definition (HYPOTHESIS): A MindCraft review counts as *SAFE-FILM* when (a) the clip is an **objective attempt trace** (not vibes), (b) ≤ 3 attentional cues / coaching points are attached, (c) the learner can request or skip replay (self-control), (d) when a miss is shown, a **paired skilled model or student’s own prior success** on a related edge is available (self-model), (e) the output is a next-attempt commitment (SAFE-AAR improve-how), (f) no public shame reel, no sports-cosplay belonging tax.

LV.3 Applied models of observation — who / what / when / how

FACT: Ste-Marie, Law, Rymal, Jenny, Hall & McCullagh (2012, *International Review of Sport and Exercise Psychology*, 5(2), 145–176, doi:10.1080/1750984X.2012.665076) advance an **Applied Model for the Use of Observation (AMUO)**: practitioners should assess observer and task characteristics, then vary **who** is observed, **what** is observed (and instructional features), **when** observation occurs, and **how** information is delivered — not treat “show

video” as a single intervention.

FACT: Ste-Marie et al. (2020, *Research Quarterly for Exercise and Sport*, doi:10.1080/02701367.2019.1693489) revisit AMUO across 2011–2018 applied-task articles; observation remains effective across a wider task base, but **guidelines still require intentional Who/What/When/How design** — effectiveness is not automatic from “video exists.”

HYPOTHESIS for MindCraft: Map AMUO onto practice:

- **Who:** self (attempt replay) ± skilled model (worked pathway for same archetype) ± peer only when SAFE-COMPARE allows.
- **What:** the decision frame that produced the soft-wrong — not the whole worksheet.
- **When:** after attempt, before next isomorphic item; optional delayed Notes deepen (aligns SAFE-AAR hot-wash + later deepen).
- **How:** sparse cueing + student-first naming (facilitation), not AI monologue.

Kill: “We added a replay button” as film-study. Survive: designed observation with AMUO knobs.

LV.4 Video feedback as performance analysis — learning ≠ tip retention

FACT: Pearson, Webb, Milligan & Dicks (2023/2025 print, *International Review of Sport and Exercise Psychology*, 18(1), 417–439, doi:10.1080/1750984X.2023.2235700) — integrative review of video feedback as a facet of PA (16 studies from 5,838 screened; OSF registration noted). Delivery mode (coach-led classroom vs self-led), environment, and **scheduling** shape learning; scheduling often aids **declarative knowledge retention** more clearly than transfer to on-field performance. Collaborative coach–athlete practices may enhance engagement. Authors call for better frameworks and experimental variety — PA is common; *learning-optimized* PA is under-specified.

Wound (transfer): Elite sport PA classrooms ≠ Maya’s 12-minute practice session. Treat Pearson et al. as a warning against equating “we reviewed the tape” with “performance transferred.”

Commercial implication: MindCraft must instrument **next-attempt behavior** (retry_120s, method-change codes, solo_transfer_pass) — not tip-quiz scores after coach cards.

LV.5 Feedback floods kill recall — less is more

FACT: Mason, Farrow & Hattie (2020, *International Journal of Sports Science & Coaching*; doi:10.1177/1747954120951080) — exploratory study of one-to-one post-match video feedback in an AFL club (6 coach–player dyads). Coaches delivered on the order of ~**30 feedback messages** per meeting. One week later, players recalled about **50% of summarised themes** but only about **6% of fine-grained feedback idea units**. Authors encourage a “**less is more**” approach.

HYPOTHESIS: AI tutors and “helpful” coach UIs recreate the AFL flood at machine speed — worse for anxious novices (SAFE-DD × anxiety; CLT). Cap coach-card points per clip at **1–3**. Prefer one bridge/misconception handle over a laundry list.

Product translation: Soft-wrong → single primary mis_ / ingredient cue → one advocacy–inquiry prompt (Rudolph/SAFE-AAR) → one next-attempt tweak. Kill “here are eight tips from your film.”

LV.6 Self-controlled video feedback — autonomy of *when*, not absence of coaching

FACT: van der Meer, van den Hoven, van der Kamp & Savelsbergh (2024, *Research Quarterly for Exercise and Sport*, 95(2), 537–545, doi:10.1080/02701367.2023.2275801; online 2023) — intermediate tennis players; self-controlled video feedback (request) vs yoked externally controlled schedule; coach provided video with attentional cueing and transitional tactical statements. Self-controlled group showed **larger tactical performance gains** vs pretest at posttest and **one-week retention**. Self-efficacy did not differ by group; gains were not predicted by self-efficacy / self-regulative questionnaire scores in this sample.

FACT (supporting line): Aiken, Fairbrother & Post (2012, *Frontiers in Psychology*, 3:338, doi:10.3389/fpsyg.2012.00338) — self-controlled video feedback benefits basketball set-shot learning relative to yoked schedules (motor-learning line often cited in later VBF work).

Wound: Meta-analytic debate exists on the strength of self-controlled feedback advantages (van der Meer et al. note McKay et al., 2022 vs Jimenez-Diaz et al., 2021). Do **not** market “science proves self-paced video raises ACT.” Treat as design prior: **learner-timed review + quality cueing** beats forced tip dumps *or* no review.

HYPOTHESIS for MindCraft: After a miss (or optional after a hard success), offer “Review this attempt?” rather than auto-playing a 90s AI lecture. Forced review every item → fatigue and controlling climate (SAFE-REWARD / SAFE-DD).

LV.7 Self-model pairing and feedforward — identity-safe media

FACT (pairing): Robertson, St. Germain & Ste-Marie (2018, *Journal of Motor Learning and Development*, 6(1), 18–34, doi:10.1123/jmld.2016-0027) — gymnasts learning two skills in a within-subject design: self-observation **paired with a skilled model** outperformed self-observation alone later in acquisition; error-identification **response sensitivity** was also higher for paired skills.

FACT (self-modeling / feedforward): Dowrick (1999, *Applied and Preventive Psychology*, 8(1), 23–39, doi:10.1016/S0962-1849(99)80009-2) reviews self-modeling; Dowrick (2012, *WIREs Cognitive Science*, 3(2), 215–230, doi:10.1002/wcs.1156) theorizes “learning from the future” via reconfigured component skills. Dowrick, Kim-Rupnow & Power (2006, *Journal of Special Education*, 39(4), 194–207, doi:10.1177/00224669060390040101) — video **feedforward** plus tutoring improved reading fluency; in 9/10 cases improvement rate was greatest during feedforward (small *N* — transfer carefully).

HYPOTHESIS: A miss clip should not stand alone. Pair with (1) a skilled worked pathway for the same archetype and/or (2) the student’s own prior success on a related edge. Feedforward spirit without fake editing: **show competence already owned, then the gap**.

Kill: Failure-only shame reels; success-only dopamine montages that hide productive misconceptions (SAFE-MISCON). Survive: paired clip design.

LV.8 SAFE-FILM stack (product law)

1. **Clip = objective attempt media** — steps, codes, timing; not “how did it feel?” alone (extends Keiser objective-media finding via LIV).
 2. **Less is more** — ≤ 3 cues per clip; prefer one primary soft-wrong handle (Mason et al.).
 3. **AMUO design** — decide who/what/when/how before shipping replay UI (Ste-Marie 2012/2020).
 4. **Self-controlled timing** — learner can request review; default is offer, not force-flood (van der Meer et al.; Aiken et al.).
 5. **Self-model pairing** — miss clip + skilled pathway and/or own prior success (Robertson/Ste-Marie line; Dowrick feedforward spirit).
 6. **Student names the frame first** — then advocacy–inquiry coach card (SAFE-AAR / Rudolph).
 7. **Transfer metric, not tip quiz** — instrument next attempt (Pearson wound).
 8. **Anti-cosplay** — borrow the *media pedagogy*, not locker-room brand (SAFE-RITUAL / SAFE-AAR).
 9. **No public error theater** — film study is private or tutor-dyad unless SAFE-COMPARE + consent (SAFE-CoP caution).
 10. **Claim ladder** — sport VBF evidence stays L1 domain evidence; MindCraft ACT/identity claims need FILM-* experiments.
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LV.9 Competitive audit — who runs film study?

ChatGPT-style tutors rarely replay *your* attempt structure and risk tip floods. Khan shows correctness with weak process pairing. Duo streak liturgy $\neq$ film study. Brilliant ahas rarely close a self-model loop. Human “sports-style” tutoring inherits facilitator variance (Pearson). MindCraft already has attempt traces + soft-wrong + cards + Notes — **formalize SAFE-FILM**.

Commercial implication: Position as **reviewed decisions** — cue the one hinge, try the next play — never “train like the pros.” Parent copy: replay *how* they thought. Student copy: “pick a clip to review.”

LV.10 Claim ladder

Claim	Max ladder without new data
Observation interventions should vary who/what/when/how (AMUO)	L1 (Ste-Marie et al., 2012; 2020)
VBF-as-PA often improves declarative retention more clearly than transfer; design/delivery matter	L1 (Pearson et al., 2023)
Dense coach tip floods $\rightarrow$ poor week-later fine-grained recall ($\sim 6\%$); less-is-more recommended	L1 exploratory (Mason, Farrow & Hattie, 2020)
Self-controlled video feedback can outperform yoked schedules on tactical learning (tennis)	L1 single study (van der Meer et al., 2024)

Self-skilled-model pairing can beat self-observation alone (gymnastics)	L1 (Robertson, St. Germain & Ste-Marie, 2018)
Video feedforward can accelerate skill acquisition in some educational samples	L1 small-N (Dowrick et al., 2006); theory (Dowrick, 2012)
SAFE-FILM improves MindCraft retry_120s / method-change / solo_transfer_pass	L1 product bet — needs FILM experiments
“Sports academy methods raise ACT / identity” undifferentiated	Banned
Highlight-shame reels / tip floods as pedagogy	Banned

LV.11 Experiments spawned

ID	Question	Design	Primary	Kill condition
FILM-1	≤3-cue clip card vs uncapped AI tip flood after miss	2-arm	tip recall @24h; retry_120s; method-change	Flood ≥ sparse on transfer → revisit cue quality only
FILM-2	Self-controlled “Review?” vs forced replay every miss	2-arm	completion; affect; next-item transfer	Forced ≥ choice on transfer <i>and</i> better affect → keep forced for novices only
FILM-3	Miss-only clip vs miss+skilled model vs miss+own prior success	3-arm	error ID; solo_transfer_pass	Miss-only ≥ paired → pairing theater
FILM-4	AMUO-scripted coach card (who/what/when/how) vs generic praise/explain	2-arm	SE quality; transfer	Generic ≥ AMUO → card content fail
FILM-5	Film close + SAFE-AAR four questions vs film without AAR questions	2-arm	improve-how specificity; week-later transfer	No-AAR ≥ AAR → film without close still ok; keep AAR optional
FILM-QUAL	10 Maya: which clip UX felt useful vs shamey/sportsy?	interview	coded: useful / cosplay / flood / shame	“Athlete” language → kill that copy

Pre-reg (XXXIV): FILM-* identify **review media architecture** — not “sports methods raise scores” or identity-from-athleticism myths.

LV.12 Confidence table

Claim	Label	Confidence
Ste-Marie et al. AMUO who/what/when/how	FACT	High
Pearson et al. VBF-as-PA: delivery/scheduling; transfer caution	FACT	High
Mason et al.: ~30 tips; ~6% fine-grained week recall; less-is-more	FACT (exploratory, small <i>N</i>)	Medium–High
van der Meer et al.: self-controlled VBF > yoked on tennis tactics	FACT (single study)	Medium
Robertson, St. Germain & Ste-Marie: self∪model > self alone	FACT	Medium–High
Dowrick feedforward reading gains	FACT (small <i>N</i>)	Medium
Sport VBF transfers cleanly to ACT prep for anxious teens	HYPOTHESIS / wounded	Low–Medium
SAFE-FILM is right default review layer for Minecraft	FOUNDER BELIEF / HYPOTHESIS	Medium
Sports-cosplay branding improves belonging or scores	SPECULATION / false as doctrine	Low (against)

LV.13 What this chapter kills

1. **Kill:** Sports-academy / pro-athlete / locker-room branding as efficacy or belonging claim.
2. **Kill:** Feedback floods (30 tips / uncapped AI wrap) labeled as film study.
3. **Kill:** Equating tip recall or “we watched the replay” with transfer / exam readiness.
4. **Kill:** Failure-only shame reels and success-only dopamine montages as identity engines.
5. **Kill:** Replay UI without AMUO who/what/when/how design.
6. **Wound:** Classroom PA and elite dyads under-transfer to short teen sessions; facilitator variance.
7. **Survive:** SAFE-FILM; less-is-more clips; self-controlled review timing; self∪model pairing; feedforward spirit via own prior success; FILM-1...5; copy as **reviewed decisions** — not sports cosplay.

Doctrine until data: Ship **SAFE-FILM** — objective attempt clips, sparse cues, learner-timed review, paired models, transfer metrics — stacked on SAFE-AAR’s brief–debrief spine, and never sell athlete mythology or tip floods as the transformation engine.

Part LVI — Music Pedagogy Ladders: Scales → Repertoire → Recital Identity

Chapter status: Living evidence brief — Researcher tick 2026-08-01 (UTC hour 00 ≡ Red Team slot, but ch56 never written → prefer Researcher per rotation; 6 researcher entries since synthesizer v1.7 → Researcher)

Primary question: What transferable design does instrumental music pedagogy offer for MindCraft’s learning ladder — technique drills → meaningful repertoire → public competence signal — without conservatory cosplay, minutes-practiced vanity, or recital theater as identity?

Owners: Product (Practice path / story quests / exam signal) · Coach UX · Brand & copy · Red Team

Commercial job: Ship a **SAFE-MUSIC** stack — scales-as-ingredient drills, repertoire-as-story/exam pieces with learner choice, recital-as-transfer proof not stage cosplay — while killing “practice like a virtuoso,” hours-logged as mastery, and pass-floor exam identity.

LVI.1 Why this chapter exists

The core OS already sketches a music analogy (drill → quest → “recital” signal). Parts LI (SAFE-DP) and LV (SAFE-FILM) borrowed *session architecture* and *review media* from high-performance domains. Music pedagogy is the clearest **ladder** domain: beginners do not start with concertos; they build technique, then play repertoire they care about, then face graded exams or recitals that mark identity — *if* practice quality, motivation, and self-belief cohere.

MindCraft’s product already has the three rungs in embryo: ingredient/concept drills (scales), story-wrapped and exam-tagged questions (repertoire), ACT/diagnostic public signals (recital). Competitors collapse the ladder: Duo gamifies minutes; Khan floods explanations; ChatGPT skips technique for fluent answers; Brilliant jumps to ahas without graded public proof.

FOUNDER BELIEF under audit: Mathematical identity grows when learners climb a **named ladder** — controlled technique → chosen meaningful pieces → evidence they can perform under evaluation — not when a brand dresses homework as a conservatory.

Claims we refuse as doctrine:

1. Conservatory / “virtuoso / prodigy” branding as belonging or efficacy.
 2. Practice **minutes** or streaks as achievement (quantity without formal practice quality).
 3. Scales-forever without repertoire *or* repertoire without technique scaffolds.
 4. Recital/stage theater as identity without solo transfer evidence.
 5. Treating a bare “pass” exam outcome as motivational success.
-

LVI.2 Constructs (product language)

Construct	Music meaning	MindCraft analogue	Failure mode if misused
Scales / technique	Isolated skill building with immediate feedback	Ingredient drills; format reps; bridge micro-skills	Endless drill without meaning → boredom / controlling motivation
Repertoire	Whole pieces worth playing; often student-chosen	Story quests; archetype instances; exam-tagged sets	Pretty narrative with no technique → theater
Recital / grade exam	Public or adjudicated performance identity marker	Diagnostic; timed mixed set; solo_transfer_pass	Stage cosplay; pass-floor as “done”
Formal practice	Goal-directed time with SRL + deliberate strategies	SAFE-DP session spine with goals + feedback	Naive play-through of worksheets
Ineffective strategies	Error-chasing without analysis; mindless repetition	Immediate correction spam; tip floods; blocked vanity accuracy	Hallam “ineffective strategies” factor
SAFE-MUSIC	Ladder doctrine that survives cosplay / quantity / theater kills	See LVI.8	Conservatory brand; minutes KPI

Operational definition (HYPOTHESIS): A MindCraft path counts as *SAFE-MUSIC* when (a) technique drills are **goal-tagged ingredients** with feedback (not infinite XP), (b) repertoire includes **learner-chosen or interest-aligned** problem sets when feasible, (c) public signals require **mixed / transfer** performance not blocked accuracy, (d) minutes practiced never outrank formal-practice quality metrics, (e) branding names the ladder (skills → pieces → proof) without virtuoso mythology.

LVI.3 Deliberate practice’s home domain — music, not Gladwell hours

FACT: Ericsson, Krampe & Tesch-Römer (1993, *Psychological Review*, 100(3), 363–406, doi:10.1037/0033-295X.100.3.363) grounded deliberate practice in the training of instrumental musicians: effortful, teacher-guided activities with goals, immediate informative feedback, and repetition after reflection — distinct from playful performance or mere hours logged. Music academy students’ structured solitary practice was the empirical prototype.

Wound (already in SAFE-DP / Part LI): Hambrick-line critiques and domain limits mean “10,000 hours” and undifferentiated practice time are **banned** marketing. Music supplies the *definition*, not a guarantee that ACT scores rise from branded “deliberate practice.”

Product implication: Scales = short, goalful ingredient reps with feedback; never sell “log more hours this week” as the transformation engine (aligns Bonneville-Roussy below; SAFE-HABIT / SAFE-REWARD).

LVI.4 Strategy quality rises with expertise — not all practice develops

FACT: Hallam, Rinta, Varvarigou, Creech, Papageorgi, Gomes & Lanipekun (2012, *Psychology of Music*, 40(5), 652–680, doi:10.1177/0305735612443868) surveyed **N = 3,325** young instrumentalists across nine expertise/grade levels. Factor analysis yielded seven practice-related factors, including: systematic practice strategies; organization of practice; use of recordings/metronome for feedback; analytic strategies; **ineffective strategies**; concentration; immediate correction of errors. Statistically significant linear relationships with grade level appeared for systematic strategies, recordings/metronome use, ineffective strategies, and immediate error correction — but **not** for organization, analytic strategies, or concentration. Educational implication: expertise tracks *which* strategies learners adopt, not merely “they practiced.”

HYPOTHESIS for MindCraft: Instrument strategy_quality proxies (sectioning hard items, requesting SAFE-FILM clip, naming soft-wrong before retry) — not minutes_in_app. Decline of enjoyment of practice with advancement in related Hallam work is a wound: advanced ladders need repertoire meaning, not more grind UI.

Kill: Progress bars powered only by time-on-task. Survive: strategy factors as product instrumentation.

LVI.5 Quantity is not enough — formal practice mediates time → achievement

FACT: Bonneville-Roussy & Bouffard (2015, *Psychology of Music*, 43(5), 686–704, doi:10.1177/0305735614534910; online 2014) propose an integrative framework: **formal practice** = goal-directed, focused practice that includes both self-regulation and deliberate practice strategies. In a 4-month prospective SEM study, their integrative model predicted musical achievement better than traditional time-only approaches; practice time predicted achievement **when associated with formal practice**, not as raw quantity.

Commercial implication: Parent dashboards that lead with “hours practiced” recreate the failed music predictor. Lead with formal-practice signals: goals set, feedback used, transfer passed. Copy: “focused reps on the weak bridge,” never “she practiced 4 hours.”

Aligns: SAFE-DP (goals/feedback/rep), SAFE-HABIT (cue ≠ colonization), streak ban as North Star.

LVI.6 Repertoire choice — interest upgrades practice behavior

FACT: Renwick & McPherson (2002, *British Journal of Music Education*, 19(2), 173–181, doi:10.1017/S0265051702000256) — case study of a beginning clarinetist (“Clarissa”). When practising **self-selected** repertoire vs teacher-assigned pieces, she engaged strategies typical of more advanced stages (silent fingering, silent thinking, singing), spent more time, and persevered through difficulty. Authors link choice/interest to higher-level cognitive engagement (consistent with broader motivation findings cited therein).

FACT / FOUNDER-adjacent: Evans (2015) applies Self-Determination Theory to music education: scales are often **not** intrinsically enjoyable; motivation quality depends on whether learners internalize the *value* of technique (identified regulation) vs pure external control (teacher said so / praise). External control of scale drills risks the Deci overjustification pattern already killed in SAFE-REWARD (Part XLIV).

HYPOTHESIS: MindCraft “repertoire” should mix (1) teacher/engine-assigned technique (scales) framed with **why this ingredient unlocks the next piece**, and (2) **choice among story arcs / problem sets** when mastery allows — choice upgrades strategy quality, not just delight metrics.

Kill: Forced syllabus-only grind with no choice forever. **Kill:** choice-only playground with no technique ladder. **Survive:** scales with identified value + repertoire choice windows.

LVI.7 Recital / grade exams — identity markers that can demotivate

FACT: Hallam, Papageorgi, Varvarigou & Creech (2020 print / 2019 online, *Psychology of Music*, doi:10.1177/0305735618816168) — N = 2,131 musicians aged 6–19 across expertise and exam outcomes (fail → highly commended). Those awarded **merely a pass** tended to undertake the **least** practice and responded least positively on organization, recordings/metronome use, analytic strategies, social life, and self-belief in musical ability. Those who **failed** were most likely to adopt ineffective strategies and less likely to enjoy performing, playing, lessons, and practice. Motivation models in the paper foreground musical **identity**, self-belief, goals, attributions, and supportive environment — not exam theater alone.

Wound: Graded exams are identity technology *and* stress technology (parent/teacher reputation dynamics noted in related literature the authors cite). A MindCraft “recital” that is only a score reveal can create pass-floor demotivation or fail-shame — opposite of FEI.

HYPOTHESIS: Public signals must be (a) criterion-referenced transfer proofs (solo_transfer_pass, mixed formats), (b) mastery-climate framed (SAFE TARGET / Part XXXVIII), (c) never sold as “your recital proves you’re a genius,” (d) paired with SAFE-AAR debrief + SAFE-FILM clip on the weak bar, not curtain-call XP.

Kill: Recital cosplay / standing ovation UI as identity. **Kill:** celebrating bare pass as transformative. **Survive:** adjudicated *competence evidence* with recovery path.

LVI.8 SAFE-MUSIC stack (product law)

1. **Name the three rungs** — technique (scales) → repertoire (pieces) → proof (recital/transfer).
2. **Scales = goalful ingredient drills** with immediate feedback (Ericsson definition; SAFE-DP).
3. **Repertoire includes choice** when possible — interest upgrades strategy quality (Renwick & McPherson).
4. **Frame technique value** (identified regulation) — why this drill unlocks the piece (Evans / SDT); avoid pure controlling rewards (SAFE-REWARD).
5. **Formal practice > minutes** — instrument strategy quality; never North-Star hours (Bonneville-Roussy & Bouffard; Hallam 2012).
6. **Public signal = mixed/transfer proof** — not blocked accuracy or stage theater (Hallam 2019 wound + SAFE-TRANSFER).
7. **Pass-floor ≠ win** — design recovery for bare-pass and fail states; protect self-belief (Hallam exam outcomes).
8. **Anti-cosplay** — borrow the *ladder*, not conservatory/prodigy brand (SAFE-RITUAL / SAFE-AAR pattern).
9. **Stack with prior SAFE-*** — DP session spine, FILM review of the hard bar, AAR close, CALIB confidence tiers.

10. **Claim ladder** — music evidence stays L1 domain evidence; MindCraft ACT/identity claims need MUSIC-* experiments.

LVI.9 Competitive audit — who runs the ladder?

Competitor	Scales	Repertoire	Recital	Ladder failure
Khan	Skill practice	Weak story meaning	Unit mastery quizzes	Explanation flood; weak identity proof
Duo	Micro-drills	Limited choice depth	Leagues/streaks as “recital”	Quantity/streak liturgy (banned NS)
Brilliant	Insight puzzles	Delight-as-repertoire	Weak graded public ACT proof	Aha without adjudicated transfer
ChatGPT tutor	Skips scales	Answers as repertoire	No solo proof	Fluent wrongness; no ladder
Human music-style tutor	Often strong	Often assigned-only	Real juries	Variance; stress; syllabus boredom

Commercial implication: Position MindCraft as **skills** → **pieces** → **proof** — “drill the hinge, play the piece, show you can transfer” — never “train like Juilliard” or “practice minutes that compound.” Parent copy: quality of practice and transfer proof. Student copy: “unlock the next piece” / “show what you can play cold.”

LVI.10 Claim ladder

Claim	Max ladder without new data
Deliberate practice prototype is music training with goals, feedback, repetition	L1 (Ericsson et al., 1993)
Strategy factors shift with grade level in large youth sample; time alone insufficient framing	L1 (Hallam et al., 2012)
Practice time predicts achievement when tied to formal (SRL+DP) practice	L1 (Bonneville-Roussy & Bouffard, 2015)
Student-selected repertoire can upgrade practice strategies vs assigned pieces	L1 case study (Renwick & McPherson, 2002)
Scales often need internalized value, not pure intrinsic fun	L1 theoretical/applied SDT (Evans, 2015)
Bare-pass exam outcomes associate with weaker practice/motivation profiles	L1 observational (Hallam et al., 2019/2020)

SAFE-MUSIC improves MindCraft transfer / identity metrics vs minutes-led UX	L1 product bet — needs MUSIC experiments
“Conservatory methods raise ACT / identity” undifferentiated	Banned
Hours practiced / streak as mastery or recital	Banned

LVI.11 Experiments spawned

ID	Question	Design	Primary	Kill condition
MUSIC-1	Named ladder UX (scales→piece→proof) vs flat question feed	2-arm	completion; solo _transfer_pass; perceived progress	Flat ≥ ladder on transfer → ladder is copy theater
MUSIC-2	Choice among 2–3 repertoire sets vs assigned-only	2-arm	strategy proxies; time-on-hard-items; affect	Assigned ≥ choice on transfer <i>and</i> affect → choice optional
MUSIC-3	Technique framed with “unlocks piece X” vs controlling “do your scales”	2-arm	persistence; identified reasons; next-piece start rate	Controlling ≥ identified → SDT frame fail
MUSIC-4	Parent dashboard: formal-practice metrics vs minutes-first	2-arm	parent WTP/comprehension; student pressure talk	Minutes-first better WTP without pressure → still wound educationally; don’t ship
MUSIC-5	Recital = mixed transfer set + AAR vs score-reveal only	2-arm	self-belief; retry; week-later transfer	Reveal-only ≥ AAR on belief <i>and</i> transfer → simplify
MUSIC-QUAL	10 Maya: does “scales/pieces/recital” language feel clarifying or elitist?	interview	coded: clarifying / cosplay / pressure	Conservatory words → kill that copy

Pre-reg (XXXIV): MUSIC-* identify **ladder architecture** — not “music methods raise scores” or prodigy-identity myths.

LVI.12 Confidence table

Claim	Label	Confidence
Ericsson et al.: DP rooted in music training structure	FACT	High
Hallam et al. 2012: large-N strategy factors × grade	FACT	High
Bonneville-Roussy & Bouffard: formal practice mediates time→achievement	FACT	High
Renwick & McPherson: choice upgrades strategies (case)	FACT (small <i>N</i>)	Medium
Evans SDT: scales need internalized value	FACT / applied theory	Medium–High
Hallam et al. 2019: pass-floor weakest practice/motivation profile	FACT (observational)	Medium–High
Music ladder transfers cleanly to ACT prep for anxious teens	HYPOTHESIS / wounded	Low–Medium
SAFE-MUSIC is right default path metaphor for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium
Conservatory cosplay / hours KPI improve belonging or scores	SPECULATION / false as doctrine	Low (against)

LVI.13 What this chapter kills

1. **Kill:** Conservatory / virtuoso / prodigy branding as efficacy or belonging claim.
2. **Kill:** Minutes practiced, streaks, or hours-logged as North Star mastery (quantity without formal practice).
3. **Kill:** Scales-only forever without repertoire meaning *or* repertoire theater without technique.
4. **Kill:** Recital/stage cosplay and bare-pass celebration as identity transformation.
5. **Kill:** Equating “we practiced a lot” with exam readiness or musical/math identity.
6. **Wound:** Graded public performance can demotivate (pass-floor, fail-shame, parent pressure); transfer carefully.
7. **Survive:** SAFE-MUSIC ladder; formal-practice metrics; choiceful repertoire; identified technique value; transfer-as-recital; MUSIC-1...5; copy as **skills** → **pieces** → **proof**.

Doctrine until data: Ship **SAFE-MUSIC** — goalful scales, meaningful chosen pieces, adjudicated transfer proof — stacked on SAFE-DP / SAFE-FILM / SAFE-AAR, and never sell conservatory mythology or practice minutes as the transformation engine.

Part LVII — Chess Annotation & Metacognition: Postmortem UX Without Grandmaster Cosplay

Chapter status: Living evidence brief — Researcher tick 2026-08-01 (UTC hour 03; hour%≠0; 7 researcher entries since synthesizer v1.7 → Researcher)

Primary question: What transferable design does chess *annotation* / *postmortem* offer for MindCraft’s after-attempt metacognition — student-generated explanation of critical positions, error classification into next drills, analysis of wins as well as losses — without grandmaster cosplay, engine-eval floods, or rating-as-identity?

Owners: Product (Practice debrief / Solver review / Notes) · Coach UX · ML soft-wrong → drill loop · Brand & copy · Red Team

Commercial job: Ship a **SAFE-ANNOT** stack — annotate before reveal, self-explain critical moves, classify → next drill, analyze successes not only failures — while killing “think like a GM,” engine dump ≡ coaching, and games-played/rating as learning proof.

LVII.1 Why this chapter exists

Parts LIV–LVI borrowed *session close* (SAFE-AAR), *review media* (SAFE-FILM), and *ladder architecture* (SAFE-MUSIC) from adjacent high-performance domains. Chess is the classic cognitive-science laboratory for **move choice under uncertainty** and for **post-game annotation** as deliberate learning — the closest analogue to MindCraft’s “attempt → review → next weak bridge” loop.

Competitors already fake the chess metaphor poorly: ChatGPT tutors monologue the “best line”; puzzle apps spam tactics without annotation; Duo/Khan celebrate streaks and accuracy without a student-authored postmortem. MindCraft’s soft-wrong science (SAFE-MISCON), hide-correctness diagnostic (SAFE-CALIB), and coach self-explanation doctrine (Part XL / SAFE-SE) need a **postmortem UX law** that is evidence-shaped, not chess-club branding.

FOUNDER BELIEF under audit: Mathematical identity grows when learners *annotate their own critical attempts* — name what they saw, why they chose, what principle failed — then convert that into one next drill, not when an engine or LLM floods them with better answers.

Claims we refuse as doctrine:

1. Grandmaster / “think like a master” branding as efficacy or belonging.
 2. Engine evaluation bars / tip floods labeled as “annotation.”
 3. Games played, rating, streaks, or puzzle counts as North Star mastery.
 4. Loss-only shame review (never annotate wins / successful strategies).
 5. Novice JOLs / vague “how well did you understand?” without forced restudy of critical moves.
 6. AI postmortem as automatically closing skill gaps (self-selection wound).
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LVII.2 Constructs (product language)

Construct	Chess meaning	MindCraft analogue	Failure mode if misused
Critical position	Branch point where evaluation swings	Soft-wrong item; bridge fail; timed blunder	Annotating every move → exhaustion
Annotation / postmortem	Written/verbal account of thought at critical nodes	Student-first debrief after attempt/session	Engine dump before student speech
Candidate moves	Alternatives considered before choice	Alternative strategies / formats considered	Fake multiple-choice theater
Self-explanation	Why this move / principle applies	Chi/Renkl-style why-prompts (Part XL)	“Explain” that restates the tip
Engine / AI feedback	Stockfish-style eval after human thought	Coach card / model solution <i>after</i> annotate	Reveal-first collapses generation
Template / chunk	Pattern recognition structures in LTM	Ingredient / archetype recognition	“Memorize more patterns” vanity
SAFE-ANNOT	Postmortem doctrine that survives cosplay / flood / rating kills	See LVII.9	Chess branding; eval bars as pedagogy

Operational definition (HYPOTHESIS): A MindCraft review counts as *SAFE-ANNOT* when (a) the learner generates explanation of **1–3 critical attempts before** correctness/model reveal, (b) annotation names a **principle or misconception family** (not vibes), (c) the loop ends in a **linked next drill** (motif → ingredient set), (d) wins/successes are eligible for annotation (not only failures), (e) engine/LLM text never substitutes for student generation, (f) branding borrows the *method*, not grandmaster mythology.

LVII.3 de Groot — verbal protocols are the annotation prototype

FACT: Adriaan de Groot’s classic work (*Het denken van den Schaker*, 1946; English *Thought and Choice in Chess*, 1965/1978, Mouton / Amsterdam University Press lineage) collected **thinking-aloud protocols** from players of varying strength (including masters and world champions) who were asked to choose a move in unfamiliar tournament positions *as if in a real game*, verbalizing plans, calculations, and considerations. The object was the **choice-of-move problem**, not whole-game biography.

FACT (structure): de Groot described phases of chess thought commonly summarized as orientation → exploration → investigation → proof (see secondary summaries in the chess-programming / cognitive literature citing de Groot). Expertise showed up less as “search farther on every branch” in the original macrostructure story, and more as **better initial perception and selection of what to investigate**.

Wound (citation hygiene): Bilali■, McLeod & Gobet (2008, *Cognitive Science* line / “Sixty years of citing de Groot”) document that textbooks often **misreport** de Groot as “experts and novices search identically.” Later replications find additional search-structure differences. Product implication: do **not** market “masters don’t calculate more — they just see” as a slogan that excuses skipping hard work *or* as a claim that MindCraft’s annotation UI replicates grandmaster perception.

Product implication: Annotation UX = structured verbalization at critical nodes (what did you notice → candidates → why this choice → what would falsify it), not a vibes journal. Aligns SAFE-AAR question spine and Part XL self-explanation prompts.

LVII.4 Chunks and templates — recognition scaffolds search (not “genius”)

FACT: Chase & Simon (1973) chunking theory — experts recall meaningful game positions far better than novices; advantage collapses on random boards — grounded expertise in **pattern memory**, not mystical IQ (classic result; see also Gobet & Simon extensions).

FACT: Gobet & Simon’s **template theory** (e.g. Gobet & Simon, 1996; Gobet, 1998 *Cognition* comparison of expert-memory theories) proposes that recurring chunks evolve into richer templates with slots — unifying low-level chunks with higher-level schematic knowledge. Gobet (2005) notes deliberate practice matters because unsupervised “just play” can acquire **irrelevant** chunks that fail to improve (or even hinder) performance.

Commercial implication: MindCraft “annotation” should cue **ingredient/archetype recognition** (“which pattern is this?”) before deep calculation monologues. Kill copy that says students must “think like Kasparov.” Survive: pattern labels tied to ontology IDs after student attempt.

Kill: Genius branding. Kill: unstructured blitz-volume as chunk building. Survive: deliberate attention to the patterns that matter (SAFE-DP).

LVII.5 Self-explanation beats prediction-only for chess novices

FACT: de Bruin, Rikers & Schmidt (2007, *Contemporary Educational Psychology*, 32(2), 188–205, doi:10.1016/j.cedpsych.2006.01.001) — novices learning a King+Rook vs King endgame. Conditions: observe computer moves; predict next move; **predict + self-explain**. Self-explanation condition showed better principled understanding (predictions applying endgame principles) and more often checkmated in the test phase. Prediction alone did **not** beat observation.

HYPOTHESIS for MindCraft: Postmortem prompts must force **why/principle language**, not mere “what would you play next?” prediction clicks. This is the chess-domain replication of Chi/Renkl SE doctrine already in Part XL — use it to harden coach UX after attempts.

Kill: Silent engine scrubbing. Kill: “tap the best move” prediction theater without explanation. Survive: annotate-why before reveal.

LVII.6 Novice metacognition is fragile — JOLs can fail; forced restudy helps

FACT: de Bruin, Rikers & Schmidt (2005, *Applied Cognitive Psychology*, 19(2), 167–181, doi:10.1002/acp.1109) — novice chess endgame learning with judgments of learning (JOLs) and move selection for restudy. Forced selection of moves for restudy improved learning vs free selection (even controlling for number restudied). Providing JOLs showed

better *self-regulatory behavior* on some measures but **no or negative** performance benefit vs no-JOL — authors argue JOLs may place high ineffective load on novice working memory.

FACT: de Bruin, Rikers & Schmidt (2007, *European Journal of Cognitive Psychology* / related metacomprehension skill-acquisition paper, doi:10.1080/09541440701326204) — experienced chess players showed higher metacomprehension accuracy and self-regulation than novices when learning an endgame; novice absolute accuracy near zero. Expertise affects monitoring quality in skill domains.

Product implication: Do not ship bare confidence sliders as “metacognition” for Maya-novices (also SAFE-CALIB wound). Prefer **forced critical-move restudy** + principle annotation. Gate fancy monitoring UI by expertise / after scaffolding.

Kill: “Rate your understanding 1–5” as the whole postmortem. Survive: select 1–3 critical items → explain → restudy linked drill.

LVII.7 Analyzing games beats mere volume — and wins matter

FACT (observational, large-N): Yiannakoulis (2026, *Simulation & Gaming*, doi:10.1177/10468781261443352) — ~2M online blitz games / ~2.1k active Lichess players (1600–1800 band). Computer-assisted game analysis positively associated with rating improvement; **analyzing wins** showed a clearer association than analyzing losses in main models; total games played only weakly associated. Authors stress measurement noise (analysis flag is blunt; biases toward null) and exploratory observational status — not an RCT.

HYPOTHESIS: MindCraft should prompt annotation on **successful hard items** and efficient solutions (Dowrick-adjacent feedforward / SAFE-FILM success clips), not only on failures — “study your brilliancies” as well as soft-wrongs. Loss-only review risks ostrich/shame dynamics the paper discusses via adjacent feedback literature.

Wound: Associational evidence ≠ “annotation raises ACT.” Claim ladder stays L1 for chess rating deltas; MindCraft transfer needs ANNOT experiments.

Kill: “Play more puzzles” as improvement doctrine. Kill: failure-only shame reels. Survive: reflective analysis of critical successes and failures.

LVII.8 AI / engine feedback — self-selection and fluency wounds

FACT / high-relevance wound: Riedl & Bogert (2024/2026 update, arXiv:2409.18660) — 5+ years, ~52k individuals on an online chess platform with optional AI game analysis. Motivated and higher-skilled players **self-select** into AI feedback and use it more productively; apparent learning gains can be an **illusion of AI effectiveness** once endogenous motivation is accounted for; AI access can **widen skill gaps**; centralized AI feedback can reduce intellectual diversity (supported via platform natural experiments in the paper).

Aligns: Part XXXIII AI trust/sycophancy — fluent eval bars are not pedagogy; overtrust risk. SAFE-FILM less-is-more; SAFE-AAR facilitation not lecture.

Product implication: Optional “engine” / model solution is a **second pass** after student annotation. Do not market auto-analysis as closing equity gaps. Instrument who seeks postmortem help (selection bias).

Kill: “AI reviews every game so everyone improves equally.” Kill: reveal-first Stockfish cosplay in Solver. Survive: human-first annotation → sparse model contrast → one drill.

LVII.9 SAFE-ANNOT stack (product law)

1. **Student annotates first** — 1–3 critical attempts before correctness / coach / model reveal.
 2. **Prompt for principles** — why/candidates/what would change your mind (de Groot structure + de Bruin SE).
 3. **Forced restudy selection** for novices — don’t rely on raw JOLs (de Bruin 2005).
 4. **Classify → next drill** — motif/misconception → ingredient set (close the loop; not insight theater).
 5. **Annotate wins too** — successful strategies and hard-earned corrects (Yiannakoulis).
 6. **Sparse AI second pass** — contrast, don’t replace; watch self-selection (Riedl & Bogert).
 7. **Pattern labels after attempt** — chunks/templates as ontology tags, not genius myths (Chase/Simon; Gobet).
 8. **Anti-cosplay** — borrow postmortem method, not “grandmaster academy” brand (SAFE-RITUAL / SAFE-MUSIC pattern).
 9. **Stack** — SAFE-AAR close, SAFE-FILM clip, SAFE-MISCON soft-wrong, SAFE-SE prompts, SAFE-CALIB tiers.
 10. **Claim ladder** — chess evidence = L1 domain; MindCraft ACT/identity claims need ANNOT-* experiments.
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LVII.10 Competitive audit — who runs a real postmortem?

Competitor	Annotation?	Student generation first?	Loop to next drill?	Failure mode
Khan	Hints/explanations	Often reveal/explain-first	Weak systematic soft-wrong→drill	Explanation flood
Duo	Minimal	No	Streak/XP loop	Quantity liturgy
Brilliant	Aha rationales	Partial	Weak exam transfer proof	Insight without postmortem habit
ChatGPT tutor	Fluent postmortem monologue	Usually model-first	Rare durable drill link	Sycophancy; fluent wrongness
Chess.com / Lichess	Engine analysis UX	Optional; often engine-first	Puzzle motifs if user closes loop	Eval bars; self-selection
Human tutor	Variable verbal postmortem	Often strong	Variable	Unscalable; shame risk

Commercial implication: Position MindCraft as **annotate** → **classify** → **drill** — “replay the critical move in your own words, then train the hinge” — never “AI grandmaster reviews your session.” Parent copy: reflection quality and linked practice, not games played. Student copy: “name the mistake / name the win, then one next rep.”

LVII.11 Claim ladder

Claim	Max ladder without new data
de Groot: verbal protocols reveal structured move-choice thought	L1 (classic methods)
Chase/Simon + Gobet/Simon: expertise tied to chunks/templates, not mysticism	L1
Self-explanation + prediction > prediction/observation for chess-novice principles	L1 (de Bruin et al., 2007 CEP)
Forced restudy selection helps novices; JOLs can fail/overload	L1 (de Bruin et al., 2005 ACP)
Metacomprehension accuracy rises with chess expertise in skill learning	L1 (de Bruin et al., 2007 EJCP)
Game analysis associated with rating gains; play volume weak; wins analysis salient	L1 observational (Yiannakoulis, 2026)
AI feedback gains partly selection; can widen gaps / reduce diversity	L1 observational + natural experiments (Riedl & Bogert)
SAFE-ANNOT improves MindCraft transfer / identity vs reveal-first UX	L1 product bet — needs ANNOT experiments
“Grandmaster methods” / engine dumps raise ACT undifferentiated	Banned
Rating / streak / games-played as mastery North Star	Banned

LVII.12 Experiments spawned

ID	Question	Design	Primary	Kill condition
ANNOT-1	Annotate-before-reveal vs reveal-first coach	2-arm	retry_120s; principle-coded annotation quality; week-later transfer	Reveal-first $\geq$ annotate on transfer $\rightarrow$ annotation is friction
ANNOT-2	SE-why prompts vs prediction-only taps	2-arm	principled language rate; next-drill completion	Prediction $\geq$ SE on transfer $\rightarrow$ simplify
ANNOT-3	Forced 2 critical restudies vs free JOL-only	2-arm	restudy adherence; transfer	JOL-only $\geq$ forced on transfer <i>and</i> affect $\rightarrow$ revisit

ANNOT-4	Annotate wins+losses vs losses-only	2-arm	return rate; self-belief; strategy reuse	Losses-only $\geq$ mixed on learning without shame cost $\rightarrow$ still prefer mixed ethically
ANNOT-5	AI second-pass on vs off after annotation	2-arm	overtrust incidents; transfer; time-to-drill	AI-on hurts calibration or transfer $\rightarrow$ default off for novices
ANNOT-QUAL	10 Maya: does “postmortem / annotate” language feel clarifying or chess-elitist?	interview	coded: clarifying / cosplay / shame	Chess words $\rightarrow$ plain “replay the hard step”

Pre-reg (XXXIV): ANNOT-* identify **postmortem architecture** — not “chess training raises ACT” or grandmaster-identity myths.

LVII.13 Confidence table

Claim	Label	Confidence
de Groot protocols established structured verbal move-choice analysis	FACT	High
Chunk/template accounts dominate expert chess memory explanations	FACT	High
de Bruin 2007: SE helps chess novices discover principles	FACT	High
de Bruin 2005: forced restudy > free; JOLs shaky for novices	FACT	High
Expertise improves metacomprehension in chess skill learning	FACT	Medium–High
Yiannakoulias: analysis $\leftrightarrow$ rating; play volume weak (observational)	FACT (assoc.)	Medium
Riedl & Bogert: AI feedback self-selection / gap / diversity wounds	FACT (platform)	Medium–High

Chess postmortem transfers cleanly to anxious ACT teens' identity	HYPOTHESIS / wounded	Low–Medium
SAFE-ANNOT is right default review metaphor for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium
Grandmaster cosplay / engine-first dumps improve belonging or scores	SPECULATION / false as doctrine	Low (against)

LVII.14 What this chapter kills

1. **Kill:** Grandmaster / “think like a master” branding as efficacy or belonging.
2. **Kill:** Engine-eval floods and LLM monologues labeled as annotation or coaching.
3. **Kill:** Games played, rating, streaks, or puzzle counts as North Star mastery.
4. **Kill:** Loss-only shame review; never studying successful critical decisions.
5. **Kill:** Bare JOLs / confidence sliders as sufficient metacognition for novices.
6. **Kill:** “AI analyzes every attempt so gaps close equally” (selection + diversity wound).
7. **Wound:** Textbook oversimplification of de Groot search findings; don’t sloganize perception.
8. **Survive:** SAFE-ANNOT; annotate-before-reveal; SE principles; forced critical restudy; classify→drill; annotate wins; sparse AI second pass; ANNOT-1...5.

Doctrine until data: Ship **SAFE-ANNOT** — student-generated postmortem on critical attempts, principle-tagged, linked to the next drill, stacked on SAFE-AAR / SAFE-FILM / SAFE-SE / SAFE-MISCON — and never sell chess mythology, engine bars, or volume metrics as the transformation engine.

Part LVIII — Therapy: Graded Exposure (Math Anxiety Ladders)

Chapter status: Living evidence brief — Researcher tick 2026-08-01

Primary question: How should MindCraft borrow *graded exposure* / *inhibitory learning* from anxiety treatment — without therapy cosplay, flooding, or “calm until zero” theater — so threatened learners approach math instead of avoiding it?

Owners: Affect / Anxiety · Learning Science · Product (Practice sequencing + stakes) · Growth / Positioning · Red Team

Commercial job: Turn math-anxiety avoidance into a **designed approach ladder** that parents can trust and students can finish — then market approach + expectancy violation, not “we cured anxiety” or day-one ACT sims as exposure therapy.

LVIII.1 Why this chapter exists

Parts XXIV (affect/anxiety), XLI (SAFE-DD), XLV (SAFE-RESILIENCE), and L (SAFE-CALIB) already say: anxiety taxes working memory, unbuffered desirable difficulty becomes threat, grit posters fail, and “raise confidence” is not a North Star. What they do **not** yet specify is the **approach schedule** when the threat *is the math itself* — timed items, word problems, graphs, exams, public board energy.

Competitors fake exposure poorly: Khan dumps harder items after a wrong answer; Duo streaks punish skip; Brilliant opens with delightful puzzles that never feel like the threat Maya avoids; ChatGPT tutors soothe (“you’ve got this”) while the student never approaches the feared format. Therapy apps and school worksheets sometimes cosplay CBT with breathing gifs and no expectancy violation.

FOUNDER BELIEF under audit: Identity transformation for high-math-anxiety (HMA) Maya requires **approaching feared math situations under destaked conditions until threat expectancies are violated** — not forever-easy fluency, not flooding, and not clinical cosplay.

Claims we refuse as doctrine:

1. Day-one full ACT / mixed timed exams labeled “exposure therapy.”
2. Habituation-only success (“stay until anxiety = 0”).
3. Clinical CBT / therapist branding without clinician oversight.
4. Permanent safety behaviors (hint binge, skip, parent rescue) as “support.”
5. Strict linear hierarchy forever (never vary order/context).
6. Anxiety meters / “Calm Score™” as North Star.
7. Exposure → ACT +2 σ / Bloom marketing.

LVIII.2 Constructs (product language)

Construct	Therapy meaning	MindCraft analogue	Failure mode if misused
Fear hierarchy	Ranked approach steps from least → most feared	Format/stakes ladder (untimed soft → timed mix)	Hierarchy as forever easy-mode
Graded exposure	Approach feared CS while learning safety	Planned approach to avoided item types	Random harder = fake exposure
Expectancy violation	Feared outcome does not occur (or is tolerable)	Soft-wrong + survive + retry without catastrophe	Calm breathing without approach
Safety behavior	Acts that prevent learning (escape, ritual)	Instant full reveal; skip; streak protect	“Support” that blocks learning
Inhibitory learning	New CS—no US memory competes with fear memory	Multiple contexts/formats after approach	One safe session → exam panic
SAFE-EXPOSE	Exposure doctrine that survives flooding / cosplay / meter kills	See LVIII.9	Therapy branding; flood sims

Operational definition (HYPOTHESIS): A MindCraft session counts as *SAFE-EXPOSE* when (a) the learner **approaches** a previously avoided or high-SUDS math situation, (b) stakes are **destaked** enough that WM can engage (SAFE-DD), (c) a named **threat expectancy** is checked (“I’ll freeze / blank / be stupid”), (d) outcome **violates or revises** that expectancy without catastrophe framing, (e) **safety behaviors are faded** (hints → cards → solo), (f) later practice **varies context/format** so learning is not locked to one cozy UI, (g) product never claims to *treat* clinical anxiety disorders.

LVIII.3 Why avoidance is the commercial enemy

FACT (construct + avoidance): Hembree (1990, *Journal for Research in Mathematics Education*, 21(1), 33–46, doi:10.2307/749455 / 10.5951/jresmetheduc.21.1.0033) — meta-analysis of 151 studies: mathematics anxiety relates to poorer achievement, poorer attitudes, and is **bound directly to avoidance** of math; valid treatments that reduce anxiety tend to accompany improved performance.

FACT (WM disruption): Ashcraft & Kirk (2001, *JEP: General*, doi:10.1037/0096-3445.130.2.224) — HMA individuals show reduced WM spans under math demand; anxiety acts as a **transitory dual-task load**, amplifying errors especially on carry/complex items (see also Ashcraft & Krause, 2007, *Psychonomic Bulletin & Review*).

Commercial implication: The product problem is not “Maya needs nicer explanations.” It is **avoidance of the situations that would create competence evidence**. Marketing that promises comfort without approach sells temporary relief and loses the identity loop (FEI). Marketing that floods her with timed mixed ACT on day one sells rigor theater and churn.

Kill: “More content solves anxiety.” Kill: “Just make it fun.” Survive: designed approach to avoided formats under recoverable stakes.

LVIII.4 Classical graded exposure and math desensitization lineage

FACT (lineage): Systematic desensitization (Wolpe tradition) pairs gradual hierarchy ascent with incompatible responses (often relaxation). Education/counseling literature adapted this to math anxiety via hierarchy construction + gradual approach (see secondary reviews; e.g. IntechOpen synthesis citing Wolpe 1958 and school applications — cite cautiously as review, not primary RCT).

FACT (math-anxiety treatment comparison): Zettle (2003, *The Psychological Record*, 53, 197–215, doi:10.1007/BF03395440) — Acceptance and Commitment Therapy (ACT) vs systematic desensitization for mathematics anxiety; both approaches target anxiety-related avoidance, with different mechanisms (acceptance/values vs desensitization). Product takeaway: **approach + new relationship to anxious thoughts** matters more than branding one school of therapy.

FACT (case / protocol shape): Open-access childhood MA CBT case work (e.g. Coping Cat–inspired hierarchy + graded exposure; PMC8283868) shows the clinical shape — least→most feared math situations. **Wound:** N=1 ≠ product efficacy claim.

HYPOTHESIS for MindCraft: Borrow *hierarchy + graded approach + destake*; do **not** ship therapist roleplay or “we treat anxiety disorders.”

Kill: Clinical cosplay. Survive: hierarchy of *math situations* (formats, timing, stakes), not feelings posters.

LVIII.5 Inhibitory learning upgrades habituation theater

FACT (mechanism shift): Craske, Treanor, Conway, Zbozinek & Vervliet (2014, *Behaviour Research and Therapy*, 58, 10–23, doi:10.1016/j.brat.2014.04.006; PMC4114726) — exposure optimized via **inhibitory learning**: goal is new inhibitory CS–no US associations and expectancy violation, **not** necessarily within-session fear reduction to zero. Strategies include expectancy violation, removal of safety signals, variability, multiple contexts, retrieval cues, affect labeling (among others).

FACT (clinical overview): Craske inhibitory-regulation line (doi:10.1159/000381574) — anxious individuals often show inhibitory learning/regulation deficits; strengthen inhibitory retrieval, not just “wait until calm.” Variability-in-exposure reviews (e.g. PMC6884337) argue varying hierarchy order/intensity can aid expectancy violation and cross-context retrieval — school-product transfer remains empirical.

Product implication: Success metric is **approach + violated expectancy + later approach in new context** (challenge_accept, retry_120s, format hops), **not** “anxiety slider hit zero.” Soft-wrong + survive is expectancy violation done right. Breathing-only without approach is cosplay.

Kill: Habituation-as-North-Star. Kill: “Stay in the hard problem until you feel calm.” Survive: name the feared outcome → approach → check what happened → vary next.

LVIII.6 Extinction is fragile — renewal kills one-session miracles

FACT: Bouton (2002, *Biological Psychiatry*, 52(10), 976–986, doi:10.1016/S0006-3223(02)01546-9; PubMed 12437938) — extinction does **not** erase the original fear association; it creates new, **context-dependent** learning. Return of fear via **renewal** (ABA / ABC patterns) is expected when context changes after extinction.

FACT (learning & memory bridge): Bouton (2004, *Learning & Memory*, 11, 485–494) — contexts set the occasion for CS–US vs CS–no US; mere removal from the extinction context can renew responding.

Commercial implication: One cozy MindCraft session where Maya survived word problems does **not** license “anxiety cured” copy — exam hall / parental watching / timed ACT is a **new context**. Product must plan **multi-context approach** (home app → timed → mixed → low-stakes quiz climate) and retrieve inhibitory learning with cues (SAFE-AAR “what was I afraid would happen?”).

Kill: One-and-done exposure marketing. Survive: spaced approach across formats/stakes (ties SAFE-TIMING / SAFE-TRANSFER).

LVIII.7 Brief cognitive tools that are *not* exposure — and must not replace it

FACT: Park, Ramirez & Beilock (2014, *Journal of Experimental Psychology: Applied*, doi:10.1037/xap0000013) — short **expressive writing** before a math exam reduced the HMA–LMA performance gap (N=80); among HMAs,

anxiety/cause/insight language related positively to math performance. Related classroom writing finding: Ramirez & Beilock (2011, *Science*, doi:10.1126/science.1199427).

HYPOTHESIS: Writing / affect labeling can free WM *before* approach (Craske lists affect labeling among optimizers) — but writing **without** subsequent approach is incomplete for avoidance. Product: optional 60–90s worry-dump *then* hierarchy step — not a journaling app substitute for practice.

Kill: Mindfulness theater as the whole product. Survive: brief unload → approach → expectancy check.

LVIII.8 Crosswalk to SAFE-DD / FEI / soft-wrong

Exposure principle	MindCraft already-adjacent law	Product move
Equip before strain	SAFE-DD equip	Hierarchy starts below panic; cards available then faded
Destake while approaching	Soft-wrong; mastery climate	No public rank; no streak-as-worth during expose steps
Expectancy violation	FEI fear→evidence	Prompt: “What did you expect?” → “What happened?”
Remove safety signals	Hint fade / solo_transfer	Cap hint depth; require attempt before full reveal
Variability / multi-context	SAFE-TRANSFER / VT	After approach succeeds, hop format/timing
Do not flood	SAFE-RESILIENCE growth-zone	Ban day-one full timed mix for HMA segment

FOUNDER BELIEF: SAFE-EXPOSE is the **affective schedule** for SAFE-DD — difficulty dosing without approach design is incomplete for Maya who never opens the hard door.

LVIII.9 SAFE-EXPOSE stack (surviving product rules)

1. **Build a situation hierarchy, not a vibes hierarchy.** Steps = concrete math situations (untimed symbolic → word → diagram → timed soft → short mix). Prefer student-ranked SUDS-lite once, then product defaults.
2. **Approach before polish.** The session’s job is approach + expectancy check, not explanation volume.
3. **Destake the approach.** Soft-wrong, mastery climate, no league, no streak identity (SAFE-COMPARE / HABIT / REWARD).
4. **Name expectancy → violate → label.** Short prompt triad; optional expressive writing only as WM unload.
5. **Fade safety behaviors.** Hints → structured cards → annotate (SAFE-ANNOT) → solo transfer. Permanent hint binge = failed exposure.
6. **Vary after success.** Do not lock learning to one cozy context (Bouton renewal).
7. **Measure approach, not calm meters.** Primary: challenge_accept, retry_120s, avoided-format approach rate, later transfer_pass. Anxiety self-report is covariate, not North Star.

8. **Anti-cosplay.** No “therapist mode,” no diagnosis, no clinical claims. Plain language: “practice the scary step under safe rules.”

LVIII.10 Competitive audit — who actually runs exposure?

Competitor	Hierarchy?	Destaked approach?	Expectancy check?	Failure mode
Khan	Weak (skill tree ≠ fear ladder)	Partial	Rare	Harder after miss ≈ flood/shame
Duo	No	No (streak threat)	No	Avoidance via skip; streak colonization
Brilliant	Puzzle delight ≠ feared exam format	Low stakes yes	Rare	Fun ≠ approach to avoided threat
ChatGPT tutor	Ad hoc	Emotional soothe	Often sycophantic reassurance	Soothe without approach; fluent wrongness
Human tutor	Often strong	Variable	Variable	Unscalable; trait-label risk (SAFE-EXPECTANCY)

Commercial implication: Position as **approach ladders for feared math situations** — scary step under safe rules, then prove in a new format — never “AI therapy” and never day-one elite timed drills.

LVIII.11 Claim ladder

Claim	Max ladder without new data
Math anxiety tied to avoidance; treatments that work often improve performance	L1 (Hembree, 1990)
Anxiety disrupts WM during math	L1 (Ashcraft & Kirk, 2001)
Exposure benefits from inhibitory learning / expectancy violation > calm-to-zero	L1 clinical translation (Craske et al., 2014+)
Extinction is context-bound; renewal expected	L1 (Bouton, 2002/2004)
Expressive writing can shrink HMA performance gap pre-test	L1 (Park/Ramirez/Beilock)
Zettle: SD and ACT both target math-anxiety avoidance pathways	L1 clinical trial lineage

SAFE-EXPOSE raises MindCraft challenge_accept / transfer for HMA segment	L1 product bet — needs EXP-O experiments
Flooding / day-one ACT sims / Calm Score™ / therapy cosplay	Banned as doctrine

LVIII.12 Experiments spawned

ID	Question	Design	Primary	Kill condition
EXP-O1	Hierarchy+destake vs random harder after miss	2-arm HMA segment	challenge_acce pt; exit; week transfer	Random $\geq$ hierarchy on transfer without churn cost $\rightarrow$ revisit
EXP-O2	Expectancy-viola tion prompts on vs off	2-arm	expectancy revision rate; retry_120s	Prompts add friction without learning $\rightarrow$ shorten/kill
EXP-O3	Safety-behavior fade vs permanent hints	2-arm	solo_transfer_ pass; hint depth	Permanent hints $\geq$ fade on transfer $\rightarrow$ still prefer fade ethically; redesign fade
EXP-O4	Single-context approach vs multi-format renewal plan	2-arm	format-hop pass after delay	Single $\geq$ multi $\rightarrow$ still keep multi for exam ecology
EXP-O5	Worry-write then approach vs approach-only	2-arm HMA	accuracy under load; completion	Write hurts or delays without gain $\rightarrow$ optional/off
EXP-O-QUAL	10 Maya: “exposure ladder” language vs “practice the scary step”	interview	coded: clarifying / clinical / shame	Clinical words $\rightarrow$ plain language only

Pre-reg (XXXIV): EXP-O* identify **approach architecture for anxiety** — not “we treat anxiety disorders” or ACT score guarantees.

LVIII.13 Confidence table

Claim	Label	Confidence
Anxiety–avoidance–performance cluster is real at population level	FACT	High
WM dual-task account of online math-anxiety cost	FACT	High
Inhibitory learning reframes exposure away from habituation-only	FACT (clinical science)	High
Bouton renewal / context-bound extinction	FACT	High
Expressive writing helps some HMA test performance	FACT	Medium–High
Case/protocol SD for child MA generalizes to ACT teens in apps	HYPOTHESIS / wounded	Low–Medium
SAFE-EXPOSE is right default for MindCraft HMA segment	FOUNDER BELIEF / HYPOTHESIS	Medium
Therapy branding / flooding / calm meters improve belonging or scores	SPECULATION / false as doctrine	Low (against)

LVIII.14 What this chapter kills

1. **Kill:** Day-one full timed mixed ACT / flooding labeled as exposure therapy.
2. **Kill:** Habituation-only success (“anxiety must hit zero”).
3. **Kill:** Clinical CBT / therapist / diagnosis cosplay in product or ads.
4. **Kill:** Permanent hint binge, skip, or parent-rescue as “support.”
5. **Kill:** Anxiety meters / Calm Score™ / “we cured math anxiety” as North Star.
6. **Kill:** One cozy session ≡ exam-ready (renewal wound).
7. **Kill:** Bloom 2-sigma / exposure→score miracle marketing.
8. **Wound:** N=1 clinical cases and review chapters as efficacy proof.
9. **Survive:** SAFE-EXPOSE; hierarchy of situations; destaked approach; expectancy violation; fade safety behaviors; multi-context renewal plan; EXP-O1...5.

Doctrine until data: Ship **SAFE-EXPOSE** — graded approach to feared math situations under recoverable stakes, with expectancy checks and faded safety behaviors, stacked on SAFE-DD / FEI / SAFE-CALIB — and never sell therapy cosplay, flooding, or calm meters as the transformation engine.

Part LIX — Parent WTP Experiments Design

Chapter status: Living evidence + experiment-design brief — Researcher tick 2026-08-01

Primary question: How should MindCraft *measure* what parents will pay for — and which message frames move purchase intent — without confusing stated preference for cash, or selling score theater / vibe identity?

Owners: Growth / Positioning · Behavioral Econ · Product (parent surfaces) · Red Team

Commercial job: Ship a **pre-registered WTP / conjoint / message stack** that tells founders which attributes to build and which claims to kill — before pricing pages and parent dashboards ossify around the wrong North Star.

LIX.1 Why this chapter exists

Part XXVII mapped the post-AI WTP stack (answers → explanations → diagnosis → affect → accountability → trusted outcome signal) and spawned PAR-1 / COMP-1. Part XXXII flagged parent anxiety transmission; Part XXXVII’s EVT-4 already asked competence-evidence vs minutes vs AI-unlimited. What the Constitution still lacked is the **methods contract**: how to run conjoint / discrete-choice and message tests so results can raise claim-ladder level (Part XXXIV L4 market causal) instead of producing inspirational survey quotes.

FOUNDER BELIEF under audit: Families will pay for *trusted diagnosis + transfer evidence + human accountability* more than for unlimited AI chat — **if** we prove it with trade-off designs, not founder intuition.

Claims we refuse as doctrine:

- 1. “Parents only buy score spikes” as settled fact without trade-off data.
- 2. “Identity transformation” as a purchasable SKU without FEI evidence parents can parse.
- 3. Likert “how much would you pay?” as serious WTP.
- 4. Stated WTP ≡ revenue (hypothetical bias ignored).
- 5. Streak / minutes / “AI unlimited” as the parent value prop by default.
- 6. One national average price as the pricing strategy.

LIX.2 Constructs (product language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
WTP	Max price at which a package is chosen over alternatives / outside option	Monthly plan parents select in CBC	Treat survey WTP as sticker price
Attribute	Package feature with levels	Diagnosis depth; tutor hours; report type; price	Too many attributes → noise
CBC / DCE	Choice among multi-attribute profiles	Parent chooses Plan A vs B vs neither	Rank-all-features surveys
Part-worth	Utility of a level	Relative value of FEI report vs score-only	Over-interpret small segments

Message test	Framing → intent / comprehension	Landing / email / dashboard copy	Confuse click with retention
SAFE-WTP	Experiment doctrine that survives vibe pricing	See LIX.10	Sell identity without transfer proof

Operational definition (HYPOTHESIS): A MindCraft parent-pricing study counts as *SAFE-WTP* when (a) it uses **choice-based** trade-offs (or incentive-aligned follow-ups), not open “name a price,” (b) attributes map to **shippable product surfaces**, (c) a **price** or clear outside option is present, (d) **hypothetical-bias mitigations** are pre-registered, (e) primary outcomes separate **purchase intent**, **comprehension of child’s bottleneck**, and **homework-conflict risk**, (f) results update packaging — not vanity decks.

LIX.3 Market facts — demand exists; composition is uneven

FACT (US tutoring industry geography): Kim, Goodman & West (2024, CESifo Working Paper No. 11251 / IZA DP 17178) — U.S. private tutoring *centers* more than tripled ~1997–2022 (~3,000 → ~10,000); locations concentrate in high-income, high-parent-education areas; >half in top income quintile; residual association with Asian American population share conditional on income/education. School-market structure (private/charter/magnet) is only marginally predictive.

FACT (household spend snapshot, cited therein): Using Consumer Expenditure Survey calculations, authors report ~6–7% of U.S. families with children 6–17 paid for tutoring in the past year (2022), averaging about **\$437 in months with a purchase**; top 10% / 5% of monthly spend near **\$1,200 / \$2,000**. (Treat as CES-derived descriptive, not MindCraft price ceiling.)

FACT (shadow-education framing): Bray (2010) and Bray & Lykins (2012, ADB) document private supplementary tutoring (“shadow education”) as a global, fee-based parallel to schooling — useful for competitive context, not a US price guide.

Commercial implication: There is a real paid market adjacent to MindCraft — but it is **skewed**. Pricing and positioning that assume median-America “all parents pay \$X for identity” will miss both the premium competitive segment and equity constraints. Segment conjoint (income × anxiety × exam stakes) is mandatory.

Kill: One national WTP number as strategy. Survive: segment-aware CBC + equity-aware packaging.

LIX.4 Why Likert WTP fails — use stated choice

FACT (method foundation): Louviere, Hensher & Swait (2000, *Stated Choice Methods*, Cambridge) — stated preference via controlled choice experiments recovers relative attribute utilities under random-utility assumptions; preferable to unstructured preference talk when products are multi-attribute.

FACT (RUM lineage): McFadden (1974) random-utility discrete choice — choices reveal trade-offs; with a monetary attribute, researchers can express part-worths as WTP for levels.

FACT (tutoring-adjacent DCE): Gambi & De Witte (2024, *Higher Education*, doi:10.1007/s10734-024-01358-z) — DCE + information experiment on HE students’ tutoring preferences/consumption; high stated WTP coexists with low private uptake; SES demand vs ability-to-pay tension; behavioral frictions (status quo, image) matter. **Wound for K-12 parents:** student HE sample ≠ Maya’s parent — method transfers, coefficients do not. Broader HE DCE literature (UK/EU/Polish university-attribute studies building on Louviere et al.) shows the same tool recovers monetary trade-offs for staff expertise, flexibility, and mode.

Product implication: Stop asking “Would you pay for an identity-transforming math world?” Ask: *Between these two monthly packages, which do you choose — or neither?*

LIX.5 Hypothetical bias — stated ≠ paid

FACT: Murphy, Allen, Stevens & Weatherhead (2005, *Environmental and Resource Economics*, 30(3), 313–325) — meta-analysis of hypothetical vs actual WTP; median hypothetical/actual ratio ~1.35 with severe positive skew; choice-based elicitation associated with lower bias than some contingent-valuation formats (model-sensitive).

FACT (mitigation lineage): Cummings & Taylor (1999, *American Economic Review*) — “cheap talk” scripts can reduce hypothetical overstatement (effects heterogeneous later). Schmidt & Bijmolt (2020, *JAMS*, doi:10.1007/s11747-019-00666-6) — consumer-goods meta: prefer indirect/choice over direct “what would you pay?”; incentive-aligned mechanisms help when feasible.

HYPOTHESIS for MindCraft: Parent CBC will still inflate absolute dollars; **rank-order of attributes** (FEI report vs streak dashboard vs AI chat hours vs human tutor minutes) is the decision-useful output. Validate top packages with a small **deposit / waitlist / paid pilot** before national pricing.

Kill: Survey WTP as P&L input. Survive: relative part-worths → packaging; cash tests → prices.

LIX.6 Attribute set for MindCraft parent CBC (shippable)

Design rule (FOUNDER BELIEF): Every attribute level must map to a screen or ops cost we can actually ship in 90 days.

Attribute	Levels (draft)	Ties to doctrine
Diagnosis depth	Vague “needs practice” / concept weakness list / bridge+format gap map	Ontology advantage; XXVII trust
Evidence report	Score/minutes only / FEI pack (retry_120s, challenge-seeking, transfer) / FEI + honest plateau note	PAR-1; anti-streak
Human accountability	None / async tutor note weekly / 2× live 25-min / 4× live	Scarcity layer

AI help policy	Unlimited chat / guarded pedagogy wrap / attempt-first then cards	XXXIII; SE constraints
Practice architecture	Streak gamification / mastery climate soft-wrong / SAFE-EXPOSE ladder language	XLII–XLIV; LVIII
Price	e.g. \$49 / \$99 / \$179 / \$299 per month (calibrate to segment)	Must include realistic span
Outside option	“Neither — keep free Khan/GPT/status quo”	Critical for WTP

HYPOTHESIS: Diagnosis depth + FEI report + modest human accountability will dominate unlimited AI and streak theater **in the anxious-middle segment**; score-chaser segment may overweight timed-test sims — measure, don’t assume.

Red Team: If FEI report wins stated choice but parents cannot *explain* the child’s bottleneck after viewing it, comprehension failed — packaging theater.

LIX.7 Message tests are not conjoint (run both)

Conjoint answers *what package*; messages answer *which story opens the wallet without raising homework conflict*.

Pre-registered message arms (builds EVT-4 / COMP-1):

Arm	Parent-facing frame	Risk
M1 Score relief	“Raise ACT math”	Overclaim; retention crash if flat
M2 Homework peace	“End nightly fights”	Soothing without competence
M3 Diagnosis trust	“See the exact bridge that’s stuck”	Jargon; need plain language
M4 Identity / confidence	“Become someone who doesn’t freeze”	Belief Score™ / raise-confidence kill (v1.7–v1.8)
M5 Hybrid	Diagnosis + transfer check + honest plateau	Longer copy; may win retention

Primary outcomes: click → trial start; 14-day activation; parent comprehension quiz (name bottleneck); reported homework conflict; **not** vanity “brand love.”

Kill: Identity-only ads that cannot point to a measurable FEI metric. Survive: hybrid that sells method, shows evidence.

LIX.8 Qualitative gate before fielding CBC

FACT / process: Part XXVII and QUAL-P (XXXII) already require ~10 parent interviews. CBC attributes invented in a founder Slack will mis-specify levels.

Protocol (HYPOTHESIS): Semi-structured interviews → code “what I thought I bought / what would make me cancel / words I trust / words that feel scammy” → freeze attribute glossary → then CBC. Do not reverse the order.

Segment quotas (SPECULATION until interviews): (1) HMA-Maya parents (anxiety/avoidance), (2) score-chaser / selective-college, (3) homework-peace exhausted, (4) price-sensitive remedial. Run latent-class or split models; do not average them into one “parent.”

LIX.9 Equity and pricing ethics

FACT (concentration): Kim et al. (2024) — private center supply concentrates where income/education are high. Packaging only for that segment reproduces shadow-education inequality.

FOUNDER BELIEF: Charging for diagnosis+accountability is legitimate if transfer evidence is real (XXVII.7). Illegitimate if the paid tier is streak cosmetics + GPT wrap.

Guardrails:

- Paid claims ↔ measured FEI / learning metrics.
- No “guaranteed score jump” in any CBC level or ad arm.
- Consider a **honest free tier** that still shows diagnosis (trust acquisition) vs freemium that hides the bottleneck (dark pattern).
- Report segment WTP separately; do not market top-quintile WTP as universal.

LIX.10 SAFE-WTP stack (surviving product rules)

1. **Choice over chatter.** CBC/DCE or incentive-aligned choice; ban standalone “name your price.”
2. **Ship-mapped attributes only.** No fictional features.
3. **Always include price + neither.** Otherwise WTP is fantasy.
4. **Bias protocol.** Cheap talk / certainty / consequentiality / small cash validation — pre-register which.
5. **Three outcomes.** Intent ≠ comprehension ≠ conflict. Optimize packaging on all three.
6. **Segment, don’t average.** Latent class or pre-defined segments.
7. **Messages ≠ packages.** Run COMP/EVT message tests alongside CBC.
8. **Anti-theater.** FEI report that parents cannot interpret loses even if chosen.
9. **Anti-overclaim.** No 2σ, no guaranteed ACT points, no “AI tutor replaces human” as paid promise.
10. **Update doctrine with data.** Part-worths that kill founder favorites get logged as kills.

LIX.11 Claim ladder

Claim	Max ladder without new data
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US private tutoring centers grew sharply; high-income concentration	L1 (Kim/Goodman/West 2024)
Shadow education is a real global fee market	L1 (Bray line)
Stated choice recovers relative utilities better than unstructured WTP talk	L1 (Louviere/Hensher/Swait; McFadden)
Hypothetical WTP inflates; choice formats help but do not erase bias	L1 (Murphy et al. 2005; Cummings & Taylor)
HE tutoring DCE: high WTP, low uptake, SES frictions	L1 adjacent (Gambi & De Witte 2024) — transfer wound to K-12 parents
MindCraft parents prefer FEI+diagnosis over unlimited AI / streaks	L0–L1 product bet — needs WTP-* experiments
Survey dollars = revenue forecast	Banned

LIX.12 Experiments spawned

ID	Question	Design	Primary	Kill condition
WTP-1	FEI report vs score/minutes vs streak dashboard in CBC	CBC, price + neither	part-worths; segment classes	Streak/minutes dominate FEI <i>and</i> later retention → revisit packaging (still educational wound)
WTP-2	Diagnosis depth (vague / concept / bridge+format) × price	CBC	WTP for bridge+format; comprehension	Depth raises intent but kills comprehension → simplify language
WTP-3	Human minutes vs AI-unlimited vs guarded wrap	CBC	part-worth human; churn proxy intent	AI-unlimited wins on intent <i>and</i> 30-day solo transfer → thesis wounded
WTP-4	Message arms M1–M5	RCT landing/email	trial start; conflict; comprehension	Identity arm wins clicks + raises conflict without retention → kill identity-only
WTP-5	Cheap-talk / certainty calibration on vs off	2× embedded in CBC	calibration gap	No gap → still keep cash pilot

WTP-6	Deposit / paid pilot of top CBC winner vs #2	Field	pay-through; 30-day retention	Stated winner loses cash → trust revealed preference
WTP-QUAL	10 parents: cancel triggers + trusted words	Interview	codebook → attribute freeze	Founder glossary ≠ parent glossary → rewrite levels

Pre-reg (XXXIV): WTP-* identify **parent package preference and message effects** — not “we proved identity sells” or national price optimality from N=200 survey.

Ties: EVT-4, PAR-1, COMP-1, MUSIC-4, TIM-4 become arms or covariates inside this family — do not fork duplicate surveys.

LIX.13 Confidence table

Claim	Label	Confidence
Paid tutoring demand exists and is income-skewed in the US center market	FACT	High
CES spend snapshots describe <i>purchasers</i> , not all parents' WTP	FACT	High
CBC/DCE is the right class of tool for multi-attribute tutoring packages	FACT (methods)	High
Absolute stated dollars will be biased upward	FACT / HYPOTHESIS	High
FEI+diagnosis beats unlimited AI for MindCraft's anxious-middle segment	HYPOTHESIS	Medium
Hybrid message (diagnosis + transfer + honest plateau) wins retention vs score-only	HYPOTHESIS	Medium
Identity-only parent ads are net-positive	SPECULATION / against doctrine until data	Low (against)
One price point fits all segments	SPECULATION / false as strategy	Low (against)

LIX.14 What this chapter kills

1. **Kill:** Likert / open-ended “how much would you pay?” as pricing research.

2. **Kill:** Treating survey WTP as revenue or national sticker price.
3. **Kill:** “Parents only buy scores” or “parents buy identity” as settled without CBC.
4. **Kill:** Streak / minutes / AI-unlimited as default parent value props without losing on part-worths.
5. **Kill:** Guaranteed ACT points / 2σ / Bloom in any package level or ad.
6. **Kill:** Averaging anxious-middle and score-chaser into one “parent persona.”
7. **Wound:** Importing HE or non-US tutoring conjoint coefficients as US ACT-parent truth.
8. **Survive:** SAFE-WTP; CBC with price+neither; ship-mapped attributes; bias protocol; message≠package; WTP-1...6 + QUAL; cash validation before scale.

Doctrine until data: Run **SAFE-WTP** — choice-based parent trade-offs on diagnosis, FEI evidence, human accountability, AI policy, and price — then package what wins *and* what parents can explain, not what flatters the identity thesis.

Part LX — Trust After AI Hallucination

Chapter status: Living evidence + repair-protocol brief — Researcher tick 2026-08-01

Primary question: After MindCraft’s AI is *caught* being fluently wrong (or quietly wrong and later exposed), how do we repair *calibrated* trust — and when must we escalate to a human — without apology theater, never-wrong marketing, or post-error abandonment?

Owners: AI / Product (Solver · coach) · Learning Science · Human Factors · Parent trust · Red Team

Commercial job: Ship a **SAFE-REPAIR** doctrine that turns inevitable model errors into a *positioning asset* (honest plateau + human accountability) rather than a churn event or a “AI that never fails” lie.

LX.1 Why this chapter exists

Part XXXIII owned *prevention and calibration*: pedagogy wrap, anti-sycophancy, Bastani unguarded-harm FACT, Lee & See appropriate reliance. Part L (SAFE-CALIB) owned *student* confidence after misses. What the Constitution still lacked is the **post-violation playbook**: what the product does in the *minutes after* a hallucination is detected — by checker, by student flag, by tutor review, or by exam-collapse discovery.

FOUNDER BELIEF under audit: Families and students will tolerate imperfect AI *if* the system (a) admits competence failure fast, (b) shows a concrete repair path tied to the Map/ontology, and (c) escalates to a human when stakes or repeated failure demand it — and will *punish* products that either hide errors or drown users in soft apologies without correction.

Claims we refuse as doctrine:

1. “Our AI doesn’t hallucinate” as marketing (false category; Bastani notes GPT Base hallucinations as mechanism risk).
2. Rote “Sorry!” as trust repair.
3. Empathy-only recovery that never corrects the math.
4. Permanent undertrust (“never trust the coach”) as the safety story.

5. Unlimited human escalation as the default UX (destroys unit economics and tutors).
6. Student thumbs-up / “was this helpful?” as proof the repair worked.

LX.2 Constructs (product language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
Hallucination / fluent wrongness	Confident false content	Wrong hint, invented step, affirming wrong setup	Treat only wild fabrications; ignore soft arithmetic slips
Trust violation	Event that breaks reliance calibration	Caught wrong coach move; silent wrong→exam miss	Every minor UI glitch as “trust crisis”
Competence vs integrity failure	Ability miss vs values/honesty miss (Kim et al.)	Bad algebra vs hiding known wrongness	Apologize for everything the same way
Trust repair	Actions that restore <i>appropriate</i> reliance	Admit → correct → verify → optional escalate	Maximize liking, not calibration
Misuse / disuse	Over-reliance / under-reliance (Parasuraman & Riley)	Copying Solver; abandoning coach after one miss	Chase either extreme as “engagement”
Escalation	Hand-off to human when AI insufficient	Tutor queue / session brief with error packet	Escalate everything or nothing
SAFE-REPAIR	Surviving post-error doctrine	See LX.10	Apology cosplay; never-wrong brand

Operational definition (HYPOTHESIS): A MindCraft post-error flow counts as *SAFE-REPAIR* when (a) the system **detects or accepts** the error signal, (b) labels it as a **competence** miss (not student shame), (c) issues an **explanatory** repair with a **deterministic correction path** (bank key / checker / Map ingredient), (d) invites a **solo re-attempt** before full solution dump, (e) offers **escalation** only under explicit triggers, and (f) measures repair by **re-engagement + verified next attempt**, not apology NPS.

LX.3 Prior art already in the Constitution (do not re-litigate)

FACT (high-school math field RCT): Bastani, Bastani, Sungu, Ge, Kaback & Mariman (2025, *PNAS*, doi:10.1073/pnas.2422633122) — unguarded GPT access inflated practice then **harmed** solo exam (~–17% vs never-AI); guarded “GPT Tutor” largely neutralized harm but did **not** show positive learning vs control; Base hallucinations and crutch copying are named mechanisms. Perception ≠ learning.

FACT (novice verification failure): Li, Song, Sundaram & Karahalios (L@S 2025, doi:10.1145/3698205.3729550) — most learners failed to detect chatbot factual errors; undetected errors harmed outcomes and self-efficacy.

FACT (appropriate reliance): Lee & See (2004, *Human Factors*, doi:10.1518/hfes.46.1.50_30392) — design for calibration, resolution, specificity; misuse and disuse are both failures.

FACT (use/misuse/disuse): Parasuraman & Riley (1997, *Human Factors*) — operators over-rely, under-rely, or abuse automation; trust is a primary mediator.

FACT (trust layers): Hoff & Bashir (2015, *Human Factors*, doi:10.1177/0018720814547570) — dispositional, situational, and learned trust; post-error recovery is largely a **learned-trust** problem.

Product implication: Prevention (PWC / ontology / checkers) remains primary. This chapter owns what happens when prevention fails in the wild — which it will.

LX.4 Trust repair is a designed capability, not a tone setting

FACT (HMI agenda): de Visser, Pak & Shaw (2018, *Ergonomics*, doi:10.1080/00140139.2018.1457701; “From ‘automation’ to ‘autonomy’”) — as systems become more autonomous, **trust repair** after violations becomes a first-class design requirement; competence vs integrity framing and context/risk matter; human–human repair findings transfer only partially.

FACT (human–human baseline): Kim, Ferrin, Cooper & Dirks (2004, *Journal of Applied Psychology*, doi:10.1037/0021-9010.89.1.104) — apology repairs **competence** violations better than denial; for **integrity** violations, denial of culpability can outperform apology (when evidence supports innocence). Follow-on attribution work (Kim, Dirks, Cooper & Ferrin, 2006, *OBHDP*, doi:10.1016/j.obhdp.2005.07.002) shows internal vs external blame interacts with violation type.

HYPOTHESIS for MindCraft math errors: Nearly all Solver/coach hallucinations are **competence** failures. Correct response class = **apology + ownership + concrete fix**, not “the curriculum is wrong” externalization, and not integrity-theater (“we would never...”). If the product *knew* the key and still affirmed the student’s wrong answer (sycophancy), treat that as closer to **integrity** — repair requires admitting the agreement failure, not only the math slip.

Wound (field realism): Liu, Du, Ma, Zhang & Yan (2024, *IEEE Trans. Human-Machine Systems*, doi:10.1109/THMS.2024.3434680) — after automated-driving failure in a test-track study (N=257), verbal apology+explanation+promise **did not restore damaged trust**; human-voice repair only partially mitigated other attitudes, mainly among less experienced drivers. **Transfer wound:** ADS safety failure ≠ wrong algebra hint — but the kill candidate survives: **do not assume apology scripts restore trust scores**.

Kill: Apology copy as the repair KPI. Survive: behavioral re-calibration after a verified correct path.

LX.5 What users prefer after LLM mistakes (preference ≠ recovery)

FACT (preregistered preference study): Zhang et al. (2025/26; *CHI / ACM*, doi:10.1145/3793679; arXiv:2507.02745) — N=162 Prolific; pairwise preferences among **rote**, **explanatory**, and **empathic** apologies after bias, fabrication (hallucination), and factual-error scenarios. **Explanatory** apologies generally preferred; **empathic** preferred in bias contexts; for hallucinations, users saw seriousness but showed **no clear apology-type preference** — uncertainty, not a settled recipe.

FACT (critical review): Harland et al. (2025, *Artificial Intelligence Review*, doi:10.1007/s10462-025-11305-8) — AI apology literature (2020–2023 synthesis) stresses that apology is only one element; identification, explanation, and **system behavior change** are required for meaningful repair. Purely discursive apology without corrective action is ethically and practically thin.

HYPOTHESIS: For Maya after a wrong coach step, the winning stack is **explanatory ownership + Map-linked correction + destaked re-attempt**, not empathic padding (“that must feel frustrating”) alone. Empathy may reduce shame (Part XXIV) but does not restore calibration.

Contradiction / limit: Some GenAI service-recovery work finds gratitude/appreciation or humorous styles can raise *willingness to forgive* in consumer settings (e.g. Systems 2024 mixed methods on language style × gratitude/apology — domain = service UX, not math learning). Treat forgive-intent as **engagement theater** unless paired with verified next-attempt success.

Commercial implication: Marketing may say “when we’re wrong, we show our work and hand you a human.” Product must actually do the second half.

LX.6 Escalation to human — scarce, triggered, packetized

FACT (hybrid classroom deployment): Kazemitabaar et al. (2025, arXiv:2510.14457) — programming course N=82; 673 AI hints; students rated 146 (22%) unhelpful; of those, only **16 (11%)** escalated to instructors. Instructor replies on escalated cases were incorrect/insufficient ~half the time. Escalation is **underused** by students and **not automatically high-quality** for humans.

FACT (transparency shifts strategy, not accuracy): Nagashima, Hladký & Rief (arXiv:2606.03822; ECTEL 2026) — fallibility warning increased hint-seeking without changing identical ITS behavior or immediate performance (already in XXXIII).

HYPOTHESIS — MindCraft escalation triggers (product contract):

1. **Checker conflict:** model claim ≠ bank key / symbolic checker → auto-flag; do not wait for student detective work.
2. **Repeat competence miss:** ≥2 verified wrong coach moves in one mission on same concept/ingredient → offer human.
3. **Affect spike + stuck:** high self-reported threat / long idle after soft-wrong → soft escalate (tutor ping), not more AI monologue.
4. **Integrity-class event:** sycophantic affirm of wrong answer against known key → mandatory human-visible audit packet.
5. **Parent/exam stakes mode:** optional “human-reviewed session” SKU (ties SAFE-WTP human-minutes attribute) — not free infinite chat.

Escalation packet (HYPOTHESIS): Human receives: problem ID, student attempt trace, AI utterance, contradiction evidence, ontology IDs, student affect flag — not a blank “help?” thread. Aligns with SAFE-AAR / SAFE-ANNOT (trace before coach dump).

Wound: Human-in-the-loop is not magic (Kazemitabaar instructor error rate). Escalation must preserve **tutor attention hygiene** — packetized, not “AI failed, you debug the LLM live.”

Press-adjacent note (SPECULATION until primary paper cited in-repo): Hybrid AI-draft + human-approve tutoring reports (e.g. Eedi/LearnLM coverage in education press) suggest low hallucination rates *under* human gatekeeping — useful as competitive *direction*, not as MindCraft proof. Prefer primary methods papers before marketing “near-zero hallucinations.”

LX.7 Identity stakes — why math trust violations cut deeper

HYPOTHESIS: For Maya, a fluent wrong coach move is not only a UX bug; it reactivates the identity narrative “even the helper confirms I’m lost” *or* the opposite overtrust “the AI said I’m right” (Parts XXV, XXX, L). Repair must separate:

- **System competence failure** (external, unstable, specific — Weiner-friendly)
- **Student strategy miss** (controllable after correct feedback)

Never: “you’re just not a math person” and never: “trust me completely next time.”

FOUNDER BELIEF: Soft-wrong already destakes *student* error. Post-hallucination UX must destake *AI* error without teaching helplessness toward tools (Lee & See disuse).

LX.8 Competitive positioning

Competitor move	Failure mode	MindCraft counter (HYPOTHESIS)
ChatGPT raw	Hallucinate + apologize vaguely + continue	Checker + Map correction + solo re-attempt
“AI tutor never wrong” ads	Integrity violation when caught	Honest fallibility + repair SLA
Unlimited chat, no human	Disuse after salient miss; or silent misuse	Triggered human minutes as paid scarce resource
Thumbs-up quality loop	Bastani/Bo: perception fails	repair_reattempt_pass, escalation_resolve_48h

Copy that survives Red Team: “We catch wrong steps, show the fix, and bring a tutor when it matters.”

Copy that dies: “Our AI never makes mistakes” / “Sorry you’re frustrated!” without a corrected path.

LX.9 SAFE-REPAIR stack (surviving product rules)

1. **Detect** — prefer deterministic contradiction over student-reported vibes.
2. **Classify** — competence vs integrity-class (sycophancy-against-key).
3. **Own** — short explanatory apology; no blame-the-student; no vague empathy wall.
4. **Correct** — show Map/ingredient + checked next step; ban full solution until re-attempt window unless escalate.
5. **Re-attempt** — destaked item; measure success.

6. **Calibrate UI** — temporarily lower coach confidence badge on that skill grain (SAFE-CALIB cousin).
7. **Escalate** — only on triggers; send packet; SLA visible to parent/student.
8. **Learn** — log to evaluation set; do not only “prompt harder.”
9. **Anti-cosplay** — no therapy script, no airline-captain apology theater, no never-wrong brand.

LX.10 Experiments (REPAIR family)

ID	Contrast	Primary outcome	Falsifier
REPAIR-1	Explanatory+correct path vs rote apology vs empathy-only after injected wrong hint	repair_reattempt_pss @10m; trust calibration item	Empathy-only wins reattempt → revisit stack
REPAIR-2	Auto-detect+own vs silent correct-and-continue	next-mission return; overtrust score	Silent fix retains more misuse → keep detect
REPAIR-3	Escalation offer after 1 vs 2 vs 3 verified AI misses	escalate rate; human resolve quality; cost/session	Instant escalate floods tutors without lift
REPAIR-4	Packetized tutor handoff vs blank “talk to tutor”	time-to-correct; tutor CSAT; reattempt	Packet no better → packet UX fail
REPAIR-5	Fallibility banner always-on vs post-error-only vs none	hint abuse; solo transfer	Always-on destroys use without calibration gain
REPAIR-6	Parent-visible “AI miss + repair” FEI note vs hidden	parent trust / WTP message arm	Hidden wins retention → wound transparency thesis
REPAIR-QUAL	10 Maya + 5 tutors: what “sorry” means after wrong hint	codebook → copy freeze	Founder apology ≠ user need

Pre-reg (XXXIV): REPAIR-* identify **post-error calibration and escalation efficiency** — not “apology increases ACT scores” or “human tutors eliminate hallucinations.”

Ties: AIT-, CAL-, ANNOT-, AAR-, WTP human-minutes, EXP-O affect gates.

LX.11 Confidence table

Claim	Label	Confidence
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Unguarded generative help can harm solo math learning; hallucinations contribute	FACT	High
Novices often fail to detect fluent errors	FACT	High
Appropriate reliance (not max trust) is the right design target	FACT (HF)	High
Apology scripts alone restore post-failure trust	SPECULATION / often false	Low (against) — Liu et al. wound
Explanatory repairs preferred over rote for many LLM errors	FACT (preference)	Medium–High
Preference among apology types for <i>hallucination</i> is unsettled	FACT	High
Competence-framed ownership + corrective action is right class for math slips	HYPOTHESIS	Medium–High
Triggered, packetized human escalation beats unlimited chat <i>and</i> no-human	HYPOTHESIS	Medium
Parent-visible repair notes raise WTP for hybrid SKUs	HYPOTHESIS	Medium
“Never hallucinates” is a viable brand claim	SPECULATION / false	Low (against)

LX.12 What this chapter kills

1. **Kill:** “Our AI doesn’t / won’t hallucinate” marketing.
2. **Kill:** Rote or empathy-only apology as the repair product.
3. **Kill:** Silent overwrite of wrong AI output without student-visible ownership when detection exists.
4. **Kill:** Thumbs-up / satisfaction as proof of trust repair.
5. **Kill:** Infinite free human escalation as default.
6. **Kill:** Permanent “never trust the AI” as brand (disuse).
7. **Wound:** Importing ADS trust-repair nulls or consumer GenAI “forgiveness” as ACT learning proof.
8. **Survive:** SAFE-REPAIR; competence ownership + checked correction + re-attempt; triggered packetized escalation; REPAIR-1...6 + QUAL.

Doctrine until data: After a hallucination, MindCraft sells **recoverable truth** — detect, own, correct, prove, escalate when scarce human attention is worth it — not warmth, not invincibility.

Part LXI — Measurement of Math Identity

Chapter status: Living evidence + measurement brief — Researcher tick 2026-08-01

Primary question: How should MindCraft *measure* mathematics identity — without mistaking a Likert bump for transformation, without inventing a proprietary Belief Score™, and without demand-effect theater?

Owners: Learning Science · Product analytics · Diagnostic / FEI · Parent messaging · Red Team

Commercial job: Ship a **SAFE-IDMEASURE** doctrine: validated subscale items + behavioral FEI corroboration + narrative probes — so marketing, experiments, and parent copy can claim *identity movement* only when measurement survives demand effects.

LXI.1 Why this chapter exists

The Constitution’s research question is identity transformation. Parts XXV (narrative identity / habit≠identity), XXXVII (expectancy-value), L (SAFE-CALIB), and LIX (SAFE-WTP) already ban empty confidence and survey-dollars-as-revenue. What was still missing is the **instrument layer**: which published scales are usable for Maya-aged students, what they actually factor into, how demand and social desirability corrupt them, and how MindCraft should triangulate without building a vanity Identity Meter™.

FOUNDER BELIEF under audit: Identity change is real and commercially load-bearing — but only if we measure it with instruments that have validity evidence *and* refuse to treat a post-session “I am a math person” checkbox as product success.

Claims we refuse as doctrine:

- 1. A proprietary “Math Identity Score™” as North Star.
- 2. Two HSLS-style items alone as causal proof of FEI.
- 3. Pre→post Likert lift after identity-themed copy as marketing evidence.
- 4. Competence/self-efficacy survey ≡ mathematics identity.
- 5. Ignoring recognition (others’ appraisals) because it is hard to productize.
- 6. Treating qualitative storytelling as “unscientific” *or* Likert as “the only science.”

LXI.2 Constructs (product language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
Mathematics identity	How a learner sees self <i>in relation to math</i> (thickening self-perception)	“Math person” / story world belonging without cosplay	Brand slogan without instrument
Interest	Desire/curiosity to engage math	Challenge accept; story-module return	Fun NPS ≡ identity

Recognition	Self + perceived others as “math person”	Tutor/parent brief language; peer opt-in	Forced public praise / leaderboard recognition
Competence / performance beliefs	Beliefs about understanding / doing math	Mastery graph; calibrated confidence	Self-efficacy survey alone as identity
Narrative identity	Reifying, endorsable, significant stories (Sfard & Prusak)	Notes / AAR “what changed about me as a learner”	Essay theater without transfer
Demand / SDR	Response bias toward expected / socially approved answers	Post-mission identity survey after identity ads	Survey lift sold as FEI
SAFE-IDMEASURE	Surviving measurement doctrine	See LXI.9	Identity Score™ productization

Operational definition (HYPOTHESIS): MindCraft claims *identity movement* only when (a) a **validated multi-item subscale** (interest + recognition + competence/performance) moves with preregistered effect size, (b) at least one **behavioral FEI** marker co-moves (challenge_accept, solo_transfer_pass, or destaked retry), and (c) a light **narrative probe** shows actual→designated story shift — not when any single channel moves alone.

LXI.3 What the field actually measures (not slogans)

FACT (SEM on college calculus, large N): Cribbs, Hazari, Sonnert & Sadler (2015, *Child Development*, 86(4), 1048–1062, doi:10.1111/cdev.12363) — FICSMath; more than 9,000 U.S. college calculus students. SEM of mathematics identity: competence/performance beliefs act largely *indirectly* on “math person” identity via **interest** and **external recognition**. Competence alone is **not** sufficient for identity development.

Product kill: Shipping only “I feel smarter at math” / self-efficacy meters as identity proof. Survive: design and messaging that create *interest* and *recognition*, not competence theater alone.

FACT (K–12 instrument validation): Cribbs & Utley (2023/2024, *Mathematics Education Research Journal*, 36, 767–789, doi:10.1007/s13394-023-00474-w) — N=1559 students grades 5–12 in two U.S. districts; adapted undergraduate items; expert content review + readability (Flesch–Kincaid ~grade 3); EFA/CFA support a **16-item** scale with three factors: **interest, recognition, competence/performance** (competence and performance again load together, echoing Cass et al., 2011). Fit indices in CFA: CFI≈0.97, RMSEA≈0.063, SRMR≈0.028. Negatively correlated with math anxiety (mAMAS) as concurrent validity.

Commercial implication: For Maya (high-school), the closest *validated* efficient instrument is this lineage — not an invented MindCraft rubric. Prefer licensed/adapted item sets with permission over DIY “identity KPIs.”

FACT (two-item national survey tradition): HSLS:09 and follow-ups include math-identity items of the form *I see myself as a math person / others see me as a math person* (Ingels et al., NCES HSLS documentation; Cronbach α reported ~0.84–0.88 in base/follow-up). Miller & Wang (2019, *Journal of Youth and Adolescence*, 48, 2038–2050, doi:10.1007/s10964-019-01115-x) use classroom-practice → motivational beliefs → mathematics identity mediation (N=525 sixth-graders); identity operationalized in that literature stream often collapses to short “math person”

self/other items.

Wound: Cribbs & Utley (2023) explicitly note that several large studies using NAEP/HSLs-style items **did not report full validity evidence** for the measure as used. Two items are useful for **screening and national comparison**, weak as sole **product experiment** endpoints.

FACT (science-identity ancestor): Carlone & Johnson (2007, *Journal of Research in Science Teaching*, 44, 1187–1218, doi:10.1002/tea.20237) — recognition, competence, performance as analytic lens for science identity (women of color). Hazari et al. added **interest** when moving to physics/math identity measurement traditions that Cribbs inherits.

FACT (narrative operationalization): Sfard & Prusak (2005, *Educational Researcher*, 34(4), 14–22, doi:10.3102/0013189X034004014) — identity as collections of **reifying, significant, endorsable** stories; learning as closing the gap between **actual** and **designated** identity narratives. Not a Likert substitute — a complementary analytic.

LXI.4 Demand effects, social desirability, and identity advertising

FACT (response styles): Kreitchmann, Abad, Ponsoda, Nieto & Morillo (2019, *Frontiers in Psychology*, 10:2309, doi:10.3389/fpsyg.2019.02309) — social-desirability and acquiescence biases distort Likert scores; ignoring them worsens validity; modeling SDR or forced-choice designs can help; each strategy has tradeoffs (reliability vs criterion validity).

HYPOTHESIS (MindCraft-specific demand): After a session that *markets* identity (“become a math person”), immediately asking “I am a math person” maximizes **demand characteristics** — students infer the desired answer. Risk rises when coaches, tutors, or parents use the same phrase as praise (Part XLVI expectancy ethics).

HYPOTHESIS: Separating **measurement windows** from **identity marketing copy** (e.g., measure at week boundaries, not mission-complete splash) reduces artifactual lift.

Kill: Pre/post “math person” Likert on the same screen as identity-themed celebration UI, sold as FEI evidence.

Survive: Blinded or temporally separated administration; behavioral corroboration; optional forced-choice / IRT-aware scoring later if scale volume justifies it.

LXI.5 What must not be confused with identity

Nearby construct	Why it is not identity	Constitution home
Self-efficacy	Can rise without “math person” story or recognition	XXV, Bandura
Calibration / confidence	Match of certainty to accuracy — honesty, not belonging	L / SAFE-CALIB
Mastery graph score	Performance evidence; necessary but not sufficient (Cribbs 2015)	Engine / XXI

Streak / habit	Cue-driven return $\neq$ identity thickening	XLIII / SAFE-HABIT
Anxiety reduction	Correlated (Cribbs & Utley) but distinct	XXIV, LVIII
Parent WTP	Purchasing $\neq$ student identity	LIX / SAFE-WTP

FOUNDER BELIEF (constrained): FEI still holds — fear→evidence→identity — but **identity** here means the Cribbs-factor bundle + narrative shift, not a mood.

LXI.6 Equity and recognition measurement (do not flatten)

FACT / tradition: Recognition is not only teacher praise; Martin and equity-oriented identity work stress how race, gender, and institutional positioning shape who gets to be seen as mathematical (see Part XXXVI equity audit; Carlone & Johnson recognition as gate).

HYPOTHESIS: Product “recognition” that is only AI cheerleading (“Great job, math person!”) will inflate Likert recognition subscales without changing peer/teacher/parent stories — a **false recognition** pathway.

Product rule: Prefer recognition that is (a) **earned** (tied to specific transferable performance), (b) **multi-source when possible** (tutor brief, parent note, optional peer — never forced leaderboard), (c) **non-trait** (task language per SAFE-EXPECTANCY).

LXI.7 Recommended MindCraft measurement stack

Tier A — Primary quantitative (product experiments): Adapt Cribbs & Utley (2023) **16-item / 3-factor** instrument (or a preregistered short form of the same factors) for 5–12; administer at baseline, 4 weeks, 8 weeks — **not** on every mission complete.

Tier B — Behavioral corroboration (always co-primary): challenge_accept, solo_transfer_pass, retry_120s after informative miss, destaked re-attempt after soft-wrong. Identity claim fails if Likert rises and these stay flat (XXV falsifier).

Tier C — Narrative light probe (qual, Appendix B adjacent): 2–3 prompts: actual story (“Right now, what kind of math learner are you?”), designated story (“Who are you becoming?”), recognition source (“Who has treated you as capable at math lately — and about what?”). Code for actual→designated gap closure (Sfard & Prusak), contamination→redemption themes (McAdams tradition in XXV).

Tier D — Screening only: HSLS-style two items for funnel diagnostics and parent dashboard *context* — never as sole RCT endpoint.

Explicit ban: No composite “Identity Score™” marketed to parents. Report **factor means + behavioral FEI**, or say nothing.

LXI.8 Experiments (IDM family)

ID	Contrast	Primary endpoints	Kill / survive rule
IDM-1	Cribbs 16-item at weeks 0/4/8 vs HSLS 2-item only	Factor α ; sensitivity to FEI intervention	2-item moves, 16-item flat → kill 2-item-as-proof
IDM-2	Measure at week boundary vs mission-complete splash	Δ interest/recognition; demand proxy	Splash-only lift → demand wound
IDM-3	Interest+recognition-focused UX vs competence-meter UX	Identity factors; challenge_accept	Competence UX lifts efficacy not identity → confirms Cribbs SEM
IDM-4	Earned tutor/parent recognition note vs AI “math person” praise	Recognition subscale; transfer	AI praise lifts Likert only → kill false recognition
IDM-5	Likert-only claim vs Likert+behavioral co-primary in experiment report	Pre-reg analysis plan adherence	Likert-alone “win” rejected
IDM-QUAL	10 Maya interviews: what “math person” means; demand cues	Codebook → item wording freeze	Founder phrase ≠ student phrase

Pre-reg (XXXIV): IDM-* identify **measurement validity and demand**, not “MindCraft raises identity by X% forever.”

Ties: EFF-, EVT-, CAL-*, WTP message arms (identity vs score packaging), HIST recognition equity.

LXI.9 SAFE-IDMEASURE stack (surviving doctrine)

1. **Use validated factors** — interest, recognition, competence/performance — not inventing a brand score.
2. **Triangulate** — subscale + FEI behavior + light narrative; one channel alone is insufficient.
3. **Separate measure from marketing** — no identity survey on identity celebration screens.
4. **Competence ≠ identity** — graph and efficacy gains without interest/recognition are incomplete.
5. **Recognition must be earned and multi-source-capable** — ban empty AI “math person” praise as the recognition product.
6. **Report factors, not a trademarked composite** — parent copy may say “students see themselves differently *and* take harder problems” only when both move.
7. **Guard demand/SDR** — preregister; consider timing, neutral framing, later forced-choice if needed.
8. **Longitudinal ownership** — single-session Δ is SPECULATION; 4–8 week windows are the minimum serious bar (feeds Part LXII).

LXI.10 Commercial implications (copy, product, positioning)

Marketing language that survives: “We track whether students *and* the people around them start treating them as someone who does hard math — and whether their practice behavior matches that story.”

Marketing language that dies: “+32% Math Identity Score™”; “Become a math person in one session”; “Our AI recognizes you as a genius.”

Positioning vs Khan / Duo / Brilliant / GPT tutors: Competitors can show accuracy, streaks, or fluency. MindCraft’s differentiated *claim* is identity — which **raises the measurement bar**, not lowers it. If we cannot measure better than a streak app’s mood survey, we do not get to use identity as the brand wedge.

Growth / instrumentation: Add IDM battery to research panel / experiment flags; do not ship Identity Meter on the default student HUD.

Parent WTP link (LIX): Identity packaging in conjoint is legitimate *only* if IDM-5-style evidence exists; otherwise sell FEI session proof and transfer, not identity poetry.

LXI.11 Confidence table

Claim	Label	Confidence
Competence/performance beliefs are insufficient alone for math identity; interest + recognition matter	FACT (SEM)	High
Validated 5–12 multi-item instrument exists (Cribbs & Utley 2023)	FACT	High
HSLs two-item forms are widely used but thin as sole experiment endpoints	FACT / HYPOTHESIS	High / Medium–High
Immediate post-mission identity Likert is demand-contaminated	HYPOTHESIS	Medium–High
AI praise alone produces false recognition subscale lift	HYPOTHESIS	Medium
Narrative probes add validity beyond Likert	HYPOTHESIS (theory-backed)	Medium
Proprietary Identity Score™ improves learning or WTP	SPECULATION / refuse	Low (against)
Single-session identity transformation is measurable and marketable	SPECULATION / refuse	Low (against)

LXI.12 What this chapter kills

1. **Kill:** Math Identity Score™ / Belief Score™-style composites as North Star.
2. **Kill:** Two-item “math person” pre/post as causal FEI proof.
3. **Kill:** Competence or confidence survey $\equiv$ identity.
4. **Kill:** Identity survey glued to identity marketing splash screens.
5. **Kill:** Empty AI “you’re a math person” as recognition product.
6. **Kill:** Single-session identity transformation claims.
7. **Wound:** Importing undergraduate-only instruments without 5–12 validity evidence.
8. **Survive:** SAFE-IDMEASURE; Cribbs-factor triangulation; IDM-1...5 + QUAL; behavioral co-primary.

Doctrine until data: MindCraft may *aim* at identity — but may **claim** identity movement only when validated interest/recognition/competence–performance subscales move **with** FEI behavior (and preferably narrative gap closure), under demand-aware administration.

Part LXII — Longitudinal Identity Change Models

Chapter status: Living evidence + horizon brief — Researcher tick 2026-08-02

Primary question: What does *honest* math-identity change look like at **8 / 26 / 52 weeks** — and which commercial claims survive fade-out, stability, and dimensional-comparison evidence?

Owners: Learning Science · Product analytics · Growth / retention · Parent messaging · Red Team

Commercial job: Ship a **SAFE-LONGID** doctrine: staged claim horizons so MindCraft can sell *direction and persistence*, not “math person in eight weeks” theater, and so subscription / package design matches how identity actually moves.

LXII.1 Why this chapter exists

Part LXI (SAFE-IDMEASURE) fixed *how* to measure identity. This chapter fixes *when* a movement counts as change vs noise, demand, or temporary lift. Founders and growth teams instinctively want a short horizon (“show identity by week 8 for renewals”). Developmental evidence is slower, stickier, and often **differentiating away from math** through high school — not smoothly ascending toward “math person.”

FOUNDER BELIEF under audit: Identity transformation is the product thesis — but durable identity change is a **multi-horizon** process. Selling L3 identity claims on L1 session windows (Part XXXIV claim ladder) is already banned; selling them on an **8-week Likert bump without persistence** is the next failure mode.

Claims we refuse as doctrine:

1. Eight-week subscale lift alone $\equiv$ transformation (marketable or scientific).
2. Linear “always becoming more of a math person” growth curves for Maya.
3. One-semester tutoring $\equiv$ occupational / STEM-pipeline identity.
4. End-of-program effects that ignore fade-out after support ends.

5. Stability evidence as proof that identity *cannot* change (fatalism).
6. Multi-year school-department reforms as templates for a 45-minute app session.

LXII.2 Constructs (product language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
Horizon	Follow-up window for an estimand	8w / 26w / 52w experiment clocks	Selling 52w outcomes on 8w data
Stability	Rank-order / mean identity often sticky	Baseline IDM battery	“Nothing moves → kill product” or “must move weekly”
Differentiation	Math vs verbal self-concepts diverge over grades	Challenge accept in <i>math</i> not generic hustle	Generic confidence copy
Persistence vs fade-out	Treatment effects that hold vs shrink after end	Retention after plan pause / summer	Celebrate end-of-program only
Process signal	Early FEI / interest / approach markers	challenge_accept, destaked retry	Process ≡ identity
SAFE-LONGID	Surviving longitudinal claim doctrine	See LXII.9	Forever-identity packaging

Operational definition (HYPOTHESIS): MindCraft may claim *identity movement at horizon H* only when SAFE-IDMEASURE triangulation holds **at H**, and for $H \geq 26$ weeks also shows **non-fade** relative to an earlier post-test (or preregistered maintenance criterion). Eight weeks may support *process / direction* claims, not L3 identity brand claims.

LXII.3 Developmental facts (not slogans)

FACT (dimensional comparison meta-analysis): Wan, Lauermaann, Bailey & Eccles (2021, *Psychological Bulletin*, 147(9), 867–889, doi:10.1037/bul0000340) — 142 samples, ~211k participants, 426 effect sizes. Correlations between math- and language-related motivational beliefs *decline* across grades (from roughly $r \approx 0.32$ in Grades 1–4 toward near-zero / slight negative in Grades 9–12). Students increasingly treat domains as **either/or**.

Commercial implication: High-school Maya is in the window where “I’m a words person, not a math person” consolidates. Product copy that treats identity as a generic confidence dial fights the developmental grain.

FACT (person-type trajectories across K–12): Wan, Tian, Guo & Lauermaann (2026, *Child Development*, doi:10.1093/chidev/aacag069) — four U.S. longitudinal samples ($Ns \approx 971$ –5,681; ages ~6–18). Annual classification into math person / language person / both-high / neither-high. **Both-high shrinks**; language-person increases; **math-person rises through middle school then plateaus in high school**; girls’ shift toward language is steeper. In samples with adult follow-up, high-school math-person status associates with more math-intensive occupations.

Product kill: Marketing that implies continuous year-on-year math-identity ascent through HS. Survive: **sustain and deepen** math identity in the HS plateau window — especially for students at risk of language-only sorting.

FACT (expectancy–value over years, not weeks): Simpkins, Davis-Kean & Eccles (2006, *Developmental Psychology*, 42(1), 70–83, doi:10.1037/0012-1649.42.1.70) — N=227; 5th-grade out-of-school math/science activity → 6th/10th-grade expectancies–values → high-school course taking beyond grades. Belief→choice chains run on **multi-year** clocks.

FACT (related classroom→belief→choice chain): Simpkins, Davis-Kean & Eccles / follow-on EVT longitudinal work (e.g., classroom experiences in middle school predicting later courses and aspirations; see also Wang lineage cited in LXI via Miller & Wang 2019) — identity-adjacent motivational beliefs are **path-dependent**, not session-local.

FACT (attainment value / “science person” as persistence predictor): Andersen & Ward (2014, *Science Education*, 98(2), 216–242, doi:10.1002/sce.21092) — HSLs:09 high-ability ninth-graders; science attainment value (“science person”-type identification) predicts STEM persistence plans across Black, Hispanic, and White subgroups (with utility operating differently by group). Snapshot is grade 9, but the *outcome* is planned persistence — a longitudinal *intent*.

Wound / transfer caution: HSLs items are thin (LXI); use as directional evidence that **identity-like attainment value predicts plans**, not as MindCraft’s instrument.

LXII.4 Stability is real — and so is change under the right ecology

FACT (college sample stability): Cribbs, Tassell, Hazari, Sadler & Sonnert (2022, *RIPEM*, 12(2), 68–91, doi:10.37001/ripem.v12i2.2923) — N=131 undergraduates; Wilcoxon signed-rank on global math-identity self-perceptions over ~three college years → **non-significant change** (identity reported as stable). Qualitative themes: formative experiences often involve *performing* and *teaching/sharing*; many believe one *can* become a math person while still harboring innate-ability beliefs.

Interpretation for MindCraft (HYPOTHESIS): Stability under business-as-usual schooling is the null. A tutoring product that expects weekly identity Likert jumps is fighting a sticky construct. Change requires **repeated recognition + interest + competence evidence** (Cribbs SEM 2015; LXI), not one delightful onboarding.

FACT (multi-year equitable departments): Boaler & Staples (2008, *Teachers College Record*, 110(3), 608–645, doi:10.1177/016146810811000302) — Railside longitudinal mixed-methods (~700 students across three CA high schools). Heterogeneous, reform-oriented department associated with larger achievement gains and narrowing of ethnic gaps by roughly the **end of year two** on several measures; students also reported more enjoyment / progression. Horn (2008, *Mathematical Thinking and Learning*, 10(3), 201–239, doi:10.1080/10986060802216177) — turnaround student identities emerge across **departmental worlds over years**, not a single classroom moment.

Kill: Citing Railside/Horn as proof that an edtech session architecture produces identity in weeks.

Survive: Identity ecology is **relational and sustained**; MindCraft’s analogue is tutor+parent recognition + FEI practice cadence over months — not a splash screen.

FACT (short belief interventions ≠ identity doctrine): Boaler, Dieckmann, Pérez-Núñez, Sun & Williams (2018, *Frontiers in Education*, 3:26, doi:10.3389/educ.2018.00026) — teacher-randomized MOOC (~six ~15-min sessions, Oct–Dec) shifted math *beliefs*/engagement and standardized scores vs control on a ~semester research clock. Useful as proof that **beliefs can move in months with math-specific messages** — but Constitution already bans empty mindset

posters; do **not** rebrand MOOC belief lifts as Cribbs-factor identity or FEI.

LXII.5 Fade-out: why 8-week wins die without maintenance

FACT (fade-out is normal): Bailey, Duncan, Cunha, Foorman & Yeager (2020, *Psychological Science in the Public Interest*, 21(2), 55–110, doi:10.1177/1529100620915848) — fade-out of educational intervention effects is **widespread**, often coexists with some persistence, and is substantive (not only measurement artifact). Persistence depends on skill type, institutional constraints, and **complementarities** with later environments.

FACT (earlier geometric-decline pattern): Bailey, Duncan, Odgers & Yu (2017, *Journal of Research on Educational Effectiveness*, 10(1), 7–39, doi:10.1080/19345747.2016.1232459) — post-treatment impacts commonly shrink within 1–2 years; early skill gains are poor guarantees of long-run advantage without supporting environments.

HYPOTHESIS (MindCraft-specific): An 8-week FEI + interest lift that is followed by **disengagement, exam-only cramming, or AI-crutch Solver use** will fade on identity factors by 26–52 weeks — even if week-8 IDM looks green.

Product implication: Retention design is not a growth vanity metric; it is **identity infrastructure**. Pause/summer plans need light maintenance cues (SAFE-HABIT subordinate to FEI), not streak guilt.

LXII.6 Horizon model for MindCraft (8 / 26 / 52)

Horizon	What can honestly move	Allowed commercial speech	Forbidden speech
~8 weeks	Process: challenge_accept, destaked retry, soft-wrong return; possible early interest/recognition subscale <i>direction</i> under SAFE-IDMEASURE	“Students start approaching harder problems and sticking after misses”	“Became a math person”; “identity transformed”; Identity Score™
~26 weeks	Triangulated factor movement + FEI co-primary; narrative actual→designated shift; first fade check vs week 8	“Over a semester, self-view <i>and</i> challenge behavior moved together” (only if true)	Guaranteed ACT points; occupational STEM destiny
~52 weeks	Persistence of week-26 gains; course/intent signals where observable; maintenance under reduced support	Annual renewals tied to <i>sustained</i> approach + transfer, not mood	“Forever math person after one year with us” without follow-up

FOUNDER BELIEF (constrained): Package tiers and parent CBC (Part LIX) should map to horizons: diagnostic + 8w process proof; semester identity-direction report; annual persistence review — never sell the annual claim with only an 8w dashboard.

SPECULATION (explicit): Exact week cutoffs are product conventions informed by school semesters and subscription cycles, not natural laws. Preregister windows; do not retrofit.

LXII.7 Analytic models (what “longitudinal” means in practice)

MindCraft does not need a 10-year panel to be rigorous. Minimum serious designs:

1. **Repeated measures / growth:** IDM factors at 0/8/26/52; random slopes; report **individual heterogeneity** (not only means).
2. **Latent transition / person-types (inspired by Wan 2026):** math / language / both / neither-style states — watch **both-high erosion** and math-person **plateau** risk.
3. **Joint process–identity models:** FEI events as time-varying covariates of interest/recognition (tests FEI pathway).
4. **Fade contrast:** end-of-active-support vs +8–12 weeks post-pause.
5. **Demand control:** never collect IDM on identity-celebration UI (LXI).

Pre-reg (XXXIV): LONG-* experiments identify **horizon-valid estimands**, not “MindCraft creates math people by $\Delta=X$.”

LXII.8 Experiments (LONG family)

ID	Contrast	Primary endpoints	Kill / survive rule
LONG-1	8w IDM+FEI vs waitlist / active control	Interest/recognition Δ ; challenge_accept	Factor $\uparrow$ FEI flat $\rightarrow$ kill identity claim; FEI $\uparrow$ factor flat $\rightarrow$ process-only speech
LONG-2	26w continuation vs 8w-then-pause	Factors at 26; fade from 8	Pause erases 8w lift $\rightarrow$ kill “8w locks identity”
LONG-3	Recognition-rich tutor/parent notes cadence vs AI praise cadence (extends IDM-4)	Recognition subscale; transfer; 26w hold	AI-only recognition fades faster $\rightarrow$ confirms false recognition
LONG-4	Maintenance micro-dose (1–2 FEI sessions/mo) vs full stop after 26w	52w factors + FEI	No maintenance $\rightarrow$ fade; informs annual package

LONG-5	Person-type transition coding (Wan-style) exploratory	Transition probs; gender split	Both-high collapse without math sustain → HS copy priority
LONG-QUAL	10 Maya + 10 parent interviews at 0/8/26: story change vs score obsession	Narrative codebook	Parent wants 8w identity certificate → WTP honesty constraint

Ties: IDM-, EVT-, FEI/SAFE-CRAFT, WTP package horizons, REPAIR trust persistence, HIST equity of who gets sustained recognition.

LXII.9 SAFE-LONGID stack (surviving doctrine)

1. **Stage claims by horizon** — 8w = process/direction; 26w = triangulated identity movement; 52w = persistence/maintenance.
 2. **Expect stickiness** — null is stable identity under weak ecology (Cribbs 2022); design for repeated FEI + earned recognition.
 3. **Respect differentiation** — HS students sort math vs language; sustain math identity, don't sell generic grit.
 4. **Plan for fade-out** — end-of-program green lights are incomplete; measure post-pause.
 5. **Do not launder mindset MOOCs or multi-year school reforms as app-session identity proof.**
 6. **Align packages to clocks** — CBC attributes and renewals match measurable horizons (LIX).
 7. **Heterogeneity first** — mean lifts can hide non-responders; report who moves.
 8. **Keep SAFE-IDMEASURE** — same instruments, longer clocks; no new Identity Score™.
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LXII.10 Commercial implications (copy, product, positioning, growth)

Marketing language that survives: “In the first weeks we look for *approach under safety*. Over a semester we look for whether students’ *self-story and challenge behavior* move together. Over a year we look for whether that holds when life gets busy.”

Marketing language that dies: “Math person in 8 weeks”; “Guaranteed identity transformation this semester”; “Join and permanently become STEM”; linear forever-up charts on landing pages.

Positioning vs Khan / Duo / Brilliant / GPT tutors: Competitors optimize session accuracy, streaks, or fluency — short clocks. MindCraft’s wedge is **durable self-as-math-doer**. That only works if we **own the long clock** in measurement and packaging. If we market identity on Duo’s time horizon, we lose the wedge and inherit their vanity metrics.

Growth: Instrument cohort retention *as* identity maintenance; annual plans should include LONG-4-style maintenance, not only denser content. Churn after week 6 is not merely CAC failure — it is **hypothesis failure** for L3 claims.

Vision / future: The company roadmap should treat 26–52 week panels as first-class research infrastructure (research panel flag), not a nice-to-have after feature shipping.

LXII.11 Confidence table

Claim	Label	Confidence
Math/verbal motivational beliefs differentiate across school years	FACT (meta)	High
Math-person rates plateau in HS; both-high shrinks; language-person rises	FACT (multi-sample longitudinal)	High
Global math identity can be stable over years under status-quo college experience	FACT (small N)	Medium–High
Multi-year equitable school departments can shift achievement/identity ecology	FACT (case / mixed methods)	Medium–High (external validity to edtech low)
Educational intervention effects commonly fade without supporting environments	FACT (reviews)	High
8w triangulated lift without 26w hold is insufficient for L3 identity marketing	HYPOTHESIS / doctrine	Medium–High
Staged 8/26/52 product clocks are the right commercial mapping	FOUNDER BELIEF / HYPOTHESIS	Medium
MindCraft will produce 52w identity persistence with maintenance micro-dosing	SPECULATION	Low–Medium (test via LONG-4)

LXII.12 What this chapter kills

1. **Kill:** “Math person in 8 weeks” / single-horizon identity transformation marketing.
2. **Kill:** Treating end-of-program identity Likert as durable without fade checks.
3. **Kill:** Linear always-rising math-identity curves for high school.
4. **Kill:** Equating short mindset/belief MOOC effects with Cribbs-factor identity + FEI.
5. **Kill:** Citing Railside/Horn multi-year worlds as proof of session-scale app identity.
6. **Kill:** Annual “forever STEM identity” packaging without 52w persistence evidence.
7. **Wound:** Using HSLS two-item snapshots as longitudinal product proof (re-anchor LXI).
8. **Survive:** SAFE-LONGID staged horizons; differentiation-aware HS positioning; fade-aware retention; LONG-1...5 + QUAL.

Doctrine until data: Identity is the North Star *target*, not an 8-week badge. Ship process truth early; claim identity movement only on triangulated **semester+** clocks; claim persistence only when fade-out is measured and survived.